Supporting Information for

Mapping Structured Absences in Scientific Papers: An Auditable LLM-Assisted Workflow Demonstrated on Cross-Coupling Chemistry

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How to read this Supporting Information (three layers). The SI is organised so that a reviewer can stop after the layer relevant to their question.

- **Layer 1 — reviewer-facing evidence (read first).** SI-0 (background lineage), SI-3 (full Kumada 1972 Stage 3 analysis), SI-4 (filling papers), SI-5 (full Stage 2F output), SI-6 (four-model cross-model comparison), SI-9 (retrieval audit), and SI-11.5/.7/.8/.10 (worked audit trace, ten-minute audit walkthrough, data-review checklist, reviewer navigation guide). These support the main-text claims directly.
- **Layer 2 — audit appendix (consult as needed).** SI-1 (version history), SI-2 (regression log), SI-8 (evaluation-protocol detail), and the apparatus subsections SI-11.1 (execution audit tables), SI-11.2 (hallucination defence), SI-11.9 (full vocabulary), SI-11.12 (formal schema), SI-11.13/.14 (applicability envelope), SI-11.16 (limitation mitigations).
- **Layer 3 — deposit-only (Zenodo / Tier-1 archive).** Full prompt text (SI-7 here gives the hash-verifiable manifest; bodies are in the deposit), full raw LLM outputs, retrieval logs, candidate tables, and file-level inventories. These are not reproduced in this PDF; the Data Availability statement gives the access path.

SI-0: Extended Background on the Methodological Lineage

SI-0.1: The Analysis of "What Is Not Written" in Scientific Papers

Textual analysis of scientific papers has primarily been directed at *what is written*—claims, evidence, citations, conclusions. Attempts to take as an object of analysis what the text does not make explicit, what the paper leaves unsaid, have developed independently in multiple academic traditions, each with its own level of abstraction and methodological interest. At the most macroscopic level, Kuhn's theory of paradigms[S6] conceptualised the layer of "exemplary problem solutions" shared by scientific practice—the unstated presuppositions, questions, and evaluative criteria. Foucault's *épistémè*[S7] took as its unit of analysis "the regularity of enunciation" that governs an era's entire intellectual practice—the conditions of what can be said and what can be asked. These frameworks provided points of departure for treating "what is not written" in scientific texts but do not take the evidence architecture of individual papers as a direct object of analysis. Three lineages have developed to address more concrete analytical units: symptomatic reading, agnotology, and the critique of symptomatic reading by surface reading. This SI-0 describes the details of these three lineages and makes explicit UQS's positioning with respect to them.

SI-0.2: The Lineage of Symptomatic Reading—Althusser, Macherey, Bazerman, Zwart

Althusser's symptomatic reading (*lecture symptomale*). Through his reading of Marx's *Capital*, Althusser[18] proposed a methodology of reading not only the questions a text has answered but also the questions it has not posed. Althusser's central insight is that a text's silences and absences are not products of the analyst's interests but *symptoms* of the text's own structure—the problematic that is necessary as a premise for a given concept but is not itself thematised in the text. Symptomatic reading is set in contrast to orthographic reading, which accepts the claims the text explicitly makes, and asks after the structural relation between a text's surface and its depth.

Macherey's analysis of literary texts. Macherey[19] extended Althusser's methodology to literary texts and established a framework for reading silences, gaps, and contradictions in a text as structural symptoms of ideology. Macherey's methodological contribution is the elaboration of the proposition that textual absences are not mere defects but traces of the historical conditions under which the text was produced. In this analysis, absence is treated not as a flaw but as the trace of an "unspoken history" left inside the text itself.

Bazerman's transposition to scientific text. Bazerman[20] brought the methodological stance of symptomatic reading into the analysis of scientific papers and showed that the genre conventions of scientific papers—citation practices, techniques of hedging, IMRaD structure—are not mere formalities but institutional devices that regulate what can be asserted and what must be excluded in silence. Bazerman's central thesis is that *the form of a paper constrains its content*. Under the publication conventions of a given period in a given field, certain kinds of claims are formally inadmissible, and as a result, structured absences are generated within the text. Bazerman's analysis transposed to the scientific-text domain what Macherey had done for literary texts, describing textual silences not as symptoms of ideology but as products of genre and institution.

Zwart's Lacanian extension. Zwart[21] carried out the most systematic extension of symptomatic reading since Bazerman, integrating Althusser's method with Lacanian psychoanalysis and Hegelian dialectics. Zwart's object of analysis is the *philosopheme*—the unconsciously held philosophical presuppositions of scientists—extracted from the discourse of genomics and synthetic biology. Two features characterise Zwart's unit of analysis: it is not the individual paper but the *discursive field* of an entire discipline, and the object extracted is a philosophical presupposition that cannot be read directly from the

textual surface but requires the analyst's interpretation. Zwart's methodology represents the deepest extension of symptomatic reading applied to scientific text and established a diagnostic practice for making visible the metaphysical structure of a field's discourse.

UQS inherits from Zwart's lineage while making two different choices. First, it restricts its unit of analysis to the individual paper (inter-paper comparison is performed only in Stage 2F). Second, it restricts its object to absences that are formally identifiable at the textual surface. This restriction sacrifices analytical depth in order to make the records comparable across papers and checkable by other analysts; it does not by itself secure agreement between analysts (the blind evaluation of SI-8.7 found low agreement on the finer categories). The two are not mutually exclusive but complementary practices addressing different layers of analysis. If individual-paper-level analysis by UQS accumulates, it could provide empirical input for Zwart-style diagnoses of field discourse.

A note on the legal-hermeneutic precursor. A still older structural analogue exists in the medieval scholastic legal tradition. The *glossatores* of the Bologna school (eleventh through thirteenth centuries, culminating in Accursius's *Glossa Ordinaria*, c. 1230) and the subsequent *commentatores* (fourteenth century, centred on Bartolus de Saxoferrato and Baldus de Ubaldis) developed a technical apparatus for reading Justinian's *Digest* whose individual operations have a striking structural parallel to UQS: *distinctio* (refining a textual case into sub-cases to surface the sub-case the text does not address) parallels D-4 candidate generation at Stage 2; *quaestio* (a hypothetical case constructed to probe what the existing text can decide) parallels the Evidence Pack as a structured under-specified question; *casus omissus* (the case the legislator omitted) and *silentium legis* (the silence of the law) terminologically anticipate the notion of *structured absence*; and the synthetic *summa* parallels the cartographic move from per-paper Stage 3 to cross-paper Stage 2F integration.[S8] The parallel is structural rather than genealogical: UQS draws its direct methodological lineage from the symptomatic-reading tradition described above and from agnotology (SI-0.3); the glossator–commentator tradition is noted here as a longer-standing analogue in an adjacent text-interpretation craft, and a fuller historical treatment of this precursor is reserved for companion methodological work.

SI-0.3: Agnotology—The Taxonomy of Absence

Proctor's concept of strategic ignorance. Agnotology—the theory of ignorance—pioneered by Proctor and Schiebinger[22] is a research domain in the history and sociology of science that investigates "the manufacture of ignorance." Proctor's central cases are the suppression of epidemiological evidence regarding the causal link between smoking and lung cancer by the tobacco industry, and the strategic amplification of scientific uncertainty in climate-change denialism. Proctor termed these cases "strategic ignorance" or the "deliberate manufacture of doubt," conceptualising ignorance not as the mere absence of knowledge but as a positive social product. Agnotology's object of analysis is *intentionally manufactured ignorance*—a description of the structure by which industrial and political forces constrain the generation and circulation of scientific knowledge.

Croissant's taxonomy of absence. Croissant[23] refined agnotology methodologically and presented a framework treating absence as a taxonomic object with attributes such as granularity, temporality, and intentionality. Croissant's contribution is the establishment of a methodology that treats absence not as "a single object" but as "structured data with multiple attributes." UQS's classification system—A/B/C/U, synchronic/retrospective, and inference tiers T1/T2/T3—is structurally kin to Croissant's taxonomy of absence. The two share the basic methodological stance of *treating absence as formally describable data*.

The object domain, however, differs qualitatively. What agnotology addresses is *intended absence*—cases in which industrial or political forces intentionally manufacture ignorance. What UQS addresses is the ordinary evidence architecture of ordinary peer-reviewed papers, where no intentional suppression is presupposed. What authors did not write is a product of dynamics qualitatively different from intentional suppression: page limits, scope of interest, period-specific norms, division of labour within a field, publication incentives. While agnotology analyses manufactured ignorance, UQS analyses the *shape* of absences structurally generated within the ordinary practice of science. The two share a taxonomic methodology while addressing different regimes of absence.

SI-0.4: The Surface-Reading Critique of Best and Marcus

Best and Marcus[24] raised a critique of the methodological stance of symptomatic reading. Their proposed *surface reading* pointed out the danger that the posture of probing for depths hidden behind the text can fall into paranoid preconception and neglect the detailed data and formal structure that the text presents at its surface. In the context of scientific text, this critique

carries legitimate force. Reading a paper so symptomatically that its concrete evidence and experimental results are dismissed as mere reflections of ideology damages the quality of the analysis itself.

UQS has structurally incorporated this critique. The staged design—in which the positive record of a paper (what was claimed, what was taken as grounds, what was concluded) must be completed first (Stage 0), and only on that foundation does extraction of absences proceed (Stage 1)—integrates the precise description of the surface and the structural analysis of absence as a sequential relation rather than as an opposition. In our preliminary analyses, this sequential relation was observed to be not merely a design principle but a substantive condition determining the quality of the analysis. The resolution of the positive record in Stage 0—how precisely the text's surface is grasped—directly determines the resolution of absence detection in Stage 1. If the surface reading is coarse, absence detection is also coarse. Surface and depth are not in opposition; the precision of the surface governs the precision of the depth.

SI-0.5: UQS's Positioning Within the Lineage

UQS can be positioned with respect to the above lineages along the following four axes:

- **Object of analysis**: UQS takes as its object the evidence architecture of the individual paper. Althusser and Macherey's object is ideological contradiction; Bazerman's is the institutional apparatus of genre; Zwart's is the philosopheme of disciplinary discourse; agnotology's is socially manufactured ignorance. UQS addresses an object distinct from any of these—the structured absences in a paper's evidence architecture.
- **Unit of analysis**: UQS takes the individual paper as its unit of analysis, with inter-paper comparison executed in an explicitly separated step (Stage 2F). Zwart's unit is the discursive field of an entire discipline; agnotology's is institutional and industrial systems.
- **Depth of analysis and formalisation**: UQS restricts its object to absences formally identifiable at the textual surface. This restriction sacrifices analytical depth in order to make the records comparable across papers and checkable by other analysts; it does not by itself secure agreement between analysts (the blind evaluation of SI-8.7 found low agreement on the finer categories). Zwart-style analysis reaches philosophical presuppositions in the depth, but relies more heavily on the analyst's interpretation.
- **Contrast with the Argumentative Zoning tradition**: While the AZ tradition from Teufel[9] onward classifies "what is written" in a paper's text, UQS takes as its classification object "what is not written"—absences that become visible only as differences from the field's normative knowledge. The two are distinguished by difference of object rather than by difference of precision, and are complementary approaches that could in principle be executed in parallel.

UQS is neither hierarchically above nor below these lineages but is a methodology operating at a different layer. Its distinctive feature is that it inherits the methodological stance of symptomatic reading while possessing kinship with the AZ tradition and agnotology in its formalisation and comparability. This cross-cutting positioning is the structural basis on which UQS can be implemented as a methodology mediated by LLMs.

SI Fig. S1: Quantitative Improvements in UQS Refinement Cycle (withdrawn in R1)

R1 correction. The submitted Fig. S1 plotted per-version totals and averages of U, B and D-4 items across the refinement cycle (e.g. a v15 total of 4 U items rising to 38, +850%). These aggregates could not be reproduced from the regression log: Entry 003 records 10 U items for one paper (hom1) at v15 alone (SI-2.B). The figure and its percentages are therefore withdrawn, and no replacement average is given. The per-entry judgements of the refinement cycle are in SI-2.

SI-1: Detailed Version History of UQS Prompts (v0 through v17)

Scope note (2026-05-10): This section documents the development history through the v17 main-pipeline baseline. The v18.4 production version used in this study is documented in Methods 6 and SI-7; the v18.x development cycle is documented in the changelogs `changelog_v18*.md` (Tier 7 of the deposit).

This supplementary section documents every versioned change to the UQS prompt system from the prototype v0 to the current v17, including brief motivations, per-version change lists, papers on which each version was tested during development

(development checks, not validation), and regressions detected (with cross-references to the full log in SI-2). The narrative is consolidated from the internal `docs/version_history.md` record (dated 2026-04-17) and the `improvements/changelog_v17.md`. Stage 2F, Stage 3, and Stage 4 are versioned independently and are summarized in SI-1.V at the end.

SI-1.A Overview of the Developmental Phases

Phase	Versions	Character
Exploratory prototype	v0	Single prompt file; no stage structure; "motivation back-inference" skeleton.
Three-stage prototype	v1–v2	Stage 0/1/2 established; A/B/C/D/E/F categories fixed in prototype.
Single-file consolidation	v3–v12	One file per stage; turn-based multi-agent scheme introduced at v12.
File-split turns	v13–v14	Stage 1 Turns 1–4 split into separate files.
Rule formalization	v15	A/B/C/U classification and `cross_field_temperature` formally defined.
Batch improvement	v16	Fourteen proposals implemented in one revision.
Regression-repair	v16.1	Three regressions detected in dev-set repair.
Refinement-set-driven	v17	Eight targeted improvements (C1–C8) from the full 10-paper refinement set.

SI-1.B Per-Version Detail

Version v0 (prototype)

Motivation for the change: Initial exploration of "the question that was not asked." Not yet a protocol — a template library of four questioning formats.

Specific changes from v(-1): N/A (initial version).

Content: Single file `prompt_逆行推論.md` ("retrograde reasoning"; original filename in Japanese) containing four Japanese questioning formats: (1) motivation back-inference (sections I–III: dissection of premises, hierarchical analysis of motive, traces of failure); (2) motivational genealogy (three operators: externalization of metric, reversal of premise, emergence conditions of a question); (3) critical dissection of question structure; (4) internal-structure analysis with categories A (deferral), B (traces of rejection), C (uncited citations), D (motivational stratification), E (scope question), F (inter-paper difference). The conceptual skeleton of the later Stage 1/Stage 2 was already present here.

Papers on which this version was tested during development: None (prototype; no formal validation set).

Regressions: None recorded.

Version v1 (three-stage establishment)

Motivation for the change: Convert v0's single-file template library into a sequential protocol with distinct Stage 0 (positive record), Stage 1 (structural extraction), and Stage 2 (interpretation).

Specific changes from v0:

- Separated `stage0.md`, `stage1.md`, `stage2.md` plus `stage1_answer.md` and `stage2_answer.md` (Murai 1993 sample outputs distributed as reference).
- Stage 0: six-field paper summary (S0-1 through S0-6); English fixed; inference forbidden (explicit-only).
- Stage 1: A/B/C/D/META/GAP blocks; subtypes for A (Forgotten / Deferred / Suppressed / Mixed) and B (Methodological limitation / Intentional narrowing / Mixed).
- Stage 2: D-1 declared motivation, D-2 inferred motivation (single item), D-3 institutional motivation, D-4 unquestioned motivation (Japanese), E scope question (Japanese candidate → English propositional statement), F within-group comparison.

Papers on which this version was tested during development: Murai (1993) as included sample.

Regressions: None recorded.

Version v2 (precision and evidence tightening)

Motivation for the change: Strengthen evidence requirements and analytical granularity.

Specific changes from v1:

- Stage 0 retained as-is.
- Stage 1: text-position markers introduced (`[PAGE N]` + `L1, L2,...` line markers; `loc: P1:L5` form); B subtype criteria formally articulated ("impossible without experimental infrastructure" vs "excluded while possible within the infrastructure"); META section separated; GAP requests in three types with priority.
- Stage 2 D-1: layered-novelty structure analysis added (technical → performance → conceptual claim layers; resistance to single-evidence refutation).
- Stage 2 D-2: expanded from 1 to 5 items — competitive positioning, timing pressure, format exploitation, audience targeting, institutional visibility. An "external factual claim requiring external verification" flag rule was added.
- Stage 2 D-3: "holy-grail narrative" analysis added.
- Stage 2 D-4: viewpoint list introduced as starting points (but "not a ceiling"); "unaskable question" category separated; each candidate required `grounded_in: [item_id list]` and `key_quote` annotations; text-free observations labelled `grounded_in: none (retrospective)`.
- Stage 2 E: split from one format into two — `confirmed_english_propositional` ($\leq$20-word proposition, for embedding similarity) and `confirmed_english_question` (question form, for lineage tracking). `derived_from` field required.
- Stage 2 F: each pattern requires `evidence_chain: [paper_id : item_id]`; paradigm-competition analysis newly added.

Papers on which this version was tested during development: Informal dev exploration only.

Regressions: None recorded.

Version v3

Motivation for the change: Initial versioned release with four-stage composition (S0, S1, S2, S2F — Stage 2F separately introduced for figure/scheme interpretation).

Specific changes from v2: Stage 2F added as a dedicated stage. Minor reorganization of S1 categories (A: deferred questions, B: traces of rejection, C: uncontinued citations).

Regressions: None recorded.

Version v4

Motivation for the change: Minor tidying.

Specific changes from v3: Small reorganization of S1 classification; slight expansion of Stage 2F; general wording adjustments.

Version v5

Motivation for the change: Introduce the anchoring rule (explicit statements only; inference excluded).

Specific changes from v4:

- Stage 0: anchoring rule formalized.
- Multi-paper-PDF boundary rule added (analyze only the target paper).
- Output discipline tightened: structured fields only; no explanatory prose.
- Stage 1 B: emphasized "negative results treated as mechanistic evidence" pattern.
- BIB field re-numbered (BIB-1 through BIB-7).
- Note: Stage 2F omitted from v5 (integrated back into Stage 2 temporarily).

Versions v6–v9 (2026-03-26 – 03-29)

Motivation for the change: Incremental clarification; no conceptual additions of note.

Specific changes: Field descriptions progressively refined; Stage 2F interpretation prompt gradually expanded; wording and formatting adjustments.

Versions v10–v12 (2026-03-28 – 03-29): turn-based multi-agent scheme

Motivation for the change: The largest architectural change in the system. Introduce sequential extraction from distinct disciplinary vantages.

Specific changes (v12 is the first fully implemented version):

- Stage 1 restructured into four turns:
  - Turn 1 (V1): primary-field expert of the paper's topic.
  - Turn 2 (V2): adjacent-field expert with duplicate re-evaluation.
  - Turn 3 (V3): methodology/statistics expert with duplicate re-evaluation.
  - Turn 4: integration into A/B/C/U plus `cross_field_temperature` metadata.
- Agent-count guidance added (3 agents for communications, 3–4 for full papers).
- Per-turn output discipline formalized.
- Multi-paper-PDF boundary rule extended across all turns.

Regressions: None recorded.

Version v13 (2026-03-30)

Motivation for the change: Workflow clarity.

Specific changes from v12:

- Stage 1 split into per-turn files: `stage1-v13-turn1.md`, `turn2.md`, `turn3.md`; Turn 4 synthesis kept as `stage1-v13-20260330.md`.
- No conceptual additions.

Version v14 (2026-03-31)

Motivation for the change: Cleanup of field-naming consistency.

Specific changes from v13: Continued per-turn file structure; field-naming consistency fixes; light wording adjustments.

Version v15 (2026-03-31 – 04-13): formalization of A/B/C/U

Motivation for the change: Formalize the core concepts and discipline that define the current methodology.

Specific changes from v14:

- File naming standardized: `prompt_stage0_v15.md` … `prompt_stage2_v15.md`.
- A/B/C/U classification formalized in Turn 4:
  - A: deferred questions (explicitly marked for future work).
  - B: boundary conditions (methodological limitations + intentional narrowing); new `reporting_omission` subtype added.
  - C: uncontinued citations.
  - U: unclassified (items that fit none of A/B/C).
- `cross_field_temperature` metadata introduced in Turn 4 (convergence across agents plus severity gap).
- D-4 prioritization via high-convergence-high-gap.
- `rhetorical_function` field introduced (evidence / reservation / suppressed concern tracking).
- Self-contradictory absence rule introduced.
- Exhaustive extraction rule introduced.
- DFT reporting checklist for computational-chemistry papers (six items: energy species, dispersion correction, solvation, conformer search, reference-state clarity, temperature consistency).
- Scope anchor rule in Stage 2: substrate scope must be described without mechanistic clauses.

Papers on which this version was tested during development: Kumada 1972, Murai 1993 (dev-set) plus preliminary exploration.

Regressions detected in v15 and later: five regressions detected in v15→v16 comparison on Kumada (see regression log Entry 001, SI-2); these were resolved at v16.1.

Version v16 (2026-04-14): batch implementation of fourteen improvements

Motivation for the change: Incorporate 14 improvement proposals derived from analysis of the first four test papers (app1, app2, comp1, comp2).

Specific changes from v15 (organized by file):

Turn 1 (prompt_stage1_turn1_v16.md):

- **A-1 Errata handling**: when S0-7 reports errata, treat the corrected version as canonical; do not generate questions against uncorrected passages.

Turn 2 (prompt_stage1_turn2_v16.md):

- **A-2 V2 preemption declaration**: if V2 selects a field from Turn 3's default set, mark `[preempts_turn3_default: yes]`.
- **A-2 Independent evaluation of selectivity** regardless of field choice (er/dr/yield ratio).
- **A-2 DFT checklist formalized**: six-item list, each with an independent question if the item is absent.

Turn 3 (prompt_stage1_turn3_v16.md):

- **A-3 Preemption check rule**: if V2 issued a preemption declaration, V3 selects the next most important methodological vantage; alternatives (NMR / kinetics / analytical chemistry) listed for DFT-preempted cases.

Turn 4 (prompt_stage1_turn4_v16.md):

- **A-4 Antagonistic-reference exclusion rule**: references that are the direct experimental target of refutation or confirmation are excluded from C items.
- **A-4 `reporting_omission` subtype**: methodological data that is conventionally required but entirely absent is marked as a B subtype.
- **A-4 Duplicate-count clarification for convergence**: V2-OVn and V3-OVn entries count toward convergence.
- **A-4 Normalization of empty `high_convergence_high_gap`**: empty is the default; fallback rules are the standard D-4 selection path.

Stage 2 (prompt_stage2_v16.md):

- **A-5 Propositional description rule for mechanism papers**: when S0-1 states a mechanistic discovery, `confirmed_english_propositional` describes only the transformation (no mechanistic clauses).
- **A-5 `key_quote` source restriction**: ESI/SI content cannot serve as `key_quote`; use adjacent main-text passages labelled T2/T3.
- **A-5 Detection-limit check for negative evidence**: when "not detected" is used as mechanistic evidence, require detection-limit verification.
- **A-5 Pivot-structure search**: require explicit identification of A↔B pivots (an A item that avoids explaining a B item) and Same-Root Groups (SRGs).

Papers on which this version was tested during development: Kumada 1972 (dev) — revealed five regressions (see SI-2 Entry 001).

Regressions detected: R1 (U-item generation failure), R2 (C-item density loss), R3 (D4-3 catalyst-identity loss), R4 (D4-7 β-H elimination loss — unresolved), R5 (E-section scope shift). Remedies in v16.1 address R1/R2/R3.

Version v16.1 (2026-04-14 – 15): regression fix

Motivation for the change: Repair three regressions detected in the v15 vs v16 dev-set comparison on Kumada 1972.

Specific changes from v16:

Regression	Fix
R1 — U items not generated	Added the **self-contradictory absence rule** to the Turn 4 U-item definition.
R2 — C items over-consolidated	Added the **exhaustive extraction rule** to the Turn 4 C-item definition.
R3 — Stage 2 did not reference Turn 4 U items	Added explicit **U-item-reference reinforcement** in Stage 2 D-4.

Papers on which this version was tested during development: dev-set (Kumada, Murai) + refinement set (10 papers). SI-2 judges 10 of the 11 v15 → v16.1 entries non-regressive and records a partial regression in one (008, app2).

Notable improvements: v16.1 increased U-item output; the per-version means given in the submitted version (0.4 → 3.4, +750%) are withdrawn because they do not reflect Entry 003, which records 10 U items for hom1 at v15 (SI-2.B). See regression log Entries 001–012 (SI-2).

Version v17 (2026-04-15): eight targeted improvements from full refinement-set analysis

Motivation for the change: After completion of all 10 refinement-set analyses (through scr2 — the third improvement checkpoint), incorporate eight targeted improvements (C1–C8). No structural redesign; v16.1 base design fully preserved. All changes are additive or exception-defining.

Specific changes from v16.1 (labeled C1–C8 per `improvements/changelog_v17.md`):

- **C1 Author-attribution verification for scope failures (Turn 1, OI-009) — required**. For 0%-yield / 0%-ee entries in scope tables, verify whether the author-offered explanation (electronic / steric / conformational) is backed by experimental or computational evidence; attribution without evidence is recorded as a B-item candidate with `reason_evidence_absent`.
- **C2 Maintain mechanistic evaluation independence after DFT checklist (Turn 2, OI-010) — required**. DFT checklist applies only to computational methodology. Evaluation of proposed catalytic cycles and reaction mechanisms proceeds independently (kinetics, isotope effects, intermediate capture). Prevents V2 from collapsing mechanistic B-item coverage when DFT is present.
- **C3 Three additions to D-4 candidate selection (Stage 2, OI-007+008) — required**:
  - (a) Detection limit of negative evidence when "not detected" is used mechanistically.
  - (b) Practical-condition extrapolation when catalyst loading / concentration / temperature deviate substantially.
  - (c) Matched-pair control symmetry for comparisons whose experimental pairs are not symmetric in substrate/reagent/conditions.
- **C4 Addition of "Dropped Quantitative Benchmark" C-item subtype (Turn 1 + Turn 4, OI-013) — required**. When a prior-work numerical value is cited in the introduction but the paper's results are never compared against it on the same axis, record a C item with this subtype and a `reason_severed` note.
- **C5 ESI delegation vs experimental absence distinction (Turn 1, OI-012) — required**. When ESI/SI references are attached, the experiment is treated as performed (partial disclosure), clearly separated from "experiment absent." Attention to partial-negation modifiers (e.g., "not entirely") is added.
- **C6 Sequential-protocol confounding check (Turn 1, OI-011) — recommended**. For sequential/staged protocols (e.g., ramped conditions, multi-step flow), check whether prior-phase state changes contaminate later phases.
- **C7 B-item consolidation exception for mixed yield bases (Turn 4, OI-004) — recommended**. When different yield bases are mixed in one paper (starting-material vs product basis, GC yield vs isolated yield) and constitute internal inconsistency, retain the item as an independent B entry (do not consolidate). Marked with `yield_basis_exception`.
- **C8 Explicit PDF direct-reading (all 6 prompts) — required**. Instructs analysts to read the PDF via the multimodal Read tool (no pdftotext). ≤20 pages read whole; >20 pages via page ranges. Figures, schemes, energy diagrams, and structural formulas are explicitly available as inputs.

File-by-file mapping:

File	Changes applied
prompt_stage0_v17.md	C8
prompt_stage1_turn1_v17.md	C1, C4, C5, C6, C8
prompt_stage1_turn2_v17.md	C2, C8
prompt_stage1_turn3_v17.md	C8
prompt_stage1_turn4_v17.md	C4 (C-definition), C7, C8
prompt_stage2_v17.md	C3, C8

Papers on which this version was tested during development: dev-set (Kumada, Murai) + refinement set (10 papers). Per Results Part 4 of the main manuscript, v17 is judged better overall than v16.1 in 11 of the 12 entries and equivalent in one, with partial regressions recorded in entries 014, 015 and 017 (SI-2.B). Two of eight targeted improvements did not fire on their target papers (C1 on app2; C2 on scr1) — see Results Part 4 for the structural analysis of those non-firings.

Regressions detected: See SI-2 Entries 013–024. All are resolved or tracked as low-severity continuation items in `improvements/open_issues.md` (project record, not deposited).

SI-1.V Independent Versioning of Stage 2F, Stage 3, Stage 4

These later stages are versioned separately from the S0–S2 pipeline. The current production files are in `prompts/stage2f3/`.

Stage	Version	File	Notes
Stage 2F	v6	`prompt_stage2f_v6.md`	Base version. Extended propositional framework + figure/table interpretation.
Stage 2F	v7	`prompt_stage2f_v7.md`	Output structure and evaluation perspectives expanded.
Stage 2F	v7.1	`prompt_stage2f_v71.md`	Minor fixes / clarifications to v7. Used for the SI-5 integrative analysis of Kumada 1972.
Stage 3	v1	`prompt_stage3_v1.md`	Initial. Used for Kumada 1972 Stage 3 (SI-3).
Stage 3	v2	`prompt_stage3_v2.md`	Revised output structure and perspectives. Used for the eight case-study paper analyses (SI-4).
Stage 4	v1	`prompt_stage4_v1.md`	Initial. Filling-paper-candidate derivation.
Stage 4	v2	`prompt_stage4_v2.md`	Revised output structure and prioritization. Used for the eight case-study paper analyses.

SI-1.C Summary of Quantitative Change across the v15→v17 Cycle

R1 note: the per-version means in this table are approximate values from an internal summary (`docs/version_history.md`) and could not be reproduced from the regression log (SI-2.B); they are withdrawn and kept only as a record of the submitted text.

Metric	v15	v16.1	v17 (projected)
A items (per-paper mean)	~9	~10	~10
B items (per-paper mean)	~12	~13	~20 (C1–C7 effect)
C items (per-paper mean)	~3.5	~4.5	~4
U items (per-paper mean)	0.4	3.4	~3
D-4 candidates (per-paper mean)	~5.5	~6.5	~8 (C3 effect)
Same-Root Groups (per-paper mean)	~3.5	~4.5	~6
Papers with ≥1 U item (of 10)	2	10	10

Source: `docs/version_history.md` §"Quantitative improvement summary."

SI-1.D Open Issues Status at v17

Open-issue ID	Severity	Topic	Status
OI-001	medium	β-H elimination pattern detection	open (continued monitoring)
OI-004	medium	bilateral-yield-mismatch propagation	partially addressed by C7
OI-005	medium	catalyst-loading anomaly loss	out of v17 scope; monitored
OI-007	high	detection limit of negative evidence	addressed by C3(a)
OI-008	high	matched-pair control asymmetry	addressed by C3(c)
OI-009	high	scope-failure author-attribution	partially addressed by C1
OI-010	high	DFT-checklist tunnel vision	addressed by C2
OI-011	medium	sequential-protocol confounding	addressed by C6
OI-012	high	ESI delegation vs experiment absence	addressed by C5
OI-013	medium-high	Dropped Quantitative Benchmark	addressed by C4

Full open-issue definitions and current status are in `improvements/open_issues.md` (project record, not deposited).

SI-2: Regression Log (24 Entries)

Scope note (2026-05-10): This regression log covers the v15→v16→v16.1→v17 development cycle. The v18.x cycle (v18.0 through v18.4) is documented in the changelogs `changelog_v18*.md` (Tier 7 of the deposit); v18.4 is the production version used in this study (Methods 6, SI-7). The log holds 30 version-pair entries for the 12 papers of this study: the 24 entries of the v15→v17 cycle reproduced below, and six later entries for the development set (Kumada 1972: 026, 031, 052, 056; Murai 1993: 055, 057) that cover v17→v18.4 and are in the deposited log (`regression_log.md`, Tier 7). Entries for other papers in the same log belong to other corpora and are not used here.

This supplementary section reproduces the outcomes of the twenty-four version-comparison audits recorded in the regression log (`regression_log.md`, Tier 7 of the deposit) during the v15 → v16 → v16.1 → v17 improvement cycle. Each entry corresponds to a per-paper comparison at a version transition. The full narrative log (including score cards, identification of regression points R1–R6, and auxiliary notes) is in the working-directory file; this SI presents the compact tabular summary plus the headline judgment for each entry, along with aggregate patterns.

SI-2.A Tabular Summary of All 24 Entries

Legend for the judgment column: **non-regressive** = the newer version met or exceeded the older version on all tracked metrics (`v(new) > v(old)` or `v(new)` $\{\geq\}$ `v(old)`); **partial regression** = the newer version improved overall but lost one or more specific analytic features (`v(new)` $\{\geq\}$ `v(old)` with R-flags); **neutral** = mixed results of comparable analytic value; **regressive** = the newer version degraded overall on the tracked metrics (none in the log).

#	Version pair	Paper ID	Metric(s) evaluated	Judgment	Notes
001	v15 / v16 / v16.1	Kumada 1972 (dev)	U-item generation, C-item density, D-4 candidate breadth, E-section scope	v16.1 > v15 > v16 (non-regressive for v16.1)	Triple comparison. R1/R2/R3 resolved at v16.1; R4 (β-H elimination) unresolved → OI-001; R5 (E scope) movement interpreted as anchoring-discipline alignment.
002	v15 / v16.1	Murai 1993 (dev)	A/B/C/U count (15 → 26); D-4 pivot structure; `self-contradictory absence rule` firing on U-1, U-2	non-regressive	R1/R5 bilateral-yield-claim dispersion → OI-004. R4 run-5 catalyst-loading anomaly loss → OI-005.
003	v15 / v16.1	hom1 (Wang et al., Chem. Sci. 2020)	A-3 deferral of insertion-step role; norbornene surrogate D-4 candidacy; `high_convergence_high_gap` detection; scope_type judgment	non-regressive	Six R-items, all low–medium impact. OI-006 resolved by v16.1's A-3 explicit capture.
004	v15 / v16.1	hom2 (Shirai et al., J. Org. Chem. 2024)	A/B/C/U reclassification (33 → 19 with U growth 1 → 4); AQ-removal suppressed classification; SRG-3 integration; D4-5 room-temperature necessity	non-regressive	v16.1 restructures A (10→2) with growth in U (1→4). High-impact A-2 `suppressed` classification on AQ removal is v16.1-specific.
005	v15 / v16.1	comp2 (Palladino et al., ACS Catal. 2023)	D4-4 detection-limit of negatives; D4-6 20 mol% practical-extrapolation; DFT self-consistency U items	non-regressive	R1/R2 → OI-007. `self-contradictory absence rule` on DFT barrier values (U-1, U-2, U-3, U-4) v16.1-specific.
006	v15 / v16.1	comp1 (Race et al., Chem. Sci. 2016)	Schemes 3A/3B matched-pair asymmetry; TEMPO-Pd(0) alternative; p-cymene `self-contradictory absence`	non-regressive	R1 Schemes 3A/3B asymmetry → OI-008. SRG-7 quantitative-omission cluster v16.1-specific.
007	v15 / v16.1	app1 (Zhang, Tutkowski et al., Chem. Sci. 2017)	Catalytic-cycle mechanistic questions (SRG-1 6 items in v15); D4-3 stereochemical assignment; D4-4 NBO / C-H···O coupling	non-regressive	R1 DFT-checklist-triggered mechanism loss → OI-010. R2/R3 mechanistic D-4 dissolution into Bs. U-item `self-contradictory absence rule` first fires (U-1, U-2).
008	v15 / v16.1	app2 (Meck et al., JACS 2025)	N-Ph electronic attribution high_convergence_HIGH_gap; base-required ee suppressed observation; mephenytoin ee absence	partial regression on R1 (R1 high-impact)	v15's unique 3/3-convergence HIGH_gap item (N-Ph electronic) lost entirely → OI-009.
009	v15 / v16.1	het2 (Newton et al., ACS Catal. 2024)	Sequential-ramp confounding (B-9); α-hydride elimination `self-contradictory absence` (U-2); $Pd(OAc)_2$ vs Pd(0) assumption (U-3); Muzart/Clark citations	non-regressive	R1 flow-reactor specific → OI-011 (new). R2/R3 medium-high impact U items v16.1-specific.

#	Version pair	Paper ID	Metric(s) evaluated	Judgment	Notes
010	v15 / v16.1	het1 (Yuan et al., JACS 2018)	U items (0 → 5 including Pd(0) non-detection, 60°C activity attribution, Cl^- origin); heterogeneity test D-4 candidacy; Wacker mass-balance	non-regressive	R2 heterogeneity test to D-4 promotion, R3/R4 Wacker / detection-limit — all resolved by v16.1's rules and OI-007 chain.
011	v15 / v16.1	scr2 (Templ et al., RSC Mechanochem. 2025)	ESI-delegation vs experiment absence (A-5); ArF over-claim detection (B-3); milling-off control recognition; series_signal; C-item density	non-regressive	R1 ESI-delegation high-impact → OI-012. R2 ArF over-claim added.
012	v15 / v16.1	scr1 (Fodor et al., ACS Sustain. Chem. Eng. 2020)	U-items (0 → 5 including DFT homogeneity premise U-4); Dropped Quantitative Benchmark (C-1, C-2); Et_3N matched-pair; fitting error `5.3% vs 5% criterion`	non-regressive	R1/R2 high-impact — U-4/U-5 and Amatore/Dupont numerical-benchmark citation loss → OI-013.
013	v16.1 / v17	Kumada 1972 (dev)	B-12 yield-basis-mixing; D4-3 Table I reliability; D4-5 kinetic vs thermodynamic selectivity; SRG subdivision (3 → 6)	non-regressive	Minor R-1/R-2 on exothermic-entry (Table I footnote f) `self-contradictory absence` loss → OI-014 (low severity).
014	v16.1 / v17	Murai 1993 (dev)	Silyl hidden-requirement D-4; chelation-assistance negative control D-4; cyclometallation intermediate isolation D-4; $Ru(CO)_2$ vs mechanism U-1	v17 ≥ v16.1, with partial regression on R1–R4 (medium / low)	R1 silyl bias → OI-015. R3 high_convergence_HIGH_gap disappearance continues OI-014 pattern. v17 gains specific new angles (B-5 mass-balance; B-11 excess-olefin contradiction; B-17 tether length).
015	v16.1 / v17	app2 (Meck et al., JACS 2025)	N–O OA stoichiometric PhOH prediction; aza-Wacker parallel ee ceiling; "unprotected nitrogen" vs footnote 23	partial regression (v17 > v16.1 overall; R1–R3)	C items first appear in app2 via C4 (Dropped Quantitative Benchmark). R1 PhOH prediction loss → OI-016.
016	v16.1 / v17	het2 (Newton et al., ACS Catal. 2024)	Pseudosteady-state D-4 candidacy; α-hydride elimination concrete by-product prediction (U-2); PdEnCat Pd(II)/Pd(0) contradiction (U-3)	non-regressive	B-2 sequential-ramp confounding now an independent B (OI-011 resolved). Non-additivity D-4 candidate (313 vs 455 ppm) new. R2 same pattern as OI-016.
017	v16.1 / v17	app1 (Zhang, Tutkowski et al.)	Cyclopropyl "confirms" rhetorical D-4; GC-yield + dr + SD + 2-eq integrated D-4	partial regression (v17 > v16.1 overall; R1–R2)	C items first appear in app1 (C-1, C-2 Dropped Quantitative Benchmark). R1 "confirms" rhetoric retention → OI-020 (new).
018	v16.1 / v17	comp2 (Palladino et al.)	Kinetic vs thermodynamic selectivity D-4; RDS attribution ambiguity (U-4); Ra/Rb DFT ordering D-4; DFT consolidation (3/3-convergence B-1)	non-regressive	OI-007 R1 (detection limit "M never detected") resolved via new D-4-3. C-1 Mårtensson Dropped Quantitative Benchmark new.

#	Version pair	Paper ID	Metric(s) evaluated	Judgment	Notes
019	v161 / v17	comp1 (Race et al.)	Radical clock triple-problem specificity; TEMPO-Pd(0) alternative; estrone multi-reliability; matched-pair D-4 candidacy	non-regressive	D-4-3 Pd-free-condition control, D-4-4 matched-pair, D-4-8 detection-limit all new. OI-008 resolved in same paper. A-item systematization (2 → 7).
020	v16.1 / v17	scr1 (Fodor et al.)	A-1 base-assisted HI elimination DFT deferral; homogeneous-premise D-4; Czernecki ρ / Amatore-Jutand Ea Dropped Quantitative Benchmarks; Eyring plot B	non-regressive (R1 partial, R2 not recovered)	R1 A-1 deferral loss → OI-022 (new). R2 continues OI-010. C-1/C-2 Dropped Quantitative Benchmark new.
021	v16.1 / v17	scr2 (Templ et al.)	A/B boundary-rule normalization (A 10 → 1, B 5 → 27); hot-filtration/Hg-test U-2; recyclability U-4; ArF over-claim (lost)	non-regressive	OI-012 and OI-013 re-confirmed in same paper. R1 ArF over-claim continues → OI-023 (new).
022	v16.1 / v17	het1 (Yuan et al.)	Pd(0) direct-observation D-4; heterogeneity-test D-4 (hot filtration / Hg poison); Cl^- origin D-4	≈ (R1, R2 substantive losses offset by v17 gains)	R1/R2 → OI-018 and OI-019 (new). v17 gains B-3 reproducibility, B-7 sequential-ramp (OI-011 re-confirmed), SRG-5/SRG-6 new groups.
023	v16.1 / v17	hom1 (Wang et al.)	KIE-quasireversibility `self-contradictory absence` text citation; C2-product 3b C3 deuteration D-4; critical / high_convergence_HIGH_gap; scope_type	non-regressive (R1 partial, R2 not recovered)	R2 (C3-D of 3b D-4) → OI-017 (new). C-1 Dropped Quantitative Benchmark new.
024	v161 / v17	hom2 (Shirai et al.)	AQ cost D-4 (synthetic overhead); room-temperature necessity D-4; dual-catalyst `self-contradictory absence` (D-4 candidate 1); NMR internal-standard reliability (D-4 candidate 3); turnover / deactivation (D-4 candidate 7)	non-regressive	R1 AQ cost integrated D-4 loss → OI-021 (new). B items 7 → 14; SRG 3 → 8 (large granularity gain).

SI-2.B Aggregate Statistics

Total entries: 24.

Distribution of judgments: *(Re-counted in R1 from the judgment lines of `improvements/_prompts/regression_log.md`; the submitted counts did not add up to 24.)*

- Non-regressive without a recorded partial regression: 19 entries.
- Partial regression recorded while the newer version is judged better overall: 4 entries (008 app2; 014 Murai; 015 app2; 017 app1).
- Equivalent (≈): 1 entry (022, het1).
- Regressive (v(new) < v(old)): 0 entries.

Patterns:

1. **v15 → v16.1 is non-regressive in 10 of the 11 entries** (002–012; partial regression in 008, app2). The new rules in v16.1 (self-contradictory absence, exhaustive extraction, U-item reference reinforcement) increase U-item output (the submitted per-version means are withdrawn: Entry 003 records 10 U items for hom1 at v15, which they did not reflect), and the five regressions detected in the v15 → v16 dev-set comparison (entry 001) are resolved at v16.1 or tracked as continued open issues (OI-001, OI-004, OI-005).
2. **v16.1 → v17 is judged better overall in 11 of the 12 entries and equivalent in one** (022, het1); entries 014, 015 and 017 record partial regressions. Improvements centre on C-item growth (Dropped Quantitative Benchmark subtype — C4 — fires across app2, app1, comp2, scr1, scr2, hom1, het2, hom2), D-4 candidate breadth (C3 negative-evidence/practical-extrapolation/matched-pair additions), and A/B boundary-rule discipline (notably in scr2, where v17 reclassifies 9 of 10 erroneous A items as B). Two of the eight targeted improvements (C1 on app2; C2 on scr1) did not fire on their primary target paper (entries 015 and 020; one possible reading is given in SI-6.8).
3. **No cumulative regression path was recorded** (this statement was not re-checked against every metric in R1). There is no case of `v(v15) > v(v17)` in any metric where the intermediate v16.1 was better than v17; i.e., the cycle does not incur net loss against the starting baseline in any tracked feature. Features preserved throughout include the A/B/C/U classification scheme, `rhetorical_function`, pivot-structure search, SRG identification, scope anchor rule, and the anchoring discipline (T1/T2/T3).
4. **Open-issue tracking is live**. The regression log generates 15 open-issue IDs (OI-001 through OI-023, not contiguous). Each is tracked in `improvements/open_issues.md` with severity, trigger for re-evaluation, and current status. Nine issues are resolved by v17 changes; five remain open for future cycles; one is marked out-of-scope.
5. **Resolution timing**. Of the 10 open issues that triggered v17 (OI-004, 007, 008, 009, 010, 011, 012, 013, plus two older — 001 from v15/v16 and 015 from Murai), seven are addressed at v17 (OI-007, 008, 010, 011, 012, 013 fully; OI-009 partially). OI-004 is partially addressed by C7. OI-001 (β-H elimination) and OI-015 (silyl hidden requirement) remain open — both require chemistry-domain sensitivity that does not yield to prompt-level instruction additions.

SI-3: Complete Stage 3 Analysis of Kumada 1972

This supplementary section presents the Stage 3 analysis of Tamao, Sumitani, and Kumada's 1972 *J. Am. Chem. Soc.* Communication on Ni-catalyzed Kumada–Corriu coupling. This section has two sources, and they are not interchangeable. **SI-3.2 to SI-3.10 reproduce one Stage 3 record**, in English translation and reorganised by Evidence Pack: `papers/dev/kumada/kumada_v17_sonnet_s3.md` (v17 Stage 0–2 outputs, Stage 3 v1 prompt `stage3-v1-20260402`, Claude Sonnet 4.6, run on 2026-04-15; deposited as `tier_3_stage_outputs/dev_set/kumada/kumada_v17_sonnet_s3.md`, with an identical copy at `tier_3_stage_outputs/stage2f_integration_outputs/kumada1972_v17_sonnet_s3.md`). The six SRG names, their members and the two CLAs are those of that record and of its Stage 1 input (`kumada_v17_sonnet_s1.md`). The per-EP "case-specific implementation" and "search term derivation" blocks in SI-3.7 were written for this SI from the SRG-level implementations of that record; they appear in no run output in that form. **The multi-run subsection below** (Top-3 EP SET, Stage 4 re-count, 11-paper re-count) is derived from the v18.4 + Stage 3 v2 + Sonnet 4.6 records of 2026-05-10 (`kumada_v184_sonnet_s3v2_en.md`, `_run2`–`_run6`, and the Japanese cell). Those runs group the same absences under different SRG names and numbers (SI-11.5, step 3) and are not the source of the SRG list or of the EP definitions.

Methodology evolution and cross-vendor anchors

v17 baseline (run 2026-04-15; prompt tag `stage3-v1-20260402`): The initial Stage 3 analysis used the Stage 3 v1 prompt (`stage3-v1-20260402`) on v17 Stage 0–2 outputs, producing an eight-EP dictionary (EP-01a, EP-03a–EP-09a) sourced from `papers/ep/kumada1972/kumada1972_v17_sonnet_s3.md`. The v17 single-run record is preserved for traceability and is the primary reference in SI-6's four-model cross-vendor comparison (Haiku 4.5, Sonnet 4.6, Opus 4.7, GPT-5.4 Thinking).

EP-10a addition (2026-04-19): A ninth EP, EP-10a *Kinetic verification pack*, was added on 2026-04-19 after five runs had assigned it or the related pattern PRI-11: Sonnet on Breitenfeld 2014, Mazet 2021 and Pengfei 2024; GPT-5.4 Thinking and Opus 4.7 on Kumada 1972. Two later GPT-5.5 Thinking runs (S0–S2 v18.2, Stage 3 v3, 2026-04-26/27) on Kumada 1972 and Murai 1993 also assigned it. **Corrected in R1:** these were described as independent derivations, but EP-10a was already a provisional label in the Stage 3 prompt from version 1, and PRI-11 was described in versions 2 and 3, so only the first Sonnet run on Breitenfeld 2014 (Stage 3 v1) was unexposed (SI-6.5); EP-10a was formally codified in v18.3 (`improvements/changelog_v183.md` C18.5). The Yamamoto 1978 *BCSJ* analysis, also run under Stage 3 v2, applied PRI-11 and EP-10a as well (SI-3.7); it is not an independent confirmation either.

EP-02a addition (2026-04-19): A tenth EP, EP-02a *Identity / purity substantiation pack*, was added after the GPT-5.4 Thinking and Opus 4.7 runs on Kumada 1972 assigned it; the label was already in the prompts they were given (SI-6.5). The original v17 Sonnet Stage 3 v1 run had subsumed this absence under EP-01a; the Sonnet v17 Stage 3 v2 run of 2026-04-17 (`kumada_v17_sonnet_s3v2.md`, §5) listed EP-02a as an auxiliary pack, and the two cross-vendor runs of 2026-04-19 provided the basis for promotion to a separate identifier. The identifier itself is one of the ten common labels defined in the Stage 3 prompt (v1 and v2). Neither addition rests on independent derivation.

ChemOntology pilot companion deliverable (2026-04-19): A mapping linking the Kumada 1972 EPs to ChemOntology-AFIR (Nath et al. 2026) reaction-path-search parameters is deposited at `application_direction_chemontology_layer_a/` (it is not part of the main text). YAML input specifications for the primary $NiCl_2$(dpe) system and three variant cases are kept in the project repository (`manuscript_A/chemontology_pilot/layer_B_input_specification/`) and are not part of the deposit. None of these materials has been executed — the pilot is planned, not run — and the Discussion 3 (Applicability) hedges are preserved throughout. See SI-3.11.

v18.4 EN multi-run (2026-05-10): Six Stage 3 v2 runs per language cell were added to test the run-stability of the Top-3 EP SET and of the Stage 4 filling labels (subsection below). They did not replace the v17 Stage 3 v1 record as the source of the SRG list and the EP dictionary; SI-3.4 to SI-3.10 still reproduce that record.

Note on Stage 3 v1 vs v2 EP attributions: the EP identifiers used throughout this SI are common to Stage 3 prompt versions v1 and v2, but the SRG→EP mapping shifts slightly between the two versions. The SRG→EP mapping in SI-3.4 and SI-3.7 is that of the Stage 3 v1 record; the v18.4 Stage 3 v2 runs attach the same EP identifiers to differently named SRGs (for EP-08a: SRG-5 "Yield data analytical quality" in place of SRG-2/SRG-6/SRG-3; SI-11.5). The deposited ChemOntology-AFIR mapping table narrows EP-04a to the cis/trans stereo-specific sense and re-attributes the cross-vs-homo mass-balance content to EP-05a, consistent with the v2 prompt framing preferred for ChemOntology-AFIR mapping. Readers who need a strict SRG→EP cross-walk should consult `application_direction_chemontology_layer_a/ep_to_chemontology_mapping.md` (deposit) §Mapping methodology.

Multi-run aggregation and Top-3 anchor (v18.4, N=6 per language cell)

The v18.4 multi-run aggregation protocol (`improvements/paperA_phase1_5_multirun_report_20260510.md` (project record, not deposited)) ran N=6 independent Stage 3 v2 runs per language cell (Kumada JA and EN, Murai JA and EN) to distinguish stable structural findings from single-run rank-order stochasticity. Key findings:

Extraction rule (revised in response to review). The Stage 3 v2 prompt defines no "Top-3" field; §5 of each output lists an EP investment order per objective function (prompt Step 5a–5b). The values first reported for this section were compiled by hand and could not be reproduced from the per-run records. They are replaced here by a mechanical re-count of all 24 per-run records (Kumada and Murai × JA and EN × six runs) under three extraction rules stated before counting: **FIRSTOBJ**, the first three EPs of the first route block in §5 (the prompt's first objective function, one VQ level above the current one); **VQ3**, the first three EPs of the route to VQ-3; **FIRST3**, the first three distinct EPs mentioned anywhere in §5. All three are reported; none is designated primary. Script, per-run table and extraction trace are deposited (`multirun_top3_per_run.csv`, `inclusion_frequency_per_cell.csv`, `extraction_trace.md`).

Kumada 1972. Top-3 principal reinforcement EP SET = {EP-03a, EP-04a, EP-06a}. Single-run rank order varies; SET membership is stable in both language cells under VQ3 and FIRST3 (every value ≥83%). Under FIRSTOBJ, EP-03a falls to 67% in the English cell because run 4 opens §5 with a route to VQ-2.

Cell (n = 6)	Rule	EP-03a	EP-04a	EP-06a	Runs with all three
Kumada JA	FIRSTOBJ	5/6 (83%)	5/6 (83%)	5/6 (83%)	5/6
Kumada JA	VQ3	6/6 (100%)	6/6 (100%)	6/6 (100%)	6/6
Kumada JA	FIRST3	5/6 (83%)	5/6 (83%)	6/6 (100%)	5/6
Kumada EN	FIRSTOBJ	4/6 (67%)	5/6 (83%)	5/6 (83%)	4/6
Kumada EN	VQ3	5/6 (83%)	6/6 (100%)	6/6 (100%)	5/6
Kumada EN	FIRST3	5/6 (83%)	5/6 (83%)	5/6 (83%)	5/6

Murai 1993. The same SET is stable in the English cell (every value ≥83% under all three rules) but not in the Japanese cell, where EP-04a appears in 2/6 (33%) runs under FIRSTOBJ and FIRST3 and 3/6 (50%) under VQ3 — below the failure criterion stated in SI-11.1 (inclusion frequency below 50%) under two rules, and at that boundary under the third. The SET is therefore not claimed to be stable for Murai 1993, and the earlier statement that {EP-03a, EP-04a, EP-06a} is a paper-class-level stable structure for Ni-catalysed cross-coupling communications is withdrawn.

Cell (n = 6)	Rule	EP-03a	EP-04a	EP-06a	Runs with all three
Murai JA	FIRSTOBJ	3/6 (50%)	2/6 (33%)	3/6 (50%)	2/6
Murai JA	VQ3	6/6 (100%)	3/6 (50%)	6/6 (100%)	3/6
Murai JA	FIRST3	4/6 (67%)	2/6 (33%)	3/6 (50%)	2/6
Murai EN	FIRSTOBJ	5/6 (83%)	5/6 (83%)	5/6 (83%)	5/6
Murai EN	VQ3	6/6 (100%)	6/6 (100%)	6/6 (100%)	6/6
Murai EN	FIRST3	5/6 (83%)	5/6 (83%)	5/6 (83%)	5/6

Stage 4 re-count of the VQ-hierarchy inversion (revised in response to review). The value first reported ("detected in 24/24 Stage 4 runs") counted runs whose own text diagnosed an inversion. The Stage 4 v2 prompt (Step 5, item 4) describes the inversion pattern and asks the model to record it when present, so such diagnoses are prompt-induced and are not used here. Instead, each run's per-EP filling label (§B, "Nature of filling") was classified under definitions fixed before counting: the label source is §B only (an EP absent from §B is unfilled; a mixed label takes its weakest component); four thresholds for "filled" — Complete or Partial (primary), Complete only, Complete/Partial/Indirect, and Complete or Partial excluding Low-confidence labels; Attempted and Paradigm-shift never count as filled; EP groups follow Table 3 (VQ-3 = EP-03a/04a/05a/06a/10a; VQ-4 = EP-08a/09a). A run shows the inversion when both VQ-4 EPs are unfilled and at least one VQ-3 EP is filled. Per-EP labels (240 rows, raw text included) and per-run judgements are deposited (`stage4_labels_per_ep.csv`, `stage4_inversion_per_run.csv`).

Threshold for "filled"	Inversion	EP-08a unfilled	EP-09a unfilled	Run text mentions an inversion
Complete or Partial (primary)	12/24	14/24	18/24	24/24
Complete only	4/24	23/24	23/24	24/24
Complete, Partial or Indirect	2/24	9/24	2/24	24/24
Complete or Partial, excluding Low confidence	16/24	20/24	18/24	24/24

At the primary threshold the inversion holds in 1/6 (Kumada JA), 3/6 (Kumada EN), 4/6 (Murai JA) and 4/6 (Murai EN) runs. Under the strictest threshold both VQ-4 EPs are almost always unfilled, but VQ-3 EPs rarely reach Complete either, so the asymmetry appears or disappears with the threshold. The inversion is therefore not claimed to be run-stable. For Kumada 1972, the non-detection of EP-08a and EP-09a rests on the locked WoS audit (main text Fig. 2; SI-9.8); in the Stage 4 runs these two EPs are labelled Partial or better in 7 and 5 of the 12 Kumada runs. In 12 EP entries a run labels an EP as filled in §B yet lists it as unfilled in §E; these are flagged in the deposited table.

The 11-paper v18.4 EN corpus (revised in response to review). The same definitions were applied to the single EN Stage 4 run of each of the 11 papers (Kumada 1972 and Murai 1993, run 1; kumada1976b, monnereau2009, breitenfeld2014, cherney2015, molander2016, molken2016, mazet2021, pengfei2024, tamao2002). The inversion holds in 6/11 papers when Complete or Partial counts as filled, 0/11 at Complete only and 1/11 when Indirect also counts; in four of the five papers without it, a VQ-4 EP is labelled Partial or better. The earlier statement that the inversion recurs in all 11 papers is withdrawn. Years from publication to the first filling were taken from the §B "Filling Onset" citation: of 20 VQ-3 fillings with a year, 13 fall 0–11 years after publication (median 2; Kumada 1972: 4–8, Murai 1993: 1–11), while 7 cite work published before the analysed paper and

were excluded after inspection because earlier work cannot fill a later paper's gap. These citations are the model's recall, often marked `tentative_citation`, and were not bibliographically verified. Only Kumada 1972 (54 years), kumada1976b (50) and Murai 1993 (33) are old enough for a 30–54-year interval; in the single EN run used for this count, none of the three has a VQ-4 EP at Partial or better, but each has one labelled Indirect; across the 12 Kumada runs, the labels vary (see above). Per-EP labels are deposited (`corpus11_labels_per_ep.csv`, `corpus11_inversion_per_paper.csv`).

Cross-paper findings supporting SI-3 specifically: The v18.4 EN analyses of the nine EP-filling-context papers (May 2026; `improvements/paperA_phase2_completion_report_20260510.md` (project record, not deposited)) was the source of an earlier statement that the VQ-hierarchy inversion extends to all 11 papers; that statement is withdrawn (see the 11-paper re-count above). PRI-11 (Mechanism-kinetics decoupling, the pattern anchoring EP-10a) was applied in five papers (Breitenfeld 2014, Pengfei 2024, Mazet 2021, Cherney 2015, Tamao 2002). Only the proposing Stage 3 v1 run on Breitenfeld 2014 was unexposed to the pattern through the prompt (SI-6.5), so these are applications, not independent confirmations; promotion readiness is not claimed.

SI-3.1 Paper Metadata and Stage 0 Summary

Bibliographic record: Tamao, K.; Sumitani, K.; Kumada, M. "Selective Carbon–Carbon Bond Formation by Cross-Coupling of Grignard Reagents with Organic Halides. Catalysis by Nickel–Phosphine Complexes." *J. Am. Chem. Soc.* **1972**, *94* (12), 4374–4376. (Communication, three journal pages.)

Scope (Stage 0): Ni–diphosphine-complex-catalyzed cross-coupling of Grignard reagents with aromatic and vinyl halides, reported as a "new and useful preparative method" (T1 quote).

Authorial claims (Stage 0): (1) selective cross-coupling without homocoupling by-products in high yield, including tolerance of β-H-containing Grignard reagents; (2) catalytic activity of $NiCl_2$(dpe)-type bidentate diphosphine complexes; (3) activity toward aryl chlorides (prior work was Br/I-dominant), with a "cis-geometry requirement" hypothesis from the ligand-activity ordering; (4) ligand-activity ranking [$NiCl_2$(dpp)] > [$NiCl_2$(dpe)] > … > [$NiCl_2(PEt_3)_2$]; (5) utility for the one-step synthesis of linear n-alkyl arylbenzenes (e.g., m-di-n-butylbenzene) that are "hardly accessible by conventional methods."

Domain classification: Homogeneous transition-metal catalysis / cross-coupling; communication format, 1972 JACS.

SI-3.2 Stage 1 Output — Structured Absences (summary cross-referenced from Stage 3 §2)

Stage 3 §2 references the following Stage 1 items (see `papers/dev/kumada/kumada_v17_sonnet_s1.md` for the full itemized output; the item numbers below are those of the v17 record and differ from the v18.4 record used in SI-11.5 — e.g. the unquantified selectivity claim is B-1 here and B-6 there, and the absent replicates are B-9 there). Categories are A (deferred), B (traces of rejection), C (uncontacted references), U (unclassified / self-contradictory absences). Inference tiers (T1/T2/T3) and `evidence_chain` refer to the Stage 1 record.

Key A items: A-1 (cis/trans selectivity future work, high convergence low gap with V1=medium, V2=high), A-2 (scope limitation presented as a positive feature).

Key B items: B-1 (selectivity claim without by-product quantification, 3/3 convergence), B-2 (mechanism cycle presented as working hypothesis without intermediate observation, high convergence low gap), B-3 (ligand-activity ranking without matched-pair control, 2/3 convergence), B-4 (glpc conditions not reported), B-5 (scope boundary "Csp^2 only" without failed Csp^3 examples), B-6 (Cl/Br/I direct comparison absent), B-7 (Et_2O "definitely superior" to THF without quantitative substantiation), B-8 (reaction times absent), B-9 (optimization criteria not disclosed), B-11 (cis/trans 43:57 ratio without replication), B-12 (yield-basis mixing across entries), B-13 (characterization reported with "and/or" disjunction), B-14 (dipy-Ni analogy conflated with the dpe system).

Key C items: C-1 (Tamura-Kochi Fe-catalyst prior work — benchmark not made), C-2 (Corey/Whitesides stoichiometric reagents — utility benchmark not made), C-3 (Corriu/Masse Ni(acac)$_2$ — Note Added in Proof, no quantitative comparison).

Key U items: U-1 (R–R homocoupling predicted by eq 3 but not reported — `self-contradictory absence`), U-2 (Table I footnote f "exothermic reaction, black mixture" implying unstable species without characterization), U-3 (cis/trans 43:57 kinetic vs thermodynamic product not tested), U-4 (bidentate-diphosphine necessity not tested against a monodentate control).

SI-3.3 Stage 2 Output — Interpretation (from Stage 3 §1 and its Stage 2 context)

D-1 Declared motivation: present a selective, practical cross-coupling method using Ni–diphosphine complexes (layered novelty: selectivity + reactivity of aryl chlorides + ligand design + synthetic utility).

D-2 Inferred motivation: competitive positioning against Corriu–Masse ($Ni(acac)_2$) via the bidentate-ligand hypothesis; timing pressure of the pre-Suzuki cross-coupling race.

D-3 Institutional motivation: JACS communication format exploits brevity to register priority for a broad claim whose details (analytical protocols, scope limits) are reserved for later Full Paper disclosure — a typical 1972 genre strategy.

D-4 Unquestioned motivations (eight candidates, per §5 short-path routing):

- D4-1: Quantitative basis of selectivity claim (anchored in SRG-2; top-priority per §5).
- D4-2: Catalytic cycle without experimental basis (anchored in SRG-1).
- D4-3: Table I data interpretability and analytical reliability (anchored in SRG-3).
- D4-4: Experimental basis of the scope limit — were sp^3 halides tried? (anchored in SRG-4).
- D4-5: Kinetic vs thermodynamic character of cis/trans selectivity (anchored in SRG-6).
- D4-6: Catalyst identity — homogeneous or heterogeneous, as the "black mixture" suggests (anchored in SRG-1).
- D4-7: Connection to a synthetic target — teleological justification of "useful" (anchored in SRG-4 partially; CLA-1 in Stage 3).
- D4-8: Bidentate-necessity cross-check with Corriu/Masse $Ni(acac)_2$ (anchored in SRG-1 partially; CLA-2 in Stage 3).

(Numbering and content follow the v17 Stage 2 record `kumada_v17_sonnet_s2.md`, D4-1 to D4-8, with its `same_root_group` fields; the CLAs are Stage 3 labels. Corrected in R1: the earlier list swapped D4-2 and D4-3, placed D4-7 under D4-6, and listed a "β-H elimination competition" candidate that the record does not contain.)

E (scope question): `tool_question` (the same label in the v17 Stage 2 record and in all twelve v18.4 Stage 2 runs). The recorded forms in the v17 Stage 2 record are: propositional, "Nickel-phosphine complexes catalyze selective cross-coupling of Grignard reagents with vinyl or aryl halides."; question, "Can a nickel-diphosphine complex catalyst enable selective cross-coupling of a Grignard reagent with a vinyl or aryl halide?"

F (within-group comparison): with Murai 1993 (C–H activation), the structural parallel is the labeling mismatch — both papers present a mechanistic "working hypothesis" at the end of the main text while narrating intermediate steps in declarative language.

SI-3.4 Stage 3 — Root-Issue Codebook (Six SRGs)

SRG-1 Mechanistic cycle without experimental evidence in the dpe-Ni system

- **Primary RI**: RI-06 Imported model. **Secondary RI**: RI-03 Unobserved explanatory.
- **Evidential requirement (ER)**: at least one stoichiometric step of the proposed cycle (the individual elementary reactions of eq 2–4) must be experimentally demonstrated in the $NiCl_2$(dpe) system by isolation, spectroscopic observation, or independent synthesis of the intermediate. Analogy from dipy-Ni does not constitute evidence of structural isomorphism.
- **Evidence Pack (EP)**: EP-06a Analogy-validation pack; EP-03a Direct-observation / state-assignment pack.
- **Contributing Stage 1 items (5)**: B-2 (high convergence low gap V1=high, V2=critical), B-14 (high convergence low gap V1=medium, V2=high), U-1, U-2, U-4. D4-2 and D4-6 both derive from this SRG. Joint highest item count among the SRGs (with SRG-3).

SRG-2 Selectivity claim without by-product data

- **Primary RI**: RI-04 Unreplicated discriminator. **Secondary RI**: RI-08 Unquantified reliability, RI-09 External superiority.
- **ER**: to support the "selective cross-coupling" claim, at least one Table I entry must have quantified yields for all four product channels (cross-coupled, homocoupled R–R, protodemetalated R–H, β-eliminated olefin) with mass balance closed.
- **EP**: EP-04a Discrimination / replication pack; EP-08a Replication / uncertainty pack; EP-09a Internal benchmark pack.
- **Contributing Stage 1 items (3)**: B-1 (3/3 convergence, high convergence low gap), U-1, C-1. D4-1 derives from this SRG. Top priority in Stage 3 §5 short-path routing.

SRG-3 Analytical quality of Table I yield data

- **Primary RI**: RI-01 Reporting integrity. **Secondary RI**: RI-08 Unquantified reliability.
- **ER**: to allow independent reproduction of Table I, glpc conditions (column, stationary phase, detector, temperature program, internal-standard identity and concentration), reaction time for each entry, and the yield calculation basis (halide-based vs Grignard-based) must be reported uniformly across all 11 entries.

- **EP**: EP-01a Reporting-coherence pack; EP-08a Replication / uncertainty pack.
- **Contributing Stage 1 items (5)**: B-4, B-8, B-9, B-12, B-13. B-4 and B-8 are high-convergence low-gap items. D4-3 derives from this SRG. Resolution of RI-01 is a necessary condition for VQ-2.

SRG-4 Scope restriction basis

- **Primary RI**: RI-07 Unsupported boundary. **Secondary RI**: RI-09 External superiority.
- **ER**: the "Only Csp^2 halides can be used" boundary claim requires a representative sample of Csp^3 halides (primary / secondary alkyl chlorides) tested under the same conditions and confirmed inactive or poorly selective. The "chlorides offer one of the most remarkable features" claim requires quantitative comparison (yield or rate) against Br and I on the same substrate.
- **EP**: EP-07a Boundary-test pack; EP-09a Internal benchmark pack.
- **Contributing Stage 1 items (3)**: B-5, B-6, A-2 (convergence = 1/3). D4-4 derives from this SRG; D4-7 partially derives from it.

SRG-5 Solvent and condition effects without quantitative or mechanistic basis

- **Primary RI**: RI-05 Qualitative overread.
- **ER**: "Diethyl ether is definitely superior to THF" requires yield measurements (at least one Table I entry) with THF as solvent. The adverb "definitely" requires quantitative support.
- **EP**: EP-05a Quantitative-ranking substantiation pack.
- **Contributing Stage 1 items (2)**: B-7, B-9. B-7 carries the adverb "definitely" as a canonical RI-05 marker.

SRG-6 Cis/trans selectivity observation

- **Primary RI**: RI-04 Unreplicated discriminator. **Secondary RI**: RI-08 Unquantified reliability.
- **ER**: the reported cis/trans ratios (80:20; 43:57) require explicit reporting of analytical method (glpc area or NMR integration), instrument settings, replicate count, and standard deviations. The near-unity 43:57 ratio in particular requires uncertainty quantification.
- **EP**: EP-04a Discrimination / replication pack; EP-08a Replication / uncertainty pack.
- **Contributing Stage 1 items (3)**: A-1 (high convergence low gap V1=medium, V2=high), B-11, U-3. D4-5 derives from this SRG.

SI-3.5 Stage 3 — Claim-Level Adjuncts (Two CLAs)

CLA-1 "Useful preparative method" — absence of teleological connection to a synthetic target

- **Primary RI**: RI-09 External superiority. **Secondary RI**: RI-07 Unsupported boundary.
- **ER**: the declaration "a new and useful preparative method" requires quantitative comparison with Corey/Whitesides stoichiometric reagents (C-2) on the same substrate (yield, step count), and the "hardly accessible by conventional methods" boundary claim requires an attempt with an alternative method confirming the difficulty.
- **EP**: EP-09a Internal benchmark pack; EP-07a Boundary-test pack.
- **Contributing Stage 1 items**: B-9, A-2, C-2. C-2 (Corey/Whitesides) is attributed only to this CLA — not to any SRG. Connects directly to D-1 claim 5.

CLA-2 Corriu/Masse ($Ni(acac)_2$) — necessity of the bidentate diphosphine

- **Primary RI**: RI-05 Qualitative overread. **Secondary RI**: RI-06 Imported model.
- **ER**: deriving a structural "cis-geometry requirement" from the ligand-activity ranking $[NiCl_2(dpp)] > [NiCl_2(dpe)] > \ldots > [NiCl_2(PEt_3)_2]$ requires independent variation of bite angle, electronic effect, and steric effect across a ligand set, together with direct comparison against monodentate / non-chelating systems including $Ni(acac)_2$.
- **EP**: EP-05a Quantitative-ranking substantiation pack; EP-06a Analogy-validation pack.
- **Contributing Stage 1 items**: C-3, B-3, U-4. B-3 is attributed only to this CLA (overlap with SRG-1 only via U-4). C-3 (Corriu/Masse Note Added in Proof) is also CLA-only. Connects directly to D-1 claim 4.

SI-3.6 Stage 3 — VQ Evaluation Matrix

The VQ matrix evaluates each root issue against each VQ level (VQ-1 observation, VQ-2 methodology, VQ-3 mechanism, VQ-4 quantitative reliability).

Root issue	VQ-1 (Obs.)	VQ-2 (Method)	VQ-3 (Mech.)	VQ-4 (Quant.)
SRG-1 (cycle without evidence)	pass (RI-06 Non-fatal at VQ-1)	not attained	not attained (RI-03, RI-06 fatal)	not attained
SRG-2 (selectivity)	pass (RI-04 Non-fatal)	not attained	not attained (RI-04 fatal)	not attained (RI-08, RI-09 fatal)
SRG-3 (Table I data)	pass	not attained (RI-01 fatal)	not attained (RI-01 fatal)	not attained
SRG-4 (scope boundary)	pass	not attained (RI-07 fatal)	not attained	not attained
SRG-5 (solvent superiority)	pass	Non-fatal qualification (RI-05 residual)	qualified (RI-05)	not attained
SRG-6 (cis/trans ratios)	pass (RI-04 Non-fatal)	not attained (RI-04 fatal)	not attained	not attained (RI-08)
CLA-1 (utility)	pass	RI-09 residual	qualified	not attained (RI-09 fatal)
CLA-2 (bidentate necessity)	pass (RI-05 Non-fatal)	RI-05 residual	qualified (RI-05)	not attained

Note on the VQ-1 name. The Stage 3 prompt names VQ-1 "Observation"; its requirement is RI-02 (identity and purity of the product or catalyst), which is how Methods 1.2 defines VQ-1. Model outputs quoted in this SI keep the prompt's name.

Aggregate current judgment: **VQ-1 (Observation)** — the paper passes VQ-1 because the eleven Table I products are known compounds whose physical properties (bp, $n^{20}D$) match literature values, and because the 1972 JACS Communication genre permits glpc retention-time comparison plus "and/or" characterization as sufficient identification for known compounds. VQ-2 is not attained because RI-01 (SRG-3, five items) and RI-07 (SRG-4) remain fatal; these must be resolved before VQ-2 can be granted. VQ-3 and VQ-4 are not attained as consequences.

Note on VQ-3 kinetic-verification branch: the VQ-3 path for SRG-6 (cis/trans selectivity) depends not only on replicated discrimination (RI-04, addressed by EP-04a) but also on distinguishing kinetic from thermodynamic product through time-course measurement; similarly, SRG-4's claim that "chlorides offer one of the most remarkable features" requires same-condition rate data to separate reactivity from selectivity. Both needs motivate EP-10a (see SI-3.7) as a complement to EP-04a and EP-07a; EP-10a is not independently fatal for any RI at VQ-2 but is required for VQ-3 closure on the kinetic-versus-thermodynamic dimension.

SI-3.7 Stage 3 — Evidence Pack (EP) Derivation

Ten Evidence Packs were derived — one identifier per reinforcement pack, sufficient to close one or more RI dependencies. The definitions below are those of the v17 Stage 3 v1 record (SRG-level implementations rewritten per EP for this SI). The same eight identifiers (EP-01a, EP-03a–EP-09a) are also assigned in the v18.4 Stage 3 v2 English runs, attached to differently named SRGs: all eight occur in five of the six runs, and EP-05a is absent from run 3; two further EPs (EP-02a *Identity / purity substantiation pack*; EP-10a *Kinetic verification pack*) were added in April 2026 from labels already listed in the Stage 3 prompt (SI-6.5) and are documented below alongside the original eight. The provenance of each addition is recorded in the SI-3 header notes and in the per-EP "Provenance" line below:

EP-01a Reporting-coherence pack

- **VQ level**: VQ-2.
- **Source RI / SRG**: RI-01 via SRG-3.
- **Case-specific implementation**: (i) provide a revised Table I as supporting information with glpc conditions (stationary phase, detector, internal standard, temperature programme) and reaction time for each of the 11 entries; (ii) unify the yield basis (halide- vs Grignard-based) across entries and present both-basis conversions; (iii) for at least three entries, report $n \geq 3$ replicate yields with standard deviations.
- **Search term derivation (for later filling-paper search)**: "Tamao Kumada Full Paper" + "Bull. Chem. Soc. Jpn 1976 49 1958" (prior-knowledge anchor) + "Ni phosphine Grignard Csp^2-halide glpc protocol 1976–1980."

EP-02a Identity / purity substantiation pack (added April 2026)

- **VQ level**: VQ-1 residual.

- **Source RI / SRG**: RI-02 via SRG-3 — extends EP-01a's reporting-coherence focus to the per-entry product-identity dimension as a separate residual at VQ-1, not subsumed by the yield-table reporting hygiene that EP-01a covers.
- **Provenance**: Assigned by the GPT-5.4 Thinking and Opus 4.7 Stage 3 v2 runs on Kumada 1972 (2026-04-19), from the common EP table of the prompt they were given. The earlier v17 single-run analysis (Stage 3 v1, Sonnet) had grouped this absence under EP-01a. The submitted version called the two assignments independent; they are not (SI-6.5).
- **ER**: to accept the paper's claim that the listed coupling products were obtained, per-entry product-identity confirmation by GC-MS, ^{1}H NMR, or authentic-sample co-injection must be provided independently of the parent retention-time / yield reporting in EP-01a.
- **Case-specific implementation**: (i) for each of the 11 Table I entries, provide one of GC-MS retention time + mass spectrum, ^{1}H NMR chemical-shift table, or authentic-sample co-injection; (ii) for products where a literature authentic sample is unavailable, provide elemental analysis or HRMS as identity substantiation; (iii) bring the verification basis into the SI alongside the EP-01a yield-table revision.
- **Search term derivation**: "Tamao Kumada Full Paper 1976" + "Ni Grignard cross-coupling product identification GC-MS NMR 1976–1980" (overlaps with the EP-01a search anchor — the filling-paper expectation is that the same Tamao 1976 *Bull. Chem. Soc. Jpn.* full paper that addresses EP-01a will partially address EP-02a as well).
- **Historical filling**: same anchor as EP-01a — Tamao et al. *Bull. Chem. Soc. Jpn.* 1976 (Full Paper I) front-matter scope reporting provides Partial filling at the 1976 inflection point. A complete VQ-1 residual closure for the original Kumada 1972 11-entry table has not been independently demonstrated within our retrieval window.

EP-03a Direct-observation / state-assignment pack

- **VQ level**: VQ-3.
- **Source RI / SRG**: RI-03 via SRG-1.
- **Case-specific implementation**: (i) follow the stoichiometric reaction of $NiCl_2$(dpe) with PhMgBr (eq 2 analog) by low-temperature NMR or ESI-MS to observe the putative intermediate Ph_2Ni(dpe) (1); (ii) verify that the isolated intermediate or a model complex (e.g., Me_2Ni(dpe)) reacts with PhCl to give eq 3 coupling; (iii) exclude Ni(I)/Ni(III) radical pathways with a radical-clock substrate (cyclopropylmethyl-Grignard) or EPR measurements.
- **Search term derivation**: "Ni(II) diphosphine intermediate NMR observation 1975–1985" + "Ni(III) aryl alkyl isolation" + "Ni dpe stoichiometric low-T NMR."

EP-04a Discrimination / replication pack

- **VQ level**: VQ-3.
- **Source RI / SRG**: RI-04 via SRG-2 and SRG-6.
- **Case-specific implementation**: (i) for n-BuMgBr + PhCl, quantify by glpc internal-standard method the n-Bu–n-Bu (homocoupling), n-Bu–H (protodemetallation), and 1-butene (β-elimination) yields and construct the mass balance; (ii) add the measured yield of R–R (predicted by eq 3) to Table I; (iii) run parallel experiments with Fe catalyst (Tamura-Kochi ref 6) and with $Ni(acac)_2$ (Corriu/Masse) to obtain internal benchmarks.
- **Search term derivation**: "Kumada cross-coupling homocoupling mass balance" + "Ni Grignard byproduct GLC determination."

EP-05a Quantitative-ranking substantiation pack

- **VQ level**: VQ-3.
- **Source RI / SRG**: RI-05 via SRG-5 (solvent superiority) and CLA-2 (bidentate necessity).
- **Case-specific implementation**: (i) run the Table I Example 1 analog (n-BuMgBr + m-$Cl_2C_6H_4$) in THF and compare yield and selectivity to Et_2O; (ii) for all six ligands, measure yield on a uniform substrate (n-BuMgBr + PhCl) and correlate with bite angle and σ-donor strength; (iii) parallel experiment with $Ni(acac)_2$ vs $NiCl_2$(dpe) on the same substrate and conditions.
- **Search term derivation**: "Ni catalyst bite angle ligand effect Grignard cross-coupling" + "dppe Kumada Ni(0) catalytic cycle."

EP-06a Analogy-validation pack

- **VQ level**: VQ-3.
- **Source RI / SRG**: RI-06 via SRG-1 and CLA-2.
- **Case-specific implementation**: stoichiometric mechanistic study of each proposed elementary step (eq 2, 3, 4) in the $NiCl_2$(dpe) system, independent of dipy-Ni analogy.

- **Search term derivation**: "Ni(II) phosphine transmetalation mechanism stoichiometric 1975–1985" + "Kumada coupling catalytic cycle experimental verification."

EP-07a Boundary-test pack

- **VQ level**: VQ-2 (for scope boundary).
- **Source RI / SRG**: RI-07 via SRG-4.
- **Case-specific implementation**: (i) test n-BuCl, sec-BuCl, $PhCH_2Cl$ (Csp^3 halides) under the same conditions ($NiCl_2$(dpe) / Et_2O) with PhMgBr and present inactive / poorly selective results as a "failed scope" row of Table I; (ii) test PhBr and PhI with the reference substrate (n-BuMgBr) under the same conditions as PhCl for direct yield comparison.
- **Search term derivation**: "Kumada coupling Csp^3 halide scope" + "Ni phosphine Cl Br I comparison."

EP-08a Replication / uncertainty pack

- **VQ level**: VQ-4.
- **Source RI / SRG**: RI-08 via SRG-2, SRG-6, SRG-3.
- **Case-specific implementation**: (i) for all 11 Table I entries, perform $n \geq 3$ replicates with standard deviations on yield; (ii) for cis/trans-dichloroethylene substrates, $n \geq 3$ replicates with glpc and ^{1}H NMR cross-method determination; (iii) time-course (1 h, 5 h, 20 h) to distinguish kinetic vs thermodynamic product. (This block combines the SRG-6 implementation and §5 step 8 of the Stage 3 v1 record; the record itself gives "glpc or ^{1}H NMR" and "$n \geq 3$ for at least three entries" at the SRG level and "all entries, $n \geq 3$" in §5.)
- **Search term derivation**: "Kumada cross-coupling standard deviation replicate measurement" + "cis/trans stilbene kinetic thermodynamic product."

EP-09a Internal benchmark pack

- **VQ level**: VQ-4.
- **Source RI / SRG**: RI-09 via SRG-2, SRG-4, CLA-1, CLA-2.
- **Case-specific implementation**: parallel reactions with Tamura-Kochi (Fe) and Corriu/Masse ($Ni(acac)_2$) and Corey/Whitesides stoichiometric reagents on the same substrate, yielding direct numerical yield, selectivity, and operability comparisons.
- **Search term derivation**: "Kumada coupling benchmark vs Corriu Masse" + "$Ni(acac)_2$ vs $NiCl_2$(dpe) Grignard coupling comparison."

EP-10a Kinetic verification pack (added 2026-04-19)

- **VQ level**: VQ-3.
- **Source RI / SRG**: RI-04 via SRG-6 (primary: time-resolved discrimination of kinetic vs thermodynamic cis/trans product); secondary contribution to SRG-4 (same-condition rate comparison for Cl/Br/I halides) and SRG-2 (time-course of cross- vs homo-coupling selectivity). Connects to the cross-paper PRI candidate that Breitenfeld 2014 §6 proposed as PRI-11 (Mechanism-kinetics decoupling): rate law not algebraically derived from the proposed catalytic cycle.
- **ER**: to accept the paper's mechanistic / selectivity claims at VQ-3, the following must be established: (i) the proposed cycle (eq 2–4) entails a rate law that can be derived algebraically and compared to measured order in [RMgX] and [R'X' ']; (ii) the observed cis/trans-stilbene ratios (80:20; 43:57) must be demonstrated to be formation-selectivity rather than post-formation isomerization, requiring time-course measurements at multiple intermediate points; (iii) the "remarkable" chloride reactivity must be separated into rate and selectivity components via same-condition initial-rate kinetics on PhCl/PhBr/PhI.
- **EP**: EP-10a Kinetic verification pack (newly introduced). Complements rather than replaces EP-04a (which provides replicate-quality statistics at a single time point); complements rather than replaces EP-07a (which provides scope-boundary endpoints but not rate profiles).
- **Case-specific implementation**: (i) rate-law derivation from the proposed catalytic cycle plus log-log rate-order plots for [PhMgBr] and [PhCl] independently varied ($n \geq 5$ points × $n \geq 2$ replicates each axis); (ii) time-course of cis-/trans-1,2-dichloroethylene + PhMgBr at t = 1 h, 5 h, 20 h, 40 h to distinguish formation-selectivity from post-formation isomerization and establish plateau behavior; (iii) Eyring analysis of PhCl + n-BuMgBr at T = 0 °C, 25 °C, 40 °C with $\Delta H‡$ and $\Delta S‡$ extracted, directly compared to PhBr and PhI under matched conditions — yields rate-separated-from-selectivity resolution of SRG-4's "remarkable chloride reactivity" claim; (iv) log-log plot of initial rate vs [catalyst] at three loadings to establish catalyst-order and detect off-cycle species.

- **Search term derivation**: "Kumada coupling rate law kinetic verification" + "Ni-phosphine Grignard transmetalation Eyring" + "cis trans stilbene kinetic thermodynamic time course."
- **Historical filling (from stage4_literature_update_20260419.md)**: Kiso, Tamao, Kumada *J. Organomet. Chem.* 1973, 50, C12–C14 (starting point: halide-dependent isomerization as local kinetic proxy); Chen et al. *J. Am. Chem. Soc.* 2019, 141, 5835–5855 (establishing point: unified computational + experimental study of chemoselectivity and stereospecificity in the benzylic-ether Kumada system, DOI: 10.1021/jacs.9b00097).

Principal-reinforcement route (updated to ten EPs; steps 3–5 reflect the Top-3 principal EP SET {EP-03a, EP-04a, EP-06a} for Kumada 1972 at v18.4 multi-run aggregate, N=6, sensitivity table above):

1. EP-01a (maximal RI-01 coverage; five SRG-3 items simultaneously) — VQ-2 gate. 1b. EP-02a (RI-02, parallel to 1, VQ-1 residual closure for product-identity confirmation; performed alongside the EP-01a yield-table revision since both share SRG-3 source).
2. EP-07a (RI-07, SRG-4) — VQ-2 gate, parallel to 1.
3. EP-06a (RI-06, high information gain for SRG-1 + CLA-2 dual-contribution) — begins VQ-3.
4. EP-04a (RI-04 static dimension; SRG-2 + SRG-6 six items) — VQ-3.
5. EP-10a (RI-04 temporal dimension; kinetic-vs-thermodynamic discrimination for SRG-6; rate-vs-selectivity decomposition for SRG-4; rate-law closure for the proposed cycle) — VQ-3, executed in parallel with EP-04a on SRG-6 entries to maximize information gain.
6. EP-03a (RI-03, SRG-1 residual) — VQ-3.
7. EP-05a (RI-05, CLA-2 residual) — VQ-3 qualified → VQ-3 full.
8. EP-09a (RI-09) — VQ-4.
9. EP-08a (RI-08) — VQ-4.

EP-10a is placed at step 5 (parallel to EP-04a on SRG-6) rather than after EP-03a because the time-course experiments it requires are logistically independent of the state-assignment experiments EP-03a requires; running EP-10a in parallel saves experimental resources and preserves the total VQ-3 closure time. The placement does not alter VQ-3 gate ordering — EP-03a, EP-10a, and EP-06a together must close RI-03, RI-04 (temporal), and RI-06 before VQ-3 is granted.

Update (2026-04-21) — PRI-11 and EP-10a applied in two further papers (Stage 3 v2, whose prompt describes both; not independent confirmations, SI-6.5). Two retrieval-diagnostic papers recovered via the SI-9 WoS × PDF cross-check pathway (Yamamoto-Hayashi-Yamamoto 1978 *BCSJ* 51, 2091 and Yamamoto-Kohara-Yamamoto 1981 *BCSJ* 54, 2010) received full v17 + Stage 3 v2 + Stage 4 v2 UQS self-analysis on 2026-04-21 (archived at `papers/ep/kumada1972/yamamoto1978/` and `../yamamoto1981/`). The Yamamoto 1978 analysis produced two Codebook-level confirmations relevant to this subsection. First, its SRG-3 (Fig. 3 time-course kinetic profile used to support Scheme 1 without algebraic rate-law derivation from the proposed cycle) matches the **PRI-11 (Mechanism-kinetics decoupling)** definition, as a further application of PRI-11 under a prompt that describes it. Second, Yamamoto 1978's EP dictionary binds rate-law absence to **EP-10a**, making this the first application of EP-10a to a paper other than Kumada 1972. The structural match (mechanism proposal + time-course data + no algebraic rate-law derivation) supports EP-10a's generality beyond the single-case Kumada 1972 derivation reported in this subsection. Neither result alters the Kumada 1972 Stage 3 output above; both are cross-paper applications of the same Codebook machinery.

SI-3.8 Era-Norm Annotation

Communication-era tolerated:

- **B-4** (glpc conditions not reported): in 1972 JACS Communications, the detailed glpc conditions (stationary phase, temperature programme, internal standard) were not expected to be reported in the main text; they were conventionally reserved for Full Papers.
- **B-8** (reaction time not reported for every entry): Communications typically report detailed procedure for 2 representative entries; abbreviation for the remaining 9 entries was within the conventional envelope.
- **B-13** (spectroscopic data not reported for every product): for known compounds, bp/n^{20}D + glpc retention-time comparison + abbreviated characterization ("characterized by …") was sufficient in 1972 Communication format.

- **B-1** (selectivity claim without mass-balance quantification): in an initial methodological Communication, the qualitative claim of "selective" was acceptable without by-product mass balance; the contrast with the Grignard-all-inactive prior art established context for a qualitative claim.
- **B-2** (catalytic-cycle "working hypothesis"): in 1972 organometallic chemistry, presenting an analogy-based "working hypothesis" in a Communication without stoichiometric verification was within the genre; the authors themselves mark it explicitly as "working hypothesis" in the closing sentence — honoring the era's evidence standard.

Modern-era deficient:

- **B-1** (selectivity without quantitative basis): modern JACS / *Nat. Chem.* require by-product profiles (cross:homo ratios, mass balance) for selectivity claims; glpc yield alone is insufficient. Further, modern standards would flag as a self-contradictory absence (U-1) the failure to quantify R–R predicted by eq 3.
- **B-4** (glpc conditions): modern experimental-reporting standards require disclosure of internal-standard identity and concentration and calibration curve; without these, independent yield verification is impossible in principle.
- **B-13** (per-entry characterization data): modern JACS requires supplementary-information NMR/MS for each compound; blanket "and/or" is not accepted.
- **B-3 / U-4** (ligand-ranking causal attribution to cis geometry): modern catalysis papers require DFT, systematic bite-angle variation, and independent steric / electronic probes to derive structural conclusions from a single ranking.
- **B-7** ("definitely superior" solvent without quantitative basis): modern methodology papers would have an editor or reviewer flag the use of confident adverbs without supporting data.

Both tags (tolerated then, deficient now):

- **B-4**: tolerated in 1972 Communication format; required for independent verification now.
- **B-13**: tolerated for known-compound characterization in 1972; full supplementary-information data expected now.
- **B-1**: qualitative selectivity claim was tolerated in an initial Communication; quantitative mass balance is required now, and eq 3's prediction of R–R makes the era-dependence particularly conspicuous.

SI-3.9 Contribution to Institutional Design (Stage 3 §6)

Stage 3 records four institutional-design contributions derived from the Kumada analysis:

1. **RI-06 "labeling mismatch" subtype candidate**: the paper presents a textbook example of an imported-model analogy (dipy-Ni + halogen-Ni transmetalation → dpe-Ni cycle) that appears in declarative language throughout the body while the hypothesis label is restricted to one closing sentence. This is recorded as a candidate RI-06 subtype — "**RI-06b: hypothesis explicitly labelled at the closing sentence but body narrated in declarative language**" — for future Codebook inclusion.

2. **RI-07 "positive reframing of restriction" pattern**: the combination of B-5 (no failed Csp^3 examples) and A-2 (scope restriction presented as the method's "most remarkable feature") shows a characteristic pattern in methodology Communications: the restriction is reframed positively rather than demonstrated empirically, and scope extension is externalized to future work. Recommended Codebook note: "RI-07 typically auto-resolves at Communication → Full Paper transition — pair with A-items."

3. **CLA identification procedure validation**: D4-7 (with C-2 as its sole independent anchor) and D4-8 (with C-3, B-3 as sole independent anchors) demonstrate that the CLA identification procedure successfully captures D-4 candidates that do not fit any SRG. Both CLAs connect directly to specific claims in D-1 (CLA-1 ↔ claim 5; CLA-2 ↔ claim 4), and both contribute EPs that strengthen the §5 route (EP-06a dual-contribution: SRG-1 RI-06 + CLA-2 RI-06 closed by a single experimental pack).

4. **VQ-1 / VQ-2 boundary Era-Norm dependence**: RI-02 (object identification) is effectively satisfied under 1972 JACS Communication format when the products are known compounds whose physical properties can be cited. Under modern standards the same characterization level would be insufficient. This boundary case is recorded as "RI-02 severity is Era-Norm dependent; known-compound property-value matching is VQ-1 Non-fatal in 1970s Communication format."

SI-3.10 Summary Judgment

- **Current VQ level**: VQ-1 (Observation) pass; VQ-2, VQ-3, VQ-4 not attained.
- **Total SRGs**: 6 (SRG-1 through SRG-6).
- **Total CLAs**: 2 (CLA-1 "utility"; CLA-2 "bidentate necessity").

- **Total EPs**: **10** (EP-01a, **EP-02a**, EP-03a, EP-04a, EP-05a, EP-06a, EP-07a, EP-08a, EP-09a, **EP-10a**). Revised 2026-04-19 from the original 8 EPs through addition of EP-10a and EP-02a, both labels already listed in the Stage 3 prompt (SI-6.5). The v17 single-run record (`papers/ep/kumada1972/kumada1972_v17_sonnet_s3.md`) retains the 8-EP form and is the traceability anchor for SI-6 cross-vendor comparison; this SI reflects the updated 10-EP dictionary; the eight original identifiers recur in the v18.4 multi-run records, whose SRG grouping differs (SI-11.5). EP-02a rests on two runs. EP-10a was assigned in four runs, and Sonnet runs on three filling papers applied the related PRI-11 pattern; all of these ran under prompts that already listed the label or described the pattern (SI-6.5), so neither EP is presented as independently derived.
- **VQ-2 minimum reinforcement**: EP-01a + EP-07a (unchanged).
- **VQ-3 minimum reinforcement**: EP-06a + EP-04a + **EP-10a** + EP-03a (updated to include EP-10a for temporal / kinetic-versus-thermodynamic resolution).
- **Historical filling pattern**: SI-4.9 (added 2026-04-19) records the filling-paper candidates named by a GPT-5.4 Thinking Stage 4 run (record: `01_response/r33b_candidates/records/stage4_literature_update_20260419.md` in the deposit). Key points:
  - **EP-01a / EP-07a**: self-filled by Tamao et al. *Bull. Chem. Soc. Jpn.* 1976 (the author group's Full Paper I).
  - **EP-03a / EP-06a**: Morrell–Kochi 1975 (aryl-alkyl Ni intermediates) is recorded as an indirect filling of EP-03a and a partial filling of EP-06a; Dawson 2020 (spectroscopic resting-state assignment for a Kumada system) is the run's establishing point for EP-03a, also indirect. Chen 2019 is named for EP-06a only in the run's recommendation list (SI-4.9, note). The system-matched Partial filling of EP-03a comes only with Mazet 2021, 49 years after the communication (Results Part 2).
  - **EP-04a (static) / EP-10a (temporal)**: Tamao-Zembayashi-Kiso-Kumada *J. Organomet. Chem.* 1973 directly addresses the stereospecific vs nonstereospecific cis/trans stilbene anomaly (partial filling of EP-04a); Kiso-Tamao-Kumada *J. Organomet. Chem.* 1973 provides the halide-dependent isomerization kinetic proxy (indirect filling of EP-10a); Chen 2019 is the run's establishing point for EP-10a, also indirect.
  - **EP-08a / EP-09a**: in the v17 Stage 4 runs of Sonnet 4.6 and GPT-5.4 Thinking, neither run named a filling paper for EP-08a (replication / uncertainty); for EP-09a (strict same-axis benchmark) the runs named only partial or indirect candidates (Tamao 1976 in the GPT-5.4 run; Corriu–Moreau 1975 in the Sonnet run), none a same-axis benchmark. In the twelve v18.4 Stage 4 runs they are labelled Partial or better in 7 and 5 runs (SI-3, multi-run subsection). The non-detection claim therefore rests on the locked WoS audit (SI-9.8), not on agreement among Stage 4 runs.
- **VQ-3 anticipated-filling-hierarchy note**: the original eight-EP analysis anticipated that EP-04a would be difficult to fill ("cis/trans kinetic vs thermodynamic test — incentive-structure mismatch"); the 2026-04-19 update revises this prediction — the 1973 self-filling paper exists and fills EP-04a partially. EP-10a's temporal dimension (rate-law closure, Eyring analysis) is addressed in the run's candidates only in 2019 (Chen), consistent with the original prediction's spirit; the run records this candidate as an indirect filling at confidence Medium.

SI-3.11 EP-ChemOntology mapping (companion-paper basis)

The structural correspondence between UQS's Evidence Pack list and ChemOntology's Elementary Reaction Process Ontology (ERPO) — outlined briefly in Discussion 3 (Applicability) of the main text — is recorded here in tabular form as the basis for a planned companion methodology paper. The mapping covers nine of the ten EPs of the Kumada 1972 dictionary (EP-01a through EP-09a). EP-10a (Kinetic verification pack), added on 2026-04-19 (see SI-3.7), is registered for inclusion in a revised mapping in subsequent methodological extensions. The full Layer A parameter mapping with `allowed_changes` rules per structural unit, calculation-level prescriptions, expected output formats, and per-EP limitation analysis is deposited at `application_direction_chemontology_layer_a/`.

EP ID	EP short name	VQ level	Addressability	ChemOntology components	Notes
EP-01a	Reporting-coherence pack	VQ-2	not directly computable	—	Reporting-hygiene issue in the experimental literature; computed yields can serve as independent reference
EP-02a	Identity/purity substantiation pack	VQ-1 residual	indirect	EXPECTED_PRODUCTS_WITH_CIDS enumeration	Computational "identity checklist" of reachable products; not a substitute for spectroscopic confirmation
EP-03a	Direct-observation / state-assignment pack	VQ-3	direct	CoordinationComplex + OA + Transmetalation + RE + BHE node labels	Reaction-network nodes for intermediates 1, 2, 3 of Kumada 1972 eqs 2–4; DFT refinement required for Ni d^{10}/d^{8} redox
EP-04a	Discrimination / replication pack	VQ-3	partial	OA + Transmetalation + RE with stereo-differentiated RCs; RCMC kinetic navigation	Branch-point energies for cis vs trans stilbene pathways across 200/300/400 K
EP-05a	Quantitative-ranking substantiation pack	VQ-3	partial	AssociationOfTwoComponents(C1R1,C1R3) for cross; BHE branch; EXCLUDE_FORMATIONS_OF	Branch-point energies for cross vs homo vs β-elim; Kumada-specific product labeling (C1R1–C1R3) absent in SI Table S3 of Nath et al. 2026
EP-06a	Analogy-validation pack	VQ-3	direct	Full ERPO cycle executed on target (dpe-Ni) system	Addresses RI-06 (cycle-level imported model) by computing the target system; most demanding EP; ReductiveElimination ERPO requires new definition or decomposition
EP-07a	Boundary-test pack	VQ-2	partial	OxidativeAddition barrier scan across PhCl/PhBr/PhI/n-BuCl/c-hex-Cl	Predictive substrate-scope map; sp^3 halides may require SET rather than concerted OA^{S1} (outside default ERPO set)
EP-08a	Replication / uncertainty pack	VQ-4	not directly computable (different concept)	Multi-level-theory robustness spread	Computational robustness (xTB vs DFT vs CCSD(T)) is conceptually different from experimental replication; would be classified as Computational-Complete in the Stage 4 *Computational* filling-type extension developed in the companion paper
EP-09a	Internal benchmark pack	VQ-4	direct	Parallel run with $Ni(acac)_2$ at matched level	Corriu/Masse benchmark (CLA-2); Δ-metric between $NiCl_2$(dpe) and $Ni(acac)_2$ at identical ERPO set and calculation level

Mapping pattern. The EPs that map cleanly are those at VQ-3 (mechanism layer), which is the evidential level computational path search is best positioned to support. EPs targeting reporting hygiene and experimental replication (EP-01a, EP-02a, EP-08a) are not directly addressable by ChemOntology-AFIR. The distribution itself is informative: it suggests that LLM-driven literature analysis (UQS) and explicit knowledge-driven path search (ChemOntology-AFIR) cover **different but complementary slices of the evidential complementarity space**, with VQ-3 mechanism filling as the natural overlap zone. The full operational form of the bridge — including a $NiCl_2$(dpe) pilot study, the Stage 4 *Computational* filling-type sub-classification (Computational-Complete / Computational-Partial / Computational-Indirect), and the bidirectional intellectual contribution analysis (UQS structuring AFIR's truncation criterion; AFIR reaction-network graphs as a new form of evidence_chain) — is developed in the companion methodology paper.

SI-4: Stage 3 Judgments for Eight Historical Case-Study Papers

v18.4 EN baseline note (2026-05-10): This SI has been updated to anchor on the v18.4 EN cascade outputs generated for this paper in May 2026. Each of the nine papers (8 EP-filling + tamao2002 comparator) was re-executed at v18.4 EN (S0 → Turn 1-4 → S2_en → S3v2_en → S4v2_en) using Claude Sonnet. The per-paper Stage 3 findings are consistent with the v17 JA baseline; the source references in the reconciliation block now point to the `_v184_sonnet_s3v2_en.md` outputs. Cross-paper synthesis in §SI-4.10 was expanded in this phase with n=11 corpus-level statements; the VQ-hierarchy-inversion statement among them is withdrawn in the revision (SI-4.10, SI-3). Full execution record: `improvements/paperA_phase2_completion_report_20260510.md` (project record, not deposited).

This SI documents the Stage 3 root-issue analysis and EP-filling judgment for each of the eight papers selected as historical fillers of the Evidence Packs (EPs) derived from the Kumada (1972) analysis. Each paper was independently analyzed with the UQS v18.4 EN protocol (Stage 0 through Stage 2, Stage 3 v2, Stage 4 v2); EP-filling judgments cross-reference the EP dictionary established in SI-3 (EP-01a through EP-09a from the initial Stage 3 run, plus EP-02a and EP-10a, added in April 2026 from labels already listed in the Stage 3 prompt; SI-6.5).

Cross-reference to the EP-to-ChemOntology-AFIR parameter mapping: the deposited mapping table (`application_direction_chemontology_layer_a/ep_to_chemontology_mapping.md`; not part of the main text) categorizes each EP by its **computational addressability** under ChemOntology-AFIR: **direct** (EP-03a, EP-06a, EP-09a), **partial** (EP-04a, EP-05a, EP-07a), or **indirect / not directly computable** (EP-01a, EP-02a, EP-08a). The historical-literature filling judgments recorded in this SI and the computational addressability classification in that table are complementary: a paper can fill an EP experimentally (reported here) even when the EP is not computationally addressable, and an EP can be computationally tractable even when no single historical paper fills it (EP-08a / EP-09a; see SI-4's unfilled-status discussion). Readers comparing this SI with the mapping table should note that the SI uses the Stage 3 v1 EP attribution (EP-04a includes cross-vs-homo mass balance) while the mapping table uses the Stage 3 v2 attribution (EP-04a restricted to cis/trans-stilbene; EP-05a holds cross-vs-homo); the reconciliation note in SI-3's preamble explains both framings.

Two important caveats apply to the cross-walk. (The per-paper EP tables below are this cross-walk; Table 4 of the main text gives the exploratory historical Stage 4 labels, a separate layer, so the two can differ for the same paper — e.g. EP-07a for Tamao 1976.) First, each paper's own Stage 3 and Stage 4 outputs describe the *paper's own* EPs and filling map (what later work filled *its* absences), not the EPs of Kumada 1972. The question "which Kumada-1972 EPs does this paper fill" is therefore addressed here by an inference layer that reads the filling paper's experimental content against the Kumada-1972 EP requirements (as reformulated in SI-3), using Results Part 2 characterizations as corroboration. Second, the eight papers in the `papers/ep/kumada1972/` folder overlap only partly with the filling-paper candidates recommended in the Kumada 1972 Stage 4 map (Tamao 1976, Hayashi 1976, Corriu 1975, Kochi 1979, Kumada 1980; only Tamao 1976 is in both); the actual eight-paper set spans a broader window (1976–2024) and includes modern paradigm-shift fillers that the Kumada Stage 4 analysis could not anticipate.

Note on EP-07a "boundary" wording. Where the per-paper analyses below describe the Kumada-1972 "Only Csp^2 halides" boundary as *dissolved*, *crossed*, or *broken*, this denotes reaching catalytic Csp^3-electrophile scope under a new mechanistic regime — not the crossing of a fundamental boundary. The elementary feasibility of alkyl–alkyl (Csp^3–Csp^3) coupling on nickel was already documented in the Uchino/Yamamoto 1970 precedent cited by Kumada 1972 (reductive elimination of Et–Et from (bpy)Ni$(Et)_2$ on reaction with PhCl; SI-10). These EP-07a entries are accordingly catalytic scope extensions along a continuum and shifts in mechanistic regime, not boundary breakthroughs; the "only Csp^2 halides" statement bounds Kumada 1972's own catalytic protocol rather than asserting that alkyl–alkyl coupling is infeasible on nickel (Limitations L9).

v18.4 EN source files (2026-05-10): The canonical source for per-paper Stage 3 judgments is now `{paper}_v184_sonnet_`

`s3v2_en.md` in each paper's folder under `papers/ep/kumada1972/`. The reconciliation block at the end of this file retains the original v17 path entries as historical provenance; the v18.4 EN outputs are fully consistent with those entries and do not require revision of the per-paper findings.

SI-4.1 Tamao et al., Bull. Chem. Soc. Jpn. 1976 (Full Paper I)

Paper role in field development. This is the authors' own "Full Paper I" of the series announced by the 1972 JACS communication, expanding the preliminary report to a ~60-entry substrate scope table (Table 7) with catalyst screening across [Ni(dppp)Cl_2], [Ni(dmpe)Cl_2], and [Ni(PPh_3)$_2Cl_2$], plus halobenzene reactivity data, solvent effect data, and spectral characterization of products. It occupies the canonical "communication → Full Paper auto-resolution" slot identified in the Kumada 1972 Stage 3 §6, closing several of the reporting and scope absences that were structurally inevitable in the 1972 communication format.

Stage 3 judgment summary (EP-filling against Kumada 1972):

EP filled (Kumada 1972)	Filling type	Evidential basis (from this paper's Stage 4 §B and Stage 3 §4)	Boundary / partial-filling notes
EP-01a Reporting-coherence pack	Partial	Full Table 7 with consolidated GLPC methodology and reaction-time reporting; catalyst comparison Table 1; §4 Era-Norm flags B4/B5/B9 as 1976-tolerated but modern-deficient	Yield-basis heterogeneity (footnote t entries on Grignard basis vs. halide basis) persists as SRG-4 in this paper's own Stage 3; EP-01a is therefore partially filled with a residual reporting-integrity gap
EP-07a Boundary-test pack	Partial	Scope table extends to ~60 entries across primary/secondary alkyl, aryl, and alkenyl Grignard partners; includes explicit "most serious limitation" (alkyl Grignard + dihaloethylenes)	Sp^3 halide substrate testing still absent; the "Only Csp^2 halides" boundary of Kumada 1972 is *preserved* rather than dissolved; the paper introduces its own unsupported negative boundary claim (SRG-2 of this paper's Stage 3), which itself becomes a new absence
EP-05a Quantitative-ranking substantiation pack	Partial	Ligand ordering [Ni(dppp)Cl_2] > [Ni(dmpe)Cl_2] > [Ni(PPh_3)$_2Cl_2$] reported with substrate-class-specific recommendations; PhI > PhCl > PhBr "anomalous" halide reactivity order noted	Bite-angle/σ-donor separation not attempted; the Kumada-1972 CLA-2 question "is chelating diphosphine a *necessary* condition?" remains open

Extended analysis. SRG-1 of this paper's own Stage 3 (the "cycle-level import" of Scheme 1 from Yamamoto and Kochi reference reactions) mirrors Kumada 1972's SRG-1 almost directly—the mechanistic proposal remained a "working hypothesis" expanded into a fuller cycle rather than experimentally anchored within the dpe/dppp-Ni system. The paper's SRG-6 (no matched-pair benchmarking vs. Fe/Ag/Cu or Pd-phosphine systems) preserves the Kumada-1972 EP-09a absence in its entirety. Critically, the paper's SRG-2 (quantitative absence of data behind "most serious limitation" for alkyl-Grignard + dihaloethylene) constitutes a *new* RI-07 absence that did not exist in the 1972 communication—an instance of the "communication → Full Paper transition incompletely auto-resolves" pattern noted in the paper's Stage 3 §6. Thus, while Kumada 1972 EP-01a and EP-07a are partially filled here, new EPs (EP-07a in its "negative boundary claim" sub-form, v0.2a extension) are simultaneously opened.

SI-4.2 Monnereau et al., Adv. Synth. Catal. 2009

Paper role in field development. A Ni/calix[4]arene-diphosphine catalyst (cationic complex 2) achieves TOFs up to 21 250 h^{-1} in Kumada–Tamao–Corriu coupling, explicitly outperforming the benchmark [$NiCl_2$(dppp)] system in conversion under identical conditions (50.9% against 47.1%, a difference of 3.8 percentage points; the paper describes the benchmark's conversion as about 10% lower, a relative figure). The paper's role in the Kumada-1972 filling landscape is narrow but specific: it provides a *matched-pair diphosphine benchmark* on a single substrate (4-bromoanisole + PhMgBr) that Kumada 1972 explicitly did not provide (SRG-4 / CLA-2 of Kumada 1972).

Stage 3 judgment summary (EP-filling against Kumada 1972):

EP filled (Kumada 1972)	Filling type	Evidential basis	Boundary / partial-filling notes
EP-09a Internal benchmark pack	Partial	Direct head-to-head comparison of complex 2 vs. [$NiCl_2$(dppp)] on 4-bromoanisole / PhMgBr under identical conditions (Table 3 entry 1, 50.9% conversion, against the [$NiCl_2$(dppp)] run reported in the text on p. 1387, 47.1%)	One-to-one comparison only; dppb, dppf, DPEphos unaddressed (this paper's own SRG-2); no replicate measurements (n=1) so the 3.8-percentage-point gap is not statistically separated from measurement error; Kumada 1972's Corriu/Ni(acac)$_2$ comparison remains unfilled
EP-05a Quantitative-ranking substantiation pack	Partial	ArCl vs. ArBr activity ordering quantified (Table 2 entries 8–9); ligand-bite-angle vs. reactivity argument introduced via "fanning motion" hypothesis	Bite-angle–rate correlation is imported from Pd/Rh systems (this paper's SRG-3 RI-06: cycle-level import); not independently validated on Ni; the quantitative ranking required by Kumada-1972 CLA-2 (bite angle *vs.* yield for the full Ni ligand set) is not constructed
EP-03a Direct-observation pack	Attempted	X-ray structure of 2; variable-temperature ^{1}H and ^{31}P NMR dynamics study of the calix fan motion; solid-state bite angle 102.8°	Active catalytic species at 100 °C in dioxane is not identified (^{31}P species at 41.8 ppm uncharacterized); Ni oxidation state at turnover not determined; catalyst resting state unobserved

Extended analysis. This paper sharpens, rather than closes, the Kumada-1972 EP-06a absence: its proposed dynamic bite-angle mechanism is itself an instance of the cycle-level imported-model pattern (Pd/Rh data → Ni system) flagged in this paper's Stage 3 SRG-3, and its Stage 3 §6 records PRI-10 (kinetic overclaim) and the "labeling mismatch" pattern. For the Kumada-1972 filling question, the unambiguously filled EP is EP-09a at Partial level (first matched-pair dppp-vs.-new-ligand benchmark in the 37-year interval since Kumada 1972); EP-05a is filled only for the ArCl vs. ArBr ordering, not for the full ligand-activity ranking question that CLA-2 of Kumada 1972 raises.

SI-4.3 Breitenfeld, Wodrich & Hu, Organometallics 2014

Paper role in field development. Mechanistic study of Nickamine ([(MeN_2N)Ni-Cl])-catalyzed alkyl–aryl Kumada coupling, establishing a *bimetallic radical oxidative addition* mechanism through radical-clock probes, initial-rate kinetics, and DFT (PBE0-dDsC//M06). Transmetalation rather than oxidative addition is identified as the turnover-determining step. In the Kumada-1972 filling landscape, this paper corresponds to the Csp^3-substrate extension highlighted in Results Part 2 as one of the field's most consequential boundary breakthroughs.

Stage 3 judgment summary (EP-filling against Kumada 1972):

EP filled (Kumada 1972)	Filling type	Evidential basis	Boundary / partial-filling notes
EP-07a Boundary-test pack	Substantial (Paradigm-shift-adjacent)	Alkyl iodides (including secondary alkyl) coupled with ArMgCl under Nickamine catalysis; Csp^3 electrophiles absent from Kumada 1972's catalytic protocol are coupled under a different catalyst system (EP-07a wording note above; SI-10)	Extension operates under a *different mechanistic regime* (Ni(I)/Ni(III) radical pathway vs. Kumada 1972's proposed Ni(0)/Ni(II) two-electron cycle); the catalytic Csp^3 scope is reached but not within the original two-electron mechanism; per Results Part 2, this is characterized as Csp^3-scope extension via mechanism-switching
EP-06a Analogy-validation pack	Partial	Catalytic cycle (Figure 8) proposed for alkyl–aryl Kumada and related to the authors' earlier alkyl–alkyl mechanism (ref 20); radical clock excludes cage-rebound; kinetic orders measured	Cycle intermediates 10–16 uncharacterized (this paper's SRG-4: RI-03 + RI-06 co-occurrence); the validation is provided for *Nickamine* system, not the dpe/dppp-Ni system of Kumada 1972; thus filling is Indirect when evaluated against Kumada 1972's SRG-1

EP filled (Kumada 1972)	Filling type	Evidential basis	Boundary / partial-filling notes
EP-03a Direct-observation pack	Attempted	Isolated/NMR-characterized products 8 and 9; catalyst 1 structure established in ref 12	Catalytic cycle intermediates not observed within the system; Mirica 2014 (JACS 136, 6499) and Biswas/Weix 2013 and Lipschutz/Tilley 2014 provide independent Ni(III) aryl-alkyl observations that this paper's Stage 4 §B flags as bundled indirect fillers

Extended analysis. This is the first of the eight papers to cleanly fill the Csp^3-scope component of Kumada-1972 EP-07a. The paper also introduces PRI-11 (Mechanism-kinetics decoupling) in its own Stage 3 §6—the rate equation for the proposed bimetallic transmetalation is not algebraically derived from the cycle—which becomes a recurring pattern across the dataset (confirmed again in Pengfei Li 2024 and Mazet 2021; see SI-4.7 and SI-4.8). For Kumada-1972 EP-06a the filling is necessarily Partial because the Nickamine pincer system is mechanistically distinct from the 1972 dpe-Ni system: extending the radical-mechanism cycle to Nickamine does not validate the two-electron cycle originally proposed by Kumada.

SI-4.4 Cherney, Kadunce & Reisman, Chem. Rev. 2015

Review-exclusion status. Stage 0 records this paper as `format: review` (BIB-7), and the Stage 4 filling-paper methodology principle documented in methods_en.md Methods 2 explicitly excludes review articles from the filling-paper designation: "review articles are excluded from the filling-paper designation as a matter of principle." Cherney et al. 2015 is therefore *not* counted as a filling paper for any Kumada-1972 EP. The authoritative author-retrospective mini-review cited in Results Part 2 and Stage 2F is instead Tamao 2002 (J. Organomet. Chem. 653, 23–26), which lives in `papers/ep/discussion/tamao2002/`. Cherney et al. 2015 is retained in the eight-paper set as a *context document* for the enantioselective/enantiospecific taxonomy that frames several of the other Csp^3-extending fillers (Breitenfeld 2014, Molander 2016, Morken 2016, Pengfei Li 2024).

Stage 3 judgment summary. Per the review-exclusion principle, no Kumada-1972 EP is credited as filled by this paper. The paper's own Stage 3 analysis (see `cherney2015_v184_sonnet_s3v2_en.md`) identifies a CLA (D-4-5) concerning the internal inconsistency of the "enantiospecific" category definition (es ≡ 100%) with measured es values (65–99%), and flags RI-06 "cycle-level import" as manifested both in the DYKAT model (SRG-1) and in the Ni/Ir photoredox cycle (SRG-4). These are properties of the review's own architecture and do not constitute filling events for Kumada 1972.

Extended analysis. The review's value for the Kumada-1972 cross-walk is twofold: (i) it provides the systematic literature classification against which individual filling papers (Breitenfeld 2014, Molander 2016, Morken 2016) can be located within the "stereogenic C–C bond-forming event" taxonomy, and (ii) its own coverage gaps—particularly the 1972–1974 Kumada-series KR/DYKAT ambiguity flagged as B-8—corroborate the Kumada-1972 SRG-6 absence (cis/trans selectivity with no kinetic/thermodynamic discrimination) as a 40-year unfilled question. Three filling-paper instances within the eight-paper set (Breitenfeld 2014 alkyl-aryl; Morken 2016 cyclic-sulfate enantioselective; Molander 2016 paradigm-shift) are all framed in Results Part 2 against the enantioselective/enantiospecific classification this review consolidates.

SI-4.5 Tellis, Kelly, Primer, Jouffroy, Patel & Molander, Acc. Chem. Res. 2016

Paper role in field development. This Account presents the "single-electron transmetalation via photoredox/Ni dual catalysis" paradigm, the Results-Part-3 archetype of the *paradigm-shift* filling of Kumada-1972 EP-07a. By inverting the $Csp > Csp^2 > Csp^3$ reactivity hierarchy of the classical two-electron regime (making Csp^3 *most* reactive under single-electron activation), it crosses the Kumada-1972 boundary via mechanism-switching rather than within the original two-electron framework.

Stage 3 judgment summary (EP-filling against Kumada 1972):

EP filled (Kumada 1972)	Filling type	Evidential basis	Boundary / partial-filling notes
EP-07a Boundary-test pack	Paradigm-shift	Substrate scope across organotrifluoroborates and alkylbis(catecholato)silicates; the "Only Csp^2 halides" Kumada-1972 restriction is not merely crossed for Csp^3-electrophiles but structurally inverted for Csp^3-*nucleophiles* via radical generation	Filling operates on a different mechanistic axis (single-electron) than Kumada 1972 intended (two-electron); the Kumada-1972 EP-07a as originally formulated (sp^3-halide testing *within* the two-electron cycle) remains formally unfilled

EP filled (Kumada 1972)	Filling type	Evidential basis	Boundary / partial-filling notes
EP-06a Analogy-validation pack	Attempted	Photoredox/Ni catalytic cycle proposed with DFT (SMD-water-(U)M06/6-311+G(d,p)//UB3LYP/6-31G(d))	Mechanistic conclusions (pathway B dominance, DKR operation) derived on benzyl-radical + bromobenzene model systems; this paper's Stage 3 SRG-2 records RI-06 "cycle-level import" for the extrapolation to secondary unactivated alkyl systems; Kumada-1972 EP-06a (validation of the *1972* Ni(0)/Ni(II) cycle) is not touched
EP-03a Direct-observation pack	Attempted	Cyclic voltammetry of oxidation potentials; computed potential-energy surface (Figure 9)	No EPR/in situ UV-Vis of Ni(I) or Ni(III); no Stern-Volmer; no quantum yield; this paper's own SRG-1 flags RI-03 as fatal for mechanistic VQ-3

Extended analysis. Results Part 2 locates this paper as the canonical paradigm-shift filler: "Molander 2016's photoredox/Ni dual catalysis transformed the paradigm itself."† The Stage 4 §B record, however, shows that almost all the *mechanistic* EPs proposed for the Molander system itself (EP-03a, EP-06a, EP-09a) remain at best Partially filled through distributed follow-up work (Primer et al. 2015 JACS 137, 2195; Shields & Doyle 2016 JACS 138, 12719; Gutierrez et al. 2015 JACS 137, 4896). In the Kumada-1972 cross-walk the cleanly filled EP is EP-07a; the Kumada-1972 EP-09a (matched-pair benchmark against prior Pd-Suzuki/Kumada on common substrates) is *not* filled, as this paper's own SRG-4 (Dropped Quantitative Benchmark) preserves precisely the same absence.

SI-4.6 Eno, Lu & Morken, JACS 2016

Paper role in field development. Ni-catalyzed asymmetric Kumada coupling of *symmetric (prochiral) cyclic sulfates* with aromatic Grignard reagents ($Ni(acac)_2$ / (R,R)-L5 PyBox). The paper constructs tertiary hydrocarbon stereocenters at up to 96:4 er; mechanistic probes (negative radical clock, 90% diastereospecificity with isotopically labeled substrate) are used to argue for an SN2-like stereoinvertive oxidative addition. In the Kumada-1972 filling landscape, Results Part 2 places it in the Csp^3 expansion cluster but identifies its *primary* Kumada-1972 filling contribution as EP-08a at Minimal level—the paper's reported practice of averaging two experiments per scope entry is the first explicit report-of-uncertainty gesture in the sequence.

Stage 3 judgment summary (EP-filling against Kumada 1972):

EP filled (Kumada 1972)	Filling type	Evidential basis	Boundary / partial-filling notes
EP-08a Replication / uncertainty pack	Minimal	SFC er values reported as "average of two experiments" (as characterized in Results Part 2); this is the first explicit replicate practice among the filling papers	Only n=2; no standard deviation; no per-entry individual values; SFC analytical conditions (column, mobile phase, UV wavelength) are not reported at all (SRG-5 of this paper's Stage 3), so even the minimal replicate practice lacks traceable analytical framing
EP-07a Boundary-test pack	Partial (Csp^3 extension)	17-entry scope of prochiral cyclic sulfates with aryl Grignards; construction of tertiary hydrocarbon stereocenters	"Current catalyst system is ineffective at transforming substrates that would deliver tertiary alcohol derivatives or quaternary centers"; ortho-ArMgBr and alkyl/alkenyl Grignards give "lower yield and nearly racemic products"—these failure boundaries are asserted qualitatively, without the kind of "failure-substrate quantification" that the v0.2a extension of RI-07 would require

EP filled (Kumada 1972)	Filling type	Evidential basis	Boundary / partial-filling notes
EP-06a Analogy-validation pack	Attempted	Scheme 4 catalytic cycle; assumption that C–C reductive elimination proceeds with retention of configuration (Hayashi/Kumada 1976, ref 1c); proposed SN2-like oxidative addition	This paper's Stage 3 SRG-2 records this as RI-06 with a labeling-mismatch pattern: "Assuming…" hedge is collapsed into the fact-narrative "Mechanistic experiments point to…" in the Abstract; the Kumada-1972 EP-06a absence (stoichiometric validation of the dpe-Ni cycle) is not touched

Extended analysis. The primary Results-Part-3 characterization—"Morken 2016 is one example of Csp^3 expansion from the same period, but its core novelty lies in enantioselectivity; the Stage 2F integrative analysis judged Morken 2016's primary filling contribution as a Minimal contribution to EP-08a (replicate measurement)"†—is consistent with the Stage 4 judgment that the analytical-reporting substrate layer (SRG-5, RI-01/RI-08) remains the strongest residual absence. The 44% mass-balance deficit in the radical-clock experiment (SRG-1, B-1/U-1) instantiates the "negative-evidence incompleteness" RI-04 pattern and thus *worsens* the EP-04a absence of Kumada 1972 rather than filling it. Cross-walked against Kumada 1972 the filling is narrow (EP-08a Minimal, EP-07a Partial for the Csp^3 tertiary-center domain only) but historically pivotal as the first instance of explicit replicate-count reporting in the sequence.

SI-4.7 Poisson, Tran, Besnard & Mazet, ACS Catal. 2021 (dppe- vs. dmpe-Ni comparative mechanism)

Paper role in field development. Comparative mechanistic study of [(dppe)$NiBr_2$] and [(dmpe)$NiBr_2$] as precatalysts for Kumada vinylation of enol phosphates. Uses stoichiometric organometallic synthesis, single-crystal X-ray of three intermediates (7c, 7d, 9c; CCDC 2116218–2116220), low-temperature ^{31}P NMR monitoring, radical-clock and TEMPO trapping, and in situ IR-based VTNA for reaction-order determination. Both catalysts are placed on Ni(0)/Ni(II) manifolds (Ni(I)/Ni(III) ruled out), and the dppe cycle is "established" while the dmpe binuclear transmetalation cycle is offered as a hypothesis. Results Part 2 identifies this paper as the 49-year partial fill of the Kumada-1972 mechanistic EPs (EP-06a, EP-03a).

Stage 3 judgment summary (EP-filling against Kumada 1972):

EP filled (Kumada 1972)	Filling type	Evidential basis	Boundary / partial-filling notes
EP-06a Analogy-validation pack	Partial	Ni(0)/Ni(II) cycle for dppe-Ni experimentally established via stoichiometric intermediate characterization; ion-exchange step (phosphate/bromide) identified as distinctive for dppe; Ni(I)/Ni(III) radical path excluded by radical-clock + TEMPO + absence of paramagnetic Ni(I) signal	Filling applies specifically to dppe-Ni vinylation of enol phosphates, not to the dpe-Ni system of Kumada 1972 with aryl chlorides; cycle-level validation for the *exact* 1972 system remains absent; dmpe proposal remains a hypothesis with RI-03/PRI-11 residuals (this paper's SRG-2)
EP-03a Direct-observation / state-assignment pack	Partial	X-ray structures of Ni(0)-butadiene, oxidative-addition, and product-bound Ni(0) intermediates (7c, 7d, 9c); low-temperature ^{31}P NMR identification of the dppe catalyst resting state	Dmpe resting state "not met with success" (this paper's SRG-2, RI-03 fatal for VQ-3 of the dmpe claim); η^1-O vs. η^2-O,O' coordination mode in 9c not reported in main text despite being extractable from CCDC 2116220 (U2 "Available-but-unutilized characterization" sub-pattern)
EP-04a Discrimination pack	Partial	Radical-clock substrate 1g (negative result); TEMPO trapping (negative); VTNA-derived kinetic orders distinguish dppe (substrate zero-order, catalyst first-order) from dmpe (catalyst apparent-second-order)	No within-condition replicates, no goodness-of-fit test, no correction for catalyst decomposition—this paper's SRG-1 (RI-04 + RI-08 co-occurrence); discrimination is not statistically decisive

Extended analysis. Results Part 2 characterizes this paper as: "Mazet 2021 experimentally established the Ni(0)/Ni(II) catalytic cycle for the first time in the dppe-Ni system, providing direct experimental grounding for the original two-electron catalytic cycle that Kumada 1972 proposed as a 'working hypothesis.' "† The 49-year delay reflects not neglect but the structural

preconditions (low-temperature NMR, in situ IR with VTNA, refined X-ray methods, appropriate model-system selection). The run assigned PRI-11, a pattern the Stage 3 prompt already described; this is an application of a prompt-supplied label, not an independent confirmation, and it is not used to promote PRI-11 (SI-6.5). For the Kumada-1972 EP-06a cross-walk the filling is Partial rather than Complete because the substrate and ligand systems differ (enol phosphates, dppe/dmpe) from the 1972 aryl-chloride + dpe-Ni system.

SI-4.8 Li et al. (Qiu group), Nat. Catal. 2024 (Ni-electrocatalysed Csp^3–Csp^3 eXEC)

Paper identification. Results Part 2 refers to this paper as "Li 2024"; the Stage 0 record confirms the first author as Pengfei Li, with Youai Qiu as corresponding author, so the folder naming (`pengfei2024/`) indexes the first author's given name while the main-manuscript short form uses the surname. Published as Nat. Catal. 7, 412–421 (2024); DOI: 10.1038/s41929-024-01118-3.

Paper role in field development. First general Ni-electrocatalysed cross-electrophile coupling (eXEC) of *unactivated* alkyl halides, covering primary–primary, primary–secondary, primary–tertiary, and secondary–secondary combinations with >90 scope substrates. Prior eXEC (Lin group) was limited to activated alkyl halides via SN2; prior reductive XEC (Gong) required stoichiometric reductants. Results Part 2 pairs it with Breitenfeld 2014 as a Csp^3 extender of the Kumada-1972 scope boundary, with the added paradigm-shift of electrochemical single-electron activation removing the stoichiometric reductant.

Stage 3 judgment summary (EP-filling against Kumada 1972):

EP filled (Kumada 1972)	Filling type	Evidential basis	Boundary / partial-filling notes
EP-07a Boundary-test pack	Paradigm-shift	Unactivated Csp^3 halides coupled across all four combinatorial classes; Csp^3 electrophiles are coupled via electrochemical single-electron activation (EP-07a wording note; SI-10)	Filling operates under Ni(I/II/III) radical cycle, not the Kumada-1972 two-electron cycle; analogous in kind to the Breitenfeld 2014 mechanism-switching extension but via an electrochemical rather than photoredox or radical-ligand channel
EP-03a Direct-observation / state-assignment pack	Attempted	CV and square-wave voltammetry of catalyst and substrates; DFT-computed redox potentials; TEMPO, cyclopropylmethyl ring-opening, and radical-substrate racemization	Operating cathode potential during 5 mA constant-current electrolysis not measured (SRG-2, RI-03); Path I (Ni(0)→radical) vs. Path II (Ni(II)→Ni(I)→Ni(III)) not discriminated (SRG-3); DFT vs. SWV 360-mV discrepancy not discussed (SRG-4, RI-04)
EP-06a Analogy-validation pack	Attempted	Proposed Ni(I/II/III) cycle with DFT-computed reductive-elimination barriers	This paper's SRG-4 records RI-06 + RI-04 co-occurrence: DFT method (functional/basis/dispersion/solvation) is not fully specified and not validated against the target Ni-terpyridine/DMF system; GitHub-released kinetic-simulation code exists but no algebraic rate law is derived from the proposed cycle (PRI-11, a pattern described in the Stage 3 prompt; not an independent instance, SI-6.5)

Extended analysis. Per Results Part 2: "The reaction that Kumada 1972 restricted to Csp^2 halides was extended to Csp^3 halides at a Substantial level by Breitenfeld 2014 and Li 2024…"† For the Kumada-1972 cross-walk the clean filling is EP-07a at paradigm-shift level (different mechanism, different activation mode, but the scope boundary dissolves). The paper's own Stage 3 analysis shows that the mechanistic EPs proposed for the *electrochemical* system (Path I/II discrimination, operating potential measurement, DFT validation) are themselves not filled, repeating the SRG-4 structure of Kumada 1972 (unsupported ligand/mechanism ranking) with modern reagents. The paper also illustrates the "self-preceded filling" pattern: the authors' own earlier paper (Li et al., Nat. Commun. 2022, 13, 3774) provides primary-halide radical evidence that could have filled this paper's SRG-5 but was not ported across, showing that EP-filling opportunities within an author's own prior work can remain structurally unexplored.

SI-4.9 Filling-Paper Candidates (1973–2020) from a GPT-5.4 Thinking Stage 4 Run (2026-04-19)

This subsection documents filling-paper identifications that were obtained by running the UQS v17 pipeline on Kumada 1972 with GPT-5.4 Thinking (2026-04-19, archived at `papers/dev/kumada/gpt54thinking/`). The GPT-5.4 run covered all six SRG roots of the Sonnet record (SI-6.10) and, unlike the Sonnet Stage 4 runs, named specific historical filling papers with DOIs. The run had web search enabled (Methods 2, "Retrieval and verification"), so its candidates are not parametric recall alone, and whether the difference reflects the models' coverage of the 1970s–1980s literature was not tested.

The filling-paper candidates below (five of them pre-1985) **supplement** the eight-paper set analyzed in SI-4.1–SI-4.8 (Tamao 1976 in SI-4.1 and the seven Layer-2 papers in SI-4.2–SI-4.8). They are not analyzed as standalone Stage 3 subjects in this SI. They are candidates proposed by one model run, and the table records that run's assignments; it is not a verified filling map. Candidate-level checks of some entries are recorded in SI-9.10 (record sets B3 and B4; the B4 record is `01_response/r33b_candidates/records/stage4_literature_update_20260419.md` in the deposit).

Filling-paper candidates (as assigned by the GPT-5.4 Thinking run; filling type as recorded in its output, except as noted below the table):

Paper	Kumada-1972 EPs filled	Filling type	Year
Tamao-Zembayashi-Kiso-Kumada, *J. Organomet. Chem.* 1973, 55, C91–C94	EP-04a (cis/trans kinetic vs thermodynamic), EP-10a (kinetic verification)	Partial	1973
Kiso, Tamao, Kumada, *J. Organomet. Chem.* 1973, 50, C12–C14	EP-10a (halide-dependent isomerization, kinetic proxy)	Indirect	1973
Morrell & Kochi, *J. Am. Chem. Soc.* 1975, 97, 7262–7270	EP-03a (arylalkyl-Ni intermediates), EP-06a (analogy-validation for imported organonickel cycle)	Indirect (EP-03a); Partial (EP-06a)	1975
Tamao et al., *Bull. Chem. Soc. Jpn.* 1976, 49, 1958–1969	EP-01a, EP-02a, EP-07a (all self-filling Full Paper I), EP-09a (partial catalyst-class benchmark)	Partial	1976
Tamao et al., *Tetrahedron* 1982, 38, 3347–3354	EP-02a (identity), EP-07a (boundary, heterocyclic extension)	Partial	1982
Chen, Lucas, Greene, Zhang, Tollefson, Erickson, Taylor, Jarvo, Hong, *J. Am. Chem. Soc.* 2019, 141, 5835–5855	EP-06a (chemoselectivity and stereospecificity unified via combined computational + experimental study), EP-10a (kinetic verification)	Indirect	2019
Dawson et al., *Chem. Eur. J.* 2020, 26, 3044–3048	EP-03a (spectroscopic state-assignment for Kumada system), EP-06a	Indirect	2020

Note on filling types. The run records one filling type for each EP in its per-EP section (§B). Three pairings in the table — EP-10a for Tamao et al. 1973 (*J. Organomet. Chem.* 55, C91), and EP-06a for both Chen 2019 and Dawson 2020 — appear only in the run's recommendation list (§C), which gives no filling type; the type shown for these papers is the one the run records for their other EP (EP-04a, EP-10a and EP-03a respectively). The recommendation list also pairs Kiso et al. 1973 with EP-04a; that pairing is not tabulated.

Confidence labels of the GPT-5.4 run compared with the original Sonnet Stage 4 (labels as recorded by the models; they are not verification results):

- EP-01a: Medium → **High** (Tamao 1976 BCSJ identified specifically).
- EP-02a: Low → **Medium** (Tamao 1976 + Tamao 1982 *Tetrahedron* chain).
- EP-03a: Low → **High** (Morrell-Kochi 1975 + Dawson 2020).
- EP-04a: Unidentified → **High** (Tamao-Zembayashi-Kiso-Kumada 1973 *J. Organomet. Chem.*; a 1973 self-filling paper that the original Stage 4 declared unfillable on incentive-structure grounds but GPT-5.4 located).
- EP-06a: Low → **High** (Morrell-Kochi 1975 + Chen 2019).
- EP-07a: Medium → **High** (Tamao 1976 BCSJ title explicitly "General Scope and Limitations").
- EP-10a: new EP not in original Stage 4 list — assigned by GPT-5.4 Stage 3 from the provisional label in its prompt, tagged to PRI-10 (Kinetic overclaim); related to the PRI-11 pattern first proposed on Breitenfeld 2014 (SI-6.5). Starting point Kiso-Tamao-Kumada 1973; establishing point Chen 2019.

EPs for which neither run named a filling paper:

- EP-08a (Replication / uncertainty pack, RI-08) — neither run named a filling paper (the GPT-5.4 run labelled the filling type Partial, with low confidence, but named no paper). The label is not stable across runs: in the twelve v18.4 Kumada Stage 4 runs EP-08a is labelled Partial or better in 7 (SI-3, Stage 4 re-count). Agreement between two single runs is therefore not used as evidence; the non-detection of EP-08a rests on the locked WoS audit (SI-9.8).
- EP-09a (strict same-axis Ni(acac)$_2$ vs NiCl$_2$(dpe) benchmark) — Tamao 1976 BCSJ provides partial catalyst-class-level benchmarking; no same-axis head-to-head paper was named by either run (the non-detection rests on the WoS audit, SI-9.8).

Relation to Fig. 2 and Results Part 2. Fig. 2 dates the first Partial filling of the system-matched mechanistic EPs (VQ-3) to Mazet 2021, 49 years after Kumada 1972. Morrell–Kochi 1975 concerns organonickel intermediates in a related system and is therefore not counted as system-matched filling; the 2019–2021 papers (Chen, Dawson, Mazet) are the system-specific filling of EP-03a / EP-06a.

Reader note: the eight-paper case-study set analyzed in SI-4.1–SI-4.8 (Tamao 1976 plus seven Layer-2 papers) emphasizes the paradigm-shift-heavy 2009–2024 slice (photoredox, radical, enantioselective, electrochemical extensions). The pre-1985 papers listed above are the "conventional-lineage" filling of the original Kumada-1972 scope-boundary / mechanistic / reporting absences. Results Part 2 originally characterized the case-study set as the entire filling landscape; a more accurate framing after this update is that the landscape has two layers — a 1973–1982 conventional self-filling cluster (the paper set of SI-4.9) plus a 2009–2024 paradigm-shift-driven cluster (the papers of SI-4.2–SI-4.8) — with a gap of about 27 years (1982–2009; "~30 years" in Results Part 2) during which organometallic cross-coupling work diversified toward Pd (Suzuki, Negishi, Stille) and relatively little new Ni-specific EP filling occurred.

SI-4.10 Cross-Paper Findings from the v18.4 EN Baseline (2026-05-10)

This subsection consolidates corpus-level patterns that emerged from running the full v18.4 EN cascade across all 9 papers (8 EP-filling + tamao2002 comparator) in May 2026, combined with the v18.4 multi-run outputs for Kumada 1972 and Murai 1993 (dev set). Full execution record: `improvements/paperA_phase2_completion_report_20260510.md` (project record, not deposited).

VQ-hierarchy inversion across the 11-paper corpus (revised in response to review)

The submitted version of this subsection stated that all 11 papers exhibit the same VQ-hierarchy inversion, that EP-08a was unfilled in all 11 papers at the Stage 4 level, and that this n = 11 convergence was the primary empirical anchor of the incentive reading. **These statements are withdrawn.** Classified from each paper's per-EP Stage 4 filling labels under definitions fixed before counting (SI-3), the inversion holds in 6 of the 11 papers when Complete or Partial counts as filled, in none at the Complete-only threshold, and in 1 when Indirect fillings also count; in four of the five papers without it (cherney2015, molander2016, molken2016, tamao2002), a VQ-4 EP is labelled Partial or better. Outside the development set each paper was analysed in a single run, and two of the 11 are reviews. For Kumada 1972, the non-detection of EP-08a and EP-09a rests on the locked Web of Science audit (SI-9.8), not on this corpus-level pattern.

PRI-11 (mechanism-kinetics decoupling): five papers on which the pattern was applied

PRI-11 — the pattern in which a catalytic cycle is proposed with a turnover-determining step assignment but no algebraic rate law is derived from the cycle to verify consistency with the measured kinetics — was applied in the Stage 3 analyses of the following papers. *(R1 correction: these were described as independent confirmations. Except the proposing run on Breitenfeld 2014 under Stage 3 v1, they ran under Stage 3 prompts that describe PRI-11, naming Breitenfeld 2014 and Pengfei 2024 (SI-6.5), so they are applications of the pattern, not independent confirmations.)*

Paper	Stage 3 locus	System
Breitenfeld 2014	SRG-4 §6 (proposing paper)	Nickamine Ni(I)/Ni(III) radical TDS = transmetalation
Pengfei Li 2024	CLA-2 + §6	Ni-electrocatalytic Ni(I/II/III) eXEC
Mazet 2021	SRG-2 (dmpe proposal)	dmpe-Ni binuclear transmetalation hypothesis
Cherney 2015	SRG-1 (DYKAT cycle)	Ni/photoredox dual-catalysis cycle
Tamao 2002	SRG-1 (retrospective)	Ni(0)/Ni(II) 2-electron cycle (historical)

The pattern fits five papers across mechanistic, electrocatalytic and retrospective contexts. Whether it would be found without

the prompt's description is untested. Codification in a later prompt version is a possible next step; it is not part of the prompts used in this paper.

PRI-12 candidate (self-reinforcing computational loop / field-consensus overread): n=2 anchor

PRI-12 — the pattern in which a computational model or paradigm-defining review becomes a self-referential authority that suppresses alternative evidentiary paths — is anchored at n=2 in this corpus:

Paper	Manifestation
Cherney 2015	Review-as-paradigm-crystallization: the enantiospecific/enantioselective taxonomy becomes a definitional authority that forecloses re-examination of the 1972–1974 Kumada-series KR/DYKAT ambiguity
Molander 2016	PES re-use across 9 substrate classes without isomorphism test: the photoredox/Ni dual-catalysis PES computed for benzyl radical + PhBr is imported to secondary unactivated alkyl systems without independent validation

PRI-12 requires n=3 independent confirmations for promotion. Current status: n=2 anchor, **candidate** only. Potential n=3 source: additional review papers in the corpus or a Stage 3 run on a Pd-paradigm paper showing analogous field-consensus crystallization.

Pattern RP-5 (imported model without isomorphism): v18.4 EN surfaces more cleanly than v17

A new pattern identified in the v18.4 EN runs but not prominent in the v17 JA baseline: RP-5 (imported mechanistic model without isomorphism demonstration) was detected in 3 of 4 G3 papers (monnereau2009, molander2016, molken2016) in the Stage 2F sub-group integration. The v18.4 prompt elaborations for RI-06 appear to elicit more granular identification of model-import instances than v17, producing a cleaner signal for this pattern. This is a potential v18.4 sensitivity gain requiring cross-language verification (not done here).

SI-4.11 Tamao 2002: the ten characteristic features and the VQ levels (R1)

Results Part 2 compares the VQ hierarchy with the author group's own retrospective. Section 2 of Tamao's 2002 mini-review (*J. Organomet. Chem.* 653, 23–26) lists ten characteristic features of the nickel–phosphine-catalysed Grignard cross-coupling. The authors assigned each feature to the VQ level of the documentation it reports (paraphrased; the numbering is Tamao's):

#	Feature (paraphrase)	What is documented	VQ level
1	The 1972 communication proposed the catalytic cycle (now generalised as oxidative addition – transmetallation – reductive elimination)	A proposed cycle, restated; no verifying evidence is given	VQ-3 topic, proposal only
2	First use of transition-metal–phosphine complexes for Kharasch-type Grignard reactions ("molecular catalysis")	Method	VQ-2
3	Activity depends on the phosphine; a general order of the $NiCl_2$(diphosphine) catalysts	Activity order stated qualitatively, without the ranking data that EP-05a requires	VQ-3 topic, observation only
4	Alkyl Grignard reagents with β-hydrogens couple selectively	Method / scope	VQ-2
5	sec-Alkyl Grignard reagents isomerise to primary, depending on the phosphine and the halide	Selectivity reported as an observation, without discrimination or mechanistic evidence	VQ-3 topic, observation only
6	Asymmetric coupling is possible with optically active phosphines	Method	VQ-2
7	Chlorobenzenes are the halides of first choice; fluorobenzene reacts slowly; vinyl chloride is very reactive (industrial use)	Method / scope	VQ-2

#	Feature (paraphrase)	What is documented	VQ level
8	Alkenyl-halide geometry is retained	Stereoselectivity reported as an observation, without discrimination or mechanistic evidence	VQ-3 topic, observation only
9	Ni catalysts are generally more reactive than Pd toward C(sp^2)–halides, which allows stepwise coupling of dihalides	Qualitative comparison, no quantitative data	VQ-2
10	First practical route to heteroarene homo- and mixed oligomers	Method / application	VQ-2

Six features report methods or scope (VQ-2). Four touch VQ-3 topics — the 1972 catalytic-cycle proposal (feature 1), an activity order (3) and two selectivity observations (5, 8) — but each as a proposal or an observation, without the discriminating or mechanistic evidence that VQ-3 documentation requires. None concerns quantitative reliability or replication (VQ-4). No feature concerns identity or purity (VQ-1). The assignment is the authors' reading of the paper, made against the VQ definitions of Methods 1.2 (VQ-1 identity/purity, VQ-2 methodology/scope, VQ-3 mechanism/selectivity, VQ-4 quantitative reliability/replication); it was not produced by a UQS run. The UQS records of Tamao 2002 are in `papers/ep/discussion/tamao2002/` (Tier 3 of the deposit, `ep_historical_case_study/tamao2002_minireview/`).

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† Quotations marked with † were originally generated in Japanese by the UQS pipeline and have been translated into English for this Supporting Information. The original Japanese outputs are preserved in the archived analysis files.

SI-5: Full Stage 2F Integrative Analysis

> **v18.4 baseline note (2026-05-10).** This section has been updated to anchor on the v18.4 EN Opus 4.7 Stage 2F baseline run for this paper in May 2026. Primary source files: `papers/ep/kumada1972/kumada1972_{g1,g2,g3,all}_v184_opus47_s2fv71.md` (all-group integration and three sub-group outputs). Model: Opus 4.7 (was Opus 4.6). Prompt baseline: v18.4 EN-only Sonnet cascade (S0 → Turn 1–4 → S2_en → S3v2_en) as sub-group input (was v17 JA-mixed). Complete execution record: `improvements/paperA_phase2_completion_report_20260510.md` (project record, not deposited). IP count expanded from 4 (v17 all-group) to 7 (v18.4 EN all-group integration mode). v6 and v7 Stage 2F outputs are retained as historical traceability; v7.1 is the primary cited version for all empirical claims.

This section documents the Stage 2F integrative analysis across the nine-paper corpus formed by Kumada 1972 and the eight EP-filling papers. The primary source is the all-group Stage 2F output (`kumada1972_all_v184_opus47_s2fv71.md`), with sub-group Stage 2F outputs (G1, G2, G3) providing the raw per-sub-group evidence. Terminology in this SI follows Methods: Stage 2F is the cross-paper comparative analysis stage; Independent-occurrence Patterns (IPs) are recurring structural features that emerged independently in each sub-group; Canonical Recurring Roots (CRRs) are the same_root_groups consolidated at the corpus level; the `group_unasked_question` is the single integrated question that the sub-group-level `group_unasked_questions` were consolidated into. The `recurring_roots` terminology used in Stage 2F outputs is retained verbatim where quoted from source.

SI-5.1 Subgroup Structure

The nine-paper corpus was partitioned into three sub-groups so that Stage 2F could detect patterns that emerge *independently* in each sub-group rather than being seeded by the shared reference paper (Kumada 1972) alone. Each sub-group was defined by a single intrinsic organizing axis so that the patterns that survive across all three partitions can be attributed to the field rather than to one thematic lens. Kumada 1972 is placed as a shared reference paper in all three sub-groups.

G1 — Direct lineage (3 papers). The diphosphine–Ni system developed by the Kumada group and its direct mechanistic heirs.

- Kumada 1972 (JACS Communication): method discovery, $NiCl_2$(dpe) catalyst.
- Kumada 1976b (BCSJ Full Paper): scope and limitations of the same $NiCl_2$(dppp)/$NiCl_2$(dppe) system.
- Mazet 2021 (ACS Catal.): comparative mechanistic study of Ni(dppe) vs Ni(dmpe) cycles.

Rationale: the three papers operate within a single catalytic manifold (Ni(0)/Ni(II) with a chelating diphosphine ligand and a Grignard nucleophile), spanning 49 years but not switching activation modes. Patterns that recur in G1 reflect persistence of gaps within a single paradigm.

G2 — Boundary breakthrough (4 papers). Papers that extended Kumada-type coupling across the $Csp^2 \rightarrow Csp^3$ boundary that Kumada 1972 explicitly excluded.

- Kumada 1972 (JACS Communication): foundational paper stating the Csp^2 restriction.
- Breitenfeld 2014 (Organometallics): Ni-pincer-catalyzed alkyl(sp^3)–aryl Kumada coupling via radical bimetallic oxidative addition.
- Cherney 2015 (Chem. Rev.): enantioselective/enantiospecific cross-coupling review documenting Csp^3 as the frontier.
- Pengfei 2024 (Nature Catalysis): Ni-electrocatalyzed Csp^3–Csp^3 cross-electrophile coupling of unactivated alkyl halides.

Rationale: these papers span the paradigm shift from two-electron Ni(0)/Ni(II) catalysis to one-electron radical Ni(I)/Ni(II)/Ni(III) or electrochemical catalysis. Patterns that recur across this shift indicate field-level conventions that are not tied to a specific activation mode.

G3 — Methodological expansion (4 papers). Papers that extended Kumada-type coupling along axes of ligand, substrate, and condition without necessarily shifting paradigm.

- Kumada 1972 (JACS Communication): foundational paper with dppp/dppe.
- Monnereau 2009 (Adv. Synth. Catal.): calixarene-diphosphine Ni catalyst for high-TOF Kumada coupling.
- Molander 2016 (Acc. Chem. Res.): photoredox/Ni dual catalysis for Csp^3–Csp^2 cross-coupling.
- Morken 2016 (JACS Communication): Ni-catalyzed asymmetric Kumada cross-coupling of prochiral cyclic sulfates.

Rationale: unlike G2, the three expansion axes in G3 coexist rather than succeed each other. Patterns that recur in G3 reflect structural gaps that persist regardless of which expansion axis a paper pursues.

Total unique papers across the three sub-groups: 9 (Kumada 1972, Kumada 1976b, Mazet 2021, Breitenfeld 2014, Cherney 2015, Pengfei 2024, Monnereau 2009, Molander 2016, Morken 2016).

SI-5.2 Independent-Occurrence Patterns (IPs)

The all-group Stage 2F v7.1 integration output reports **seven IPs** at the integration level. An IP is defined as a structural pattern identified independently in all three sub-groups (or in at least two of the three, for IP-5 and IP-7) without being seeded by the shared reference paper. Their independent emergence across sub-groups with different thematic foci (lineage, boundary, expansion) is consistent with field-wide structural conventions; it comes from single integration runs and is not tested on other corpora. The seven IPs reported in `kumada1972_all_v184_opus47_s2fv71.md` are reproduced below, supplemented by the per-sub-group citations that gave rise to each IP assignment. (The v17 Opus 4.6 all-group output reported four IPs; the expansion to seven reflects the v18.4 EN prompt elaborations that surface imported-model and DFT-discriminant patterns more cleanly than v17.)

Note on framing. SI-5.2 below preserves the **absence-centric framing** used by the Stage 2F source file ("without validation," "underreported," "self-contradictory absence," "asserted without matched-pair")—that framing is the native language of the UQS methodology, which identifies absences as its primary analytical output. Results Part 3 of the main manuscript describes the *same* patterns using a **developmental framing** ("stepwise deepening," "evolution of analytical infrastructure," "handoff to the next researcher," "opportunity for systematic benchmarking") that fits the publication-ready narrative of how the field has moved forward. The two framings are complementary: the absence-centric framing makes visible the structural gaps that UQS detects, and the developmental framing shows how those gaps have served as implicit hand-off points between generations of research. Readers who want the rigorous UQS framing should use SI-5.2; readers who want the developmental narrative should use Results Part 3.

IP-1: Catalytic cycle proposed without elementary step validation. All nine papers propose a Ni-based catalytic cycle at some level of specificity, yet in none of the nine is a complete set of elementary steps experimentally validated within the actual operating catalytic system under catalytic conditions. The pattern appeared as a repeated pattern in every sub-group (RP-1 in G1, RP-1 in G2, RP-1 in G3), which is why it is listed as the first IP.

- G1: RP-1 covers Kumada 1972 (intermediates 1/2/3 in eqs 2–4 neither isolated nor observed; cycle assembled from dipy-Ni stoichiometric analogy), Kumada 1976b (Scheme 1 acknowledged as unverified, external precedents from different Ni systems), and Mazet 2021 (dppe cycle partially supported by intermediate characterization but dmpe cycle resting on acknowledged unreliable VTNA and an unidentified resting state).

- G2: RP-1 covers Kumada 1972, Breitenfeld 2014 ("complete catalytic cycle" with intermediates 10–16, none observed; bimetallic OA vs escape-rebound rests entirely on unvalidated DFT), Cherney 2015 (DYKAT/SET invoked without krac/kproductive or spectroscopic verification), and Pengfei 2024 (two pathways acknowledged as undiscriminated).
- G3: RP-1 covers Kumada 1972, Monnereau 2009 (41.8 ppm activated species with unassigned Ni oxidation state; RDS assignment from electronic trend only), Molander 2016 (Ni(I)/Ni(III) intermediates spectroscopically unobserved; no Stern-Volmer quenching), and Morken 2016 (pyboxNi(Ar) and Ni-sulfate intermediates neither trapped nor characterized; Ni oxidation states unspecified).

The pattern therefore appears in all nine papers across all paradigms (2-electron Ni(0)/Ni(II), radical Ni(I)/Ni(III), photoredox/Ni, electrochemical SET) and all publication formats (communication, full paper, review, Account). The all-group file names it "the single most universally recurring pattern in the corpus."

IP-2: Quantitative analytical basis for primary data underreported. Every paper in the corpus grounds its primary quantitative claims (yields, conversions, kinetic orders, ee values) in instrumental measurements whose reliability cannot be independently assessed from the published data. The pattern appeared as RP-2 in all three sub-groups.

- G1: glpc with no column/detector/standard (Kumada 1972), GLPC column temperature/carrier gas/detector absent and a mathematically impossible yield > conversion entry (Kumada 1976b), IR band not specified for VTNA, no within-condition replicates, NMR yields without uncertainty, qualitative "excellent mass balance" (Mazet 2021).
- G2: radical-clock fit with unexamined negative intercept and consecutive non-randomized runs (Breitenfeld 2014), no PRISMA-style search methodology (Cherney 2015), > 90 substrate examples with isolated yields only and no conversion data (Pengfei 2024).
- G3: GC response factors unverified for 17 biaryls, no replicates, TOF from single time-point (Monnereau 2009), HPLC conditions for ee absent and all scope yields single-experiment (Molander 2016), SFC conditions absent for 17 er values and "average of two experiments" without individual values or SD (Morken 2016).

The specific analytical instrument changes across the 54-year span (glpc → GC → VTNA → SFC → CV), but the structural gap that measurement conditions are insufficient for independent replication is invariant. The all-group file records this pattern in all nine papers.

IP-3: Self-contradictory absence — model predicts observables that are not tested. All nine papers contain at least one instance where a proposed model generates a specific, testable prediction that the paper itself does not test and does not acknowledge as untested. This pattern appeared as RP-4 in G1, RP-3 in G2, and RP-3 in G3, and is the pattern most densely recurring as U-items (items that the A/B/C classification cannot naturally accommodate) in every sub-group. The integration output gives 28 distinct U items across 9 papers (kumada1972: 2; kumada1976b: 4; mazet2021: 4; breitenfeld2014: 4; cherney2015: 8; pengfei2024: 7; monnereau2009: 3; molander2016: 7; morken2016: 4). *R1 note: the per-paper counts sum to 43; the total of 28 is the integration model's figure, whose de-duplication rule is not recorded, and it is not used elsewhere.*

- G1: R–R homo-coupling predicted by eq 3 but unreported (Kumada 1972 U-1); homogeneous cycle proposed but black mixture unanalyzed (Kumada 1972 U-2); β-hydride elimination suppression claim contradicted by extensive secondary→primary isomerization (Kumada 1976b U2); VTNA acknowledged as unreliable yet mechanistic conclusions drawn unchanged (Mazet 2021 U3).
- G2: Complete catalytic cycle with predicted but uncharacterized intermediates 10–16 (Breitenfeld 2014 U-1); DET mechanism predicting conversion-independent ee not tested, SET cycle predicting IrII and NiIII intermediates neither observed (Cherney 2015 U-1, U-3); DFT Ni(I)/Ni(0) prediction vs SWV measurement disagreeing by 360 mV undiscussed (Pengfei 2024 U-2).
- G3: TOF denominator contradicted by incomplete activation yet retained (Monnereau 2009 U-1); NiI/NiIII intermediates spectroscopically unobserved yet cycle "established" (Molander 2016 U-2); reductive-elimination retention assumed yet untested, rendering stereochemical assignment indeterminate (Morken 2016 U-3).

The all-group file records this pattern in all nine papers and observes that its universal recurrence confirms the G1/G2/G3 assessments that it is a genre convention rather than an individually chosen gap. Its structure — "model predicts observable → observable not tested → model retained" — is described in the sub-group outputs as not fitting naturally into the A/B/C framework and as reflecting the narrative rather than predictive function of catalytic cycle diagrams in cross-coupling papers.

IP-4: Selectivity / scope / boundary claims without complete byproduct or negative-result accounting. All nine papers make positive-feature claims (selectivity over homocoupling, scope coverage, substrate class boundaries) without providing the negative-result data that would convert assertion into demonstration. The pattern appeared as RP-3 in G1, RP-4 in G2, and RP-6 in G3.

- G1: “complex mixture, homo coupling predominating” for alkyl + dichloroethylene entries never characterized (Kumada 1976b A-2/U-4); Figure 9 ~75–85% mass-balance gap, scope failures never revisited (Mazet 2021 B-1/B-6).
- G2: linear:cyclized ratio without total yields/conversions (Breitenfeld 2014 B-7); frontier prioritization without inventory of inaccessible targets (Cherney 2015 B-3/B-11/CLA-2); general-strategy claim with caption-footnote ligand bifurcation, no head-to-head with Lin/Gong prior art (Pengfei 2024 B-3/B-6/B-8).
- G3: bromomesitylene 780 h^{-1} outlier under “high activities” without explanation (Monnereau 2009 A-3); amine-silicate “performed poorly” with no yields (Molander 2016 B-4/B-8); tertiary/quaternary “ineffective” with no values (Morken 2016 A-1/B-3/B-4).

The all-group file records the field-wide structure: positive-feature framing of selectivity/scope/boundary without the negative-result data that would convert assertion into demonstration, in all nine papers.

IP-5: Benchmark superiority asserted without matched-pair comparison on shared substrate. Methodology papers in the corpus claim improvement over prior methods without providing common-substrate, common-conditions head-to-head comparison with the named prior art. This pattern was named explicitly in G3 (RP-4) and observed partially in G1 and G2 (RP-4 partial / implicit).

- G1: Kumada 1976b “exothermicity of reactions” qualitative ranking; Mazet 2021 dppe vs dmpe two-point %V_Bur from non-catalytic cod complexes.
- G2: Pengfei 2024 no head-to-head with Lin 2022 (Pengfei 2024’s ref 60) or Gong 2011/2013 (Pengfei 2024’s refs 29–30) on shared substrates.
- G3: Monnereau 2009 dppp comparison restricted to one substrate at non-headline loading; Molander 2016 Figure 2 prior-Pd-Suzuki depiction without yield comparison; Morken 2016 Fu/Zhong-Bian/Nakamura/Jarvo cited symmetrically without er/yield comparison.

The all-group file reports that this pattern appears in at least 9 of 9 papers once the integration view aggregates G1 (implicit) + G2 (partial) + G3 (named). The closest structural analogue routinely receives the same cite-and-contrast treatment as more distant references.

IP-6: Imported model — mechanistic framework adopted from non-isomorphous source. All eight non-anchor papers adopt a mechanistic framework from an analogy system without demonstrating that the analogy is isomorphous with the target system. This pattern appeared as a sub-pattern of RP-1 in G1, as RP-5 in G2 and G3. It was not identified as a separate IP in the v17 all-group output; v18.4 EN prompt elaborations surface it as a distinct field-wide signal.

- G1: Kumada 1976b imports (dipy)Ni precedent (Jolly-Wilke) without excluding the Felkin-Corriu Ni(0) alternative (C-7); Mazet 2021 dmpe bimetallic TS lacks DFT corroboration.
- G2: Breitenfeld 2014 PBE0-dDsC/TZ2P//M06/def2-SVP without benchmarking; Cherney 2015 Lin et al. Ni^{I}/Ni^{III} DFT applied across 427 references without EPR/CV cross-reference; Pengfei 2024 2.8 kcal mol^{-1} Ni(III)-vs-Ni(II) preference with functional/basis set/solvation/spin states undisclosed.
- G3: Monnereau 2009 Pd diphosphine bite-angle precedent applied to Ni without addressing Ni(I)/Ni(III) accessibility; Molander 2016 Gutierrez 2015 SMD-water PES ($\varepsilon \approx 80$) applied to dioxane/acetone/EtOAc/DMF ($\varepsilon = 2.2–38$) and to nine substrate classes; Morken 2016 generic Ni(0)/Ni(II) manifold + organic SN2 geometry without PyBox-Ni DFT.

The all-group file observes that in 7 of 8 non-anchor papers, IP-1 (cycle without elementary-step validation) and IP-6 (cycle imported from non-isomorphous source) are the same gap viewed from two angles: the cycle is unobserved because it was never the target of system-specific measurement, and it was never the target of measurement because it was already accepted by analogy. Stage 3 v18.4 EP-06a (analogy-validation pack) is the canonical experimental remediation for this coupled gap.

IP-7: Computational model deployed as sole pathway discriminant without protocol benchmarking. Post-DFT papers in the corpus use DFT as the primary (often sole) discriminant between competing mechanistic pathways without reporting functional/basis-set benchmarking, spin-state sensitivity, or zero-point/solvation uncertainty estimates. This pattern was named explicitly in G2 (RP-5) and G3 (RP-5), with G1 partially contributing through Mazet 2021. It was not identified as a separate IP in the v17 all-group output; it is emergent in the v18.4 EN integration and is the only IP whose absolute magnitude scales with era.

- G1: Mazet 2021 (DFT corroboration explicitly absent for bimetallic TM step).
- G2: Breitenfeld 2014 + Cherney 2015 (Lin et al. Ni^{I}/Ni^{III} DFT applied across three reaction types without EPR/CV cross-reference) + Pengfei 2024 (2.8 kcal mol^{-1} pathway preference without method disclosure or uncertainty).
- G3: Monnereau 2009 single ^{31}P shift; Molander 2016 PES without $\langle S^2 \rangle$/ZPE/solvation sensitivity; Morken 2016 stoichiometric assumption without DFT for PyBox-Ni geometry.

The all-group file notes that IP-7 severity increases with year — Pengfei 2024 reaches 3/3 high multi-agent convergence (the strongest single-paper signal in the corpus) — reflecting that the field expectation for computational benchmarking crystallizes faster than disclosure practice.

IP couplings (integration-only view). The seven IPs do not appear independently. The all-group file identifies four tight couplings whose joint occurrence is the actual signature of the field convention: C-α (IP-1 ↔ IP-6: cycle unobserved ↔ cycle imported, remediated by EP-03a + EP-06a); C-β (IP-3 ↔ IP-1: self-contradictory absence is the generation rule by which IP-1 manifests as a paper-level observable, remediated by EP-04a); C-γ (IP-2 ↔ IP-4: analytical underreporting and selectivity-without-accounting are operationally inseparable, remediated by EP-08a + EP-09a + EP-01a); C-δ (IP-5 ↔ IP-7: benchmark superiority and DFT-as-discriminant co-occur under the same post-2010 evidentiary regime, remediated by EP-05a + computational benchmarks). The four couplings reduce "seven independent field gaps" to "four functional gap-clusters," each with a coherent experimental closure package and a characteristically asymmetric filling rate (C-α partial, C-β zero, C-γ zero, C-δ zero).

SI-5.3 Recurring Roots Taxonomy

The all-group Stage 2F v7.1 output cross-references `recurring_roots` (same_root_groups promoted to cross-paper recurrence) across the three sub-groups. The table below lists the canonical recurring roots (CRRs) with the IP affiliation stated by the all-group file, updated to reflect the v18.4 EN seven-IP structure.

CRR ID	Description (as recorded)	IP affiliation	G1	G2	G3	Classification	Papers	Years
CRR-1	Catalytic cycle assembled from analogy, not validated by system-specific experiment	IP-1, IP-6	Yes (root #1)	Yes (root #1)	Yes (root #1)	Field-wide	All 9 papers	1972–2024
CRR-2	Quantitative measurement basis underreported	IP-2	Yes (root #2)	Yes (root #2)	Yes (root #2)	Field-wide	All 9 papers	1972–2024
CRR-3	Selectivity claim without mass balance	IP-4 (partial), IP-3	Yes (root #3)	Yes (root #3)	Yes (root #3; RP-6 subtype “scope failures undiagnosed”)	Field-wide	All 9 papers	1972–2024
CRR-4	Benchmark asserted, not verified by matched-pair comparison	IP-5	Implicit	Partial (RP-4)	Yes (root #4)	Field-wide (promoted by the integration)	All 9 papers	1972–2024
CRR-5	DFT as sole pathway discriminant	IP-7	—	Yes (root #4)	Yes (within RP-1)	Paradigm-specific (post-DFT papers)	7 of 9 (post-DFT; the output does not list them)	2009–2024
CRR-6	Ligand effect attribution from insufficient experimental basis	— (not IP-elevated in the all-group output)	Yes (root #5)	—	Yes (within RP-1/RP-5)	Paradigm-specific (diphosphine ligand papers)	Kumada 1976b, Mazet 2021, Monnereau 2009, Breitenfeld 2014, Morken 2016 (5 of 9)	1976–2021
CRR-7	Imported model without isomorphism demonstration	IP-6	Yes (RP-1 sub-pattern)	Yes (RP-5)	Yes (RP-5)	Field-wide (all non-anchor papers)	All 8 non-anchor papers (IP-6 evidence chain)	1976–2024
CRR-8	Self-contradictory absence around mechanism-derived predictions	IP-3	Yes (root #4)	Within RP-3	Recurring U-1	Field-wide	All 9 papers (28 U items)	1972–2024

The all-group file explicitly flags IP-1 (cycle validation), IP-2 (analytical underreporting), IP-3 (self-contradictory absence), and IP-6 (imported model) as the field-wide patterns that are independent of paradigm. CRR-5 / IP-7 (DFT as discriminant) and CRR-6 (ligand-effect attribution) are reported as paradigm-specific rather than field-wide. IP-4 (selectivity without accounting) and IP-5 (benchmark superiority) are field-wide within methodology papers.

SI-5.4 Subgroup-Specific Analyses

The Stage 2F integration promoted a pattern to IP status only if it emerged independently across sub-groups. Seven sub-group-specific patterns (SG-G1-1 through SG-G3-3) that satisfied the preservation criteria — (a) chemically concrete gap, (b) multi-paper persistence or uniqueness within the sub-group, and (c) testable hypothesis lost if absorbed into an abstract IP — were recorded in the all-group file without being promoted.

G1 (direct lineage). G1-specific findings concern chemically concrete properties of the dpe/dppe/dmpe-Ni system. SSP-1 (MgX_2 byproduct fate and catalytic effect, shared with G3) concerns the Lewis-acidic magnesium halide by-product formed in each Kumada turnover (formally MgBrCl for RMgBr and an organic chloride, as in Kumada 1972; $MgBr_2$ in the Morken 2016 system): Kumada 1972 implicitly predicts its formation but does not characterize it; Kumada 1976b flags its fate as an uncharacterized prediction (U1) and defers it (A6); Mazet 2021 lists it as D-4-7 and blames "catalyst deactivation and product inhibition" for same-excess VTNA non-overlay without testing $MgBr_2$ addition; Morken 2016 B-11 reports that $MgBr_2$ reacts directly with cyclic sulfate 1 to furnish a 3-bromopropanol derivative, demonstrating it is not inert. SSP-2 (homogeneous vs heterogeneous — the "black mixture" question) persists across Kumada 1972 (B-10, U-2) and Kumada 1976b; no mercury-drop test, hot filtration test, or TEM characterization is reported, and the black-mixture observation is not revisited in 1976. SSP-3 (E/Z selectivity origin in dpe-Ni vinyl halide coupling) spans Kumada 1972 (A-1, B-11, U-3, cis-enrichment of stilbene from both cis- and trans-dichloroethylene, deferred), Kumada 1976b (Table 7 E/Z data without mechanistic interpretation), and Mazet 2021 (temperature-dependent E/Z reversal, A1, B12). SSP-4 (β-HE suppression claim contradicted by isomerization) is unique to Kumada 1976b but structurally relevant to G2 radical pathways. Each G1-specific finding identifies a testable experiment ($MgBr_2$ spike, mercury-drop, Eyring analysis / bite-angle ligand series, RE/β-HE competition kinetics) that remains unperformed.

SG-G1-1: Genre-induced labeling mismatch. Hypothesis vocabulary at one rhetorical location with declarative / near-factual verbs in surrounding mechanistic narrative. Visible in all three G1 papers (Kumada 1972 "working hypothesis" final sentence + factual eqs 2–4; Kumada 1976b "can be rationalized by" + substantive declarative conclusions; Mazet 2021 "we propose" / "we establish" rhetorical separation). Not foregrounded in G2/G3 outputs. Worth preserving because it identifies a chemically concrete linguistic mismatch that persists across journal/era/language community and aligns with Kumada Stage 3 §6 institutional codification candidate (RI-06 co-occurrence note).

G2 (boundary breakthrough). G2-specific findings concern how the $sp^2 \rightarrow sp^3$ boundary was crossed and the new gaps the crossing created. The G2 analysis describes the shift as paradigm succession: Paradigm A (two-electron Ni(0)/Ni(II), Kumada 1972) does not explain why Csp^3 halides are excluded; Paradigm B (one-electron Ni(I)/Ni(II)/Ni(III) radical, Breitenfeld 2014 and Pengfei 2024) was developed because the two-electron pathway fails for Csp^3. G2 shares with G1 the absence of mass balance quantifying cross vs homo selectivity — Kumada 1972 reports no R–R; Pengfei 2024 reports > 90 isolated yields without homo-coupling data beyond the model reaction.

SG-G2-1: Csp^2/Csp^3 boundary as paradigm-succession driver. The boundary itself is the organizing structural feature of G2; the 1972 paper's positive-feature framing of sp^3 exclusion (B-4, no failed sp^3 datum) becomes the founding problem of the radical paradigm (Breitenfeld 2014 onward). Identifies the structural mechanism by which paradigm succession occurs — a positive-feature framing in one paradigm becomes a negative-result need in the next. Specific to G2's group composition; would not appear in lineage-only or expansion-only frames.

SG-G2-2: Review-as-paradigm-crystallization (Cherney 2015 specific). The review functions as the documentation moment at which the Ni^{I}/Ni^{3} + precoordination + DKR/KR/DYKAT framework becomes legible as a coherent paradigm. This role is unique to the review-genre paper in G2. Worth preserving because it identifies a distinct gap class (cross-study aggregation without normalization; frontier prioritization without target inventory; mechanistic-framework crystallization across independent reaction types) not present in primary papers. Cherney 2015 accumulates 8 U items — the corpus maximum — attributable to genre amplification: the review aggregates primary-paper proposals as if established, multiplying self-contradictory absences at field level.

SG-G2-3: Anode characterization absent in electrocatalytic paper (Pengfei 2024 U-3). The anodic half-reaction is structurally absent from the proposed cycle despite a 55-pp yield difference between Fe and Zn anodes. A chemically concrete gap (specific reagent comparison, specific yield difference, specific cycle component absent) that would be lost if absorbed into IP-1 (general mechanism-validation absence). G2-specific because no other paper in the corpus uses electrocatalysis.

G3 (methodological expansion). G3-specific findings organize around three coexisting (not successive) paradigms — ligand design for two-electron catalysis (Kumada 1972, Monnereau 2009); single-electron activation via photoredox/Ni dual catalysis (Molander 2016); and enantioselective desymmetrization via Ni/chiral ligand (Morken 2016). G3 also documents nucleophile-type divergence (DV-1: Grignard vs trifluoroborates/silicates), success-metric divergence (DV-2: yield, TOF, paradigm + scope, er), and format-driven evidentiary-depth divergence (DV-3: communication vs full paper vs Account). Despite these divergences, RP-1 (cycle validation) is shared across all three paradigms — the evidentiary basis for promoting it to IP-1 at the integration level.

SG-G3-1: Single-^{31}P-shift identification of "active species" (Monnereau 2009 U-1). A chemically concrete gap: an oxidation state assignment based on a single chemical shift (δ = 41.8 ppm) without corroborating EPR / UV-Vis / multinuclear NMR. Distinct from generic IP-1 because it identifies a specific characterization step that was performed but truncated. Sub-group-specific to G3.

SG-G3-2: PES re-use across nine substrate classes (Molander 2016 U-7). A specific testable hypothesis: a PES computed in SMD-water for benzylic/PhBr cannot, without re-computation, predict $\Delta\Delta G‡$ for α-CF_3/alkylsilicate/azaborine classes. The hypothesis is testable; the prediction is computable; the absence is documented. Worth preserving as a concrete instance of IP-6 (imported model) that would be lost if absorbed into the abstract field-wide statement. Aligns with PRI-12 candidate pattern (self-reinforcing computational loop) (candidate pattern; SI-4.10).

SG-G3-3: ArMgBr vs Ar_2Mg active-species ambiguity (Morken 2016 CLA-2). Chemically specific Schlenk-equilibrium gap: the cycle is drawn with ArMgBr but the aging protocol implies Ar_2Mg dominance, and the active transmetalating species is unidentified. Unique to G3 within the corpus because no other Kumada-style paper in G2/G1 foregrounds aging protocols.

These seven sub-group findings are preserved in the field-level summary as concrete instances; they should not be absorbed into the abstract IP-1/IP-3/IP-6 statements without losing their chemical specificity.

SI-5.5 Integrated group_unasked_question Derivation

The all-group Stage 2F v7.1 output synthesizes the three sub-group-level `group_unasked_questions` into a single integrated `group_unasked_question`. The three sub-group questions, reproduced verbatim from `kumada1972_all_v184_opus47_s2fv71.md`, are:

- **G1 `group_unasked_question`:** "What is the complete, time-resolved, quantitatively closed account of all species (including byproducts and intermediates) and their concentrations across an entire turnover cycle of Ni-diphosphine-catalyzed Kumada coupling under one defined set of catalytic conditions, to a level that allows the proposed catalytic cycle's algebraic rate-law predictions to be directly compared with measurement?"
- **G2 `group_unasked_question`:** "What direct experimental evidence (under operating catalytic conditions, not by DFT or stoichiometric analogy) establishes the elementary steps and electronic-structure assignments of the proposed catalytic cycles for Ni-catalyzed cross-coupling — across the 2-electron Csp^2 regime, the radical Csp^3 regime, the stereoconvergent secondary-alkyl regime, and the electrochemical Csp^3–Csp^3 regime — and how does the quantitative selectivity of cross-coupling over homocoupling change as the substrate hybridization, the proposed Ni oxidation-state manifold, and the activation mode shift across these regimes?"
- **G3 `group_unasked_question`:** "What is the minimum set of experimental and computational evidence required to certify that a proposed Ni-catalyzed cross-coupling mechanism (whether 2-electron Ni(0)/Ni(II), 1-electron radical, photoredox NiI/NiIII, or asymmetric stereoinvertive) is the operative mechanism rather than a narrative consistent with the data — and has any paper in this group met that standard for its own central mechanistic claim?"

The all-group file assesses the three sub-group questions as *facets of a single underlying problem* (the absence of a quantitatively closed, mechanistically validated, paradigm-discriminated account of any Ni-catalyzed cross-coupling) and derives the integrated formulation as a 4-facet consolidation:

> "Across 52 years and four mechanistic paradigms (closed-shell Ni(0)/Ni(II), radical Ni(I)/Ni(III), photoredox NiI/NiIII, electrochemical Ni(I/II/III)) of Ni-catalyzed cross-coupling, what is the quantitatively closed, time-resolved, paradigm-discriminating evidentiary package — comprising elementary-step intermediate characterization under turnover conditions, replicate-based primary-data uncertainty, complete byproduct mass balance, matched-pair benchmark comparison against the closest prior method, and a falsifiable test of the proposed cycle's own predictions — that no paper in the corpus has assembled for its own central mechanistic claim, and what publication-incentive structure has preserved this absence as a stable feature of the field rather than a paper-level oversight?"

The four facets of this integrated question are: (lineage facet = within-system completeness, G1 question); (boundary facet = cross-paradigm discrimination, G2 question); (expansion facet = certification standard, G3 question); plus the publication-

incentive question ("and what … structure has preserved this absence") — the fourth facet that integration mode adds by observing that the absence is stably reproduced across paradigm transitions, format changes, and tool generations. This four-facet structure supersedes the three-facet (a)(b)(c) formulation reported in the v17 all-group output.

The relationship of the sub-group questions to the integrated question is: G1 addresses the completeness facet (within-system time-resolved account); G2 addresses the discrimination facet (selectivity mechanism across paradigms); G3 addresses the certification facet (what evidence would be sufficient). The integration-mode synthesis adds the incentive facet, making the interlock explicit: answering any one facet demands satisfying the others, and no paper in the 9-paper corpus simultaneously achieves all three experimental facets. Mazet 2021 comes closest (intermediate characterization for dppe as partial completeness; distinct mechanisms for two catalysts as partial certification) but does not provide complete product distribution and its VTNA data are compromised by decomposition artifacts.

Consolidation procedure. Each sub-group `group_unasked_question` was derived from that sub-group's `recurring_roots` and its strongest-convergence (3/3) items on the cross_field_temperature metric: in G1, the speciation question derives from Kumada 1972 B-1 and Kumada 1976b B2; in G2, the selectivity-mechanism question from Kumada 1972 B-1, Breitenfeld 2014 B-1, and Pengfei 2024 B-1; in G3, the certification-standard question from Monnereau 2009 B-7 (TOF denominator), Molander 2016 U-7 (DFT extrapolation), and Morken 2016 B-1/U-1 (radical-clock mass balance). The all-group file records that in 9 of 9 papers the highest-convergence Stage 1 items concentrate exactly on the gap between the central mechanistic claim and its experimental support, confirming the integrated question as the field-level diagnostic.

SI-5.6 Temporal Trajectory Analysis

The corpus spans 52 years of continuous development (1972–2024 for the experimental and review papers; 2026 is the calendar year in which the Stage 2F output itself was produced, giving the 54-year span referenced in Part 4 of the main text). The all-group file and the sub-group files each record `temporal_trajectory` classifications for each recurring pattern. The composite picture is that different patterns follow different trajectories — some narrow, some shift, some remain time-invariant, and one is strictly emergent.

1972 — single-paradigm foundation. Kumada 1972 proposes a two-electron Ni(0)/Ni(II) cycle for Csp^2 coupling with Csp^3–X excluded, assembled from a dipy-Ni stoichiometric analogy (B-14). The paper instantiates every CRR for the first time: cycle without validation (CRR-1/CRR-7), analytical underreporting (CRR-2), selectivity without mass balance (CRR-3), benchmark assertion without matched comparison (CRR-4), and ligand-effect attribution from insufficient data (CRR-6).

1976 — consolidation of the same paradigm. Kumada 1976b addresses some 1972 gaps (GLPC instrument stated, Table 5 solvent data) while introducing new ones (U3 yield > conversion; U2 β-suppression contradicted by isomerization). G1 classifies CRR-2 here as "narrowing" and CRR-3 as partial narrowing (mass balance still not achieved).

2009 — high-TOF expansion within the same paradigm. Monnereau 2009 extends the diphosphine-Ni paradigm with a calixarene architecture and a novel rate-acceleration hypothesis (SG-G3-1), introducing CRR-6 at a new level of specificity while leaving CRR-1/CRR-2 at 1972–1976 levels.

2014 — paradigm shift to one-electron radical activation. Breitenfeld 2014 is the first corpus paper outside Ni(0)/Ni(II), proposing radical bimetallic OA for sp^3–sp^2 coupling. This is the transition point for CRR-1 (classified "shifting" in both G1 and G2) and the emergence of CRR-5/IP-7 (DFT as sole discriminant; "emergent" with transition ~2010–2015). The sp^2 boundary of Kumada 1972 is substantially filled by paradigm shift rather than tested within the original paradigm.

2015, 2016 — paradigm coexistence. Cherney 2015 documents the paradigm landscape and introduces a new class of gap (mechanistic classifications becoming authoritative through review endorsement, SG-G2-2). Molander 2016 introduces photoredox/Ni dual catalysis and SG-G3-2 (DKR benzyl → secondary alkyl extrapolation). Morken 2016 extends the enantioselective-desymmetrization axis and introduces the confirmed $MgBr_2$ direct reactivity observation (SSP-1, B-11) while retaining the RE retention assumption (U-3).

2021 — mechanistic investigation of the original paradigm. Mazet 2021 deploys modern mechanistic tools (in situ NMR, VTNA, X-ray, radical clock) to the dppe-Ni system. G1 classifies CRR-1 at 2021 as transitioning from "no tools available" to "tools deployed but producing incomplete or ambiguous results."

2024 — electrochemical extension with DFT–experiment discrepancy. Pengfei 2024 extends sp^3–sp^3 cross-electrophile coupling to electrochemistry, retains CRR-1 through CRR-4 and CRR-7 intact, intensifies CRR-5/IP-7 (DFT-to-SWV 360 mV discrepancy), and adds SG-G2-3. Pengfei 2024 reaches 3/3 multi-agent convergence for IP-7 — the strongest single-paper signal in the entire corpus — reflecting that field expectation for DFT benchmarking crystallizes faster than disclosure practice.

Aggregate temporal-trajectory pattern. CRR-1, CRR-7 (IP-6), and IP-7 are reported as *shifting* / *emergent* with clear transition points (~2010–2015 for CRR-1 and IP-7; 1976–2009 for CRR-7 source shift from stoichiometric to Pd-DFT precedent). CRR-2 is *narrowing* in G1 and *time-invariant* in G2 and G3 (instruments change, structural gap unchanged). CRR-3 is *narrowing-then-stalling*. CRR-4, CRR-5 are *time-invariant* or *emergent* respectively. The all-group integration observes that the field exhibits a coherent two-band structure: incentive-locked gaps (IPs 2–5, which no methodological development closes) and tool-locked gaps (IPs 1, 6, 7, which methodology has nominally unlocked but practice has not yet adopted).

SI-5.7 EP-Filling × VQ-Hierarchy Cross-Reference (preserved model output, not used as evidence)

Authors' note. This subsection preserves the reading that the v18.4 all-group integration (Opus 4.7, Stage 2F v7.1; `papers/ep/kumada1972/kumada1972_all_v184_opus47_s2fv71.md`) produced. It is kept so that the record of what the integration produced is complete. It is not the authors' analysis, and none of it is used as evidence in the main text. The quoted paragraphs below are an English paraphrase of that output, not a verbatim copy. It contains the following errors and unsupported readings. (i) It reverses the VQ levels of Table 3: EP-07a is VQ-2 and EP-08a/EP-09a are VQ-4 there, so 'highest-VQ' and 'lowest-VQ' below are inverted. (ii) dpe and dppe are the same ligand (Scheme 1), so 'dppe not dpe' is wrong. (iii) The filling percentages are the model's estimates, not audited values; the audited classification of EP-08a and EP-09a is in SI-9.8, and the filling map of the main text (Fig. 2, Table 4) differs from the table below. (iv) The causal and predictive statements — that publishability decides which EPs are filled, that Stage 3 can identify in advance which EPs will remain unfilled, that the corpus demonstrates incentive-driven absence, and that EP-04a shows the gap to be incentive-driven rather than technically constrained — are the model's. The data cannot separate incentive from experimental burden, technique availability or retrieval bias; the main text treats incentive as one of several accounts (Discussion 1).

Preserved model output. The table and the text that follows are reproduced from the integration output.

> The Kumada 1972 Stage 3 v18.4 output (`kumada_v184_sonnet_s3v2_en.md`) allocates eight EPs (EP-01a through EP-09a, enumerated from Stage 3 SRGs and CLAs) representing the minimal experimental investments needed to close the founding paper's structural gaps. Mapping the corpus's 52-year filling landscape to these EPs reveals a systematic anti-correlation between VQ hierarchy (the urgency-priority ranking assigned by Stage 3) and the observed filling rate:

EP	What kumada1972 needs	Filled by which non-anchor papers?	Filling depth
EP-04a (discrimination/replication: R–R quantification)	Quantify Et-Et byproduct from EtMgBr/PhCl entry	**NEVER FILLED** in any of the 8 non-anchor papers across 52 years	0%
EP-03a (direct observation: black-mixture characterization, mercury-drop test)	Distinguish homogeneous Ni(II) from heterogeneous Ni^0	**NEVER FILLED**; no paper performs mercury-drop/hot-filtration/XRPD on any Kumada system	0%
EP-08a (replication/uncertainty: independent runs of Examples 1, 2)	Replicate preparative examples; report range	**NEVER FILLED** in any descendant paper	0%
EP-09a (internal benchmark: shared-substrate yield comparison vs Tamura-Kochi or Corriu-Masse)	Single head-to-head comparison on one shared substrate	**NEVER FILLED**; closest approach is Monnereau 2009 single dppp comparison at one loading	0%
EP-01a (reporting coherence: Table I footnote with method disclosure)	Internal-standard identity, response factors, full analytical method	**NEVER FILLED** (retroactive remediation structurally outside publication landscape)	0%
EP-05a (quantitative ranking/TON for six-catalyst comparison)	Per-catalyst yield in supplementary table	**NEVER FILLED**; descendant papers expand ligand classes without within-system numerical ranking	0%
EP-06a (analogy validation: independent (dpe)Ni intermediate synthesis)	Synthesize and characterize [dpe-Ni(n-Bu)(Cl)] and [dpe-Ni(n-$Bu)_2$]	**PARTIALLY FILLED**: Mazet 2021 X-ray of OA complex 9c (analogue only; non-catalytic isolation)	~25%
EP-07a (boundary test: failed sp^3 entry)	Run n-BuMgBr + n-BuCl under standard conditions, report 0% / <LOD	**PARTIALLY ADDRESSED** via paradigm-succession counter-example (Breitenfeld 2014, Pengfei 2024 demonstrate Csp^3 is feasible under different catalysts); dpe-system negative-result remains unperformed	~40%

> **Anti-correlation finding.** The VQ hierarchy and filling order are anti-correlated: the highest-VQ EPs (mechanism-validation: EP-06a, EP-07a) achieve the highest partial-filling rates (25% and 40%) because their resolution is structurally publishable (novel intermediate characterization; paradigm-shifting boundary demonstration). The

lowest-VQ EPs (EP-04a, EP-03a, EP-08a, EP-09a, EP-01a, EP-05a — analytical/replication/benchmark/reporting-coherence type) achieve 0% filling because their resolution yields only audit-type contributions that are structurally outside the publication-incentive landscape.

Mechanistic reading of the anti-correlation. The all-group file identifies three distinct filling mechanisms: (i) technology-driven partial filling (EP-06a: X-ray structures + in-situ NMR progress enables intermediate characterization in cognate systems); (ii) paradigm-succession counter-example (EP-07a: the boundary is addressed by demonstrating that it is system-dependent, not by performing the within-system negative-result experiment); (iii) structurally zero filling (EP-04a/03a/08a/09a/01a/05a: experiments that yield at most an audit-type result with no novel claim, and thus do not enter the publication-incentive landscape at any point in the 52-year window). The third mechanism is the most consequential: it implies that Stage 3 EP allocation can identify in advance which EPs will remain unfilled, based on whether their closure is structurally publishable in the current journal-incentive equilibrium.

Correspondence to IP coupling analysis. The EP filling landscape maps directly onto the four IP couplings identified in SI-5.2: coupling C-α (IP-1 ↔ IP-6, EP-03a + EP-06a) achieves ~12% average filling; coupling C-β (IP-3 ↔ IP-1, EP-04a) achieves 0%; coupling C-γ (IP-2 ↔ IP-4, EP-08a + EP-09a + EP-01a) achieves 0%; coupling C-δ (IP-5 ↔ IP-7, EP-05a) achieves 0%. Only the coupling whose remediation aligns with technological progress (C-α) shows any movement. This correspondence between IP-coupling structure and EP-filling rates is the integration-mode finding most relevant to this paper's "incentive-misaligned structural absences" thesis: the corpus demonstrates empirically that even when Stage 3 EP allocation precisely specifies the experiment, the experiment is reproduced rather than closed across 52 years.

A-4 anomaly. The most striking integration anomaly: EP-04a (R–R discrimination run, quantify Et-Et byproduct from EtMgBr/PhCl via one glpc measurement with an authentic standard) is 0% filled across 52 years and 8 papers. This is the simplest experiment in the entire EP catalogue and the most reliably unperformed — the strongest single-EP demonstration that the filling gap is incentive-driven rather than technically constrained.

SI-6: Four-Model v17 Comparison on Kumada 1972 — Claude Haiku 4.5, Claude Sonnet 4.6, Claude Opus 4.7, and GPT-5.4 Thinking

This supplementary section documents the inter-model comparison that grounds the reasoning-oriented post-training hypothesis discussed in Discussion 2 of the main manuscript. Four frontier reasoning models — three within the Claude family (Haiku 4.5, Sonnet 4.6, Opus 4.7) and one cross-vendor reasoning model (GPT-5.4 Thinking) — were run on the same input (Tamao, Sumitani, Kumada, *J. Am. Chem. Soc.* **1972**, *94*, 4374–4376) under **model-pinned, uniform-prompt-version (v17)** execution. All four models completed Stage 0 through Stage 4 including the v2 Stage 3 Root-Issue Codebook and the v2 Stage 4 EP filling map.

Scope note: an earlier three-model comparison run with UQS v15 (Claude 3.5 Haiku, Sonnet, Opus) is archived at `analysis/analysis_model_comparison_v15.md` (project record, not deposited) and is retained as a historical record. The current SI-6 supersedes it with the four-model v17 matrix, which eliminates the v15/v17 prompt-version mismatch caveat and provides a properly-controlled comparison. A trial run of a Google Gemini model on Kumada 1972 via Antigravity (auto-reassigned to Gemini 1.5 Pro rather than the intended Gemini 3 Flash) was retained as an internal methodology record outside publication scope; see `papers/dev/kumada/gemini3flash/README.md`. A model-pinned Gemini 3 Flash v17 run is a planned future extension.

SI-6.1 Comparison Methodology

- **Input**: identical PDF of Tamao, Sumitani, Kumada, *J. Am. Chem. Soc.* **1972**, *94*, 4374–4376 (Communication, three journal pages).
- **Prompt version**: UQS v17 uniformly across all four models. This resolves the v15/v17 mismatch caveat that appeared in earlier drafts of SI-6. Note that the *production* configuration used elsewhere in this paper is v18.4; the four-model comparison is therefore at v17, while the Top-3 reinforcement EP SET = {EP-03a, EP-04a, EP-06a} is shown stable across the v17 → v18.4 transition (Methods 1.4; Results Part 4). A v18.4-matched N × 4-model rerun is registered as Phase 2 verification path (b) below (Phase 2 denotes the planned follow-up study of the method, here and elsewhere in the SI).

- **Session integrity**: each model's run was performed in a single session so Stage 2 receives full Stage 1 Turn 1–3 context, per Methods 6. Stage 3 v2 and Stage 4 v2 prompts were run on the output of Stage 2 within an extended-context session.
- **Web search**: the Claude runs used no external search (Stage 4 is parametric recall); the GPT-5.4 Thinking run had web search enabled (Methods 2, "Retrieval and verification"), so its Stage 4 candidates and DOIs are not evidence of parametric coverage.
- **Multimodal PDF reading**: Claude via the Read tool (native multimodal PDF input); GPT-5.4 via the ChatGPT PDF upload path. Diagnostic tests (Scheme 1 intermediate geometry; Table I footnote-f attribution to three vinyl substrates; eq 3 R–R byproduct mass-balance absence) confirmed multimodal parsing across all four models; pdftotext equivalence was ruled out.
- **Run dates**: Claude Haiku 4.5 run on 2026-04-19 11:43–11:51; Claude Opus 4.7 run on 2026-04-19 11:45–12:12; Claude Sonnet 4.6 run archived at 2026-04-15/16/17 with SI-3 reproducing the Stage 3 output; GPT-5.4 Thinking run on 2026-04-19 09:09–09:13.
- **Archive**: `papers/dev/kumada/kumada_v17_{haiku,sonnet,opus47}_*.md` and `papers/dev/kumada/gpt54thinking/kumada_v17_gpt54thinking_*.md`.

SI-6.2 Comparison Metrics

Metric family	Operational definition
Output volume	Total word/character count across Stage 0 + Stage 1 + Stage 2 + Stage 3 + Stage 4. Cost-efficiency proxy.
Question-generation breadth (Stage 1)	Per-vantage question counts (V1/V2/V3) and new-versus-overlap splits. Stage 1 Turn 4 integration counts for A/B/C/U items.
Turn 4 classification density	Counts for A/B/C/U items. Higher U counts indicate active self-contradictory absence rule firing; higher C counts indicate exhaustive citation analysis including the Dropped Quantitative Benchmark subtype (v17 C4).
Stage 2 D-4 candidates	Number of D-4 candidates plus completeness of candidate metadata (`grounded_in`, `key_quote`, `temperature_note`, `same_root_group`). Fallback rule application noted where `high_convergence_high_gap` is empty.
Stage 2 E-section	`confirmed_english_propositional`, `confirmed_english_question`, `scope_type` accuracy. The correct judgment for Kumada 1972 is `tool_question` (the paper answers "can this catalyst do this?").
Stage 3 Root-Issue Codebook density	Number of SRGs and CLAs. Finer decomposition indicates higher analytical resolution.
Stage 3 VQ judgment	Per-VQ (VQ-1/2/3/4) judgment. Single-layer vs dual-layer (era-norm / modern-era) distinction is itself a capability dimension.
Stage 4 filling paper identification	Specificity of proposed filling papers (DOI-level vs topic-level). Two runs naming the same historical paper is recorded as agreement between runs, not as verification (SI-6.7).
Position-citation (`loc`) fidelity	Stage 1 items' page/line markers. Claude models preserve via multimodal Read; GPT-5.4 returns `loc: not recoverable`.
Output discipline	Absence of prompt-forbidden content (concluding summaries, follow-up offers).

Note on count-based metrics. The Stage 1 per-vantage question counts (*breadth*) and the Turn 4 A/B/C/U counts (*density*) operate under a per-vantage generation budget (10–15 questions per vantage in Turns 1–3; SI-7), so these counts are reported as descriptive indicators of how each model populates that budget, not as a completeness measurement of a paper's absences. The cross-model count differences in SI-6.3 therefore partly reflect budget-population and consolidation behaviour rather than an intrinsic detection ceiling; the load-bearing cross-model signals in this comparison are the categorical and structural convergences (`scope_type`, VQ judgment, SRG decomposition, EP re-derivation, `loc` fidelity), not the raw counts.

SI-6.3 Results by Model — Quantitative Summary

Dimension	**Haiku 4.5 (v17)**	**Sonnet 4.6 (v17)**	**Opus 4.7 (v17)**	**GPT-5.4 Thinking (v17)**
Stage 1 A items	4	~6	**10+**	2
Stage 1 B items	~20	~14	~25	15
Stage 1 C items	~5	~4	~6	7
Stage 1 U items	~4	~4	~4	2
D-4 candidates	**8**	8	8	6 (+ 1 retrospective)

Dimension	**Haiku 4.5 (v17)**	**Sonnet 4.6 (v17)**	**Opus 4.7 (v17)**	**GPT-5.4 Thinking (v17)**
SRGs	5	6	**10**	6
CLAs (explicit)	0 (absorbed in SRGs)	1 (ligand-activity order and cis-geometry claim, with the Corriu–Masse comparison)	2 (Corriu-Masse comparison; "selective" operational definition)	0 (absorbed in SRGs)
`confirmed_english_propositional`	generated	generated	generated	generated
`confirmed_english_question`	generated	generated	generated	generated
`scope_type`	tool_question (correct)	**tool_question**	**tool_question**	**tool_question**
VQ judgment	VQ-2 (qualified)	**VQ-1**	**era-norm VQ-2 / modern VQ-1 qualified (dual-layer)**	VQ-1
Era-Norm Annotation	present	present	**dual-layer (tolerated / deficient / both)**	present
Stage 3 completion (v2)	✓	✓	✓	✓
Stage 4 completion (v2)	✓	✓	✓	✓
Stage 4 specific-paper identification	general topical candidates	general topical candidates	**specific papers with DOIs (7+ papers)**	**specific papers with DOIs (7 papers)**
Position-citation `loc`	present (verbal: `P1:Rcol:25∗`)	present (`P2:L61∗`)	present (`P4375:Lcol–Rcol`)	**absent (`loc: not recoverable`)**
Discipline violations	none	none	none	none

SI-6.4 Per-Model Qualitative Observations

Claude Haiku 4.5 (v17) — the "Haiku fails at D-4" narrative is invalidated

Haiku 4.5 running the v17 prompt produces a **complete structural output**: 8 D-4 candidates with `grounded_in` and `key_quote` metadata, 5 SRGs, 4 U items (self-contradictory absence rule firing), both `confirmed_english_∗` forms, `scope_type = tool_question`, and full position-citation markers in Stage 1 (verbal form: `P1:Rcol:25∗`). This is a **categorical departure from the v15 Haiku run**, where D-4 was absent and discipline violations (forbidden concluding prose) were present. Stage 3 judgment is VQ-2 qualified — more lenient than Sonnet's VQ-1 and closer to Opus 4.7's era-norm VQ-2. Stage 4 is completed but with topical filling candidates rather than specific DOIs.

With the v17 prompt, the smallest Claude tier (Haiku 4.5) produced the complete output structure; this says nothing about the correctness of its content. The framing of early SI-6 drafts — "Haiku fails at D-4 as evidence of a capability threshold" — is therefore not supported. One reading is that the v15 failure was prompt-specific (no self-contradictory absence rule, no explicit Era-Norm, an underspecified D-4 fallback); but the v15 run also used an older model, Claude 3.5 Haiku, so prompt and model changed together and the two cannot be separated.

Claude Sonnet 4.6 (v17) — reference baseline for SI-3

This comparison uses Sonnet's Stage 3 v2 record (`kumada_v17_sonnet_s3v2.md`), as for the other three models. The detailed Stage 3 analysis of SI-3 comes from the Stage 3 v1 record of the same Stage 0–2 run (`kumada_v17_sonnet_s3.md`), which has the same six SRGs but two CLAs (utility; bidentate necessity). Structural outputs of the v2 record are complete: 8 D-4 candidates with full metadata, 6 SRGs, 1 CLA (ligand-activity order and cis-geometry claim, with the Corriu–Masse comparison), `scope_type = tool_question`, full position-citation (`P2:L61∗` format), Era-Norm Annotation in single-layer form. VQ judgment is VQ-1 with explicit reasoning that RI-01 (SRG-3) and RI-07 (SRG-4) are fatal at VQ-2 under modern standards. Stage 4 provides topical filling candidates at confidence = Low–Medium; specific historical papers were not DOI-level identified by Sonnet alone.

R1 correction: the 'unique Sonnet insight (D4-7)' is withdrawn. The submitted text attributed to Sonnet a D-4 candidate reading the 22-percentage-point yield differential between Ph-Et (98%) and Ph-n-Bu (76%) in Table I as a signal of β-hydride elimination. In the v17 Sonnet Stage 2 record D4-7 is a different candidate (the connection to a synthetic target; SI-3.3), and none of the v17 Sonnet outputs we examined (Stages 1, 2, 3 and 3 v2) makes the yield-differential argument. The rest of this paragraph is kept as submitted for the record and is not used as evidence. This is the diagnostic pattern that, in earlier drafts of SI-6, was flagged as a Sonnet-family-specific capability. The finding persists after the four-model v17 comparison: Haiku 4.5, Opus 4.7, and GPT-5.4 Thinking all have B/A items that mention β-H elimination or the Ph-alkyl scope, but none promotes the numerical 22-pp differential to a D-4 candidate.

Claude Opus 4.7 (v17) — highest analytical resolution with dual-layer Era-Norm

Opus 4.7's v17 run produces the **most granular Root-Issue Codebook**: 10 SRGs (SRG-1 through SRG-10) plus 2 CLAs (Corriu-Masse comparison; the "selective" operational definition). Stage 1 Turn 4 produces 10+ A items with full position-citation in a unique format (`P4375:Lcol–Rcol`, referring to the 1972 JACS column layout). Stage 2 includes 8 D-4 candidates with fallback-rule compliance (`high_convergence_high_gap: none – fallback rule applied†` is explicitly noted and the fallback procedure applied).

Unique Opus 4.7 capability (dual-layer VQ judgment): Opus 4.7 explicitly separates the VQ judgment into `era–norm` and `modern–era` layers. For Kumada 1972 the judgment is **era-norm VQ-2 / modern VQ-1 qualified**. This is the most internally consistent VQ judgment in the four-model matrix: Sonnet's VQ-1 corresponds to Opus 4.7's `modern VQ–1 qualified`; Haiku's VQ-2 qualified corresponds to Opus 4.7's `era–norm VQ–2`. In other words, one interpretation is that the VQ divergence across the four models reflects an implicit choice of era-norm context rather than a difference in capability; repeated runs show that the level is not stable even within one model (SI-6.10), so this interpretation is not established. Opus 4.7 is the model that makes this choice explicit.

Stage 4 specific paper identification: Opus 4.7 identifies eight specific historical papers by full citation: Tamao et al. BCSJ 1976, Morrell & Kochi JACS 1975, Yamamoto-Kohara-Yamamoto BCSJ 1978, Parshall JACS 1974, Kochi "Organometallic Mechanisms and Catalysis" (1978) + Whitesides group Ni-dialkyl thermolysis, Terao-Watanabe-Kambe JACS 2002, Anderson-Jones-Vicic JACS 2004/2006, Phapale-Cárdenas Chem. Eur. J. 2009. This list partially overlaps with GPT-5.4 Thinking's list (Morrell-Kochi 1975; Tamao 1976 BCSJ) and adds Yamamoto 1978 (later retracted as a misattribution; SI-6.7), Parshall 1974, Terao-Kambe 2002, Anderson-Jones-Vicic 2004/2006 — so the union of the two runs names more candidate papers than either GPT-5.4 or Sonnet alone.

GPT-5.4 Thinking (v17) — cross-vendor robustness comparison plus Stage 4 specificity

GPT-5.4 Thinking's v17 run (2026-04-19) produced the same output structure as the Sonnet record of SI-3: VQ-1 judgment as in the Sonnet v17 run, 6 SRGs, `scope_type = tool_question`, both `confirmed_english_∗` forms, 6 D-4 candidates + 1 retrospective. This shows that the prompt's output structure carries across vendors. It is not independent confirmation of the analysis: both runs used the same prompt, and the models' training data overlap (Limitations L2).

GPT-5.4 also assigns **EP-10a Kinetic verification pack**, a provisional label already listed in the Stage 3 prompt it was given; this is not an independent derivation (SI-6.5).

Stage 4 specific paper identification: GPT-5.4 Thinking identifies seven papers with DOIs: Tamao-Zembayashi-Kiso-Kumada *J. Organomet. Chem.* 1973 (55, C91; olefinic-halide stereochemistry), Kiso-Tamao-Kumada *J. Organomet. Chem.* 1973, Morrell-Kochi *JACS* 1975, Tamao BCSJ 1976, Tamao *Tetrahedron* 1982, Chen *JACS* 2019, Dawson *Chem. Eur. J.* 2020. This list partially overlaps with Opus 4.7's list (Tamao 1976; Morrell-Kochi 1975) and adds the 1973 Tamao-Zembayashi-Kiso-Kumada paper on olefinic-halide stereochemistry (proposed as a partial filling of EP-04a, which Sonnet predicted as unfillable), the 1973 Kiso-Tamao-Kumada halide-isomerization Communication, Tamao 1982 *Tetrahedron*, Chen 2019, and Dawson 2020. The union of the two runs names more 1973–2020 Kumada-lineage candidates than either run alone (the GPT-5.4 run had web search enabled).

Weakness: `loc: not recoverable` throughout Stage 1. This is a tool-level rather than capability-level limitation (GPT's PDF intake does not surface per-page position tokens in the same way Claude's multimodal Read does), but it affects auditability of the `evidence_chain` — co-author verification must rely on quoted text strings rather than position markers.

SI-6.5 Runs that assigned EP-10a, and the prompt they used (corrected in R1)

The submitted version of this section presented EP-10a (Kinetic verification pack) as "independently re-derived" across five, later seven, sources, and called this "the strongest single cross-case validation signal". That description is withdrawn. The deposited Stage 3 prompts show that the runs were not independent of the label:

- The common EP table of the Stage 3 prompt lists `EP–10a | Kinetic verification pack（暫定）` ("provisional") from version 1 (`prompt_stage3_v1.md`, tag `stage3–v1–20260402`), together with PRI-10 *Kinetic overclaim* (provisional). EP-02a is in the same table.
- Versions 2 and 3 (`stage3–v2–20260416`, `stage3–v3–20260425`) also describe PRI-11 *Mechanism-kinetics decoupling*, naming Breitenfeld 2014 as its origin and Pengfei 2024 as a confirmation.
- The Opus 4.7 output on Kumada 1972 reproduces the label verbatim, including 「（暫定）」.

Source	Run date	Stage 3 prompt	What the run did
Sonnet 4.6 on Breitenfeld 2014	2026-04-16	v1 (no PRI-11 in the prompt)	Proposed a PRI-11 candidate, *mechanism-kinetics decoupling*, for this paper. **The only source not exposed to the pattern through the prompt.** EP-10a not assigned.
Sonnet 4.6 on Mazet 2021 and on Pengfei/Li 2024	2026-04-16	v1	Proposed different PRI-11 candidates (matched-pair control; analytical-to-operating extrapolation), not mechanism-kinetics decoupling.
Sonnet 4.6 on Breitenfeld 2014, Mazet 2021, Pengfei/Li 2024	2026-04-17	v2 (describes PRI-11)	Applied PRI-11 as described in the prompt.
GPT-5.4 Thinking on Kumada 1972	2026-04-19	v2	Assigned EP-10a (tagged to PRI-10).
Opus 4.7 on Kumada 1972	2026-04-19	v2	Assigned EP-10a, reproducing the prompt label.
GPT-5.5 Thinking on Kumada 1972 and Murai 1993	2026-04-26/27 (after EP-10a was added)	v3 (S0–S2 v18.2)	Assigned EP-10a.

What the record supports is narrower: several models, on several papers, judged a prompt-supplied provisional label applicable, and one Stage 3 v1 run proposed the related pattern before the prompt described it. That is evidence that the label fits, not evidence of independent derivation. EP-10a and EP-02a remain in the dictionary (SI-3.7), but neither is used as robustness or cross-vendor evidence in the revised manuscript. Version-control history begins on 2026-06-10, so the prompt contents at run time rest on the prompt tags and on the verbatim echoes in the outputs.

SI-6.6 VQ Judgment Divergence and the Era-Norm Reading

The four-model VQ judgments for Kumada 1972 are:

Model	VQ judgment
Claude Haiku 4.5	VQ-2 (qualified)
Claude Sonnet 4.6	VQ-1 (Observation)
Claude Opus 4.7	**era-norm VQ-2 / modern VQ-1 qualified (dual-layer)**
GPT-5.4 Thinking	VQ-1 (Observation)

R1 note. The reading below is an interpretation, not a demonstrated resolution. Repeated runs (SI-6.10, Table SI-6.10-2) show that the overall VQ level is not stable even within one model. Sonnet 4.6 gave VQ-2 in 10 of the 12 runs of the paper's v18.4 configuration, VQ-1 in the three v17 runs, and VQ-1 in 9 of 10 Stage 3-only runs on other inputs; Haiku 4.5 gave VQ-2 in only 4 of its 9 Stage 3 runs, so the "Haiku's VQ-2" premise of the reading below rests on a subset of runs. The four-model difference below is therefore confounded with prompt version and input, and it cannot be attributed to the era-norm frame or to the model. Absent Opus 4.7's dual-layer framing, this could be read as two-vs-two vendor/tier disagreement. One reading of Opus 4.7's explicit separation is: **Sonnet and GPT-5.4's VQ-1 corresponds to the "modern VQ-1 qualified" layer of Opus 4.7's judgment; Haiku's VQ-2 corresponds to the "era-norm VQ-2" layer.** On this reading the four models would differ in which era-norm context they implicitly adopt rather than in capability; the confound above means the reading is not established.

If the interpretation held, it would have three implications (it is not established; see the R1 note above):

1. **For the VQ protocol**: the Stage 3 v2 prompt should be extended to require models to **explicitly report both era-norm and modern-era VQ judgments** as a standard output. Opus 4.7 does this spontaneously; other models should be prompted to. This is a recommended v18 update.
2. **For SI-3 interpretation**: the Sonnet VQ-1 judgment used in SI-3 is the "modern" layer. A complete SI-3 reading should recognize that the *era-norm* layer is VQ-2 (the 1972 Communication format passes the methodological gate when evaluated against 1972 norms). The Era-Norm Annotation in SI-3.8 already captures this distinction at item level; the VQ judgment at section level was not similarly layered in SI-3 but will be in a future revision.
3. **For Discussion 2 (post-training hypothesis)**: the apparent VQ-2 vs VQ-1 disagreement would not reflect a capability gradient; the repeated runs of SI-6.10 show that it is at least partly configuration-dependent. This removes one of the "capability divergence" signals that early drafts of Discussion 2 relied on, and instead strengthens the **methodology-robustness** interpretation of the cross-vendor data.

A further divergence not quantified in SI-6.3: filling-type assignment (Stage 4)

The quantitative table in SI-6.3 tracks structural dimensions (output volume, A/B/C/U counts, SRG / CLA counts, D-4 candidates, VQ judgment) but does **not** tabulate the **filling-type label** assigned to specific filling-paper candidates in Stage 4. An ad-hoc review of the four Stage 4 outputs (2026-04-22) surfaces a divergence pattern that complements the VQ era-norm finding above and deserves explicit acknowledgment:

- **Paradigm-shift label usage**: only two of the four models used the "Paradigm-shift" filling-type label at all. Opus 4.7 applied it to the Anderson-Jones-Vicic 2004/2006 Ni(I)/Ni(III) intermediate work and to the Kambe 2002 sp^3 scope extension; Sonnet (English output) applied it as an explicit sub-entry for Breitenfeld-Hu 2013. Sonnet (Japanese output — the submission primary) and GPT-5.4 Thinking did not use the label at any point, preferring Partial / Indirect even for the same candidate family. Haiku 4.5 did not use it either and additionally introduced its own non-five-type vocabulary ("Attempted but Insufficient," "Communication-era tolerated," "Era-norm dependent") — a pattern that may be read as **structural dissatisfaction with the five-type vocabulary's implicit one-dimensionality** (i.e., Haiku appears to be attempting to encode a second axis capturing era-norm versus data-completeness, rather than failing at compliance in a random direction).
- **Sonnet JA–EN asymmetry within a single model**: the Sonnet 4.6 Japanese output (`_s4v2.md`) and the Sonnet 4.6 English output (`_s4v2_en.md`) were produced by the same model version under the same v17 prompt family, differing only in the language track. The Japanese output does not use "Paradigm-shift"; the English output does. One plausible hypothesis — not empirically tested here — is a corpus-level hedging bias in Japanese academic prose that makes strong paradigm-change claims more costly to commit to in Japanese than in English. The JA–EN difference is registered as a study target in `future_work_pointers/deferred_items_registry.md` (deposit).
- **Self-reflexive observation**: this filling-type divergence is itself a UQS-detectable absence within our own methodology. The SI-6 comparison table was designed to measure structural dimensions amenable to quantitative reporting; the filling-type layer, which requires conceptual rather than structural judgment, was not included in the quantitative comparison and therefore remained invisible until the ad-hoc review. A UQS-style analysis of UQS would surface this silence as an RI-01 (Reporting integrity) item for the present methodology paper — the measurement frame was not uniformly applied across the available judgment layers. We record this observation explicitly rather than leaving it implicit, and the corresponding Phase 2 work (systematic cross-model filling-type agreement study with dimensional-adequacy test) is registered as `future_work_pointers/deferred_items_registry.md` §1.15.
- **Methodological interpretation**: the Partial-vs-Complete boundary is in part a data-completeness judgment (what fraction of the EP's requirements is covered by the candidate's evidence), whereas the Paradigm-shift label requires a conceptual judgment on **whether the source paper's question is being answered in the source's framework or in a different framework that supersedes it**. The former is more operationalizable from the source and candidate texts alone; the latter requires an external judgment about framework continuity. It is therefore unsurprising, in hindsight, that cross-model agreement is weaker on Paradigm-shift than on Partial. The weakness is an invitation to v18 prompt revision: **operationalizing the Paradigm-shift criterion** is the boundary between a methodology that remains LLM-dependent (where Paradigm-shift is an implicit concept-level call the model makes) and one that is transferable to human analysts (where the criterion is articulated and reproducible). This elevation of the v18 revision scope is registered in `future_work_pointers/deferred_items_registry.md` §1.8 as a first-priority item.
- **Scope extension — the same question applies to the remaining four filling types.** The ad-hoc 2026-04-22 review surfaced the Paradigm-shift divergence because the label's cross-model rarity made it visually salient. The same question, however, applies to the **Complete / Partial / Indirect / Attempted** assignments: the SI-6.3 table does not report per-type agreement rates for these four either, and Partial-vs-Complete boundary cases in particular could exhibit analogous (though quieter) divergence that the current measurement frame would not detect. The asymmetry between data-completeness and framework-continuity judgments described above makes the remaining four types less likely to show Paradigm-shift-magnitude divergence, but the absence of quantitative measurement leaves the claim unverified. The Phase 2 cross-model agreement study registered as `future_work_pointers/deferred_items_registry.md` §1.15 is therefore scoped to **all five filling types**, with κ-based agreement reporting and the dimensional-adequacy test applied uniformly rather than to Paradigm-shift alone. The self-reflexive observation from the previous bullet — that the SI-6.3 measurement frame was not uniformly applied across available judgment layers — generalizes to this full scope.

SI-6.7 Stage 4 Filling-Paper Union (Cross-Model Composite)

Combining the specific-DOI filling papers identified by Opus 4.7 and GPT-5.4 Thinking (Sonnet and Haiku provided topical candidates only) yields a composite list of candidates longer than any single run's (the GPT-5.4 run had web search enabled); it

is a list of candidates, not a verified filling map:

Year	Paper	EPs filled	Identified by	Coverage
1973	Tamao-Zembayashi-Kiso-Kumada *J. Organomet. Chem.*	EP-04a (stereochem), EP-10a (kinetic proxy)	GPT-5.4	1973 self-filling of cis/trans stilbene anomaly
1973	Kiso-Tamao-Kumada *J. Organomet. Chem.*	EP-10a (halide isomerization)	GPT-5.4	1973 halide-dependent isomerization
1974	Parshall *J. Am. Chem. Soc.*	EP-03a (Ni intermediate)	Opus 4.7	Nickel intermediate characterization
1975	Morrell-Kochi *J. Am. Chem. Soc.*	EP-03a (partial), EP-04a, EP-06a, EP-10a	GPT-5.4 + **Opus 4.7**	aryl-alkyl Ni(II) mechanism; imported-model validation
1976	Tamao-Sumitani-Kiso-Zembayashi-Fujioka-Kodama-Nakajima-Minato-Kumada *Bull. Chem. Soc. Jpn.*	EP-01a, EP-02a, EP-05a (partial), EP-07a, EP-09a (partial)	GPT-5.4 + **Opus 4.7**	"General Scope and Limitations" Full Paper I; 60-entry scope table

Year	Paper	EPs filled	Identified by	Coverage
1978	~~Yamamoto-Kohara-Yamamoto~~ ~~*Bull. Chem. Soc. Jpn.* 51, 2091~~ (retracted — citation-content separation, see below)	~~EP-03a~~	Opus 4.7	× **Triple-mismatch citation-content separation** (PDF-verified 2026-04-21, see SI-9.3 mode iv): the 1978 *BCSJ* 51, 2091 record is the Yamamoto-**Hayashi**-Yamamoto **polycondensation** paper (poly-phenylene synthesis from p-dihalobenzene + Mg with Ni/Pd catalysts), containing no trans-NiAr(X)(PEt_3)$_2$ isolation. The 1981 *BCSJ* 54, 2010 paper by Yamamoto-**Kohara**-Yamamoto is about **monoalkyl**-Ni(II) complexes (R = Me, Et) and explicitly cites the monoaryl-Ni isolation work in its refs 8–10 as distinct prior literature. Thus the LLM's citation conflates (α) authorship (Hayashi ≠ Kohara), (β) journal-year-volume (1978,51,2091 ≠ 1981,54,2010), and (γ) content (polycondensation / monoalkyl ≠ monoaryl isolation). Removed from the filling union. **UQS self-analysis completed 2026-04-21** (archived at `papers/ep/kumada1972/yamamoto1978/`): the actual 1978 Hayashi polycondensation paper **inherits** Kumada 1972's EP-03a gap rather than filling it (Scheme 1 proposes a Kumada-lineage catalytic cycle with intermediates **1** and **2** but isolates none), and applies the PRI-11 pattern and the EP-10a label, both supplied by the Stage 3 prompt (not independent confirmations; SI-6.5) (see commentary below and SI-3.7 update). Not reinstated into the filling union — retraction stands; the UQS pointer makes the methodological value of the hallucination-victim paper visible for downstream readers.

Year	Paper	EPs filled	Identified by	Coverage
1981	Yamamoto-Kohara-Yamamoto *Bull. Chem. Soc. Jpn.* 54, 2010	EP-03a (Partial, via alkyl-Ni analog; full text read by the authors)	Retrieval-diagnostic (added 2026-04-21; UQS self-analysis 2026-04-21)	Post-hoc addition from PDF verification: isolates 13 $NiR(Y)L_n$ monoalkyl-Ni(II) complexes including *trans*-$NiMe(OCOPh)(PEt_3)_2$ (complex **10**) and NiEt(Cl)(bpy) (complex **14**). A full v17 + Stage 3 v2 + Stage 4 v2 UQS self-analysis was subsequently performed on this paper (archived at `papers/ep/kumada1972/yamamoto1981/`) and its reading is consistent with the Partial verdict: the **$NiR(X)L_n$ structural framework** is empirically accessible and validated across bpy / PEt_3 ligand families for R = Me, Et, but the paper explicitly demarcates monoaryl-Ni isolation as prior work of other groups (refs 8–10 of the paper: Hidai 1971, Morrell-Kochi 1975, Uchino-Yamamoto-Ikeda 1975), uses bpy / PEt_3 rather than dpe, does not test catalytic relevance, and does not exclude Ni(I)/Ni(III) pathways. The filling depth is therefore limited to the structural-framework layer; R = aryl, L = dpe, and catalytic connection remain unfilled by this paper. Not identified by any v17 Stage 4 run; recovered only by the parametric-retrieval ↔ WoS cross-check. EP-alignment table: `papers/ep/kumada1972/yamamoto1981/STATUS.md`.
1978	Kochi "Organometallic Mechanisms and Catalysis" textbook + Whitesides Ni-dialkyl thermolysis	EP-06a, EP-04a (partial)	Opus 4.7	Textbook-level analogy-validation
1982	Tamao *Tetrahedron*	EP-02a, EP-07a	GPT-5.4	Heterocyclic halides scope extension
2002	Terao-Watanabe-Ikumi-Kuniyasu-Kambe *J. Am. Chem. Soc.*	EP-07a	Opus 4.7	sp^3 halide boundary extension (pre-Breitenfeld)
2004 / 2006	Anderson-Jones-Vicic *J. Am. Chem. Soc.*	EP-03a, EP-07a (partial)	Opus 4.7	Ni(I)-aryl intermediate direct observation
2009	Phapale-Guisán-Ceinos-Buñuel-Cárdenas *Chem. Eur. J.*	EP-04a (modern validation)	Opus 4.7	Modern benchmarking of Ni-Kumada
2019	Chen et al. *J. Am. Chem. Soc.*	EP-06a, EP-10a	GPT-5.4	Mechanism-kinetics unification for Kumada system
2020	Dawson et al. *Chem. Eur. J.*	EP-03a, EP-06a	GPT-5.4	Spectroscopic state-assignment for Kumada system

Cross-model convergence: Tamao 1976 BCSJ and Morrell-Kochi 1975 are identified by **both** Opus 4.7 and GPT-5.4 Thinking. Two runs naming the same paper is agreement between two single runs (one with web search enabled), not verification; candidate-level checks are in SI-9.10.

Opus 4.7-specific contribution: Parshall 1974, ~~Yamamoto 1978~~ (retracted; see row above), Kochi textbook 1978, Terao-Kambe 2002, Anderson-Jones-Vicic 2004/2006 (not the Jones–Vicic 2005 context paper of Results Part 2), Phapale 2009 — five remaining papers in the 1974–2009 interval (after removing the Yamamoto 1978 citation) that GPT-5.4 did not identify. This extends the "long gap" interpretation in SI-4.9 / Results Part 2: the gap of about 27 years between layers 1 and 2 (1982–2009) has internal structure, with 2002–2009 papers serving as a bridge.

Post-hoc retrieval-diagnostic addition (2026-04-21): Yamamoto-Kohara-Yamamoto 1981 *BCSJ* 54, 2010 was added via library-hardcopy PDF verification as an alkyl-Ni analog partial filler for EP-03a. It was not identified by any of the four v17 Stage 4 runs. Recovering this paper required WoS title-keyword screening followed by PDF reading — a cross-check pathway distinct from the model-parametric retrieval reported in SI-6. A full v17 + Stage 3 v2 + Stage 4 v2 UQS self-analysis was then executed on this paper (archived at `papers/ep/kumada1972/yamamoto1981/` with full provenance in `STATUS.md`); the analysis confirms the Partial-filler verdict at the structural-framework level ($NiR(X)L_n$ scaffold empirically validated for R = Me, Et across bpy / PEt_3 ligand families) while establishing that R = aryl, L = dpe, catalytic connection, and Ni(I)/Ni(III) exclusion remain unfilled by this paper. This single recovery illustrates the SI-9 pilot finding that WoS structured queries, despite a 0/5 hit rate for the LLM's DOI-specific claims, can surface structurally analogous filling candidates when title keywords align; conversely, the LLM's parametric retrieval surfaced content patterns that the strict WoS query envelope filtered out. The UQS self-analysis additionally establishes that Yamamoto 1981's own EP dictionary (EP-02a, EP-05a, EP-08a, EP-09a as unfilled; EP-03a / EP-06a as partially filled by Yamamoto 1971 / Vrieze 1975) is structurally parallel to Kumada 1972's — a genre-level absence pattern that reinforces the SI-4.9 observation about reporting and benchmarking conventions in early-1970s and early-1980s Ni organometallic literature.

Hallucination-victim paper also analyzed (2026-04-21): The 1978 *BCSJ* 51, 2091 record — recovered through the same retrieval-diagnostic pathway as the citation target of the triple-mismatch Opus 4.7 hallucination (see SI-9.3 (iv)) — received a parallel full v17 + Stage 3 v2 + Stage 4 v2 UQS self-analysis on the same day (archived at `papers/ep/kumada1972/yamamoto1978/`). Although the paper is **not** an EP-03a filler for Kumada 1972 (it is a polycondensation paper whose mechanism proposal is not backed by intermediate isolation), it proves methodologically informative: Yamamoto 1978 *inherits* Kumada 1972's EP-03a gap rather than filling it, proposing a Scheme 1 catalytic cycle in the Kumada lineage (with explicitly named intermediates **1** and **2** that parallel Kumada's intermediates) and extending it to a polymer-scale system without isolating any intermediate. Together, the three papers form an observation-level chain — Kumada 1972 proposes intermediates without isolation → Yamamoto 1978 inherits and extends the gap to polycondensation → Yamamoto 1981 provides first partial self-filling at the alkyl-analog level — that illustrates how absences can *propagate* across related publications of the same author lineage rather than being resolved. The chain is not recorded as a new filling-type category in Methods 2 (the existing Complete / Partial / Indirect / Attempted / Paradigm-shift vocabulary is left unchanged for submission stability); formalizing "gap-inheritance" as a category is registered as a Phase 2 Codebook revision candidate in `future_work_pointers/deferred_items_registry.md` §1.7.

GPT-5.4-specific contribution: Tamao-Kiso 1973 (self-filling of EP-04a), Kiso-Tamao 1973, Tamao 1982 *Tetrahedron*, Chen 2019, Dawson 2020 — five papers covering self-filling at 1973 and modern technology-driven validation at 2019–2020.

The list is kept as a record of the model runs; it is not used as a filling map in the main text.

SI-6.8 Correspondence with Reasoning-Oriented Post-Training Profiles

Discussion 2 proposes that UQS executability requires a capability profile cultivated by reasoning-oriented post-training (a vendor-neutral family of training regimes that emphasise structural prompt compliance, external-regulation parallel retention, multi-step inference stability, and suppression of prohibited output forms). The four-model v17 comparison provides the following evidence relevant to this hypothesis:

Convergence across four frontier models. All four models (Claude Haiku 4.5, Sonnet 4.6, Opus 4.7; GPT-5.4 Thinking) produce the **complete structural output template**: D-4 candidates with metadata, SRGs, U items (self-contradictory absence rule), Era-Norm Annotation, `scope_type` correctly judged, both `confirmed_english_*` forms, Stage 3 and Stage 4 completion. This convergence is consistent with the **weak form** of the post-training hypothesis — that reasoning-oriented training across vendors converges on the capability UQS needs — but four models on one paper cannot test it.

Cross-tier success within Claude family. Haiku 4.5 (smallest Claude tier) with the v17 prompt produced the complete output structure, where Claude 3.5 Haiku with the v15 prompt did not. Two combinations were observed, and both prompt and model differ between them, so the effect of the prompt (self-contradictory absence rule, Dropped Quantitative Benchmark C subtype, explicit Era-Norm, the dual-layer Stage 3 v2 structure) cannot be separated from that of the model. A joint prompt–model requirement is one hypothesis consistent with the two observations; it is not shown by them.

Analytical-depth gradient within the family. Among the four models, Opus 4.7 produces the highest Root-Issue decomposition (10 SRGs), the dual-layer VQ judgment, and the richest Stage 4 specific-paper identification. GPT-5.4 Thinking matches Sonnet/Opus on core structural outputs and contributes distinct Stage 4 specific papers from a different literature-coverage angle. Haiku 4.5 completes the structural template but with lower granularity (5 SRGs, topical rather than specific Stage 4 candidates). The gradient is **analytical-depth dependent, not structural-template dependent** — all models pass the structural gate; they vary in depth.

Residual family-specific capabilities. 10-SRG decomposition (Opus 4.7); specific-paper DOIs from different slices of the literature (Opus 4.7, parametric; GPT-5.4, with web search enabled, so not comparable as coverage). With one run per model, these differences may reflect run-to-run variation (SI-6.10) as much as model family; they are recorded, not interpreted as family-specific capabilities.

What the data do NOT support:

1. **Strong form of the post-training hypothesis** (*Anthropic safety-oriented post-training is specifically necessary for UQS*) is **weakened** by the GPT-5.4 Thinking convergence, but not decisively excluded because model-family, training-corpus, and inference-time differences remain entangled.
2. **Pure capability-threshold narrative** (*below some parameter scale, UQS fails categorically*) is **not supported** by Haiku 4.5's success under v17.
3. **Pure prompt-engineering narrative** (*any LLM with v17 can do UQS*) is **not confirmed** — we have no negative case at v17 yet; the Gemini 1.5 Pro trial was at v17 but auto-reassigned and retained as internal-only.

What the data DO support:

- *In the configurations examined, UQS ran to a complete structural output with a mature prompt (v17) and a reasoning-oriented frontier model.* That both are required is a hypothesis consistent with the four-model and Haiku v15/v17 comparison; the comparison does not vary post-training while holding the other factors fixed, so it cannot show either to be necessary.
- *Analytical depth varies across model families beyond the structural template* — a less central finding but useful for production-setup guidance (SI-6.9).

C1 / C2 non-firing: one possible reading. Two prompt-instruction additions recorded in the regression log (SI-2; C1 on app2, C2 on scr1) failed to activate during the v17 cycle; both required two-stage reasoning. One possible reading, under the joint-requirement hypothesis, is that the model's inference depth is a further threshold in two-stage-reasoning contexts that the v17 prompt does not remove; this too is an unseparated hypothesis. Disentangling this from vendor-effect awaits the prompt-version-matched N × 4 model matrix on multiple papers (Phase 2 verification path b above; see Data Availability for the Phase 2 register).

Future verification paths. Whether reasoning-oriented post-training, model scale, generational maturity, inference-time compute, or prompt maturation is the primary factor cannot be isolated from four models on one paper. Three verification paths are registered as Phase 2 work: (a) comparison with and without reasoning-oriented post-training on an identical base model; (b) prompt-version-matched N × 4 model matrix across multiple papers; (c) benchmarks mapped to UQS's capability requirements.

SI-6.9 Recommended Production Setup (Revised)

Based on the four-model v17 comparison, the production configuration recommended in Methods 6 is as follows:

1. **Primary model: Claude Sonnet 4.6** — for its balance of structural completeness, full position-citation (`loc`) preservation, and the consistency advantage that v15–v17 prompt refinement was validated on Sonnet.
2. **Recommended supplementary uses**:
    - **Claude Opus 4.7** for the deepest Root-Issue Codebook decomposition (10 SRGs vs 5–6 in other models) and the dual-layer Era-Norm VQ judgment. Recommended when Stage 3 granularity matters for downstream VQ audit or Codebook extension.
    - **GPT-5.4 Thinking** for cross-vendor robustness comparison (VQ judgment independence check) and, with web search enabled, a second source of Stage 4 candidates.
    - **Claude Opus 4.7 + GPT-5.4 Thinking union** for Stage 4 filling-paper candidates (the two runs together name more 1974–2020 Kumada-lineage candidates than either alone; every candidate still needs the checks of SI-9.10).
    - **Claude Haiku 4.5** for cost-sensitive batch runs where the structural template is sufficient and deep analytical granularity is not required. Cross-check against Sonnet recommended on flagship papers.

3. **Deprecated / research-only**:
    - Older Haiku generations with v15 or earlier prompts — incomplete structural output.
    - Gemini-family runs not under model-pinned v17 protocol — retained as internal methodology records only.

SI-6.10 SRG-root correspondence across models, and a Haiku 4.5 replication (R1 addendum)

Why this section was added. The main text previously stated that the four models "agreed on the same six SRGs". This section records the numbers of SRGs each model reports: Haiku 4.5, 5; Sonnet 4.6, 6; Opus 4.7, 10; GPT-5.4 Thinking, 6. At R1 we therefore re-derived the correspondence at the level of **which data absences surfaced**, not SRG labels or counts. We also added a pre-registered Haiku 4.5 replication (N = 3, the v17 "repro" replicates, `haiku45_repro_2026-09/`), run on 2026-09-29, before that model's retirement floor of 2026-10-15.

Method.

- *Reference roots.* The six roots are the six SRGs listed in SI-3.4, relabelled R1–R6 here (R1 = SRG-1, …, R6 = SRG-6), each defined by its central data absence. SI-3.4 reproduces the v17 + Stage 3 v1 Sonnet 4.6 record (`papers/dev/kumada/kumada_v17_sonnet_s3.md`). The "Sonnet 4.6 (reference)" row below is a different run, the v17 + Stage 3 v2 Japanese output (`kumada_v17_sonnet_s3v2.md`), which has the same six SRG names.
- *Blocks.* Every SRG/CLA block of seven Stage 3 outputs was extracted: the four April 2026 runs (one of them the Sonnet reference) and the three Haiku 4.5 replicates. That gave 46 blocks, two of which are heading-only "no CLA" blocks. Model names were removed and the blocks shuffled.
- *Rating.* Two LLM raters worked independently, with separate directories and opposite reading orders. For each block they listed the roots it addresses as a central absence. Rubric: `srg_root_mapping/rubric_srg_root_mapping.md`, frozen before rating.
- *Agreement.* The raters gave identical root sets for 44 of the 46 blocks; both disagreements are in the third Haiku replicate.
- *Author check (2026-10-05).* An author read all 46 blocks against the rater verdicts and agreed with the 44 agreed root sets. The two disagreements were left undecided. One of them (a CLA on the unstated Ni oxidation state: R1 or none) does not change the table, because the same replicate's SRG-1 already covers R1. The other (SRG-5 on scope limits and side reactions: R2 + R4 or R2 alone) decides whether that replicate covers R4; it stays marked △.
- *Replication configuration.* `haiku45_replication/preregistration_haiku45_R1.md`: v17 Stages 0–2 and Stage 3 v2 (JA), as in the April run, with the model verified as `claude-haiku-4-5`.

Table SI-6.10-1. Roots present per output (both raters agree; △ = raters disagree).

output	R1 cycle	R2 by-products	R3 yield analytics	R4 scope	R5 solvent/conditions	R6 cis/trans	SRG/CLA blocks
Sonnet 4.6 (reference, 2026-04)	✓	✓	✓	✓	✓	✓	7
Opus 4.7 (2026-04)	✓	✓	✓	✓	✓	✓	12
GPT-5.4 Thinking (2026-04)	✓	✓	✓	✓	✓	✓	6
Haiku 4.5 (2026-04)	✓	✓	✓	—	✓	✓	5
Haiku 4.5 replicate 1 (2026-09)	✓	✓	✓	✓	—	—	4
Haiku 4.5 replicate 2 (2026-09)	✓	✓	✓	—	—	—	4
Haiku 4.5 replicate 3 (2026-09)	✓	✓	✓	△	✓	✓	6

Reading.

- *Present in every output.* R1–R3 — the unevidenced catalytic cycle, the missing by-product data behind the selectivity claim, and the analytical basis of the Table I yields — appear in all seven outputs. That includes the three Haiku replicates run five months after the original analyses.
- *Other roots.* R4–R6 depend on the model and, for Haiku 4.5, on the run.
- *The earlier main-text statement.* "Four models agreed on the same six SRGs" is corrected: three of the four models covered all six roots, and Haiku 4.5 covered five, lacking the scope-restriction root.
- *Haiku 4.5 replicates.* They cover three to six roots, so single Haiku runs should not be read as a stable inventory beyond R1–R3.

Table SI-6.10-2. VQ judgment for Kumada 1972 across models and runs (read from each Stage 3 output's "current judgment").

model and configuration	runs	VQ judgment
Sonnet 4.6, v18.4 configuration (the paper's; Stage 3 v2, 6 JA + 6 EN runs, 2026-05)	12	VQ-2 in 10 of 12 (both exceptions are run 4, JA and EN: VQ-1)
Sonnet 4.6, v17 (Stage 3 v1, v2 JA and v2 EN, 2026-04)	3	VQ-1 in 3 of 3
Sonnet 4.6, Stage 3 v2 EN only, 2026-09 (the B2 → Stage 3 check, SI-11.20: 5 on pipeline repro records, 5 on single-prompt records)	10	VQ-1 in 9 of 10
Haiku 4.5, v17 (Stage 3 v2 JA): the original run of 2026-04 and three replicates of 2026-09 (`haiku45_repro_2026-09/`)	4	VQ-2 in 4 of 4 (three labelled limited or qualified; in replicate 2 the reasoning under the heading says VQ-2 is not reached)
Haiku 4.5, v17 (Stage 3 v2), 2026-09, subagent runs (`haiku45_2026-09/`, runs 2–3; run 3 given one prompt per turn)	2	VQ-1 not reached; VQ-1 qualified
Haiku 4.5, v18.4 (Stage 3 v4), 2026-09 (`haiku45_2026-09/`, runs 1–3)	3	modern baseline: VQ-1 in 2 of 3, VQ-1 not reached in 1; two runs add an era-norm qualified VQ-2
Opus 4.7	1 run of 2026-04	VQ-2 (era-norm) / VQ-1 qualified (modern)
GPT-5.4 Thinking	1 run of 2026-04	VQ-1

The same model gives different overall VQ levels under different prompt versions and inputs. The overall VQ level is therefore not used as a stable quantity in this paper. *R1 correction:* an earlier R1 draft reported "Sonnet 4.6: VQ-1 in 10 of 11" because it omitted the 12 runs of the paper's own v18.4 configuration. That statement was wrong, and it is replaced by the rows above. *R1 correction (2026-10-16):* the Haiku 4.5 row previously listed only the four v17 runs that gave VQ-2. The five further Haiku Stage 3 runs of 2026-09 deposited in `papers/dev/kumada/haiku45_2026-09/` gave lower levels and are now listed; under the same v17 configuration the two 2026-09 subagent runs gave VQ-1 or below.

Datum-selection agreement: within one model vs across models

The statistics above describe stability of *labels and counts*. They do not answer the question a referee raised directly: is the divergence **between** two models larger than the variation **inside** one model? To answer it on one scale, we reduced each Stage 1 Turn 4 output to the set of source passages it anchors to (a *datum* = a quoted location), matched datums across runs by fuzzy agreement of the quoted text (threshold 88; signatures shorter than 30 characters excluded), and computed the pairwise Jaccard index. This is measured at the level of *which passages surface*, not of how a surfaced passage is labelled, consistent with the treatment elsewhere in this SI.

All runs use the paper's v18.4 configuration on Kumada 1972. Values were computed with the deposited matcher, `tier_7_aggregation_scripts_and_methodology_evolution/classification_concordance/classification_concordance.py` (project path `scripts/classification_concordance.py`); `si6_10_datum_jaccard.py` in the same folder, run from the deposit root, reprints this table from the deposited runs (requires `rapidfuzz`). The inputs are the five files `haiku45_2026-09/kumada_v184_haiku45_run{1–5}_s1.md` and `repro_2026-07/kumada_v184_sonnet46_repro_run{1–5}_s1.md`; the sub-folder `_deviation_prompt_lookahead/` holds superseded runs and must be excluded. The values are the mean and sample SD of the per-pair Jaccard index over the listed pairs (the script's printed summary is a pooled Jaccard, which differs).

Comparison	Runs	Pairs	Mean pairwise datum Jaccard
Within Claude Haiku 4.5	5	10	0.258 ± 0.090
Within Claude Sonnet 4.6	5	10	0.297 ± 0.054
Across the two models	5 × 5	25	**0.188 ± 0.090**

The cross-model value falls below the lower of the two within-model values. Two runs of the *same* model on the *same* input agree on only 26–30% of the passages they anchor to; two runs of *different* models agree on 19%. With 10–25 pairs per estimate we do not test whether this difference exceeds the run-to-run variation; what the data do not show is a cross-model divergence larger than the variation inside either model. Accordingly, SI-6.6 no longer presents the era-norm frame as resolving the VQ divergence; it is retained only as a description of how the two outputs differ in wording, not as an explanation of why they differ.

Two limitations are material. First, this covers one paper and two models, and is descriptive only. Second, the measure depends on quoted-text matching, so a systematic difference in citation format would depress it. Such a difference is present: the five replicates use four different item-row formats (for example `A1 | P1:Lcol:37–40* | "…"` in replicate 1 and `B–2 | loc: … | quote: …` in replicate 4). The signature extractor handles them, so the comparison stands — but the fact that one model under one prompt version emits several row formats is itself an instance of the instability at issue.

What a reader can and cannot check. The arms compared here have different lifespans. The published retirement floors (Anthropic model-deprecation page, checked 2026-10-08) are 2026-10-15 for `claude-haiku-4-5`, a date now past, and 2027-02-17 for `claude-sonnet-4-6` (a floor is a "not sooner than" date, not a scheduled retirement, and retirement is announced at least 60 days in advance). **Once Haiku 4.5 is actually retired (not before the floor), this comparison can no longer be regenerated by anyone — including us.** We state this plainly because the section exists to answer a reproducibility question, and it would be misleading to answer it with a measurement that is itself unrepeatable.

Two layers are involved, as set out in SI-11 (*Rerun equivalence and model-access contingency*):

Layer	What it means here	After Haiku 4.5 is retired
Rerun	Regenerate the outputs under the archived prompt, model, and corpus configuration	**Not available** for the Haiku arm
Inspection	Recompute the reported quantities from the deposited per-run outputs, using the deposited instrument	**Available** — unchanged

Every number in the table above is an inspection-layer quantity: the ten Haiku pairs, the ten Sonnet pairs and the twenty-five cross pairs are all recomputable from the deposited Stage 1 Turn 4 outputs with the deposited matcher, by anyone, at any time. What becomes unavailable is the ability to produce a *new* Haiku run and ask whether it behaves the same way.

We therefore do **not** offer the four-model comparison as a reproducible protocol. It is a **time-stamped observation**, and the claims that rest on it are correspondingly bounded: it supports the statement that UQS executed on four models in 2026, and it does not support any statement a future reader could independently re-establish by rerunning those models. Claims intended to be re-establishable rest instead on the components that do not expire — the prompt versions (SHA-256 in SI-7), the deposited outputs and evidence chains, the scoring instruments, and the model with the longest floor.

This limit is not specific to this paper. Anthropic's own deprecation documentation states that model retirement means "researchers lose access to models for ongoing and comparative studies", and commits to long-term preservation of model weights as partial mitigation. Any methodology that treats a hosted model as part of its apparatus inherits a reproducibility horizon set by that model's lifecycle, not by the study design. Recording the floors alongside the measurements, as above, is the minimum that makes the horizon visible to a reader.

Provenance: these are the v18.4 Haiku 4.5 runs 1–5 of `haiku45_2026-09/` (not the v17 "repro" replicates used for root coverage above). Runs 1–3 were made in 2026-09; runs 4–5 were added on 2026-10-01, before that model's retirement floor (2026-10-15), to match the 10 pairs available for Sonnet 4.6. Run conditions for all five are recorded verbatim in the README of the output folder. The Sonnet 4.6 runs are the five archived reproduction runs. These are a different arm from the pre-registered v17 replication reported above.

Files: `srg_root_mapping/` (extraction script, packets, key, rubric, rater verdicts, aggregation), `papers/dev/kumada/haiku45_repro_2026-09/` (pre-registered v17 replicate outputs), `papers/dev/kumada/haiku45_2026-09/` (v18.4 Haiku replicates 1–5 and run conditions) and `papers/dev/kumada/repro_2026-07/` (v18.4 Sonnet 4.6 reproduction runs).

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† Text marked with † was originally generated in Japanese by the UQS pipeline and has been translated into English for this Supporting Information.

SI-7: UQS Prompt Manifest (v18.4 EN Production Configuration)

The v18.4 English production prompt set used for the v18.4 analyses reported in this paper, together with the earlier prompt files used where the text says so (Kumada 1972 Stage 3 EP-dictionary record: Stage 3 v1; refinement set and four-model comparison: v17), is deposited verbatim in the public archive (Data Availability Tier 1) and archived at the Zenodo deposit accompanying this paper. To keep the Supplementary Information readable while preserving full reproducibility, this section gives a **prompt manifest** — each production prompt's filename, version tag, role, length, and SHA-256 content hash — rather than reproducing the prompt bodies inline. A reviewer can confirm that any locally obtained prompt is byte-identical to the one used here by recomputing its SHA-256 hash and comparing it against the value below; the full prompt text is one click away in the deposit.

The pipeline order is Stage 0 → Stage 1 (Turns 1–4 over vantage points V1/V2/V3 and synthesis) → Stage 2 → Stage 3 → Stage 4, with the Stage 2F cross-paper integration prompt applied to corpora of completed per-paper outputs. Stage 0 and Stage 1 are language-neutral and were executed in English; Stage 2/3/4 of the English-language analyses use the English-track prompt files (`*_en.md`); the Japanese-language multi-run cells, the four context analyses and the R1 control runs used the Japanese-language Stage 2 (row 23), and the Japanese-language Stage 3 v2 and Stage 4 v2 (rows 12–13) were used by the v17 analyses (four-model comparison, refinement set) and the Japanese-language cells. The version-by-version refinement history through v17 is in SI-1; the v18.x changes are in the changelogs in Tier 7 of the deposit.

Production prompt manifest. Source directories: `prompts/v184/` and `prompts/stage2f3/` in the project repository; in the deposit (Data Availability Tier 1) they are `tier_1_prompt_versions/v184_production/` and the top level of `tier_1_prompt_versions/`. Hashes are SHA-256 of the exact deposited files.

#	Pipeline step	Prompt file	Version tag	Words
1	Stage 0 — Positive record	`prompts/v184/prompt_stage0_v184.md`	stage0-v184-20260502	969
2	Stage 1 Turn 1 — Vantage V1 (primary field)	`prompts/v184/prompt_stage1_turn1_v184.md`	v18.4	2145
3	Stage 1 Turn 2 — Vantage V2 (adjacent field)	`prompts/v184/prompt_stage1_turn2_v184.md`	v18.4	1596
4	Stage 1 Turn 3 — Vantage V3 (methodological)	`prompts/v184/prompt_stage1_turn3_v184.md`	v18.4	676
5	Stage 1 Turn 4 — Synthesis	`prompts/v184/prompt_stage1_turn4_v184.md`	v18.4	2797
6	Stage 2 — Interpretive analysis	`prompts/v184/prompt_stage2_v184_en.md`	v18.4 EN	4391
7	Stage 3 — Knowledge-status reassessment	`prompts/stage2f3/prompt_stage3_v2_en.md`	Stage 3 v2 EN	2287
8	Stage 4 — Filling-paper identification	`prompts/stage2f3/prompt_stage4_v2_en.md`	Stage 4 v2 EN	2338
9	Stage 2F — Cross-paper integration	`prompts/stage2f3/prompt_stage2f_v71.md`	Stage 2F v7.1 (internal tag `stage2f-v7-20260417`, as in the outputs)	1375
10	Stage 3 — Kumada 1972 EP-dictionary record (v17 Stage 0–2, run 2026-04-15; SI-3.4–3.10, Table 3)	`prompts/stage2f3/prompt_stage3_v1.md`	Stage 3 v1	668
11	Stage 3 — four context analyses (Uchino 1970, Terao–Kambe 2002, Jones–Vicic 2005, Martin–Buchwald 2007)	`prompts/stage2f3/prompt_stage3_v4.md`	Stage 3 v4	1383
12	Stage 3 — Japanese-language prompt: the four-model comparison (SI-6), the refinement set, and the Japanese-language cells of the multi-run (Kumada JA, Murai JA)	`prompts/stage2f3/prompt_stage3_v2.md`	Stage 3 v2 (JA)	783

#	Pipeline step	Prompt file	Version tag	Words
13	Stage 4 — Japanese-language prompt: the four-model comparison, the refinement set, and the Japanese-language cells of the multi-run	`prompts/stage2f3/prompt_stage4_v2.md`	Stage 4 v2 (JA)	633
14	Stage 4 — earlier version (v17 runs)	`prompts/stage2f3/prompt_stage4_v1.md`	Stage 4 v1	557
15	Stage 4 — four context analyses (as for Stage 3 v4)	`prompts/stage2f3/prompt_stage4_v6.md`	Stage 4 v6	1498
16	Stage 0 — v17 (refinement set, SI-6)	`prompts/v17/prompt_stage0_v17.md`	stage0-v17-20260415	969
17	Stage 1 Turn 1 — v17 (refinement set, SI-6)	`prompts/v17/prompt_stage1_turn1_v17.md`	stage1-v17-20260415	1614
18	Stage 1 Turn 2 — v17 (refinement set, SI-6)	`prompts/v17/prompt_stage1_turn2_v17.md`	stage1-v17-20260415	1247
19	Stage 1 Turn 3 — v17 (refinement set, SI-6)	`prompts/v17/prompt_stage1_turn3_v17.md`	stage1-v17-20260415	585
20	Stage 1 Turn 4 — v17 (refinement set, SI-6)	`prompts/v17/prompt_stage1_turn4_v17.md`	stage1-v17-20260415	2212
21	Stage 2 — v17 (JA/EN mixed; used by all refinement-set runs and all four SI-6 models)	`prompts/v17/prompt_stage2_v17.md`	stage2-v17-20260415	2319
22	Stage 2 — v17 EN (used only by the English-track Kumada run `kumada_v17_sonnet_s2_en.md`)	`prompts/v17/prompt_stage2_v17_en.md`	stage2-v17-en-20260419	3659
23	Stage 2 — v18.4 (Japanese-language prompt; used by the Japanese-language multi-run cells, the four context analyses, and the R1 control runs: reproducibility, decontamination, Haiku 4.5 and input ablation)	`prompts/v184/prompt_stage2_v184.md`	stage2-v184-20260502	3124
24	Stage 0 — v19 (reproducibility control, successor-version arm; Methods 6)	`prompts/v19/prompt_stage0_v19.md`	stage0-v19-20260504	969
25	Stage 1 Turn 1 — v19 (same arm)	`prompts/v19/prompt_stage1_turn1_v19.md`	stage1-v19-20260504	2145
26	Stage 1 Turn 2 — v19 (same arm)	`prompts/v19/prompt_stage1_turn2_v19.md`	stage1-v19-20260504	1596
27	Stage 1 Turn 3 — v19 (same arm)	`prompts/v19/prompt_stage1_turn3_v19.md`	stage1-v19-20260504	676
28	Stage 1 Turn 4 — v19 (same arm)	`prompts/v19/prompt_stage1_turn4_v19.md`	stage1-v19-20260504	3907
29	Stage 2 — v19 (same arm; Japanese-language)	`prompts/v19/prompt_stage2_v19.md`	stage2-v19-20260504	4011

Prompt-file hashes. SHA-256 of each prompt file, by row number of the manifest above.

#	SHA-256
1	`41cd24e8b67534e89bfb8190b056ab3493885d340f8c97db65a732f6336a5bad`
2	`078e9d954e46776959605b0087eade8ad2871e88e3f4d9e7ccb6b22573518fea`
3	`cc4788515a4e3e841810d014104c70233d1538d1901780f378c819d96e25d777`
4	`117e3a2a6a6be420c9e3c65ef504b7f1e4ef181e37670b1bac4820c54f3ff41e`
5	`8f1bf57280a48742b669ff59d8273c5e205145fd9c81091c2333188c09a8ed1e`
6	`5127fd1fccbdd3c41ff68fd564d75e40906d417fca162933264cd6a1fb950dbf`

#	SHA-256	
7	b1358ff4c0e64eba6e49c1483b85f1bb638d25098f49147e1b264880897cb170	
8	3dd957a647c70e94a57e5ad46eda6e92d28dafd7ed2cf75efa469a5c3492abdd	
9	0d959a716a441907008701da788abd3339e984355dfcdf92fb36025ed07be756	
10	4f8f1771dab236b925ca2bb249c8e8bdf74be9d25daa68a3f6e08061ced4a214	
11	0aa38a1d5882085f8afdc2e6666476a72abacd5cad4394f3476d2ae55982242a	
12	4ebae4eadc5a75506ea81dd44c7e313ae3741ede0b90c2caf68b4e5a866a5afd	
13	41815da628d29e0eab5f738225ef4c03c34d1a94eddbeca557b327c732ec1f82	
14	11138d8305e92ab0b69357b89c1590edcd151815e9bd65fd47b970ffcaed634f	
15	e47652717ba6908310b007c548c75c4a6a6d85f049260fb4324edb966976f728	
16	9e039bd4df94fbd6b72484de2e1ca8934b7f9320edef7d63d1800fcbc53933aa	†
17	6d04456b07d33d6613671748181f3101486a3d5086b37398ef181ea5016489d0	†
18	80e0da6b7983951be567f8962db763b6156a7af3b35fec2d2648b2c23dd11e4f	†
19	a6d970f8ca31b15715c6308f81b0033456a79380a8ef199f81dfb937a9865775	†
20	99f4bd0c9f24098be689c2881bb6521025220e187a0168ca3e26ed5bfb6ea1ee	†
21	1dc8bf0af60f6f3f28599cce7b120a7fc37aecb5a2abb8de7931e5e6ec0a98b0	†
22	18ce4cc0e8c7130f8fb5cdccbe9470d2019c2f11559a655a556a092f1fe7c249	†
23	0b65c8c7a5b12aa2bfa6639d3ee935f966a659d8f43df7be0264dad07273530c	‡
24	ab03e457f4fb2b34bc72c40f3e6635aa910d44e65c63e3bd64b9795395efe82e	§
25	6b582dd6d9ba544fc8114e54e6cd52068d644a5cc1028af41c37e048da6bdf01	§
26	016c4d3ab5eb63d1304eecba4872017914c86682c3246ec6a01753f5dddbff72	§
27	8962a48af82ca225c725fc0ecedfdaf9ca51ec52d4de9f265157c04589b0c02e	§
28	9a78d255441334f0ae3131b7fc08afa055fca100b8bb031e4f1286751cc540a6	§
29	42b5c03e1c515029214f790fdbc81cf15a68ea848e0754582b9b5575c66c11e3	§

† *The v17 rows were added at R1, and their hashes were computed on 2026-09-29, not recorded at the time of the runs.* The v17 analyses (the refinement set, SI-6) were run on 2026-04-15–19. That these files are identical to the ones used then is supported by two things: their filesystem modification times, which are earlier than or equal to the run dates (2026-04-15 13:46–13:54 for Stages 0–2, and 2026-04-19 08:56 for the v17 EN Stage 2); and the `prompt_version` tags, which match the headers of the archived outputs. The repository's version history begins with a snapshot on 2026-06-10, so identity at the time of the runs cannot be proven from version control.

‡ *Row 23 was added on 2026-10-16, after an independent check found that this Stage 2 prompt, which the deposit has always contained, was missing from the table; its hash was computed then. Its filesystem modification time (2026-05-02 08:15) is earlier than the runs that used it, and its* `prompt_version` *tag matches their output headers.*

§ *Rows 24–29 were added on 2026-10-16; they are the v19 prompts of the successor-version arm of the reproducibility control (Methods 6;* `papers/dev/kumada/repro_2026-07/`*, runs of 2026-07-19), and the files are now in the deposit (*`tier_1_prompt_versions/v19_successor_arm/`*). Their hashes were computed on 2026-10-16. The files were last changed before the runs (version control: 2026-07-18; filesystem: 2026-05-04 to 2026-07-08); the internal tags (…*`-v19-20260504`*) were not updated when the files were revised before the runs, and they match the output headers.*

How to retrieve and verify the full prompt text. (i) Obtain the deposit (public Tier-1 archive or the Zenodo record cited in the Data Availability statement). (ii) For any prompt above, recompute the hash — e.g. `shasum -a 256 prompts/v184/prompt_stage0_v184.md` — and confirm it matches the SHA-256 column. (iii) The deposit's `MANIFEST.csv` lists every prompt file with the same hashes, so a single manifest check confirms the whole set is intact. The prompts are plain Markdown and are read directly; no build step is required to inspect them.

Revision note (R1). In the submitted deposit, the Stage 3 v2 EN file carried one line re-indented on 2026-06-26 and therefore did not match the hash above; the R1 deposit carries the version used for the runs, which does. The six Stage 3 / Stage 4 rows below the Stage 2F row were added in R1; the corresponding files were missing from the submitted deposit.

Note on prior SI-7 versions. Earlier manuscript versions reproduced the full prompt bodies inline here (~19,000 words). The bodies were moved to the deposit and replaced by this manifest to improve SI readability without reducing reproducibility — verification is now a hash check against deposited files rather than a visual diff against pasted text.

SI-8: Evaluation Protocol Details

This SI expands Methods 7 into a step-by-step operational description of the two-tier evaluation protocol for UQS outputs, gives the full definitions of the six response categories, and describes the Tier 2 collective-pattern utilization protocol and the limited expert-verification procedure. At the time of the original submission, systematic pilot-application data under the Tier 1 protocol had not been collected, and SI-8.5 documents the planned data structure. SI-8.6 reports an unstructured pilot conducted before submission, and SI-8.7 reports the blind item-level evaluation conducted for the revision.

Per Methods 7 of the main manuscript, the UQS evaluation is designed as a two-tier structure reflecting the two-fold purpose of the methodology — complementing the researcher's perspective (individual-paper level) and extracting collective patterns (Stage 2F level). This design rejects both precision evaluation aimed at agreement with a "ground truth" and non-sustainable expert evaluation. The UQS outputs that Tier 1 evaluates are, by construction, difference detections against field-normative knowledge; consequently, conventional precision/recall against a gold standard is not an appropriate evaluation target for this methodology.

SI-8.1 Advisor Evaluation of Student Questions — Full Procedure

Per Methods 7 of the main manuscript, Tier 1 of the UQS evaluation is the *advisor evaluation of student questions*. The procedure operationalizes this evaluation in six steps. Steps 1 and 2 are automated post-processing of UQS outputs; steps 3–5 are human-facing; step 6 is record-keeping.

Step 1 — Conversion of Stage 1 outputs to student-question form. The Stage 1 A/B/C/U items carry machine-readable identifiers (`item_id`, `loc`, `quote`, `classification`; A items also carry an inference tier T1/T2/T3). Presented in this raw form, items are immediately legible as LLM output, which defeats the purpose of the advisor-framing (see Step 3). A post-processing LLM pass rewrites each item into the natural form of a question or observation that a graduate student might bring to a thesis advisor after reading the paper. Every rewrite must preserve (a) the specific object of the question (e.g., "the R–R homocoupling by-product"); (b) the anchor in the paper (e.g., "the selectivity claim in the introduction"); and (c) the absence being pointed at (e.g., "no quantitative detection limit is given"). The rewrite is a rhetorical transformation only; no analytical content is added or removed.

Step 2 — Conversion of Stage 2 D-4 candidates to student-question form. The D-4 candidates (items of *unquestioned motivation* generated in Stage 2's multi-layer motivational analysis) require a different rewriting style than the Stage 1 items. Whereas Stage 1 items point at absent evidence or unconducted measurements, D-4 candidates point at assumptions the authors did not recognize as assumptions. The rewrite should preserve this orientation: the student question takes the form of "why does the paper assume X without justifying it?" or "is X really the right frame for this problem?" rather than "where is the evidence for X?". Items from A/B/C/U and from D-4 can be mixed in the presented list (so that the advisor does not adjust judgment by category).

Step 3 — Presentation of the advisor prompt. The respondent is presented with the full rewritten list together with the prompt: *"The following is a list of questions and observations that a student brought to you after reading a paper. As an advisor, how would you evaluate each?"* The advisor is given the paper itself, not the Stage 0–Stage 2 outputs, since the evaluation should reconstruct the advisor's normal mode of reading.

Rationale for the advisor-framing — four response biases the framing mitigates. The Tier 1 protocol replaces the natural framing for evaluating UQS output — *"were you aware of this absence?"* — because that framing invites at least four response biases that contaminate honest assessment. The biases are well-known in survey methodology and cognitive psychology; they are recapitulated here because UQS evaluation sits in their joint footprint and the protocol design must address each.

Bias 1 — Post-hoc rationalization. When asked to recall a past cognitive state, respondents reconstruct rather than retrieve, and the reconstruction is colored by the present salience of the prompt. A respondent reading a UQS-surfaced absence that strikes them as important is likely to retrofit a memory of having noticed it; conversely, an absence that strikes them as trivial is likely to be retrospectively minimized. The result is that "did you notice X?" measures present judgment of X, not past awareness.

Bias 2 — Face-saving. Acknowledging that one had not noticed an absence — particularly when the absence was surfaced by an LLM — is socially costly for an expert, since it can be read as a concession of expertise. The cost is paid privately by the respondent in interview-style settings, but the expectation of paying it shapes the response itself.

Bias 3 — Cognitive dissonance. The two beliefs "I am an expert in this field" and "I had not noticed this absence" are uncomfortable to hold simultaneously. The standard resolutions — "I knew but judged it unimportant", "everyone knows this so it didn't need stating" — are retrofitted reasons rather than independent evidence about whether the absence is genuinely

important. This bias is distinct from face-saving because it operates intrapersonally and shapes the respondent's own causal account of their behavior.

Bias 4 — The cultural connotation of "have you noticed X?". In both Japanese and English, the construction *"have you noticed X?"* is conventionally used to indirectly point out an inattention or oversight. This pragmatic loading is independent of the actual content of X, and respondents react to the form before they evaluate the content; defensive responses are elicited as a side effect of the question form itself.

The advisor-framing addresses these four biases by displacement rather than elimination. It changes the respondent's role from one whose past cognitive state is being audited to one whose present professional judgment is being requested. The honest answer to "what would you say to a student who asked this?" does not require disclosing past awareness, so Biases 1–3 lose their primary anchor; the question form replaces the inattention-pointing construction with a request for advisory judgment, so Bias 4 is dissolved at the linguistic layer. None of the four biases is *eliminated* — respondents may still reconstruct past states, save face, or experience dissonance — but the structural pressure to do so within the response is reduced. The fact that the evaluator role is professionally familiar to the respondents (advising on student questions is routine work) further reduces the cognitive load that direct AI-output evaluation imposes.

Step 4 — Recording responses into the six categories. For each rewritten item, the advisor assigns one of the six response categories defined in SI-8.2. The assignment is not forced; if an item does not fit cleanly into any single category, the advisor may leave it blank and add a free-text comment. The six categories are designed so that the second category — *already studied but demonstrates mature insight* — is the core of the evaluation, and the fine-grained distinction between "basic, already-established knowledge" and "already studied but demonstrates mature insight" is the most diagnostic for UQS's analytical value.

Step 5 — Recording metadata. Alongside category assignments, three metadata fields are recorded per item: (a) optional 1–2 line free-text comment from the advisor; (b) a confidence marker (high/medium/low) indicating the advisor's confidence in the assignment; (c) the time spent on the item if practical. Per-session metadata is also recorded: respondent identifier, paper identifier, UQS prompt version used to generate the items, date of evaluation, and total number of items rated.

Step 6 — Aggregation target. The resulting table has columns {paper_id, item_id (original UQS ID), category assigned, comment, confidence, time, respondent_id, session_id, uqs_version}. Aggregation across items, categories (A/B/C/U/D-4), papers, and respondents yields the data described in SI-8.5.

SI-8.2 Six Response Categories — Full Definitions

Per Methods 7, responses are recorded in six categories (italicized below for first-mention emphasis):

(i) *Unexplored and valuable question.* *Definition.* The item points at an absence that has not yet been addressed in the literature, or has been addressed only in a very limited way, and is worth pursuing as a research project — for example, a topic suitable as a master's thesis or a proposal increment. *Distinguishing features.* Distinguished from (ii) by the absence (or near-absence) of prior literature addressing the question; distinguished from (iii) by being directly pursuable without major reformulation. *Illustrative example.* A question of the form "has the electronic structure of the dpe-Ni(0) catalyst intermediate been observed spectroscopically under turnover conditions?" would fit (i) if, at the time of the advisor's judgment, no such observation has been reported.

(ii) *Already studied but demonstrates mature insight.* (Core of the evaluation.) *Definition.* The question is already addressed by published work, but the very fact that the student arrived at the question demonstrates grasp of the argumentative structure of the field rather than mere enumeration. The advisor would praise the student and direct them to the relevant literature. *Distinguishing features.* Distinguished from (iv) by the quality of the question: (iv) items are reachable from a textbook or a major review, whereas (ii) items show that the asker has recognized a non-obvious tension in the paper's evidence structure. Distinguished from (i) by the existence of substantive prior work. *Illustrative example.* Applied to Kumada 1972's Stage 1 output, an item of the form "what is the detection limit for the R–R homocoupling by-product, and does the selectivity claim hold at limit-of-detection sensitivity?" fits (ii): the downstream literature on Ni-catalyzed cross-coupling addressed homocoupling in detail, but the student's recognition that the selectivity claim in Kumada 1972 is not anchored to a detection limit reflects grasp of the paper's evidence architecture. Marking this category correctly is the evaluation's primary signal that UQS extracts the field's argumentative structure rather than performing normative matching. *(R1 note: in the blind evaluation of SI-8.7, four organic chemists all judged the corresponding Kumada 1972 item off-target, (v), not (ii). The example is kept as submitted; see SI-8.7.)*

(iii) *Valuable but requires adjustment.* *Definition.* The substantive concern is legitimate, but the question as framed cannot be pursued directly given realistic constraints (thesis scale, equipment, data availability). The advisor would decompose the question or substitute a tractable analogue. *Distinguishing features.* Distinguished from (i) by the need for reformulation

before the question becomes a research plan; distinguished from (v) (*off-target*) by the legitimacy of the underlying concern. *Illustrative example.* A question of the form "can the cis-geometry hypothesis be tested directly by single-crystal XRD of the transition state?" has a legitimate core (mechanism verification) but cannot be pursued as stated; it would be adjusted to accessible probes.

(iv) *Basic, already-established knowledge.* *Definition.* The answer is available in a standard review or textbook; the question does not demonstrate more than a surface reading. The advisor would direct the student to the standard references before further discussion. *Distinguishing features.* Distinguished from (ii) by the absence of evidence that the student has grasped the field's non-obvious structure; a (ii) item still has an answer in the literature, but reaching the question required structural insight. *Illustrative example.* "Why does Grignard reagent react with nickel catalysts at all?" is (iv) — the answer is in any organometallic textbook.

(v) *Off-target.* *Definition.* The question is not meaningful in the paper's or the field's context, or rests on a misreading. The advisor would correct the misunderstanding in supervision. *Distinguishing features.* Distinguished from (iv) by lack of a meaningful answer (rather than the answer being elementary); distinguished from (iii) by the illegitimacy of the underlying concern. *Illustrative example.* "Why don't the authors consider the effect of photochemical excitation on this reaction?" would be (v) for a paper reporting ground-state thermal cross-coupling with no photochemical context.

(vi) *Unable to evaluate.* *Definition.* The advisor cannot assess the item without additional information or expertise outside their field. This category is retained rather than forced: forcing an uncertain assignment would pollute the aggregate distribution. *Distinguishing features.* Distinguishes genuine uncertainty from the adjacent categories; the presence of (vi) assignments is itself informative (e.g., heavy (vi) loading on a particular UQS category suggests that category requires an evaluator of different disciplinary background).

SI-8.3 Tier 2 Collective-Pattern Utilization Protocol

Per Methods 7, the intrinsic value of Stage 2F-level output is recorded not through precision evaluation but through accumulation of use cases. Stage 2F aggregates individual-paper Stage 1/Stage 2 outputs into collective structures — `recurring_roots`, `temporal_trajectory`, structural patterns correlated with incentive structure, and `group_unasked_question` — that the individual-paper level cannot produce. Reviewing these outputs for "precision" would require the reviewer to perform the cross-paper analysis themselves, which defeats the purpose of the automated aggregation.

The Tier 2 protocol therefore records *utilization events* rather than *judgment events*. Each utilization event is logged with the following fields: (a) which Stage 2F output (which `recurring_root`, which `group_unasked_question`) was shown; (b) which researcher consumed it and in what context (individual reading, research-planning meeting, graduate-supervision session, grant-proposal draft, lecture preparation); (c) whether the collective pattern elicited a qualitative reaction ("this changed how I read paper X", "this suggested an experiment", "I used this to structure my student's reading list"); (d) whether the structural similarity noted across temporally distant papers functioned as a fresh perspective rather than as a rediscovery of an already-known pattern; (e) downstream action, if any (a new experiment, a new reading assignment, a revision to a research plan).

The Tier 2 log is append-only. It accumulates as UQS is applied in practice and becomes the primary evidence base for claims about the utility of the collective-pattern extraction. Unlike Tier 1, there is no target distribution or success threshold; the Tier 2 protocol is designed to document use, not to prove precision.

SI-8.4 Limited Expert Verification Procedure (Preliminary; Phase 2 Systematic Study Planned)

Phase 2 denotes the planned follow-up study of the method (as in SI-6 and SI-11).

Alongside the two operational tiers, a *limited expert verification* procedure is defined for the initial validation stage of the methodology. Per Methods 7, its preliminary scope is one to two experts and two to three papers, and its purpose is technical quality assurance, not significance judgment. The procedure as specified here is preliminary and qualitative; systematic execution with a wider advisor panel and full inter-rater-reliability reporting is deferred to Phase 2 (see `future_work_pointers/pilot_evaluation_issues.md` §A-3).

The expert verification targets two technical properties of Stage 1 output: (a) *positional accuracy* — whether each item's `evidence_chain` correctly cites the textual location it claims (i.e., the quoted span actually appears at the stated location in the paper and the surrounding context supports the item's reading); and (b) *inference-tier accuracy* — for A-category items, which alone carry a tier (Methods 1.1), whether the T1/T2/T3 assignment is appropriate (T1 for textual direct statement, T2 for one-step inference from text, T3 for external knowledge required to notice the absence). The expert is given the paper and the

Stage 1 output (not the student-question rewrite), and checks each item against these two criteria. Findings are recorded per item as {positional: ok/miscited/ambiguous, tier (A items only): ok/should_be_T1/should_be_T2/should_be_T3, comment}.

Per Methods 7, the expert is *not* asked whether the identified absence is significant. Significance judgment is the role of Tier 1 (advisor evaluation), where the distribution of advisor responses provides a more appropriate signal than a small number of self-identified experts. This separation of technical and value judgments is a deliberate design choice: technical quality assurance is a solvable, well-scoped task; significance evaluation is distribution-based and requires the wider sampling that Tier 1 delivers.

SI-8.5 Verification Data (Preliminary Machine-Assisted Check; Pilot-Application Data Forthcoming)

Machine-assisted positional pre-check (preliminary). Pending the systematic expert verification of SI-8.4 (Phase 2), a machine-assisted positional-faithfulness check was performed on the deposited Stage 1 outputs of three v18.4 papers (Kumada 1972; Monnereau 2009; Li 2024; 82 records total). Each record's `evidence_chain` quoted span was matched against the source-PDF text by a normalised string-match procedure (NFKD normalisation of subscripts/superscripts, dash/quote folding, whitespace collapse), with figure-panel values and citation descriptors confirmed manually. All 82 records' quoted content was located in the source, with no miscitation detected (76/82 by automatic exact/normalised match; the remaining six — four citation descriptors and two figure-panel numerical values not captured by linear text extraction — confirmed by targeted manual lookup). Within these 82 records the quoted spans were found in the source. The per-item match records of this check were not retained, so it cannot be re-audited item by item and is reported as preliminary. It is an automated / LLM-assisted check, **not** the independent human-expert verification specified in SI-8.4 (independent positional sign-off and T1/T2/T3 inference-tier accuracy judgment), which remains Phase 2. Inference tiers are recorded only for A-category (*Deferred*) records (Methods 1.1); the three papers have nine, all labelled T1. B/C/U records carry no inference tier, so a tier-accuracy check applies to those nine records only, and it was not part of this preliminary check.

Pilot-application data (plan at submission). Pilot-application data for the advisor-evaluation protocol (Tier 1) had not been systematically collected at the time of the original submission; the evaluations reported in this revision are in SI-8.6 and SI-8.7, and the multi-paper evaluation described below remains future work. The authors will report such data in a separate publication, following the protocols specified in SI-8.1 through SI-8.4. The forthcoming report is expected to include: (a) per-item category assignments aggregated across respondents and across papers, with particular attention to the distribution of category (ii) as the evaluation's core signal; (b) per-UQS-category tendencies (whether A, B, C, U, or D-4 items disproportionately attract particular advisor responses); (c) inter-respondent variance on shared items, which serves as external validation of UQS's multi-vantage-point design; (d) a Tier 2 utilization log covering the first months of Stage 2F outputs being used in research-planning and graduate-supervision contexts; (e) expert-verification results on positional and inference-tier accuracy for two to three representative papers.

SI-8.6 Preliminary pilot: an experimental chemist's reading of the seven SSPs (R1 addendum)

Provenance and status. In June 2026, before submission, the list of seven SSPs was sent to two experimental organic chemists who are not authors. One replied in writing, item by item; that reply is reported below. The other replied only orally, with a comment on the list as a whole (see the end of this section). On 2026-06-01 the first chemist read the seven SSPs (Results Part 3) and answered, in their own words, "as I would answer a student who brought this proposal", without looking anything up. The framing matches the Tier 1 advisor protocol (SI-8.1), but the feedback was not collected under that protocol. It is unblinded — the expert knew the items came from UQS — and it comes from a single rater. We report it as an unstructured pilot. The expert agreed to its use, anonymised. The second chemist agreed in July 2026 to the anonymised use of their comment.

Coding into the SI-8.2 categories, by the authors. The expert did not code the items, and the coding was not sent back for confirmation; it is our reading of the expert's comments, and the expert's own overall comment, summarised below, is consistent with it:

SSP	category	the expert's reasoning (paraphrased)
SSP-1 Mg-halide by-product role	(ii) already studied, but a good point	the idea is good, but by-product effects occur to everyone and have probably been examined; Mg salts are Lewis acids and counter-anion effects are well studied
SSP-2 homogeneous vs heterogeneous "black mixture"	(v) off-target	the colour may come from a minor, intensely coloured species; not important

SSP	category	the expert's reasoning (paraphrased)
SSP-3 origin of E/Z selectivity	(i) unexplored and valuable (tentative)	knowing the origin seems meaningful, if it leads to better selectivity; prior work unknown and needs a literature search
SSP-4 β-H-elimination suppression claim	(iv) established knowledge	widely accepted in Ni catalysis; the phrase "survive a direct test" was unclear
SSP-5 DKR ordering for non-benzylic alkyls	(iii) valuable but needs reformulation	asymmetric Ni radical coupling is well developed; a radical-capture design would be the natural route rather than chirality from the Grignard side
SSP-6 operating cathode potential	(v) off-target	once the reaction is reproducible, experimentalists would not care; it is easy to measure if wanted
SSP-7 calixarene "fanning motion"	(iii) valuable but needs reformulation	could be done if unprecedented, but is unlikely to be interesting; surpassing established ligands is hard

Distribution: (i) 1, (ii) 1, (iii) 2, (iv) 1, (v) 2, (vi) 0.

The expert's overall comment (paraphrased). In experimental chemistry a research question starts from a problem — for example, that a compound must be made in quantity — and descends to the technical obstacle and its solution. Most of the SSPs read as fine-grained matters whose underlying purpose is not visible. This comment is the evidence for Limitation L10.

The second chemist's comment. On the same day, 2026-06-01, the second chemist commented orally and in person on the list as a whole. In the author's account, organic chemistry first sets a large question and then asks how existing tools can reach it, so the questions are of a different kind. This is the author's recollection, not a quotation. It contains no item-level judgement and is not coded into the SI-8.2 categories. The two chemists work at different institutions and have no known connection to each other. Both responded on the day they received the list. We therefore treat the two comments as independent, while noting that this is our judgement and was not tested.

SI-8.7 Blind item-level evaluation under the Tier 1 protocol (R1)

The evaluation was pre-registered (`01_response/d1_expert/preregistration_D1_itemwise.md`) before any item was drawn. Dated amendments added two raters before any rating was read (Amendments 1 and 2) and set how comments are reported (Amendment 3). A post-rating expertise questionnaire planned in Amendment 1 was not administered (Amendment 4).

- *Raters.* Four organic chemists who are not authors. Raters 1 and 2 were pre-registered and form the primary analysis. Raters 3 and 4 agreed to take part after the design was fixed and were added as a secondary analysis, numbered by the order in which their forms were returned. Three of the four are former collaborators of an author on separate projects; the fourth is a former office neighbour of an author. None had seen any UQS output. The two chemists who received the SSP list in June 2026 (SI-8.6) are not raters. The raters are not identified. Those who agreed to be named are thanked in the Acknowledgements together with other chemists, for discussions, so that the list does not show who rated or who commented on the SSP list. Before rating, one rater noted that organometallic chemistry is outside their own field, and another that they know much of the field's inside history. No further information on the raters' expertise was collected.
- *Items.* Ten items were drawn at random from one Kumada 1972 v18.4 run: 7 Stage 1 items and 3 D-4 candidates. Each was rewritten into student-question form by a registered prompt. An author checked each rewrite for preservation of its object and anchor.
- *Procedure.* The raters used the SI-8.2 categories on paper forms and were told not to look anything up. They rated independently, without knowing the item sources or the hypotheses. All four forms were returned complete, and no item was dropped.

Table SI-8.7-1. Items and ratings. Collapsed scheme: (i)–(iii) = worth raising; (iv)–(v) = not worth raising; (vi) = unable to evaluate.

#	type	item (short form)	R1	R2	R3	R4	worth raising
1	D-4	no independent isolation or spectroscopy of the three proposed intermediates	(i)	(iii)	(iii)	(iv)	3/4
2	D-4	high yields with β-H-bearing alkyl Grignards vs the instability the paper itself notes	(i)	(ii)	(ii)	(ii)	4/4
3	S1	the identity of the black solid is not discussed	(i)	(v)	(iv)	(vi)	1/4
4	S1	no homo-coupling quantification behind "selective cross-coupling"	(v)	(v)	(v)	(v)	0/4
5	S1	the closing sentence names no concrete next problem	(v)	(v)	(v)	(iii)	1/4
6	S1	the basis of "seldom employed in synthetic practice", given ref. 4	(v)	(iv)	(v)	(ii)	1/4
7	D-4	GC and isolated yields mixed, so entries cannot be compared	(ii)	(iv)	(iv)	(iv)	1/4
8	S1	why both yield bases are not given under the same conditions	(v)	(iv)	(iv)	(v)	0/4
9	S1	no experimental comparison with the Corriu–Masse $Ni(acac)_2$ system	(vi)	(ii)	(vi)	(ii)	2/4
10	S1	how the ligand activity order was established	(ii)	(iii)	(v)	(ii)	3/4

Table SI-8.7-2. Distribution per rater.

rater	(i)	(ii)	(iii)	(iv)	(v)	(vi)	worth raising (95% Wilson CI)	confidence high / medium / low
R1	3	2	0	0	4	1	5/10 (0.24–0.76)	10 / 0 / 0
R2	0	2	2	3	3	0	4/10 (0.17–0.69)	5 / 5 / 0
R3	0	1	1	3	4	1	2/10 (0.06–0.51)	10 / 0 / 0
R4	0	4	1	2	2	1	5/10 (0.24–0.76)	4 / 6 / 0
all	3	9	4	8	13	3	16/40	29 / 11 / 0

Table SI-8.7-3. Agreement.

pair	role	six categories: raw, Cohen's κ	collapsed: raw, Cohen's κ
R1–R2	**primary**	2/10, 0.05	7/10, 0.46
R1–R3	secondary	4/10, 0.26	7/10, 0.51
R1–R4	secondary	3/10, 0.16	4/10, −0.03
R2–R3	secondary	6/10, 0.47	8/10, 0.60
R2–R4	secondary	4/10, 0.23	6/10, 0.29
R3–R4	secondary	3/10, 0.12	4/10, 0.02
all four	secondary	Fleiss' κ 0.19	Fleiss' κ 0.28

Results.

- *Agreement.* The two pre-registered raters agreed on the six categories for 2 of 10 items (κ = 0.05). On whether an item was worth raising, they agreed on 7 of 10 (κ = 0.46). Across all four raters, κ was 0.19 and 0.28. The six-way distinction was not used consistently across raters; the coarse judgement was used more consistently, but only moderately.
- *Worth raising.* Per rater, 2–5 of the 10 items were judged worth raising (16 of 40 ratings). The intervals are wide, and no significance claim is made.
- *Category (ii).* Presented as a student's questions, 9 of 40 ratings placed an item in (ii), the category the protocol reserves for a question that shows grasp of the field's argument.
- *Item type (descriptive).* The three D-4 candidates were judged worth raising in 8 of 12 ratings, and the seven Stage 1 items in 8 of 28.
- *Unanimous items.* All four raters judged item 2 worth raising, three of them as (ii). All four judged item 4 off-target; one rater explained that with near-equimolar reagents and yields up to 98%, little homo-coupling can have formed. The illustrative example in SI-8.2, written before submission, placed a homo-coupling question of this kind in (ii). The blind ratings contradict that example. We leave the example as submitted and record the contradiction here. All four raters also judged item 8, the mixed yield bases, not worth raising.
- *Comments.* One rater wrote a comment on every item; the others wrote none. To keep the raters anonymous, the comments are summarised by theme and not linked to any rater's ratings (pre-registration Amendment 3). Several items are explained by the format of a 1972 communication. Several have a chemical explanation the item did not consider: product volatility for the mixed yield bases, and metallic Ni^0 for the black solid. One comment drew on background knowledge not in the paper.

Limits. Ten items, one paper, one run. Each item passed through a rewriting step, which may have improved or degraded it. The raters differ in how close their field is to Ni-catalysed cross-coupling, in both directions (SI-8.7, Raters); beyond their own remarks before rating, their expertise was not recorded. The ratings do not test whether UQS items are better or worse than items from another source; there is no comparison arm in this evaluation.

SI-9: Pilot Comparison of Stage 4 LLM-Parametric Retrieval against Structured Database Queries (EP-03a Case)

Paths written in repository form resolve in the Zenodo deposit as follows: `papers/dev/…`, `papers/ep/…`, `papers/test/…` → `tier_3_stage_outputs/` (full) and `tier_8_redacted_public_raw_outputs/` (redacted); `retrieval_audit/…` → `tier_5_retrieval_logs/retrieval_audit_ep03a_07a_08a_09a/…`; `prompts/…` → `tier_1_prompt_versions/`; `improvements/…` → `tier_6_audit_decisions/` or `tier_7_aggregation_scripts_and_methodology_evolution/`. `MANIFEST.csv` at the deposit root lists every file. Source-paper PDFs are not deposited.

Stage 4 filling-paper identification operates as a semantic-retrieval-like recall over the language model's parametric knowledge (Methods 2, "Computational nature of Stage 4 filling-paper identification"). A reviewer may reasonably ask whether this retrieval silently omits a substantial body of literature that a conventional keyword query would recover. This section documents a pilot experiment that probes the question directly for one Evidence Pack, by comparing the LLM-identified candidates against a structured Web of Science (WoS) Core Collection query constructed from the same reverse-engineered topic terms.

Full experiment design, queries, raw exports, and per-candidate verification records are archived at `retrieval_audit/ep03a_wos_llm/`; this section summarizes the findings for manuscript reading.

SI-9.1 Design

Target. Evidence Pack **EP-03a** from the Kumada 1972 Stage 3 dictionary (SI-3): *Direct-observation / state-assignment pack — direct spectroscopic or structural observation of Ni intermediates relevant to the Kumada catalytic cycle*.

LLM retrieval (mode 1). The candidate list is the union of EP-03a filling-paper identifications produced by two model-pinned v17 runs on Kumada 1972 (Claude Opus 4.7 and GPT-5.4 Thinking, both documented in SI-6; the GPT-5.4 run had web search enabled, so its candidates are not parametric recall alone). Six specific candidate papers emerged across the two runs; the consolidated list with ligand and characterization metadata is at `retrieval_audit/ep03a_wos_llm/llm_candidates_ep03a.md`.

Structured query (mode 2). Five WoS Core Collection query variants, built from the reverse-engineered topic terms that define EP-03a's case-specific implementation. The variants progressively relax or tighten the five AND-connected clauses (metal, ligand class, target species, operation, characterization method):

Variant	Ligand clause	Reaction-context clause	N returned
v1 (baseline)	diphosphine-only (`diphosphine OR dppe OR dpe OR dppp OR dmpe`)	none	49
A	any phosphine (`phosphine OR phosphan* OR ...`)	none	171
B	any phosphine	cross-coupling / Kumada / Grignard required	25
C	any phosphine	none (isolation-operation emphasis)	74
D	any + Ni(I)/Ni(III)	cross-coupling required	21
Union of five variants	—	—	**268 unique records (262 with a DOI)**

All queries restricted to `DT=Article` (matching Stage 4's review-exclusion policy) and to chemistry WoS categories. Queries were executed on 2026-04-20.

Overlap check. Each LLM candidate was looked up by exact DOI (Tier 1) and, where the Tier 1 DOI reconstructed by the analyst retrieved a different paper (see SI-9.5), by author-year-journal-volume triangulation (Tier 2). Existence in WoS Core Collection was confirmed for all six candidates.

SI-9.2 Overlap result

Topical-query overlap: 0 / 6. None of the six LLM candidates appears in any of the five topical-query result sets. The five variants collectively retrieved 268 unique records (262 with a DOI), none overlapping with the LLM candidate list.

Candidate existence in WoS: 6 / 6. Direct DOI lookups (Tier 1, or Tier 2 where necessary) confirm that all six candidates are indexed in WoS Core Collection. The retrieval-layer failure is therefore orthogonal to database coverage.

Candidate coverage on the LLM side. The six candidates span 1974–2020 and fall within the reliable-knowledge windows of all four Claude models documented in Methods 6 (earliest reliable-knowledge cutoff: Haiku 4.5 at February 2025). The zero-overlap result is therefore not attributable to a training-data cutoff truncation on the LLM side either. Both retrieval modes have access to every candidate; the two modes nonetheless fail to converge on any of them, which is the puzzle the five failure modes below resolve.

SI-9.3 Five diagnosed failure modes

The zero-overlap result is not a single-query artifact. A structural analysis of the query–candidate interaction identifies five distinct failure modes, each operationally characterized and directly demonstrated by the pilot data. Modes (i)–(iii) are keyword-retrieval failures; modes (iv) and (v) are LLM-output reliability failures, separated by whether the cited bibliographic record exists at all (mode (iv): the DOI resolves to a real paper, but the LLM's description does not match it; mode (v): the citation tuple itself does not resolve to any record). The two LLM-output failures are orthogonal and can co-occur in a single citation.

(i) WoS abstract-coverage gap

Affected candidates: Parshall 1974, Morrell-Kochi 1975, Yamamoto 1978, Anderson-Jones-Vicic 2004, Jones-Vicic 2006.

Pre-1991 *J. Am. Chem. Soc.* and *Bull. Chem. Soc. Jpn.* records in WoS Core Collection commonly carry title, author, and citation metadata but no indexed abstract. A topical query with five AND-connected clauses cannot be satisfied from title alone when no single-word title spans all clauses. The affected set extends, unexpectedly, to two post-1991 *J. Am. Chem. Soc.* papers whose WoS records also have empty abstract fields: Anderson-Jones-Vicic 2004 (a two-page communication, pp. 8100–8101) and Jones-Vicic 2006 (a nine-page article, pp. 13175–13183). The cause of the missing abstracts is not known; article length does not explain it.

Year distributions across the five variants confirm the structural character of the gap: the earliest record appearing in any variant is from 1991 or 1992, even though all queries admit `PY=(1972–2026)`. No 1970s or 1980s records were retrieved.

(ii) Ligand-class semantic-generalization gap

Affected candidates: Anderson-Jones-Vicic 2004, Jones-Vicic 2006.

These papers use **terpyridine** (tpy), a tridentate N-donor ligand, not a phosphine. The LLM nevertheless identifies them as EP-03a filling papers because they meet the structural criterion of EP-03a — direct spectroscopic and structural characterization of Ni intermediates relevant to catalytic cross-coupling — via a paradigm-shift Ni(I)/Ni(III) mechanism. The generalization is performed semantically: the LLM implicitly maps "Ni–phosphine intermediate" in the source EP onto "Ni intermediate in related cross-coupling context," allowing the ligand class to vary.

A keyword query that requires `phosphine OR phosphan* OR diphosphine OR dppe...` systematically excludes tpy-supported Ni intermediates as a matter of query design. No reformulation within the phosphine-keyword family can retrieve them. The filling pathway crosses ligand classes; keyword retrieval does not.

(iii) Compound-name enumeration gap

Affected candidate: Dawson 2020.

Dawson 2020 uses **Xantphos** (4,5-bis(diphenylphosphino)-9,9-dimethylxanthene). Xantphos is a diphosphine by class, but its commonly used trade name does not match the wildcard `phosphan*` (left-anchored in WoS). The abstract uses "Xantphos" throughout without the generic term "phosphine," so the paper is excluded from retrieval despite belonging to the ligand class the query intends to cover.

Enumerating diphosphine trade names (Xantphos, dppf, BINAP, DPEphos, SEGPHOS, Josiphos, Walphos, DuPhos, PHOX, …) is open-ended: new proprietary ligands continue to enter the literature. The LLM, in contrast, recognizes Xantphos as a diphosphine by class membership without requiring the generic term to appear in the text.

(iv) Citation-content separation hallucination (triple-mismatch, PDF-verified)

Affected candidate: "Yamamoto-Kohara-Yamamoto 1978 *BCSJ* 51, 2091" as claimed by Opus 4.7.

Tier 1 DOI lookup (`10.1246/bcsj.51.2091`) returns a real paper whose year, journal, volume, and page match the LLM output. Library-hardcopy retrieval of both the 1978 and 1981 *BCSJ* PDFs (obtained 2026-04-21) permits a three-layer decomposition of the mismatch:

Layer	LLM output	Actual 1978 BCSJ 51, 2091	Actual 1981 BCSJ 54, 2010
Middle author	"Kohara"	**Hayashi** (Y.)	**Kohara** (T.)
Journal-volume-page	1978, 51, 2091	1978, 51, 2091 ✓	1981, 54, 2010
Content described	trans-NiAr(X)(PEt_3)$_2$ isolation, X-ray, Kumada-lineage monoaryl-Ni intermediate	Poly-phenylene polycondensation (p-dihalobenzene + Mg with Ni/Pd/Fe/Cr/Co catalysts); Scheme 1 proposes a Kumada-style $NiX_2L_2 \rightarrow NiR_2L_2 \rightarrow NiR'(X)L_2$ cycle but **no intermediate is isolated**; cites Kumada 1972 as ref 6(a)	Isolates 13 **monoalkyl**-Ni(II) complexes (R = Me, Et), including *trans*-NiMe(OCOPh)(PEt_3)$_2$ and NiEt(Cl)(bpy); paper explicitly states that monoaryl-Ni isolation is covered in refs 8–10 as *distinct prior work*

The LLM's citation is therefore a **triple mismatch**: (α) authorship (Hayashi ≠ Kohara), (β) journal-year-volume combination (1978, 51, 2091 ≠ 1981, 54, 2010), and (γ) content description (polycondensation / monoalkyl-Ni ≠ monoaryl-Ni isolation). Neither the 1978 paper *nor* the 1981 paper matches the content the LLM attributed to "Yamamoto-Kohara-Yamamoto 1978." The LLM output appears to have fused an authorship pattern it could reconstruct (the existence of a Yamamoto–Kohara–Yamamoto isolation paper in BCSJ), a year/page pair it assembled from a nearby Yamamoto–Hayashi–Yamamoto paper in the same journal, and a content description consistent with the Parshall/Morrell–Kochi family of papers rather than with either Yamamoto paper.

This three-layer decomposition is only accessible through PDF reading. Automated citation resolution (DOI lookup, CrossRef) confirms that a paper exists at the claimed (year, journal, volume, page) tuple, and metadata-only checks satisfy author-year-journal consistency tests at one layer (the Yamamoto lead-and-senior authors are real) while failing at another (the middle author). Only content-level reading of the physical record against what the LLM described reveals that *both* candidate papers — the 1978 Hayashi paper and the 1981 Kohara paper — are content-level non-matches for the described subject.

In the present case, detection unfolded in two stages: (1) an analyst-reconstructed DOI happened to hit the correct record for BCSJ 51, 2091, and its title (polycondensation) flagged the content mismatch; (2) library-hardcopy retrieval of the 1981 paper, initially hypothesized as the "likely intended citation," disconfirmed even that fallback interpretation, replacing a *candidate-swap* hypothesis with a *fabricated-composite* diagnosis. Cross-model convergence did not assist: Opus 4.7 cited Yamamoto 1978 while GPT-5.4 did not, so the divergence was a warning signal rather than an absence of warning, but a *systematic* procedure for catching citation-content mismatches in single-model outputs is not defined by the present methodology and remains an open problem. Content verification by a human reader, or at minimum an independent second-model check followed by divergence-flagging and hardcopy-PDF fact-check, is the safest mitigation available today. One practical corollary from this case: metadata-only checks (DOI resolution, author-year matching) can be satisfied even when the described content originates from neither the cited paper nor any nearby candidate, making content-reading non-optional for substantive LLM citation claims.

(v) Cross-model citation hallucination — a paper-dependent pattern (preliminary, n=2 hypothesis)

Affected pattern: full-citation fabrication observed in Stage 4 retrieval on Kumada 1972, with sharply paper-dependent rates. This mode is structurally distinct from (iv): mode (iv) describes an LLM citation whose DOI resolves to a real paper but whose LLM-supplied description does not match it; mode (v) describes a citation tuple (authors + journal + year + volume + page) for which Crossref returns no record at all.

Symptom. During Stage 4 retrieval, LLM-parametric models can generate citations that combine real authors from different papers into non-existent author tuples, omit middle authors, or describe a paper's chemistry incorrectly while reporting an otherwise plausible bibliographic shell. Crossref-based bibliographic verification reveals these as fabrications. In the two papers examined below, the errors differed in pattern between target papers analyzed by the same model under identical prompts.

Mechanism (hypothesis). We hypothesize that the hallucination rate depends on structural properties of the target paper rather than on model capability per se; two papers cannot show this. Four contributing factors are proposed: (a) paper age (older papers, especially those preceding the digital citation era, are more vulnerable; this risk axis is distinct from training-data cutoff truncation, since all candidates considered here lie within the reliable-knowledge windows documented in Methods 6); (b) author-network density (research groups with many related publications and rich author-permutation patterns invite combinatorial confusion); (c) named-reaction status (papers that lend their name to a reaction class — e.g., Kumada, Heck, Stille,

Suzuki, Sonogashira, Negishi — anchor large derivative literatures); (d) citation-offshoot count (heavily cited foundational papers acquire many lookalike derivatives in the model's training distribution). Mode (v) is therefore an author-network combinatorial failure rather than a recall-window failure.

Observed example. Applying v18.2 of our Stage 4 prompt to two devset papers, we observed different error patterns on the two papers, as counted below. For Kumada 1972 (a 54-year-old organometallic-catalysis communication satisfying all four factors), of the three Sonnet-generated bibliographic predictions specific enough to check, two could not be found (2/3). "Tamao-Sumitani-Hayashi-Kumada *J. Am. Chem. Soc.* 1976" combines the author lists of two real papers (Tamao-Sumitani-Kiso-Kumada, *Bull. Chem. Soc. Jpn.*; Hayashi-Tajika-Tamao-Kumada, *J. Am. Chem. Soc.*), so it is a confirmed author-combination fabrication. For "Nagase-Hayashi-Kumada *Tetrahedron Lett.* 1973–74" no matching paper was found, but the log (`papers/dev/kumada/cross_model_v182_comparison.md` §5.2) does not identify where the combination came from, so it is unresolved rather than shown to be fabricated; under SI-9.10 it is counted among the 18 candidates that could not be resolved to any paper (retracted). The other three predictions were too vague to check. For Murai 1993 (a 33-year-old C–H activation communication with a more orderly Murai-Kakiuchi-Chatani author network), the same prompt generated 0/7 (0%) full author-fabrications and 1/7 (14%) partial hallucinations — the partial case described Fukuyama-Chatani-Tatsumi-Kakiuchi-Murai *J. Am. Chem. Soc.* 1998 (10.1021/ja982794b) as Rh-catalysed when the actual paper is $Ru_3(CO)_{12}$-catalysed (a mode (iv) overlap, since the DOI resolves correctly); under the retention rule of SI-9.10 this candidate is retracted. Two devset papers do not establish a population claim; we report this as a preliminary observation.

Preliminary mitigation. A web-search-augmented model run in parallel (GPT-5.5 Thinking, with chat-context web retrieval) produced 9/9 and 8/8 fully verifiable citations on the same two papers, with no observed full author-fabrications. Combined with Crossref API verification of the bibliographic shell, this cross-model verification provides a structural detection mechanism for failure mode (v): a citation that returns no Crossref hit and is generated by only one of two parallel models is a high-prior fabrication. The verification cost is concentrated on papers exhibiting one or more of the four structural risk factors; for papers with fewer of these factors (e.g., Murai 1993 in our pilot: more recent, with a less permuted author network, although it too lends its name to a reaction), the observed fabrication risk was lower. The operational form of this verification protocol is left to subsequent methodological extensions.

SI-9.3b Analyst verification-cost profile

Candidate-level verification (SI-9.10) is analyst-supervised, and its cost is concentrated on the minority of candidates exhibiting the mode-(iv)/(v) structural risk factors; most candidates resolve in minutes. The observed costs in this study were:

Candidate / issue	Failure mode	Verification action	Cost
Tamao–Zembayashi 1973 (ref. 32; middle-author substitution)	(iv) middle-author / content	single Crossref DOI lookup	minutes
Chen 2019 (ref. 42; full author-list replacement)	(iv) content separation	content-keyword + DOI cross-check	minutes
Yamamoto 1978 (triple-mismatch)	(iv)/(v), pre-digital non-indexed source	library hardcopy retrieval	days

(R1: the submitted table used the submitted reference numbers (23, 33, 34) and also listed Dawson 2020 under middle-author substitution; no verification record shows an author error for Dawson 2020, so it is removed.)

The minutes-scale cost is typical; the days-scale cost is the worst case and is confined to pre-digital, non-indexed sources. This asymmetry is an explicit practicality bound on UQS: it is an **analyst-supervised** workflow, not an unsupervised pipeline, and for older-literature targets the verification burden — not candidate generation — is the rate-limiting step. The mode-(v) risk factors (Section (v)) predict which candidates incur the days-scale cost, so the burden is anticipable rather than uniform.

SI-9.4 Interpretation

The five modes are structurally distinct. Together they explain why topical-query recall and LLM parametric recall intersect at zero for this EP, and they are not symmetric failure modes of a single underlying mechanism:

Mode	Locus of failure	Fixable by query reformulation?
(i) Abstract-coverage gap	Database index completeness	No — intrinsic to WoS back-file coverage
(ii) Ligand-class gap	Keyword–semantic mismatch	No — keyword queries cannot perform class-membership reasoning

Mode	Locus of failure	Fixable by query reformulation?
(iii) Compound-name gap	Trade-name enumeration bounds	Only by unbounded keyword enumeration
(iv) Citation-content hallucination	LLM output reliability (DOI resolves; description mismatched)	Not a database failure; mitigated at the LLM-output layer (content-level reading)
(v) Cross-model citation hallucination	LLM output reliability (citation tuple does not resolve); paper-dependent	Not a database failure; mitigated at the LLM-output layer (Crossref API + cross-model parallel retrieval)

Modes (i)–(iii) are keyword-retrieval failures that the LLM escapes because its parametric knowledge generalizes semantically and is grounded in text that extends well beyond WoS-indexed abstracts. Modes (iv) and (v) are LLM-output failures that keyword retrieval escapes because keyword retrieval returns only papers that actually exist in the queried database with matching metadata — the database cannot fabricate a citation. The two retrieval modes therefore have **different error profiles**, not different performance levels along a single axis.

This supports a complementarity framing: rather than treating one retrieval mode as a substitute for the other, the two can be combined. LLM retrieval provides recall across paradigm shifts, ligand-class generalizations, and compound-name classes that keyword queries cannot cover. Database retrieval provides metadata-level ground truth that catches citation hallucinations. A practical workflow might therefore use LLM retrieval to generate candidate lists, database queries (Crossref for tuple resolution; WoS/SciFinder/Reaxys for indexed-record presence) to confirm the metadata of each candidate, cross-model parallel retrieval to surface paper-dependent fabrications (mode (v)), and content-level reading (abstract or full text) to verify that the cited paper says what the LLM claims (mode (iv)).

Empirical observation on the cross-model verification pathway. Beyond the per-mode mitigations above, an integrated mitigation pathway has been preliminarily characterized in the v18.2 cycle. Stage 4 was executed on two devset papers (Kumada 1972 and Murai 1993) under the same prompt by two parallel models with structurally different retrieval modalities: Sonnet 4.6 with parametric retrieval only, and GPT-5.5 Thinking with chat-context web retrieval. Each candidate from each model was then resolved against the Crossref API at the (author, year, journal, volume, page) tuple level. Three observations follow. First, the GPT-5.5 retrievals returned 9/9 (Kumada) and 8/8 (Murai) candidates with Crossref-resolvable DOIs and substantively coherent content descriptions — web-anchored retrieval reproduced the bibliographic-shell verification automatically and concurrently with candidate generation. Second, the Sonnet retrievals exhibited the paper-dependent pattern documented as mode (v) — two of three checkable predictions not found on Kumada 1972 (one a confirmed author combination, one unresolved), against no full fabrication on Murai 1993 — making single-model verification unevenly necessary across target papers. Third, divergence between the two models served as a high-prior fabrication signal: Sonnet candidates that GPT-5.5 did not corroborate and that did not resolve in Crossref were the candidates retracted on inspection. The combination of cross-model parallel retrieval with Crossref-API tuple resolution thus provides a structural detection mechanism for mode (v) and an automatable bibliographic check that complements the content reading required for mode (iv). The operational form, including conditional-triggering rules calibrated to the four structural risk factors of mode (v), is left to subsequent methodological extensions; the present observation is reported as preliminary empirical signal rather than a verified protocol. Full comparison logs are at `papers/dev/{kumada,murai}/cross_model_v182_comparison.md` and the meta-analysis at `improvements/cross_model_kumada_vs_murai_meta.md`.

SI-9.5 An empirical byproduct: DOI reconstruction from metadata

During Tier 1 query preparation, the analyst reconstructed three of the six DOIs from author, year, volume, and page when the LLM output had not recorded an explicit DOI. Two of the three guesses returned papers by different authors in the same journal, and the third matched by coincidence:

Candidate	Analyst-guessed DOI	Actual DOI	Outcome
Parshall 1974	`10.1021/ja00815a017`	`10.1021/ja00815a009`	Different paper (Gleicher, quinonoid calculations)
Anderson-Jones-Vicic 2004	`10.1021/ja0488380`	`10.1021/ja0483903`	Different paper (Hong-Day-Grubbs, Ru metathesis)
Jones-Vicic 2006	`10.1021/ja063334i`	`10.1021/ja063334i`	Accidental correct match

The DOI suffixes used by ACS journals are not derivable from volume and page via any apparent arithmetic pattern; the single successful guess was coincidental. Any procedure that reconstructs DOIs from citation metadata — whether by a human analyst or by an LLM — will therefore produce unreliable DOIs unless each is verified against an authoritative index. This observation

independently supports failure mode (iv): LLM reconstruction of citation details beyond what is explicitly stored in the model's parametric memory is a distinct risk axis.

SI-9.6 Limitations of this pilot

- **n = 1 EP.** Generalization to other EPs is not claimed. The pattern observed for EP-03a — a diphosphine-specific source term whose historical fillings generalize across ligand classes and span a pre-1991 literature layer — is specific to this case. EPs with different linguistic signatures may behave differently.
- **Single database.** WoS Core Collection is one of several chemistry-literature databases. SciFinder and Reaxys, both full-text-capable for a larger subset of ACS papers, may show different coverage profiles. A Phase 2 benchmark across multiple databases would substantially strengthen the findings.
- **Single-analyst triage.** Relevance judgments on the 268 retrieved WoS records were not systematically triaged in this pilot. The zero-overlap claim relies on DOI-level comparison, which is sufficient to establish modes (i)–(iv) but does not quantify keyword-retrieval precision within the retrieved set. Mode (v) rests separately on n = 2 devset-paper comparison and is reported as preliminary.
- **Yamamoto BCSJ verification completed 2026-04-21.** The 1978 and 1981 *BCSJ* PDFs were retrieved from a university library (hardcopy; not available through the Japanese Chemical Society's online institutional channels) and read directly. The diagnosis of failure mode (iv) was strengthened from a candidate-swap hypothesis to a confirmed three-layer mismatch (see SI-9.3 (iv)). No analogous library-hardcopy gap remains for the remaining candidates.

Recovery pathway also produced methodological signal, not just diagnostic signal. An ameliorative observation on the hallucination-detection cost: once the two Yamamoto *BCSJ* PDFs were in hand for verification, full v17 + Stage 3 v2 + Stage 4 v2 UQS self-analyses were executed on both papers (`tier_3_stage_outputs/citation_integrity_cases/yamamoto1978/` and `…/yamamoto1981/`; source PDFs not deposited). The 1981 analysis confirmed the Partial EP-03a filler verdict for Kumada 1972 with methodological traceability; the 1978 analysis established that the hallucination-victim paper itself is methodologically informative as a Kumada-EP-03a-inheritor (it *extends* the gap to polycondensation rather than closing it). The 1978 Stage 3 output additionally assigned the provisional PRI-11 code (Mechanism-kinetics decoupling) and EP-10a, both supplied by the Stage 3 prompt (SI-6.5); these are applications of prompt-supplied labels, not independent confirmations, and they do not count towards the promotion threshold of Codebook v0.2a. It was the first cross-paper application of EP-10a (Kinetic verification pack) originally derived from Kumada 1972 (see SI-3.7 update 2026-04-21). This incidental yield is not generalizable — two papers do not a Phase 2 benchmark make — but it does moderate the cost calculus of hallucination mitigation: content-level PDF verification, once performed, produces Codebook-evolution signals that pure metadata checking does not.

A Phase 2 benchmark — multiple EPs, multiple databases, systematic relevance triage, multiple analysts, and multiple devset papers for the cross-model fabrication test — would be required to generalize the five failure modes into a quantitative retrieval-profile claim. The present findings are therefore reported as an n = 1 (modes (i)–(iv)) plus n = 2 (mode (v)) empirical illustration of the structural differences between retrieval modes, not as a precision/recall benchmark.

SI-9.7 Data availability

All queries, raw WoS exports, Python parsers, and per-candidate verification records are archived at `retrieval_audit/ep03a_wos_llm/`:

- `wos_query.txt`, `wos_query_alternatives.md` — five query variants
- `wos_raw_49hits.txt`, `wos_variant_{a,b,c,d}.txt` — raw tagged-format exports
- `llm_candidates_ep03a.md` — consolidated LLM candidate list with ligand and filling-type metadata
- `candidate_check_{1..6}*.txt` — Tier 1 and Tier 2 direct lookups per candidate
- `verification_log.md` — PDF-level content verification (Parshall 1974, Anderson-Jones-Vicic 2004 + erratum, Yamamoto-Hayashi-Yamamoto 1978 *BCSJ* 51, 2091, and Yamamoto-Kohara-Yamamoto 1981 *BCSJ* 54, 2010 all confirmed)
- `pdfs/` — not deposited (copyright; the directory is not in the deposit). The seven candidates verified to date were read in full text: five through ACS (Parshall 1974, Morrell-Kochi 1975, Anderson-Jones-Vicic 2004 and its erratum, Jones-Vicic 2006) and two from library hardcopy (Yamamoto-Hayashi-Yamamoto *BCSJ* 51, 2091, 1978; Yamamoto-Kohara-Yamamoto *BCSJ* 54, 2010, 1981)
- `diagnostic_report.md` — full internal analysis
- `query_anomalies_log.md` — a WoS autocorrect anomaly (`alkylnickel` → `allylnickel`) encountered during variant preparation, illustrating a separate fragility of structured keyword search

SI-9.8 Extension to Modules 2–4 (EP-08a, EP-09a, EP-07a) — 2026-05-20

The pilot reported in SI-9.1 — SI-9.7 above was extended to three further Evidence Packs in May 2026 in order to support the negative claims for EP-08a (replicate-measurement data) and EP-09a (substrate-matched head-to-head benchmark) at the WoS-bibliographic retrieval layer, and to calibrate the audit methodology against a positive-control case (EP-07a, substrate-scope extension). Filling criteria were locked **before** retrieval in `tier_6_audit_decisions/ep_audit_criteria_20260520.md`, following the recommendation that pre-retrieval criteria-locking is necessary to defend against a post-hoc adjustment concern that a sympathetic reviewer might otherwise raise.

SI-9.8.1 Modules 2 + 3 (EP-08a, EP-09a) — negative claims

Modules 2 (EP-08a) and 3 (EP-09a) each tested 4 orthogonal WoS Variants per EP, designed to test the Stage 4 finding (EP-08a / EP-09a left unfilled by a Sonnet 4.6 run and by a GPT-5.4 Thinking run with web search enabled; the label is not stable across runs, SI-4.9) against structured bibliographic retrieval. For EP-08a, an additional 2 syntax-relaxed formulations (AND co-occurrence; NEAR/3 proximity) were applied to the named-reaction control variant to rule out hyphen-syntax artifact (all returned 0 hits).

Cumulative results (22 candidate papers across 10 query formulations):

- **EP-08a**: 11 candidates total, 0 Complete + 0 Partial + 0 Minimal + 11 Not applicable per the audit criteria locked in `tier_6_audit_decisions/ep_audit_criteria_20260520.md` §4. Candidates clustered into five non-Kumada chemistry domains: stilbene-as-ligand of organometallic complexes (Wicht 1998 Pt; Nattmann 2020 Ni(0)), stilbene-as-oxidisable-substrate (Solans-Monfort 2011 Ni-cyclam; Iwamoto 2001 Mn-MCM-41), mesoporous-catalysis epoxidation (Iwamoto 2001, 2004), photochemistry-electron-donor (El-Agamey 2012), and Mg(I) main-group chemistry (Dange 2019). The keyword combination of Kumada / Ni-catalysis with reproducibility / replicate / n=3 / standard deviation returns papers that *mention* reproducibility as a problem (e.g., Joshi-Pangu 2011 $NiCl_2$ hydration; Takahashi 2021 mechanochemical Grignard) but do not *provide* per-substrate replicate data for the dppe-Ni Kumada Table I substrate set.
- **EP-09a**: 11 candidates total, 0 Complete + 0 Partial + **2 Minimal** + 9 Not applicable per audit criteria §5. The 2 Minimal candidates — Quesnelle-Snieckus 2018 (cross-reaction-class qualitative comparison between Suzuki-Miyaura, Negishi, and Corriu-Kumada protocols; no within-Ni-Kumada predecessor numerical benchmark) and Berding 2009 (verbal "highest activity in this reaction reported to date" without matched-condition data) — sit at the weakest inclusion criterion ("verbal claim of superiority without head-to-head data"). The remaining 9 candidates are catalyst-design papers introducing new Ni-pincer, NHC-Ni, dinuclear-Ni, or Co-Zr systems where named-predecessor terms appear as historical context, not as benchmark targets.

SI-9.8.2 Module 4 (EP-07a) — positive-control calibration

Module 4 was abbreviated to 2 variants and run as positive control. Variant A (Csp^3 Kumada scope) returned 9 hits including **2 Partial fillers** under identical methodology and criteria-locking protocol:

- **Breitenfeld et al. 2013** (*J. Am. Chem. Soc.*, 10.1021/ja4051923) — N2N-Ni pincer alkyl-alkyl Kumada coupling; mechanism study revealing bimetallic radical oxidative addition; Csp^3 scope extension in a related (Ni-based) but not strict dppe-Ni system, satisfying audit criteria §3 Partial criterion 3.
- **Perez Garcia et al. 2016** (*ACS Catal.*, 10.1021/acscatal.5b02324) — PNNN-Pyr-Ni pincer cross-coupling of nonactivated secondary alkyl halides; secondary Csp^3 scope extension in related Ni-pincer system.

The positive control shows that, under these query conditions, the audit can retrieve Partial fillers that exist (EP-07a). It does not establish the recall of the EP-08a/EP-09a queries, and it does not show that no such evidence exists: the EP-08a/EP-09a result is a bounded non-detection within the audited set, not an absence of the knowledge (main text, Limitation L1). Variant B of Module 4 (Cl/Br/I direct comparison) returned 0 hits owing to the narrow requirement that all three halide-name keywords co-occur in the topic field; this is a query-design boundary, not a true negative for EP-07a as a whole given the Variant A positive findings.

SI-9.8.3 Three convergent diagnostic patterns (Module 3, EP-09a)

Across EP-09a Variants A–D, three structural patterns emerged consistently in the keyword-matched candidate set:

1. **Catalyst-design papers dominate the keyword matches.** NHC-Ni, pincer-Ni, dinuclear-Ni, and Co-Zr systems generate the bulk of hits; named-predecessor terms appear as historical context rather than as benchmark targets.

2. **Within-family comparisons dominate when comparison language is used.** Where "comparison", "vs", "compared to", or related phrases co-occur with Kumada, the comparison is typically internal to the new catalyst's own ligand-variant family (e.g., Zhou 2014 Co/Zr Xyl vs Mes), not against named external predecessors.
3. **Explicit benchmark vocabulary co-occurs with cross-coupling-Grignard effectively never.** Variant B (`head-to-head` / `side-by-side` / `parallel reaction` / `benchmark` keywords) returned 1 off-target hit; Variant D (Fe-Kumada vs Ni-Kumada matched-conditions) returned 0 hits.

These three patterns describe a structural pattern in modern Ni-Kumada publishing: comparisons organise internally (within new catalyst's variants, within scope-extension paper) rather than externally (against named predecessor catalysts at matched conditions). These patterns are consistent with the publication-incentive interpretation discussed in Discussion 1, but observing how the retrieved papers organise their comparisons cannot separate that interpretation from the other accounts considered there.

SI-9.8.4 Soler-Yanes 2014 boundary observation

Soler-Yanes et al. 2014 (*Eur. J. Org. Chem.*, 10.1002/ejoc.201403007) was returned by EP-09a Variant A. The paper uses **Ni(acac)$_2$-TMEDA** — the Corriu/Masse predecessor catalyst class — as the chosen catalyst for a new substrate scope (functionalised benzyl chlorides and allyl chlorides with alkylmagnesium reagents), but does **not** benchmark dppe-Ni against Ni(acac)$_2$ at matched substrate + matched conditions. This is consistent with the publication-incentive interpretation, on which using an existing catalyst for a new substrate scope counts as publishable novelty and benchmarking two existing catalysts at matched conditions does not; it is one example and does not test that interpretation. The paper is therefore classified as Not applicable for EP-09a Complete or Partial filler conditions.

SI-9.8.5 Methodology observations supplementing SI-9.3 failure modes

The five failure modes diagnosed in SI-9.3 extend naturally to Modules 2–4. Two additional methodology observations from the present extension:

- **Strict phrase matching versus syntax relaxation.** For EP-08a Variant D ("Kumada-Tamao" / "Kumada-Corriu" named-reaction terminology), three orthogonal formulations — strict quoted phrase, AND co-occurrence (`Kumada AND Tamao`), and NEAR/3 proximity (`Kumada NEAR/3 Tamao`) — all returned 0 hits. This rules out a hyphen-syntax artefact in this variant; the result remains a non-detection under the queries run (SI-9.8.2).
- **Positive-control calibration as methodology check.** The 2 Partial fillers in EP-07a Variant A show that the audit pipeline can return Partial fillers where they exist; they do not measure the recall of the EP-08a / EP-09a queries (SI-9.8.2). The contrast — EP-07a 2 Partial fillers versus EP-08a / EP-09a 0 Partial fillers under identical methodology and pre-retrieval criteria-locking — is consistent with the EP-08a / EP-09a non-detection not being an artefact of the audit method, within the queries run; it remains a bounded non-detection, not a demonstrated absence.

SI-9.8.6 Data availability (Modules 2–4)

Queries, per-candidate analyses and hit-count summaries are archived under the per-EP directories. Raw WoS exports from 2026-05-20 are archived for EP-03a (all variants) and for EP-08a Variant A. The exports for the remaining six EP-07a / EP-08a / EP-09a variants were overwritten before deposit (each file was a copy of the EP-08a Variant A export). All twelve EP-07a / EP-08a / EP-09a queries were therefore re-run verbatim on 2026-10-05 and the six exports re-generated (`retrieval_audit/wos_reexport_20261005/`). Eleven of the twelve hit counts are unchanged and every returned record is one already classified in the May analysis; no new record appeared. The exception is EP-08a Variant A, which returned 2 records instead of 3: Zhou et al. 2020 (10.1039/d0qo00901f), classified Not applicable in May, is no longer returned under `DT=(Article)`. The EP-08a classification (no Complete, Partial or Minimal filler) is unchanged. Hit counts and classifications below are those of 2026-05-20:

- `retrieval_audit/ep07a_wos_llm/` — Module 4 positive control (2 variants, 9 + 0 hits)
- `retrieval_audit/ep08a_wos_llm/` — Module 2 EP-08a negative claim (4 strict + 2 syntax-relaxed = 6 formulations, 11 cumulative hits)
- `retrieval_audit/ep09a_wos_llm/` — Module 3 EP-09a negative claim (4 variants, 11 cumulative hits)

A consolidated cumulative summary across Modules 2–4 is at `retrieval_audit/tier1_audit_summary_20260520.md`. The pre-retrieval audit criteria (Complete / Partial / Minimal / Non-filler conditions per EP) are locked at `tier_6_audit_decisions/ep_audit_criteria_20260520.md`. The claim-lock document written before these audit modules is at `tier_6_audit_decisions/paperA_dd_target_narrative_20260520.md`.

SI-9.9 Per-EP audit summary table (Modules 2–4)

The following table consolidates the per-EP classification outcomes from Modules 2–4 ("Not applicable" is the Non-filler class of the locked criteria, as named in the main text). Total candidates: 31 across 12 query formulations (EP-08a 6 formulations, EP-09a 4 variants, EP-07a 2 variants). Detailed per-candidate rationales are available in the per-EP analysis files under `retrieval_audit/ep{NN}a_wos_llm/wos_variant_{a..d}_analysis.md`.

Module 2 — EP-08a (replicate-measurement, negative claim, 11 candidates, 0 Complete + 0 Partial + 0 Minimal + 11 Not applicable)

WoS Variant	Hit count	Candidate (DOI)	0B §4 classification	Rationale (one line)
A (strict)	3	Zhou-Xu-Zhang-Kong 2020 (10.1039/d0qo00901f)	Not applicable	Ni-electroreductive coupling highlight; not Grignard Kumada
A	3	Kurig-Cagan-Harper-Kawamata-Blackmond-Baran 2026 (10.1002/anie.9252206)	Not applicable	Reductive sp^2-sp^3 coupling with aryl-hydrazine reductant; NiII/bipy ≠ dppe-Ni
A	3	Takahashi-Hu-Gao-Gao-Pang-Seo-Jiang-Maeda-Takaya-Kubota-Ito 2021 (10.1038/s41467-021-26962-w)	Not applicable	Mechanochemical Grignard preparation; not Table I replicate data
B (Ni-specific + replicate methodology)	1	Cornacchio-Price-Jennings-McDonald-Staroverov-Jones 2009 (10.1021/jo801771d)	Not applicable	$PdCl_2$(dppf) Kumada — Pd not Ni; thienyl substrates
C (cis/trans + kinetic + Ni/Grignard)	7	Wicht-Zhuravel-Gregush-Glueck-Guzei-Liable-Sands-Rheingold 1998 (10.1021/om971000j)	Not applicable	trans-stilbene as Pt(0) ligand; not Kumada product
C	7	Nattmann-Cornella 2020 (10.1021/acs.organomet.0c00485)	Not applicable	Ni(0)(stilbene)$_3$ precatalyst; stilbene as supporting ligand
C	7	Solans-Monfort-Fierro-Hermosilla-Sieiro-Sodupe-Mas-Ballesté 2011 (10.1039/c0dt01605e)	Not applicable	Ni-cyclam oxidation chemistry; cis-stilbene as oxidisable substrate
C	7	Iwamoto-Tanaka 2001 (10.1023/A:1012257731538)	Not applicable	Mn-MCM-41 epoxidation; trans-stilbene as epoxidation substrate
C	7	Iwamoto 2004 (10.5059/yukigoseikyokaishi.62.411)	Not applicable	Review of mesoporous catalysis
C	7	El-Agamey-Fukuzumi 2012 (10.1002/chem.201201328)	Not applicable	Carotenoid radical cation photochemistry; trans-stilbene as electron donor
C	7	Dange-Gair-Jones-Juckel-Aldridge-Jones 2019 (10.1039/c9sc00200f)	Not applicable	Mg(I) main-group chemistry; trans-stilbene as 1,2-addition substrate
D (strict phrase)	0	—	—	No paper combines "Kumada-Tamao"/"Kumada-Corriu" phrase with replicate variability keywords
D′ (AND co-occurrence)	0	—	—	Same as D; rules out hyphen-syntax artifact
D″ (NEAR/3 proximity)	0	—	—	Same as D; three orthogonal syntax formulations all return 0

Module 3 — EP-09a (substrate-matched head-to-head benchmark, negative claim, 11 candidates, 0 Complete + 0 Partial + 2 Minimal + 9 Not applicable)

WoS Variant	Hit count	Candidate (DOI)	0B §5 classification	Rationale (one line)
A (strict named-benchmark)	9	Park-Buck-Rizzo 1998 (10.1016/S0040-4020(98)00786-8)	Not applicable	Carbazole synthesis via Ni-Corriu-Kumada coupling; no benchmark
A	9	Matsubara-Yamamoto-Miyazaki-Inatomi-Nonaka-Koga-Yamada-Veiros-Kirchner 2017 (10.1021/acs.organomet.6b00451)	Not applicable	Dinuclear NHC-Ni(I) mechanism study; "Kumada-Tamao-Corriu" = reaction name attribution
A	9	Sun-Li-Sun 2014 (10.1039/c4dt00461b)	Not applicable	[CNN]-pincer-NHC-Ni(II) on unactivated C-Cl bonds
A	9	Soler-Yanes-Guisán-Ceinos-Buñuel-Cárdenas 2014 (10.1002/ejoc.201403007)	Not applicable	**Uses Ni(acac)$_2$-TMEDA on benzyl/allyl chlorides — boundary observation, see Discussion 1**
A	9	**Quesnelle-Snieckus 2018** (10.1055/s-0037-1611053)	**Minimal**	Cross-reaction-class qualitative comparison (Suzuki vs Negishi vs Corriu-Kumada); no within-Ni-Kumada numerical benchmark
A	9	Wei-Abdiaj-Sambiagio-Li-Zysman-Colman-Alcázar-Noël 2019 (10.1002/anie.201906462)	Not applicable	Fe-catalysed Kumada-Corriu in flow + photoredox; not Ni
A	9	Matsubara-Fujii-Hosokawa-Inatomi-Yamada-Koga 2019 (10.3390/molecules24183222)	Not applicable	NHC-Ni(I) with fluorinated arylphosphine catalyst design
A	9	**Berding-van Dijkman-Lutz-Spek-Bouwman 2009** (10.1039/b905036a)	**Minimal**	Verbal "highest activity in this reaction reported to date" without matched-condition numerical benchmark
A	9	Li-Zhu-Shao-Li-Chang-Gao-Zhang-Wei 2016 (10.1016/j.tet.2015.10.069)	Not applicable	Pd nanoparticle Stille + Kumada on diazonium / aryl chlorides
B (head-to-head Ni-Grignard)	1	Kosar-Ayub-Gilani-Shah-Mahmood 2020 (10.1002/qua.26106)	Not applicable	DFT methods benchmark (computational), not Ni-catalyst benchmark
C (comparison + predecessor)	1	Zhou-Saper-Krogman-Foxman-Thomas 2014 (10.1039/c3dt52133h)	Not applicable	Co/Zr heterobimetallic catalyst; within-family Xyl vs Mes ligand-variant comparison
D (Fe vs Ni matched)	0	—	—	No paper combines Fe-cat + Ni-cat + Grignard/Kumada + matched-conditions language

Module 4 — EP-07a (substrate-scope extension, positive control, 9 candidates, 0 Complete + 2 Partial + 0 Minimal + 7 Not applicable)

WoS Variant	Hit count	Candidate (DOI)	0B §3 classification	Rationale (one line)
A (Csp^3 Kumada scope)	9	**Breitenfeld-Ruiz-Wodrich-Hu 2013** (10.1021/ja4051923)	**Partial**	N2N-Ni pincer alkyl-alkyl Kumada paradigm-shift; Csp^3 extension in related Ni system
A	9	Coombs-Perry-Kwon-Thomas-Ess 2018 (10.1021/acs.organomet.8b00449)	Not applicable	Co-Zr heterobimetallic Kumada DFT study; not Ni
A	9	Bauer-Cheung-Hu 2015 (10.1055/s-0034-1380136)	Not applicable	Fe-bopa pincer; not Ni
A	9	Bauer-Wodrich-Scopelliti-Hu 2015 (10.1021/om501122p)	Not applicable	Fe-bopa pincer mechanism study; not Ni

WoS Variant	Hit count	Candidate (DOI)	0B §3 classification	Rationale (one line)
A	9	**Perez Garcia–Di Franco–Epenoy–Scopelliti–Hu 2016** (10.1021/acscatal.5b02324)	**Partial**	PNNN-Pyr-Ni pincer secondary alkyl halide scope; Csp^3 extension in related Ni system
A	9	Wei-Abdiaj-Sambiagio-Li-Zysman-Colman-Alcázar-Noël 2019 (10.1002/anie.201906462)	Not applicable	Fe-catalysed Kumada (same paper as EP-09a Module 3)
A	9	Tsien-Péter-Zeng et al. 2025 (10.1002/anie.202517207)	Not applicable	Modern Ni redox-neutral radical cross-coupling with sulfonyl hydrazides; Kumada cited as inferior precedent only
A	9	Bisz-Szostak 2019 (10.1002/adsc.201800849)	Not applicable	Fe-catalysed $C(sp^2)$-$C(sp^3)$ cross-coupling; not Ni
A	9	Do-Anosike-Beng 2023 (10.1039/d3ra04690g)	Not applicable	Pd-catalysed Kumada on lactam-bromolactones; not Ni
B (Cl/Br/I direct comparison)	0	—	—	Three-halide-name simultaneous keyword requirement too narrow; query-design boundary, not true negative (Variant A positive findings already calibrate methodology)

Cumulative classification summary across Modules 2–4

EP	Module	Variants run	Hits	Complete	Partial	Minimal	Not applicable
EP-08a	2	4 + 2 syntax-relaxed = 6	11	0	0	0	11
EP-09a	3	4	11	0	0	2	9
EP-07a (positive control)	4	2	9	0	**2**	0	7
Total	—	12	**31**	**0**	**2**	**2**	**27**

The 2 Partial fillers in the EP-07a positive control (Breitenfeld 2013, Perez Garcia 2016) show that the audit pipeline can identify Partial filling literature for an EP where it exists. They do not establish the recall of the EP-08a and EP-09a queries, so the 0-Partial result for EP-08a and EP-09a is a non-detection in the WoS-indexed academic record under the criteria locked pre-retrieval, not a demonstrated absence. The 2 Minimal candidates in EP-09a (Quesnelle-Snieckus 2018; Berding 2009) sit at the weakest inclusion criterion ("verbal claim of superiority without head-to-head data" or "cross-reaction-class qualitative comparison").

Soler-Yanes et al. 2014 (EP-09a Variant A, Not applicable) is an instance of the pattern discussed in SI-9.8.4: the paper uses $Ni(acac)_2$-TMEDA (the Corriu predecessor catalyst class) as the chosen catalyst for a new substrate scope (benzyl/allyl chlorides), but does not benchmark dppe-Ni against $Ni(acac)_2$ at matched conditions. It is consistent with the publication-incentive reading but does not test it.

SI-9.10 Retained and retracted Stage 4 candidates (R1)

Rule. A Stage 4 candidate is *retained* when its bibliographic record resolves to a single paper in Crossref (after correcting author or page errors where the record identifies the intended paper) **and** the paper's content was checked and supports the claimed role. It is *retracted* when no such paper exists, when the citation points to a different paper, or when the paper does not contain what the model said. Everything else is *unclassifiable*. A lenient count, in which a Crossref match alone counts as retained, is also given.

The 37 Stage 4 analyses reported in the paper (24 Kumada / Murai multi-run analyses and 9 case-study analyses under Stage 4 v2; 4 context analyses under Stage 4 v6; Sonnet 4.6, Stages 0–2 v18.4) proposed 468 candidates. None was checked individually, so all 468 are unclassifiable. Their per-run filling labels are model judgements. They enter the paper only through the run-stability counts of SI-3, not through the filling map, whose papers were all obtained and read (Methods 2).

Verification studies. Candidate-level verdicts exist for 295 candidates from 48 other Stage 4 runs on Kumada 1972 and Murai 1993. Under the strict rule, 17 are retained, 25 retracted and 253 unclassifiable; under the lenient rule, 155 retained, 25 retracted and 115 unclassifiable. Of the 25 retractions, 18 could not be resolved to any paper, 4 cite a real paper whose content does not match, and 3 point to a different paper. Only one study, the exemplar control (Methods 6), attempted a bibliographic check of every candidate it produced: of 51 candidates, 2 were bibliographic failures (one wrong page range; one journal-volume-page coordinate at which no paper exists) and 4 could not be checked, because the runs gave incomplete details and marked them as uncertain; those 4 are counted neither as confirmed nor as errors. The check concerns the citation, not whether the paper supports the filling. These counts describe the checks that were made, not a fabrication rate.

Files. `tier_4_classification_tables/stage4_candidates_R1/candidates.csv` lists every candidate with its status, the rule applied and the record it rests on; `build_tally.py` regenerates it from the deposited run files and verification records.

SI-10. Cited dipy-Ni model source and analogy-validation in the Kumada 1972 case

This SI documents the analysis underlying the main text's treatment of Uchino, Yamamoto, and Ikeda (1970) as the cited dipy-Ni model source for Kumada 1972's proposed catalytic cycle. The structural relationship between Uchino 1970 and Kumada 1972's EP-06a (Analogy-validation pack) is documented across three layers: (i) the bibliographic note on Kumada 1972's compound Reference 11; (ii) the trace through which the Kumada UQS analysis already identified the dipy-Ni → dpe-Ni analogy internally; (iii) a targeted UQS analysis of Uchino 1970 itself and its evidence-pack cross-reference against Kumada 1972; (iv) the operational definition of "system-matched documentation" used throughout the main text; and (v) the exploratory sandbox 6-paper landscape that is documented but not used as empirical base.

Note on the role assigned to Uchino 1970. Throughout this paper, Uchino 1970 is treated as an *antecedent cited-model source* for Kumada 1972's proposed catalytic cycle, not as a member of the Kumada-anchored filling-paper set. The filling-paper analysis is defined relative to the evidential requirements generated by the 1972 communication (the EPs of EP-01a through EP-10a; see Stage 3 in Methods); papers cited by Kumada 1972 itself as precedent are by construction inputs to the EP architecture, not forward fillers of it. This is a role distinction, not an exclusion: the contribution of Uchino 1970 enters the analysis as a central source of the analogy-validation requirement EP-06a (see SI-10.4), and the system-matched filling level of EP-06a is satisfied most strongly by later documentation in the dpe/dppe-Ni / Grignard context (SI-10.5; Results Part 2).

Uchino/Yamamoto 1970 and the Csp^3-scope reading of EP-07a. Beyond serving as the dipy-Ni model source for the proposed catalytic cycle (EP-06a), Uchino, Yamamoto, and Ikeda 1970 bears directly on EP-07a. On reaction of (bpy)Ni$(Et)_2$ with chlorobenzene it gives (bpy)Ni(Ph)Cl together with butane (Et–Et) — that is, oxidative addition of the aryl chloride and reductive elimination of an alkyl–alkyl (Csp^3–Csp^3) bond from a dialkyl-Ni species. The elementary feasibility of alkyl–alkyl coupling on nickel was therefore already documented in the precedent that Kumada 1972 cites. The later Csp^3 extensions catalogued for EP-07a (Breitenfeld 2014, Molander 2016, Li 2024; SI-4) are accordingly best read as catalytic scope extensions to Csp^3 electrophiles under new mechanistic regimes, not as the crossing of a fundamental boundary: Kumada 1972's "only Csp^2 halides" statement bounds its own catalytic protocol and is not a claim that alkyl–alkyl coupling is infeasible on nickel, the supporting grounds sitting in its Reference 11. This illustrates a general point — in an early communication an apparent record-level boundary can have its resolving grounds in the cited references, which a per-paper reading anchored to the target text does not automatically incorporate (Limitations L9).

SI-10.1 Bibliographic note on Kumada 1972's Reference 11

Reference 11 in Kumada 1972[1] is a compound reference of two publications:

> (11) M. Uchino, A. Yamamoto, and S. Ikeda, *J. Organomet. Chem.* **1970**, *24*, C63; M. Abedini, *Quart. Bull. Fac. Sci., Tehran Univ.* **1971**, *2*(4), 1.

The first item — Uchino, Yamamoto, and Ikeda (1970) — was retrieved and subjected to the full UQS workflow at Sonnet 4.6 with v18.4 prompts (Stage 0 through Stage 4); the analysis is summarised in §SI-10.3 below.

The second item — Abedini (1971) — was not accessible during the present revision. *Quart. Bull. Fac. Sci., Tehran Univ.* is a Tehran University internal bulletin, and no electronic deposit was identified at the time of writing. Postal access through interlibrary loan has been requested but has not arrived; Abedini 1971 is therefore flagged here as an **unresolved antecedent source** and is **not used as evidence in the main-text argument**.

SI-10.2 Kumada v17 UQS trace: B-14 → RI-06 → EP-06a

The Kumada 1972 UQS analysis (v17 prompts on Sonnet, in `papers/dev/kumada/kumada_v17_sonnet_{s1,s3v2,s4v2}.md`) had already identified Uchino 1970 as a cited dipy-Ni model source at multiple stages. This subsection documents the internal trace, which is the structural basis for the cited-model-source framing used in the main text.

Stage 1 — observation B-14. The Stage 1 analysis on Kumada 1972 records that "eq 1 from refs 11–12 = dipy-Ni system"† — i.e., the equation introducing the catalytic cycle is sourced from the dipy-Ni chemistry cited as refs 11–12 of Kumada 1972. Ref 11 is the Uchino-Yamamoto-Ikeda 1970 communication; Ref 12 is a related Yamamoto-group entry of the same era.

Stage 1 — retrieval gap GAP-4. A retrieval-gap flag records: "Retrieve refs 11–12 (Uchino et al.) cited as the stoichiometric model for eq 1". This is the Kumada Stage 1 analysis explicitly marking the cited dipy-Ni precedent as a retrieval target.

Stage 3v2 — routing of B-14 through RI-06 to EP-06a. The Stage 3v2 analysis routes B-14 through RI-06 (Imported model) — a recurring-issue class describing "cycle-level import from a structurally non-identical system without explicit isomorphism verification" — and from there to **EP-06a (Analogy-validation pack)**, the Evidence Pack defined for "experimental demonstration of the imported catalytic cycle in the target system". EP-06a's case-specific implementation in the Stage 3v2 output calls for "stoichiometric organonickel experiments demonstrating the proposed cycle (eqs 2–4) in the $NiCl_2$(dpe) system itself, not in the cited dipy-Ni model".

Stage 4v2 — non-filling note. The Stage 4 retrospective filling-paper analysis on Kumada EP-06a explicitly notes that Uchino 1970 is **not** a filler of Kumada EP-06a, because the ligand class (dipy, N,N) differs from the target system (dpe/dppe, P,P) and the stoichiometric organonickel chemistry on dipy-Ni does not transfer directly to dpe/dppe-Ni.

Consequence for the main-text narrative. The Kumada UQS analysis was not retrospectively realigned in response to external feedback; the cited-model-source role of Uchino 1970 was already embedded in the Stage 1 / Stage 3 / Stage 4 trace from the outset. The main-text treatment of Uchino 1970 as the cited model source that generates EP-06a is therefore consistent with the UQS analysis as originally executed.

SI-10.3 Uchino 1970 — UQS analysis summary

Bibliographic information

M. Uchino, A. Yamamoto, S. Ikeda, "Preparation of a phenyl-nickel complex, phenyl(dipyridyl)nickel chloride, an olefin dimerization catalyst," *J. Organomet. Chem.* **1970**, *24*, C63–C64. DOI: 10.1016/S0022-328X(00)84475-7. Format: Preliminary communication (two journal pages, C63–C64).

Claim architecture

Uchino 1970 has a two-layer claim architecture:

- **Layer 1 (synthesis novelty):** the crystalline isolation of phenyl(dipyridyl)nickel chloride (compound II, C_6H_5(dipy)NiCl·½C_6H_5Cl) from the reaction of diethyl(dipyridyl)nickel (compound I) with chlorobenzene at room temperature under nitrogen.
- **Layer 2 (catalytic significance):** compound II is asserted to generate a "highly active" catalyst for α-olefin dimerization in combination with alkylaluminium halides; quantitative catalytic data are explicitly deferred to a separate publication.

Layer 1 is independent of Layer 2: a refutation of the catalytic claim would not undermine the isolation result, while refutation of compound II's identity would propagate to Layer 2. The Layer 2 utility claim is α-olefin dimerization, which is a **different reaction context** from the Grignard cross-coupling context of Kumada 1972. This distinction is important for §SI-10.6 below.

What Uchino 1970 establishes at the observation level

- Crystalline isolation of compound II (60% yield, melting point with decomposition at 75.5–75.7 °C);
- Elemental analysis (C, H, N, Cl) consistent with the proposed formulation;
- Formation of compound II from the reaction of diethyl(dipyridyl)nickel with chlorobenzene, accompanied by evolution of n-butane together with smaller amounts of ethane and ethylene (qualitatively reported);
- Three diagnostic chemical reactions of compound II — protonolysis, methylation, and ethanolysis — described as giving "stoichiometric quantities" of the expected organic products;
- Stability of compound II in chlorobenzene only; decomposition observed in tetrahydrofuran, acetone, and toluene at room temperature.

What Uchino 1970 does not establish

- The oxidation-state trajectory of nickel during the I → II transformation;
- The mechanism of the aryl C–Cl activation step (whether two-electron oxidative addition, σ-bond metathesis, or a one-electron radical pathway);
- The quantitative molar ratio of n-butane / ethane / ethylene evolved;
- The causal link between chlorobenzene-enhanced catalytic activity and the formation of compound II (no control experiment with chloride salt addition or with other aryl halides);
- Quantitative catalytic performance (turnover number, turnover frequency, selectivity, substrate scope) — deferred to a separate publication;
- Spectroscopic characterisation of compound II (no IR, NMR, UV-Vis, mass spectrometry, or single-crystal X-ray diffraction reported);
- Independent verification of the ½-equivalent chlorobenzene solvate composition (the formula is determined by best-fit elemental analysis);
- Mechanistic basis for the selective stability in chlorobenzene only;
- Rationale for the choice of 2,2'-dipyridyl as the supporting ligand (no comparison with other chelating or monodentate phosphine ligands).

SI-10.4 Uchino 1970 ↔ Kumada 1972 EP cross-reference

Mapping the Stage 3 Evidence Pack assignments for the two papers yields the following cross-reference. The shared incompleteness pattern reflects the common communication-era format of the two papers; the asymmetric relation through EP-06a is the structural finding documented in §SI-10.2 and §SI-10.5.

EP	Uchino 1970 status	Kumada 1972 status	Relation
EP-01a (Reporting coherence)	partial (communication format)	partial (communication format)	shared communication-era format
EP-02a (Identity / purity)	unfilled (no spectroscopic characterisation)	unfilled (VQ-1 residual)	shared format constraint
EP-03a (Direct observation / state-assignment)	unfilled (no mechanism, no oxidation-state trajectory)	unfilled (RI-03)	shared format constraint
EP-04a (Discrimination / replication)	unfilled (diagnostic reactions not quantified)	unfilled (RI-04)	shared format constraint
EP-05a (Quantitative ranking)	unfilled (Layer 2 catalytic data deferred)	unfilled (RI-05)	shared format constraint; **different reaction context** (α-olefin dimerization vs Grignard cross-coupling)
EP-06a (Analogy validation)	unfilled (Chatt–Shaw / Miller analogies asserted, not validated)	**unfilled — generated by importing the Uchino dipy-Ni model into the dpe-Ni / Grignard cycle** (RI-06; B-14 in Stage 1, see §SI-10.2)	**asymmetric: Uchino is the cited model source from which Kumada EP-06a is generated; not a filler**
EP-07a (Boundary test)	weakly partial (solvent-selectivity observed but unexplained)	unfilled (RI-07)	similar format constraint
EP-08a (Replication / uncertainty)	unfilled (elemental analysis single-point)	unfilled (RI-08)	shared format constraint
EP-09a (Internal benchmark)	unfilled (no I vs II catalytic comparison)	unfilled (CLA-2)	shared format constraint
EP-10a (Kinetic verification)	unfilled (no kinetics)	unfilled (EP label from the Stage 3 prompt; SI-6.5)	shared format constraint

The shared incompleteness on EP-01a / EP-02a / EP-03a / EP-04a / EP-08a / EP-10a reflects the common communication-era documentation norms of 1970– 1972 organonickel chemistry. The asymmetric relation on EP-06a is the structural axis on which the main-text narrative turns and is elaborated in §SI-10.5.

SI-10.5 dipy-Ni vs dpe/dppe-Ni and the operational meaning of "system-matched documentation"

EP-06a (Analogy-validation pack) calls for "experimental demonstration of the imported catalytic cycle in the target system". For Kumada 1972, the target system is **$NiCl_2$(dpe) / Grignard cross-coupling**. The Uchino 1970 model source is **diethyl(dipyridyl)Ni / chlorobenzene stoichiometric chemistry**.

Operational definition of "system-matched documentation"

Across the main text, the phrase **"system-matched documentation"** is used as the strongest filling level for EP-06a. This SI section makes the underlying criteria explicit. Documentation is treated as system-matched to the Kumada 1972 dpe/dppe-Ni / Grignard system when it satisfies **four axes simultaneously**:

1. **Ligand class** — a P,P chelate of the dpe/dppe class (1,2-bis- (diphenylphosphino)ethane or close homologue), rather than a monodentate phosphine, an N,N chelate, or a structurally distant bidentate.
2. **Ni-centred catalytic manifold** — a Ni(0)/Ni(II) two-electron catalytic cycle on the bidentate-phosphine-Ni platform, rather than a radical Ni(I)/Ni(III) manifold or an alternate transition-metal manifold.
3. **Grignard transmetalation context** — the catalytic cycle operates with R-MgX as the transmetalation reagent, with the associated Mg–X counter-ion environment, rather than with R-Li, R-Al, or R-Zn / R-B / R-Si reagents.[S2]
4. **Experimentally observed or directly constrained organonickel intermediates** — the intermediates implicated in the proposed cycle (Ni(II) aryl/alkyl species, Ni(0) regenerated state, transmetalation transition states) are either characterised directly in the same system or its close isomorph (spectroscopic observation, single- crystal structure of a viable intermediate) or constrained experimentally by kinetic analysis, radical-clock probes, isotope- effect measurements, stoichiometric-competence demonstration, or in-situ reaction monitoring — rather than imported by analogy from an off-target system. In practice, mechanism is established by combinations of these complementary modes; the criterion is satisfied when the available evidence collectively constrains the intermediates implicated by the proposed cycle to a degree that would not be obtained by analogical import alone.

Documentation that satisfies all four axes simultaneously is treated as the strongest filling level (Complete filling) for EP-06a. Documentation that satisfies one or several but not all of these axes — for example, Morrell-Kochi 1975 (monodentate-phosphine, R-Li/R-Al) — is treated as related-system Partial filling. The cited-model-source role of Uchino 1970 corresponds to a case in which several axes are off (N,N chelate; stoichiometric, not catalytic; α-olefin dimerization utility) and the "system-matched" classification therefore cannot apply.

Why Uchino 1970 fails the system-matched criterion

Mapping Uchino 1970 against the four axes:

Axis	Uchino 1970	Kumada 1972 target	Match?
Ligand class	dipy (N,N chelate)	dpe/dppe (P,P chelate)	**No**
Catalytic manifold	stoichiometric organonickel transformation	Ni(0)/Ni(II) catalytic cycle	**No**
Transmetalation context	none (no Grignard)	R-MgX with associated Mg–X chemistry	**No**
Observed or directly constrained intermediates	C_6H_5(dipy)NiCl (compound II) characterised in the dipy-Ni system; no kinetic or radical-clock evidence	Ni(II) aryl/alkyl on dpe/dppe; Ni(0) regeneration; constrained by kinetics / radical-clock / in-situ monitoring	**No**

Each axis is sufficient on its own to block direct transfer of Uchino's stoichiometric observations to the dpe-Ni / Grignard catalytic-cycle hypothesis. Their combination means that Uchino's observations cannot serve as the analogy-validation evidence Kumada 1972's proposed cycle requires. This is the chemical content of the statement that Uchino 1970 is the cited model source from which EP-06a arises rather than a filler of EP-06a.

EP-06a is satisfied most strongly by experimental demonstrations performed in the dpe/dppe-Ni system itself, on Grignard cross-coupling substrates — work that arrived progressively in the Morrell-Kochi 1975, Chen 2019, Dawson 2020, and Mazet 2021 publications. The Morrell-Kochi 1975 entry partially satisfies axes 2 and 4 but fails axes 1 and 3 (monodentate phosphine; R-Li / R-Al); the Chen / Dawson / Mazet 2019–2021 series progressively satisfies more axes, with Mazet 2021 providing the closest currently identified configuration on a bisphosphine-Ni / Grignard-context system isomorph.

SI-10.6 Transposition rather than filling: Kumada 1972's role toward Uchino 1970's deferred EPs

The Uchino 1970 UQS analysis (§SI-10.3) shows that Uchino's own evidence architecture left several EPs unfilled, notably:

- **EP-05a (Quantitative ranking)** — catalytic performance of compound II in the α-olefin dimerization context was deferred to a separate publication;

- **EP-03a (Direct observation / state-assignment)** — Ni oxidation-state trajectory during the I → II transformation was not characterised;
- **EP-10a (Kinetic verification)** — kinetic data on the I + PhCl transformation were not reported.

It would be tempting to say that Kumada 1972 "filled" these Uchino EPs by demonstrating Ni-catalysed catalytic activity, by proposing a oxidation-state trajectory in the catalytic cycle, and by providing a basis for later kinetic investigation. **This phrasing should be avoided.** Uchino 1970's utility-layer claim is α-olefin dimerization, not Grignard cross-coupling; Kumada 1972 did not present quantitative or mechanistic data on the α-olefin dimerization context Uchino had used.

The accurate description is **transposition / recontextualisation**: Kumada 1972 transposed the organonickel precedent from the dipy-Ni / olefin dimerization context into the dpe-Ni / Grignard cross-coupling context. This transposition generated a new set of evidence requirements (Kumada's own EP-03a, EP-06a, EP-10a) within the cross-coupling context; it did not fill Uchino's deferred questions within the original dimerization context.

This distinction matters because it preserves the asymmetry of the analogy-validation relation: Uchino's role toward Kumada is **source of imported model**, while Kumada's role toward Uchino is **transposition into a new reaction context**, not bidirectional EP filling.

SI-10.7 Stage 4 candidates caveat and sandbox 6-paper context

Uchino 1970 Stage 4 caveat

The full UQS workflow on Uchino 1970 was extended through Stage 4 (retrospective filling-paper search). The Stage 4 outputs identify candidate filling sources for several of Uchino's unfilled EPs (in particular EP-02a identity / purity and EP-05a catalytic-ranking candidates) drawn from subsequent Yamamoto-group publications and from α-olefin dimerization literature. These Stage 4 candidates are documented in `papers/dev/uchino/uchino1970_v184_sonnet_s4v6.md` but are **not used as evidence in the main-text argument of the present paper**, because individual filling assignments at Stage 4 carry an uncertainty / verification-required flag that has not been resolved within the scope of this revision.

Sandbox 6-paper landscape (exploratory, not used as empirical base)

A broader UQS analysis of six 1941–2020 antecedent and contextual organonickel / transition-metal / Grignard / alkyl-halide papers — Kharasch & Fields 1941 (*J. Am. Chem. Soc.* **63**, 2316); Tamura & Kochi 1971 (*J. Am. Chem. Soc.* **93**, 1485; Cu); Ward, Lawler, & Marzilli 1970 (*Tetrahedron Letters* 7, 521; CIDNP evidence for radical character in Grignard / alkyl-halide reactions); Tamura & Kochi 1971 (*J. Organomet. Chem.* **31**, 289; Fe; full paper with kinetics); Abraham & Hogarth 1968 (*J. Organomet. Chem.* **12**, 497; Co; Kharasch mechanism); and Chirik 2020 (*Organometallics* **39**, 775; Kochi historical profile) — has been **completed** in `papers/pericyclic/25_kharasch1941/` (since moved to another corpus of the project) and `papers/sandbox/69_tamura1971a/` through `papers/sandbox/73_chirik2020/` (Stage 0–4 + Stage Unpack at Sonnet 4.6 with v18.4 prompts; not deposited). Under the project's rule for sandbox analyses, these outputs are **exploratory and are not used as the empirical base for any paper, including the present manuscript**. The Kochi-Tamura *J. Am. Chem. Soc.* 1971 Ag paper (DOI 10.1021/ja00735a028) and the Tamura-Kochi *Bull. Chem. Soc. Jpn.* 1971 transition-metal survey (DOI 10.1246/bcsj.44.3063, which includes Ni) are not in the sandbox set.

These analyses are not used in the main text. They could serve a later study of whether the communication-era evidential-incompleteness pattern of §SI-10.4 extends to other 1968–1971 communications; that would require treating them as a corpus of their own, which is not part of this paper.

Post-1972 Ni context papers (development set). Three additional Kumada-context papers were UQS-analysed at Sonnet 4.6 with v18.4 prompts (`papers/dev/terao2002`, `papers/dev/jones2005`, `papers/dev/martin2007`; Stage 0–4 + Stage Unpack): Terao, Watanabe, Ikumi, Kuniyasu & Kambe 2002 (ref 44; Ni/1,3-diene Csp^3 alkyl–alkyl coupling), Jones, McFarland, Anderson & Vicic 2005 (ref 45; terpyridine-Ni(I)/Ni(II) intermediate isolation under Negishi-like conditions), and Martin & Buchwald 2007 (ref 29; low-temperature Pd coupling of Knochel-type Grignards). These three papers are **not part of the eight-paper historical case-study set and are not used as filling-map members or for any EP-filling assignment**; that set is unchanged. The three papers are cited in the main text only as concrete instances of, respectively, the earlier Ni Csp^3-scope precedent (Terao–Kambe 2002, Results Part 2), the earlier cross-system Ni(I)/Ni(III) mechanistic evidence (Jones–Vicic 2005, Results Part 2), and the reaction-class extensions that lie outside the anchor's EP dictionary (Martin–Buchwald 2007, Results Part 1). Cross-walked against the Kumada-1972 EP dictionary, Terao–Kambe 2002 addresses EP-07a (Csp^3-scope boundary) and Jones–Vicic 2005 addresses EP-03a/EP-06a, but in each case under a catalyst system distinct from the dpe/dppe-Ni / Grignard system, so neither constitutes system-matched filling — the same cross-system status already recorded for Breitenfeld 2014 (SI-4).

SI-10.8 Alternative and complementary accounts of the filling order

The publication-incentive reading of the VQ-hierarchy filling order (Discussion 1) is **not** claimed to be the only or dominant cause. At least four further structural conditions act as complementary — and in part alternative — accounts of the same filling order. Distinguishing them (incentive vs burden vs technical-platform availability vs tacit convention) would require a controlled cross-field study; the present case study establishes the *shape* of the filling order, not its dominant cause.

1. **Academic-corpus boundedness.** UQS's analytical reach is limited to the published academic record (Methods 3, *Note on academic corpus boundedness*); for industrially deployed chemistry, the kinetic robustness profiles, scale-up parameters, and matched-pair benchmarks that EP-08a and EP-09a request plausibly exist in patent and proprietary process records that fall outside this observable space.
2. **Per-study experimental burden.** Burden differs systematically across EP classes: EP-03a-class mechanism-validation closures are tractable within single-laboratory studies, whereas EP-08a-class replication closures require multi-laboratory consortia or technology-enabled platforms; the EP-08a non-detection is therefore best read as a combined signature of incentive misalignment and per-study burden.
3. **Industrial-deployment expectation.** Field attention was concentrated on deployable-chemistry evidence categories (substrate scope, ligand design, paradigm-shifting demonstrations) while replicate measurements and matched-pair benchmarking were placed under a "future challenges" frame.
4. **Tacit-knowledge convention.** Some low-VQ relations (e.g., the formation and behaviour of MgX_2 byproducts) are left unwritten because they were taken for granted within the contemporaneous field rather than judged unimportant (Methods 4, boundary case β), so the absence can reflect disciplinary common knowledge rather than incentive misalignment alone.

A clean empirical illustration of the case-level pattern is Soler-Yanes et al. 2014 (*Eur. J. Org. Chem.*; SI-9.8.4): it employs $Ni(acac)_2$-TMEDA — the Corriu/Masse predecessor catalyst class — for a new substrate scope without benchmarking dppe-Ni against $Ni(acac)_2$ at matched conditions; the predecessor class is in active use, so what is structurally absent within the academic corpus is the head-to-head numerical comparison.

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† Text marked with † was originally generated in Japanese by the UQS pipeline and has been translated into English for this Supporting Information.

SI-11: Operational and Audit Details

Paths written in repository form resolve in the Zenodo deposit as follows: `papers/dev/…`, `papers/ep/…`, `papers/test/…` → `tier_3_stage_outputs/` *(full) and* `tier_8_redacted_public_raw_outputs/` *(redacted);* `retrieval_audit/…` → `tier_5_retrieval_logs/retrieval_audit_ep03a_07a_08a_09a/…`; `prompts/…` → `tier_1_prompt_versions/`; `improvements/…` → `tier_6_audit_decisions/` *or* `tier_7_aggregation_scripts_and_methodology_evolution/`. `MANIFEST.csv` *at the deposit root lists every file. Source-paper PDFs are not deposited.*

This section collects the per-stage operational detail, execution-audit tables, hallucination-control specification, and structured retrieval-audit procedure that support the claims made in the main text Methods. The main text Methods states the methods' logic and the load-bearing claims; the full audit apparatus is deposited here and, at finer granularity, in the Zenodo archive (Data Availability Tiers 1–9). Moving this material out of the main text preserves auditability — every item below remains inspectable — while keeping the main text at the level a reader needs to follow the method's logic.

SI-11.1: LLM execution audit tables

These tables expand the summary given in Methods 6 (*LLM Setup and Execution Environment*). They make explicit, at the per-stage and per-model level, what the language model generates, what structured verification is applied, what the analyst adjudicates, and the configuration a third party would need to attempt a SET-level rerun.

Analyst intervention map. UQS is an LLM-assisted workflow with analyst oversight at every stage; analyst judgment is neither hidden nor diffuse. The table below makes explicit, at the per-stage level, what the LLM generates, what structured verification is applied, what the analyst adjudicates, the downstream output-dependence of each judgment, and the future expert-validation status registered for Phase 2.

Step	LLM-generated component	Structured verification applied	Analyst adjudication	Output-dependence (downstream)	Future expert-validation status (Phase 2)
Stage 0	Scope / authorial-claim / domain record	Anchoring rule (no inference) enforced by prompt	Confirms anchoring, flags scope ambiguity	All later stages depend on this baseline	Tier 1 advisor verification (Methods 7)
Stage 1	A/B/C/U absence records + evidence chain (location, quotation); T1/T2/T3 tier for A items	Per-record evidence_chain to specific loci; cross_field_temperature across V1/V2/V3	Confirms tier judgment and evidence-chain validity for boundary cases	Drives Stage 2 / 2F / 3 inputs	Tier 1 advisor evaluation of Stage 1 + D-4 outputs (SI-8)
Stage 2	D-1 to D-4 motivational interpretation + E scope propositions	D-4 candidates require evidence_chain	Confirms D-4 anchoring; classifies boundary D-1/D-2 splits	Feeds Stage 2F integration	Tier 1 advisor protocol pilot
Stage 2F	Recurring_roots, IPs, SSPs	Independence: per-paper outputs unchanged; integration on independently produced records	Confirms SSP source-paper anchors and subgroup membership; rejects unanchored SSPs	Primary research-question-brief output (SSPs)	Tier 1 advisor evaluation of SSP usefulness
Stage 3	SRG / RI / EP dictionary + VQ matrix + Era-Norm Annotation	EP case-specific implementation required; N=6 multi-run aggregation (Principle 4)	Tests SET-level inclusion frequency against a heuristic ≥75% threshold (met for Kumada 1972 under two of the three extraction rules, with one EP at 67% under the third; not met for the Japanese cell of Murai 1993; SI-3); accepts SET claims, rejects rank claims	Drives Stage 4 retrieval and Stage 2F integration	IRR study on EP derivation across independent analyst panel
Stage 4	Candidate citation + content description + per-candidate confidence (High/Medium/Low)	Not checked per candidate in the 37 reported analyses. A paper enters the filling map only after its full text is obtained and analysed, or it is retrieved in the WoS audit (Methods 2). Candidate-level checks (Crossref resolution, content check, cross-vendor run for high-risk cases) were made in separate verification studies (SI-9.10)	Adjudicates filling status (Complete / Partial / Minimal / Non-filler) against pre-locked criteria; retracts fabrications; designates Accounts-type bundled-filler exceptions	Determines filling-map findings	Independent third-party rerun under disjoint retrieval mode
Negative-claim warrant	—	Structured WoS audit with EP-07a positive-control comparator; pre-locked filling criteria	Confirms zero-Complete / zero-Partial result against positive-control non-zero result	Anchors EP-08a / EP-09a non-detection at retrieval-audit layer	Broader-database audits + citation-network analytics

The analyst role across all stages is **adjudication against pre-specified rules**, not generation of substantive claims. Substantive evidential content (absence records, EPs, filling-paper candidates, SSP question briefs) is generated by the LLM under prompt constraint and verified against the deposited prompts, evidence chains, and retrieval logs that constitute the auditability of the workflow.

Configuration by analysis (R1: one table for all prompt versions and models). Each row is one group of analyses reported in this paper. Prompt versions are the `prompt_version` tags and models the `model_info` lines in the deposited outputs, read on 2026-10-05; tag dates are the prompt versions' dates. Run dates are file dates, which can be later than the run (copies), not earlier. They are given to the month or day, as the record allows.

#	Analysis (where reported)	Papers	Stages 0–2	Stage 3	Stage 4 / Stage 2F	Model	Run dates
1	Kumada 1972 EP-dictionary record (SI-3.2–3.10; Table 3)	Kumada 1972	v17	v1 (`stage3-v1-20260402`)	—	Sonnet 4.6	2026-04-15
2	Four-model comparison (SI-6) and the GPT-5.4 Stage 4 candidates of SI-4.9 (web search enabled)	Kumada 1972	v17	v2 (`stage3-v2-20260416`); Sonnet also v2 EN (`stage3-v2-en-20260419`)	Stage 4 v2 (`stage4-v2-20260416`)	Sonnet 4.6; Opus 4.7; Haiku 4.5; GPT-5.4 Thinking	2026-04-17 to 04-19
3	Refinement set (Results Part 4; SI-11.17)	ten Heck papers	v17	v2	Stage 4 v2	Sonnet 4.6	2026-04-15 (S0–S2) to 04-17 (S3, S4)
4	Stage 2F, v17 inputs (the seven SSP briefs and Fig. 4 of Results Part 3; SI-5)	9 papers	v17	—	Stage 2F v6 / v7 / v7.1	Opus 4.6	2026-04-16 to 04-17
5	Development-set multi-run (SI-3; Table 6)	Kumada 1972, Murai 1993	v18.4 (`stage0-v184-20260502`)	v2 (JA cell) and v2 EN (EN cell), N = 6 each	Stage 4 v2 (JA) / v2 EN (EN)	Sonnet 4.6	2026-05-02 (S0–S2) and 2026-05-10 (multi-run S3, S4)
6	EP-context analyses (Results Part 2; SI-4)	8 case-study papers and Tamao 2002	v18.4	v2 EN	Stage 4 v2 EN	Sonnet 4.6	2026-05-10
7	Stage 2F, v18.4 inputs (IP-1–IP-7 of Results Part 3; SI-5)	9 papers	v18.4	—	Stage 2F v6 / v7 / v7.1	Opus 4.7	2026-05-10
8	Four context analyses (Methods 6)	Uchino 1970, Terao–Kambe 2002, Jones–Vicic 2005, Martin–Buchwald 2007	v18.4	v4 (`stage3-v4-20260427`)	Stage 4 v6 (`stage4-v6-20260504`)	Sonnet 4.6	2026-05 to 06
9	Haiku 4.5 replications, Kumada 1972 (SI-6.10; R1)	Kumada 1972	(a) v17, runs 2–3; (b) v18.4, runs 1–3; (c) v18.4, runs 4–5; (d) v17, pre-registered replication, N = 3	(a) v2; (b) v4; (c) not run; (d) v2	(a) Stage 4 v2; (b) Stage 4 v6	Haiku 4.5	(a), (b) 2026-09-14; (c) 2026-10-01; (d) 2026-09-29
10	Held-out set and single-prompt baselines (Methods 6; SI-11.20; R1)	4 post-cutoff Heck papers; Kumada 1972	v18.4 (pipeline arm)	v2 EN	—	Sonnet 4.6	2026-09-29
11	Reproducibility control (Methods 6)	Kumada 1972	v18.4 and v19, N = 5 each	—	—	Sonnet 4.6	2026-07
12	Decontamination control (Methods 6)	Kumada 1972	v18.4 with exemplars removed, N = 5	—	—	Sonnet 4.6 (GPT runs in the same folder are not reported)	2026-07
13	Input-ablation control (SI-11.19)	six papers, document vs bibliographic stub	v18.4, 36 runs	—	—	Sonnet 4.6	2026-08
14	Stage 4 exemplar control (Methods 6)	Kumada 1972	fixed S0–S3 record	—	Stage 4 v6, 5 + 5 runs	Sonnet 4.6	2026-08-25
15	GPT-5.5 runs that assigned EP-10a (SI-3; SI-6.5)	Kumada 1972, Murai 1993	v18.2	v3	Stage 4 v3	GPT-5.5 Thinking	2026-04-26/27

#	Analysis (where reported)	Papers	Stages 0–2	Stage 3	Stage 4 / Stage 2F	Model	Run dates
16	D1 item rewriting (SI-8.7; R1)	10 items from row 11 (v18.4 run 1)	—	—	—	Sonnet 4.6	2026-09-29

Stages 0–2 "v18.4" means the v18.4 Stage 0–2 prompt suite tagged 2026-05-02. Stages 0 and 1 are language-neutral; Stage 2 exists in an English and a Japanese version (SI-7, rows 6 and 23), and rows 5 (Japanese-language cells), 8, 9 (v18.4 runs), 11, 12 and 13 used the Japanese one, as the `prompt_version` headers of their outputs record. Stage 3 and Stage 4 have their own version tracks (SI-7). Rows 1–4 used v17 and predate v18.4. Earlier versions of this table dated v18.4 runs to April 2026 and listed a v18.4 Opus 4.7 Stage 3 arm. Both were wrong: no v18.4 run predates 2026-05-02, and the Opus 4.7 Stage 3 run is row 2 (v17).

No single per-execution manifest is deposited. The execution details at this granularity (stage, prompt version, model identifier, execution date, N, output location) are recorded in the header of each deposited output, in the deposit's `MANIFEST.csv` (one row per file, with which version of the deposit holds it) and `zenodo_manifest.md`, and in the SI tables cited for each analysis (SI-1, SI-6.10, SI-7, SI-11.18–SI-11.20). Anthropic and OpenAI system-card knowledge cutoffs and training-data cutoffs are recorded in Methods 6. No execution reported in the main text is missing from the manifest; the manifest is the authoritative record for the configuration-pinned SET-level rerun consistency claim made in Methods 6 and Results Part 4.

Model-use ledger (per-model role separation). Six LLMs produced the analyses reported here. Five have a row in the ledger below; the sixth, Claude Opus 4.6, was used only for the v17 Stage 2F runs (row 4 of the table above). Claude 3.5 Haiku appears only in the earlier v15 development comparison cited in SI-6 and Discussion 2, which is not among the analyses of the table. Their use is summarised here to make the GPT-5.4 vs GPT-5.5 separation explicit and to ensure every cross-model claim in the manuscript resolves to a specific model + role pair.

Model	Version	Role	Prompt version	Date	N	Output location	Claim role
Claude Sonnet	4.6 (2026-02-17)	Primary Stage 0–4 baseline	v17 (rows 1–3 above) and v18.4 (rows 5, 6, 8, 10–13; English or Japanese Stage 2 as in SI-7)	2026-04 (v17) + 2026-05 (v18.4, including the N=6 multi-run) + 2026-09 (R1)	1 (primary); 6 (multi-run)	`papers/{corpus}/{paper}/{paper}_v184_sonnet_s*.md` (v17 outputs for the 12 development and refinement papers; v18.4 outputs for the development set, the eight case-study papers, Tamao 2002 and the four context papers); SI-1, SI-3, SI-4	Internal robustness via N=6 multi-run aggregation; baseline for configuration-pinned SET-level rerun consistency
Claude Opus	4.7 (2026-04-16)	Stage 2F group integration (primary) + Stage 3 cross-model arm	Stage 2F v6 / v7 / v7.1 (v18.4 inputs) + v17 (Stage 3, row 2)	2026-04, 2026-05	1 per cell (4 groups × 3 versions = 12 Stage 2F outputs) + 1 Stage 3 cross-model	SI-5, SI-6, SI-7	SSP cross-paper integration; Stage 3 cross-model robustness arm
Claude Haiku	4.5 (2025-10-15)	Cross-tier cross-model arm (SI-6 four-model comparison)	v17 (cross-model run set); v18.4 and v17 replications (row 9)	2026-04; 2026-09	1 (2026-04); 2–5 per arm (row 9)	SI-6	Cross-tier robustness within Claude family (lower-tier comparison)
GPT-5.4 Thinking	(2026-03-05)	**SI-6 four-model cross-vendor comparison** (the main robustness check)	v17 (cross-model run set)	2026-04	1 per cell	SI-6	Cross-vendor robustness arm of the four-model comparison; **part of the main internal-robustness warrant**
GPT-5.5 Thinking	OpenAI o-series	**EP-10a assignment + Stage 4 cross-check only**	S0–S2 v18.2, Stage 3 v3, Stage 4 v3 (row 15)	2026-04-26/27	per-instance	SI-3.7 (EP-10a provenance); Stage 4 verification protocol (Methods §2)	**Not part of the main SI-6 four-model comparison**; ad-hoc provenance source for the EP-10a dictionary expansion (April 2026) and a cross-check parallel-retrieval source for high-prior-fabrication-risk Stage 4 candidates

GPT-5.4 vs GPT-5.5 reconciliation. The four-model robustness comparison reported in SI-6 uses Claude Haiku 4.5, Claude Sonnet 4.6, Claude Opus 4.7, and **GPT-5.4 Thinking** as the four models. GPT-5.5 Thinking is **not** part of that four-model set; it appears in two distinct ad-hoc roles only: (i) as one of the later runs that assigned EP-10a under a prompt that lists it (SI-6.5); and (ii) as a cross-model parallel-retrieval source for high-prior-fabrication-risk Stage 4 candidates in the Stage 4 verification protocol. All other "cross-vendor", "cross-model", or "four-model" claims in the manuscript refer specifically to the GPT-5.4-based SI-6 four-model set, not to GPT-5.5.

Corpus-count reconciliation. The corpus counts referenced across Methods 3 and 6 resolve as follows: **8 case-study papers** (seven filling papers and one review, Cherney 2015); **9 Stage 2F integration papers** (the 8 case-study papers + Kumada 1972, the shared anchor); the **11-paper v18.4 EN production-run baseline** (Kumada 1972 + Murai 1993 at N=6 multi-run, plus the 8 case-study papers + Tamao 2002 at N=1); the **12-paper prompt-refinement and regression corpus** (2 development + 10 refinement); and **21 deposited records** (12 refinement + 8 case-study + 1 retrospective Tamao 2002). Stage 2F runs cover four groups (G1 / G2 / G3 / all) × three prompt versions (v6 / v7 / v7.1). The 21-paper deposit adds per-paper records and mirrors, not a further evaluation corpus.

Rerun equivalence and model-access contingency. *Configuration-pinned SET-level rerun consistency* means recovery of the same qualifying output SET under the archived prompt, model, corpus, and criteria configuration. Text-identical wording and single-run rank order are not required and are not claimed. If exact model access becomes unavailable (model deprecation, API change, training-data refresh), the **deposited-record audit** (inspection of the per-stage outputs, evidence chains, retrieval logs, criteria documents, and manifest already in the deposit) remains reproducible as an *inspection task*, independent of rerun access; **rerun consistency** in that contingency becomes limited to available same-configuration reruns or nearest-available-model reruns, reported separately. The rerun-equivalence checklist below specifies the per-cell fields a third party needs to attempt SET-level rerun consistency.

Rerun-equivalence checklist field	Description
Input file identity	DOI of the source paper (corpus manifest, Tier 2); source PDFs are not deposited for copyright reasons and are obtained from the DOI
Prompt version	Verbatim filename in `tier_1_prompt_versions/` (SHA-256 in SI-7)
Model identifier + release + cutoff	Exact name and identifier (e.g., Claude Sonnet 4.6, `claude-sonnet-4-6`, released 2026-02-17); reliable-knowledge + training-data cutoffs from model card
Settings	Temperature / top_p / max tokens / system-prompt scaffold (where exposed by the API)
Run date	UTC date stamp; archived in `MANIFEST.csv`
Output SET expected	Top-3 reinforcement EP SET = {EP-03a, EP-04a, EP-06a} for Kumada 1972 (N=6 per language cell; 67–100% per EP across three extraction rules); per-paper values in SI-3
Allowed deviations	Text-identical wording; single-run rank order within Top-3 SET; Layer-1 single-shot retrieval differences resolved by cross-model aggregation
Failure criteria	Top-3 SET inclusion frequency drops below 50%; cross-vendor convergence on EP derivation breaks at the SRG / RI level; Stage 4 verification chain fails Crossref or content-match

SI-11.2: Hallucination patterns (A–E) and the seven-step defence

This section expands the hallucination-control summary in Methods 2. The seven steps below are safeguards across the whole pipeline; several apply to Stages 0–3 rather than to Stage 4 candidates. They are not a per-candidate procedure. How a paper enters the filling map, and which candidate-level checks were actually made, is stated in Methods 2 and SI-9.10. (The submitted version described a four-step chain applied to every Stage 4 candidate, and said that steps 2–4 below corresponded to it; neither was accurate.) UQS treats LLM hallucination not as a binary defect but as a graded phenomenon along a verification-cost spectrum (the "high-signal imperfect recall" framing developed in Discussion 3). The pipeline applies a seven-step defence against five identified hallucination patterns to convert raw LLM output into auditable findings. This empirical taxonomy and verification chain operationalise, for the chemistry-citation setting, an active 2023–2026 literature on citation-hallucination detection: consistency-based self-detection of fabricated references,[S3] failure-mode taxonomies of fabricated citations,[S4] and multi-source bibliographic verification against CrossRef / Semantic Scholar / OpenAlex.[S5]

Five hallucination patterns observed across vendors (Patterns A–E). A taxonomy of LLM citation-hallucination modes was developed empirically across four models from two vendors (Sonnet 4.6, Opus 4.7, Haiku 4.5, GPT-5.4 Thinking) on Kumada 1972 retrievals (SI-6, SI-9):

- *Pattern A — Author-name swap*: a real author's name is replaced with another real author's name from the same field, with high prior similarity (e.g., middle authors substituted in surrounding co-author networks). Detection: cross-model parallel retrieval; correction: DOI lookup against Crossref.
- *Pattern B — Author-name front/back omission*: a multi-author paper is reported with a truncated author list (e.g., "Tamao 1976" reported correctly, "Tamao, Sumitani, Kumada 1972" sometimes truncated to a subset). Detection: full author-list verification against Crossref.
- *Pattern C — Year/volume mismatch*: a real paper is reported with a slightly wrong year, volume, or page (typically off by one or two units). Detection: DOI lookup; correction: bibliographic shell verification.
- *Pattern D — Cross-paper hybrid*: features of two real papers by the same authors in the same year are merged into one fabricated reference (e.g., title from Paper X with journal/page from Paper Y). Detection: DOI lookup against Crossref combined with content match against the resolved DOI's actual content. The Yamamoto 1978 triple-mismatch was a Pattern D case (SI-9.3 (iv); SI-6.7).
- *Pattern E — Non-anglicised given-name omission*: Japanese and Chinese given names are omitted or anglicised inconsistently when retrieving cross-language author networks (e.g., "Sakabe Kiwako" reduced to "Sakabe K." or omitted entirely). Detection: explicit non-anglicised author preservation in prompts; corroboration: the v18.4 EN run preserved Japanese given names across nine source papers (Tamao Kohei, Hayashi Tamio, Kumada Makoto, Sumitani Koji, Kakiuchi Fumitoshi, Chatani Naoto).

These five patterns map onto the general failure-mode taxonomy of fabricated citations.[S4] Patterns A–C are chemistry-specific refinements of *partial attribute corruption* (real metadata blended or perturbed); full-citation non-resolution corresponds to *total fabrication*; and the cross-paper hybrid (Pattern D) additionally overlaps *identifier hijacking*. Pattern E (non-anglicised given-name omission) is a cross-language sub-mode specific to multilingual author networks that is not separately enumerated in the general-domain taxonomies, and is contributed here as a chemistry-corpus-derived addition relevant to any corpus spanning non-Latin-script author names.

Seven-step defence applied across the pipeline. The steps below are pipeline-wide safeguards, not a procedure applied to every candidate: the 468 candidates of the 37 Stage 4 analyses were not checked one by one, and the candidate-level checks that were made are tallied in SI-9.10. What controls the filling map is that every paper in it was obtained in full text and analysed, or retrieved in the locked Web of Science audit (Methods 2):

1. *Key-quote anchoring*: every absence carries an evidence_chain to specific textual loci (Methods 1.4 Principle 3); claims with no textual anchor are not valid output.
2. *DOI verification against Crossref*: a candidate citation that is checked is resolved against Crossref, and failure to resolve (full-citation fabrication) results in retraction; this was done for the candidates of the verification studies (SI-9.10), not for every Stage 4 candidate. Resolving candidates against authoritative bibliographic databases is the core mechanism of dedicated multi-source citation-verification tools.[S5]
3. *Author verification protocol*: full author lists, including non-anglicised given names, are verified against Crossref records; Patterns A, B, and E are detected at this step.
4. *Cross-vendor convergence*: retrieval of the same EP across two or more vendors (Sonnet, Opus, Haiku, GPT-5.4 Thinking) flags vendor-specific idiosyncrasy. This cross-model agreement check generalises, across vendors, the single-model self-consistency signal shown to discriminate hallucinated from genuine references.[S3] Convergence is the strongest internal evidence available in this audit package supporting retrieval-audit-grade documentary warrant, with the partial-independence caveat documented in Limitations L2.
5. *Multi-run aggregation at the SET level*: at fixed prompt and model, $N \geq 6$ runs convert single-run rank stochasticity into stable SET membership claims (Methods 1.4 Principle 4).
6. *Unanchored items are not reported*: an absence with no textual anchor (step 1) is not reported as a finding. A three-valued `anchor_status` field (`anchored` / `latent_reconstructable` / `unanchored`) is a planned extension; it is not part of the prompts used in this paper.
7. *Model tier for primary execution*: primary execution uses the Sonnet / Opus tier, with Haiku reserved for cost-sensitive secondary checks. (The submitted SI justified this by an inverse correlation between model size and hallucination rate "observed" in the four-model SI-6 comparison. SI-6 records no per-model hallucination rates, so that justification is withdrawn; the choice of primary model is explained in Methods 6.)

Layered detection hierarchy. The seven defences operate across three detection layers. Layer 1 (automatic verification) covers DOI lookup, key-quote anchoring and the exclusion of unanchored items, and catches Pattern C and most of Pattern D. Layer 2 (semi-automatic verification) covers cross-vendor convergence and multi-run SET stability, and surfaces Patterns A and B at vendor-specific failure rates. Layer 3 (human-required verification) covers author-list cross-checks for non-anglicised names, content match for Pattern D edge cases, and library-hardcopy verification for pre-DOI papers; Pattern E and the Yamamoto

1978 triple-mismatch case were resolved at this layer (SI-9.3 (iv)).

Residual error profile. The seven-step defence reduces but does not eliminate hallucination risk. Three residual classes remain: (a) vendor-shared bias, since all four models, from two vendors, share substantial training-corpus overlap; correctable jointly only by truly independent verification, registered for long-horizon methodological studies; (b) subtle misinterpretation of Stage 0 / Stage 1 content, mitigated by single-session context persistence but not eliminated; (c) multimodal limits on figure / scheme content, since Pattern D–style errors in claims that depend on visual evidence are harder to detect at Layer 1. The Limitations chapter (L2, L3, L6) makes these residual error classes explicit; they are foregrounded rather than concealed.

SI-11.3: Structured Web of Science retrieval audit and the two-layer case-study structure

This section expands two procedural items from Methods 3 (*Selection of Analyzed Papers*): the full structured Web of Science retrieval-audit procedure that stress-tests the EP-08a / EP-09a negative claims, and the two-layer composition of the historical case-study set.

Structured bibliographic retrieval audit. To stress-test the negative claims for EP-08a (replicate-measurement data) and EP-09a (substrate-matched head-to-head benchmark against named predecessor catalysts) beyond the LLM-parametric retrieval reported in Stage 4, a structured Web of Science Core Collection retrieval audit was performed in May 2026 across three Evidence Packs (EP-08a, EP-09a as negative claims; EP-07a substrate-scope extension as positive control). Filling criteria (Complete / Partial / Minimal / Non-filler) were pre-specified and locked before retrieval in an internal audit-criteria document. WoS query variants (four each for EP-08a and EP-09a, with two further syntax-relaxed formulations for EP-08a; two for EP-07a), designed to triangulate the LLM-parametric finding under different keyword combinations, returned 22 candidate papers across EP-08a + EP-09a (0 Complete + 0 Partial + 2 Minimal + 20 Not applicable per the pre-locked criteria) and 9 candidate papers for EP-07a (0 Complete + 2 Partial + 0 Minimal + 7 Not applicable). The 2 Partial fillers identified in the positive-control EP-07a (Breitenfeld 2013, Perez Garcia 2016, both Ni-pincer scope-extension papers) show that the audit pipeline can identify filling literature under the same query-and-classification protocol where it exists. They do not measure the recall of the EP-08a and EP-09a queries, so the 0-Partial result remains a non-detection in the WoS-indexed record under the pre-locked criteria, not a demonstrated absence (SI-9.8.2). The full per-EP candidate tables, classification rationales, query strings, and raw WoS exports are reported in SI-9.8 and archived under the corresponding `retrieval_audit/ep{NN}a_wos_llm/` directories listed in Data Availability Tier 5.

Two-layer structure of the historical case study set. The eight historical case-study papers comprise one Layer-1 paper (Tamao et al., *Bull. Chem. Soc. Jpn.* 1976 — the author group's Full Paper I, retained as the canonical Layer-1 auto-resolution anchor) plus seven Layer-2 papers (2009–2024; six paradigm-shift papers and one review, Cherney 2015) that extend Kumada 1972's evidence architecture across mechanism-paradigm boundaries (photoredox, radical Ni(I)/Ni(III), enantioselective, and electrochemical single-electron channels). The fuller *conventional self-filling layer* (1973–1982) — Tamao 1976 plus companion stereochemistry and kinetics papers (Tamao-Zembayashi-Kiso-Kumada 1973; Kiso-Tamao-Kumada 1973), related mechanistic work by the Kochi group (Morrell & Kochi 1975), and the heterocyclic-halide scope-extension paper (Tamao et al., *Tetrahedron* 1982) — was named by one GPT-5.4 Thinking Stage 4 run (web search enabled) as filling the same EPs from within the original two-electron paradigm; among these, only Tamao 1976 is included in the eight-paper case study set as the representative Layer-1 anchor. Inclusion of the 2009–2024 layer as the case study set reflects a deliberate choice to privilege the broader temporal slice of evidential complementarity, capturing both the author-group auto-resolution pattern (Tamao 1976 as the representative filler) and the later paradigm-shift fillers. The layer-1 papers (1973–1982) are listed in SI-4.9 as candidates from that run (record: `01_response/r33b_candidates/records/stage4_literature_update_20260419.md` in the deposit); they were not analysed as primary case study subjects because their filling pattern is structurally homogeneous (author group Full Papers and closely adjacent mechanism studies) and adds limited paradigm diversity beyond Tamao 1976. The interval of about 27 years between the two layers (1982–2009; "~30 years" in Results Part 2) is itself an analytically informative feature and is discussed in Results Part 2 and Discussion 1.

SI-11.4: In-flight 3-signal monitoring — numeric ranges

This section gives the per-signal numeric derivations summarised in Methods 5. Once execution begins, three operational signals indicate whether UQS is performing within calibrated range or has drifted into a domain where pipeline output should be treated cautiously.

Signal 1 — Stage 2 anchor pass rate. Stage 2 requires every D-4 candidate to carry an evidence_chain to specific textual loci. The fraction of Stage 1 candidates that pass the anchor requirement at Stage 2 measures how much of the paper's structure consists of anchorable claims rather than narrative speculation. A range of 40–70% was observed informally on the twelve-paper

refinement corpus; no deposited table records it, so the range is a heuristic rather than a calibration. Below 30% suggests the target paper's evidence architecture is too sparse for UQS's anchoring rule.

Signal 2 — Stage 4 cross-vendor convergence. When Stage 4 is run with two or more LLM vendors, the fraction of EPs whose filling-paper candidates converge across vendors measures whether retrieval is operating on shared structural signals rather than vendor-specific idiosyncrasy. The submitted SI gave a calibrated range of 60–85% on Kumada 1972 across the four-model comparison. We could not trace that range to any deposited record, and it is withdrawn; the signal is a heuristic without a calibrated range. Below 30% would indicate that the paper's filling pattern is not robustly encoded across training corpora.

Signal 3 — Multi-run categorical stability. When Stage 3 is run N $\geq$ 6 times at fixed prompt and model, the fraction of EPs whose SET membership stabilises across runs ($\geq$75% inclusion frequency) measures whether categorical extraction is stable or stochastic. Observed rate at the v18.4 baseline: for Kumada 1972, 67–100% per EP of the Top-3 SET {EP-03a, EP-04a, EP-06a} across three extraction rules; for Murai 1993, $\geq$83% in the English cell but 33–50% for EP-04a in the Japanese cell (SI-3). The $\geq$75% figure is a heuristic threshold, not a calibrated rate. Single-run rank order is stochastic but SET membership is stable; when even SET membership becomes unstable (below approximately 50% inclusion), the target is borderline.

SI-11.5: Worked audit trace (EP-08a)

Worked audit trace (EP-08a as exemplar of bounded non-detection inspection). To make the chain from source text to deposited audit material concrete, the following trace walks EP-08a from its textual locus in Kumada 1972 to its retrieval-audit decision and the deposited evidence files. The same trace pattern applies to every EP in the dictionary; an analogous positive-evidence trace for EP-03a (the strongest retrieval-audit positive case) is given in SI-9.

Trace step	Concrete content for EP-08a	Deposited file / SI location
1. **Source textual locus**	Kumada 1972 Table I reports product yields by GLPC for individual entries without explicit reporting of replicate measurements (no n=2/3 entries, no range/SD, no per-entry replicate column)	Kumada 1972 Table I (DOI 10.1021/ja00767a075 in the Tier 2 corpus manifest; the source PDF is not deposited, for copyright reasons)
2. **Stage 1 record**	Stage 1 records a B-category item, B-9 in the deposited final Stage 1 output, classified *reporting_omission* and anchored to Table I (all entries). Its note states that no replicate data or standard deviations are reported for any entry and that all yields are single determinations. It was raised from the methodological vantage (V3; convergence 1/3).	`papers/dev/kumada/kumada_v184_sonnet_s1.md` → B-9 (Tier 3)
3. **SRG / CLA group**	Stage 3 groups B-9 with B-2, B-7 and B-11 into SRG-5 *Yield data analytical quality*, one of six SRGs in the Kumada Stage 3 output. Its evidential requirement is replicated yields with reported spread, or enough analytical metadata to judge their reliability, and its EPs are EP-08a and EP-01a.	`papers/dev/kumada/kumada_v184_sonnet_s3v2_en.md` §2 → SRG-5 (Tier 3; the Japanese-cell output `kumada_v184_sonnet_s3v2.md` has the same SRG-5)
4. **EP definition**	EP-08a *Replication / uncertainty pack*. The dictionary entry (SI-3.7), reproduced verbatim in the pre-locked audit criteria (the (i)–(iii) wording is the SI's English recomposition of that record's SRG-6 implementation and §5 step 8), gives the case-specific implementation: (i) for all 11 Table I entries, n $\geq$ 3 replicates with standard deviations on yield; (ii) for the cis/trans-dichloroethylene substrates, n $\geq$ 3 replicates with glpc and ^{1}H NMR cross-method determination; (iii) a time-course (1 h, 5 h, 20 h). The criterion levels Complete / Partial / Minimal / Non-filler were locked before retrieval. The dictionary entry derives EP-08a from SRG-2, SRG-6 and SRG-3 of the SI-3.4 record, whose SRG numbering differs from the v18.4 run in step 3	`tier_3_stage_outputs/dev_set/kumada/kumada_v17_sonnet_s3.md` §2 SRG-2 / SRG-3 / SRG-6 and §5 step 8 (the Stage 3 v1 record, in Japanese: RI-08 assignment and replicate / time-course implementations) + the English dictionary text as quoted in `tier_6_audit_decisions/ep_audit_criteria_20260520.md` §4
5. **Stage 4 retrieval criteria**	WoS query variants (four each for EP-08a and EP-09a; two for EP-07a) designed to triangulate the EP-08a finding; EP-08a additionally tested under two syntax-relaxed formulations (hyphen-syntax artefact ruled out); EP-07a parallel positive-control audit under identical protocol	query strings in `tier_5_retrieval_logs/retrieval_audit_ep03a_07a_08a_09a/ep08a_wos_llm/wos_variant_{a–d}_analysis.md` (§Query) and in the corresponding EP-07a files (positive control)

Trace step	Concrete content for EP-08a	Deposited file / SI location
6. **Retrieval decision**	For this EP implementation, no retrieved academic-record candidate met the pre-locked Complete/Partial criteria; one Minimal-level contribution from the case-study set (Morken 2016, "average of two experiments"; outside the WoS candidate set, so SI-9.9 lists no Minimal candidate for EP-08a) is reported separately and does not constitute Complete/Partial filling under the locked criteria; this is a **bounded non-detection within the 1972–2026 published academic-record window under pre-locked filling criteria** (full warrant in Results Part 2)	per-candidate classifications in `tier_5_retrieval_logs/retrieval_audit_ep03a_07a_08a_09a/ep08a_wos_llm/wos_variant_*_analysis.md` and counts in `hit_counts.md`; SI-9.8 (full per-EP candidate tables). No separate retained/retracted log was kept
7. **Deposited audit material**	The reviewer can reproduce step 5 by running the deposited query strings against WoS Core Collection, compare the result with the deposited raw exports (`wos_variant_*.txt`, and the re-exports of 2026-10-05 in `wos_reexport_20261005/`; see `WOS_EXPORT_STATUS.md`), and verify per-candidate classifications against the pre-locked criteria document. The EP-07a positive-control parallel audit (2 Partial fillers identified under identical protocol) shows that the protocol can return Partial fillers; it does not measure the recall of the EP-08a / EP-09a queries (SI-9.8.2)	`tier_5_retrieval_logs/retrieval_audit_ep03a_07a_08a_09a/ep08a_wos_llm/` (query strings, classifications and counts in `wos_variant_*_analysis.md` and `hit_counts.md`; raw exports `wos_variant_*.txt`) + `tier_5_retrieval_logs/retrieval_audit_ep03a_07a_08a_09a/ep07a_wos_llm/` parallel; SI-9.8

R1 correction. Steps 2 and 3 previously named a Stage 1 item "B-08" and an "SRG-08 *Reporting-granularity hygiene*", and gave an inference tier. None of these exists in the deposited outputs, and Stage 1 B items carry no tier field. Step 4 also quoted an implementation ("replicate (n≥2 per entry) GLPC or HPLC yields …") that appears in neither the Stage 3 output nor the dictionary. All three steps are corrected above to what the deposited files actually contain.

This trace operationalises the audit map row for EP-08a / EP-09a: each manuscript claim has a deposited chain of evidence linking source text to retrieval decision to deposit file, and the reviewer's audit action is to walk the chain in reverse to verify each link.

SI-11.6: Frozen-criteria and dictionary-expansion chronology

To make the temporal sequence of EP derivation, prompt versions, EP dictionary expansion, criteria lock, retrieval audit, manuscript integration, and deposit freeze explicit, the chronology below maps each event to its date and to the deposited artefact in which it is anchored. EP-02a and EP-10a are flagged as **dictionary expansions under the documented cross-model expansion rule**, not part of the originally frozen single-run Stage 3 output for Kumada 1972.

Event	Date	Deposited artefact
Kumada Stage 3 EP derivation (initial single-run Sonnet 4.6 dictionary EP-01a / EP-03a … EP-09a; 8 EPs; v17 Stage 0–2 + Stage 3 v1, prompt tag `stage3-v1-20260402`)	2026-04-15 (run)	`tier_3_stage_outputs/dev_set/kumada/kumada_v17_sonnet_s3.md` (Tier 3)
Prompt versions of the v18.4 configuration (tag dates: Stage 3 v2 2026-04-16; Stage 2F v7.1 file 2026-04-17, tag `stage2f-v7-20260417`; Stage 3 v2 EN and Stage 4 v2 EN 2026-04-19; Stage 0 / 1 / 2 v18.4 2026-05-02)	2026-04-16 to 2026-05-02	`tier_1_prompt_versions/`; SI-1 (version history), SI-7 (hashes)
EP-02a dictionary expansion (Identity / purity substantiation pack) under the **cross-model expansion rule** — assigned by GPT-5.4 Thinking + Opus 4.7 Stage 3 runs on Kumada 1972, from a label already in their prompt (SI-6.5)	2026-04-19	SI-3.7 (provenance) + SI-6.5 (cross-model record)

Event	Date	Deposited artefact
EP-10a dictionary expansion (Kinetic verification pack) under the same expansion rule — assigned by GPT-5.4 and Opus 4.7 on Kumada 1972, while Sonnet runs on three filling papers applied the related PRI-11 pattern; the label was in the Stage 3 prompt from version 1, so these were not independent (SI-6.5)	2026-04-19	SI-3.7 (full provenance)
EP-10a applied after the expansion — two GPT-5.5 Thinking runs (S0–S2 v18.2, Stage 3 v3) on Kumada 1972 and Murai 1993 also assigned it; they postdate the expansion and are not part of its basis	2026-04-26/27	SI-3.7
EP audit criteria lock (pre-locked Complete / Partial / Minimal / Non-filler criteria per EP)	2026-05-20	`tier_6_audit_decisions/ep_audit_criteria_20260520.md` (Tier 6)
WoS structured retrieval audit (EP-07a positive control + EP-08a / EP-09a negative claims; four Variants per EP, EP-08a + two syntax-relaxed)	2026-05	`tier_5_retrieval_logs/retrieval_audit_ep03a_07a_08a_09a/ep{07a,08a,09a}_wos_llm/`; SI-9.8
Manuscript integration (cumulative revisions; auditability inventory + worked trace + 10-min walkthrough + model-use ledger)	2026-05-24	the submitted manuscript; working records of this step are kept in the project repository and are not part of the deposit
Zenodo deposit freeze (submitted version shared with the editor through a preview link of an unpublished record; the revised deposit published with DOI 10.5281/zenodo.23437448, files restricted until acceptance)	submitted version packaged 2026-06-29; revised version R1 packaged 2026-09-28, final R1 version 2026-10-17	deposit `README.md`, `REVISION_R1.md`

Dictionary-expansion rule (verbatim). EP-02a and EP-10a are dictionary expansions under the documented cross-model expansion rule, not part of the originally frozen single-run Stage 3 output. The rule, locked at the v18.4 prompt-freeze date above, specifies that an EP may be added to a per-paper dictionary if it is independently surfaced by $\geq$2 vendors (or by $\geq$1 vendor + $\geq$1 cross-paper run) at Stage 3, and the addition is documented with provenance (source models, source papers, dates) in SI-3.7. EP-02a and EP-10a appeared to meet this rule, but the runs that met it had both labels in their prompts, so the rule's "independently surfaced" condition was not in fact satisfied (SI-6.5; corrected in R1). The rule itself cannot be shown to predate the additions: the v18.4 prompts are tagged 2026-05-02, after the 2026-04-19 additions; no other EPs were added post the initial Sonnet 4.6 Stage 3 run. All EP-08a / EP-09a / EP-07a status claims pertain to the eight originally-derived EPs + EP-02a/EP-10a after expansion was complete, under the criteria locked on 2026-05-20 (before the WoS audit was performed in 2026-05).

SI-11.7: Ten-minute reviewer audit walkthrough

The following seven-step path lets a reviewer inspect the central auditability claim end-to-end. All paths are relative to the root of the Zenodo deposit (version R1); the deposit's `README.md` §0 gives the same path with the re-count commands, and `RUNNING.md` describes how the analyses were executed. (The walkthrough in the submitted version named several files that the deposit did not contain; it is rewritten here against the deposit as revised.)

10-minute reviewer audit path:

1. **Check integrity.** Run `shasum -a 256 -c CHECKSUMS.sha256` at the deposit root; `MANIFEST.csv` lists every file with its SHA-256, size and top-level folder.
2. **Open the Stage 1 output** `tier_3_stage_outputs/dev_set/kumada/kumada_v184_sonnet_s1.md` and inspect the worked-example record U-1 of Methods 1.1 (category U, location and quotation; U items carry no inference-tier field). Tier 3 carries the verbatim key-quote excerpts (reviewer access); the redacted public version is under `tier_8_redacted_public_raw_outputs/`.
3. **Open the Stage 3 record** `tier_3_stage_outputs/dev_set/kumada/kumada_v17_sonnet_s3.md` (§2, §5) for the SRG→RI→EP assignment of EP-08a, and `tier_6_audit_decisions/ep_audit_criteria_20260520.md` §4 for its English dictionary text and the Complete / Partial / Minimal / Non-filler criteria; `kumada_v184_sonnet_s3v2_en.md` §2 SRG-5 shows the same absence in the v18.4 run.
4. **Open the locked criteria** `tier_6_audit_decisions/ep_audit_criteria_20260520.md` and confirm that they were fixed before retrieval.
5. **Open the EP-08a Web of Science audit** `tier_5_retrieval_logs/retrieval_audit_ep03a_07a_08a_09a/ep08a_wos_llm/`: each `wos_variant_{a–d}_analysis.md` gives the query string and the per-candidate classification under the locked criteria, `hit_counts.md` the counts, and `wos_variant_*.txt` the raw exports (facts

only). `WOS_EXPORT_STATUS.md` in the parent folder records which raw exports are being re-run.

6. **Open the EP-07a positive control** `…/ep07a_wos_llm/wos_variant_a_analysis.md` and confirm that the same procedure returned two Partial fillers.
7. **Open one Stage 2F file** `tier_3_stage_outputs/stage2f_integration_outputs/kumada1972_all_v17_opus_s2fv71.md` (the v17 Opus 4.6 integration from which the SSPs are taken; Results Part 3) and trace an SSP (e.g., SSP-1, MgX_2 effect) from the Results Part 3 table back to its source-paper anchors.

Optional: re-derive the multi-run numbers of SI-3 with the three scripts in `tier_4_classification_tables/reaggregation_R1/` (Python 3 only), which read the Tier 3 outputs and should reproduce the deposited CSV files byte for byte.

SI-11.8: Data-review artefact checklist

The checklist lists what the deposit provides for verifying its integrity (rather than its content). (The version submitted specified a manifest schema, a public / reviewer-only partition and per-PDF checksum files that the deposit did not provide; it is replaced here by what the revised deposit contains.)

Field	What the deposit provides
File manifest	`MANIFEST.csv` at the root: `path`, `sha256`, `size_bytes`, `top_level`, one row per file
Checksums	`CHECKSUMS.sha256` at the root; verify with `shasum -a 256 -c CHECKSUMS.sha256`
File counts by tier	read from `MANIFEST.csv` (`top_level` column)
Model and run provenance	`zenodo_manifest.md` §1 (model identifiers as recorded in the outputs, roles), §2 (prompt × model × stage), §3 (execution-date ranges); the `model_info` line of each output; `papers/_audits/paperA_quote_gate_R1_2026-09/model_identity_audit.log`
Public vs reviewer-only material	Tier 3 holds the full outputs with verbatim key-quote excerpts (reviewer access); Tier 8 holds the same outputs with quotations replaced by location references (public version). Source-paper PDFs, OCR full texts, page scans and Web of Science abstract / keyword / funding fields are not deposited
Source papers	identified by DOI in `tier_2_input_metadata/paperA_21_paper_corpus_manifest.csv` (Crossref-verified); obtained by the reader from the DOI
Access	private reviewer link during review and public release as stated in the Data Availability section
Reviewer audit action	the walkthrough in SI-11.7; `README.md` §0; `RUNNING.md`

SI-11.9: Full categorical vocabulary

This section gives the complete categorical vocabulary of UQS with rationale; the compact one-line definitions in Methods 1.2 are sufficient to read the Results and Discussion.

The vocabulary defined here is used by every stage of the pipeline and recurs throughout the rest of the paper.

A/B/C/U — absence categories. The four categories reflect structurally distinct sources of absence: **A** (*Deferred*) absences are absences the authors themselves recognise and signal as "for future work"; **B** (*Traces of Rejection*) absences mark scope boundaries that the authors actively declined; **C** (*Uncontinued Citations*) absences are works cited but not substantively built upon; **U** (*Unclassified*) is the residual category — absences that resist A/B/C placement, often the structurally most informative kind because their resistance to standard categorisation signals an unusual evidential pattern (the U-1 worked example above is one such case).

T1/T2/T3 — inference tiers. Three tiers (rather than a binary explicit / implicit judgment) capture the *inference distance* between the text and the identified absence: **T1** (textual direct statement) — the paper itself signals the deferral or scope boundary; **T2** (one-step inference) — the absence is reachable by a single inference step from the text; **T3** (external knowledge) — the absence requires field-norm knowledge beyond the text. The three-tier structure makes the inference distance explicit and supports differential weighting in downstream audits.

evidence_chain. A required citation list pointing to the specific textual loci that anchor the absence. The evidence_chain converts absence-claims into auditable identifications by forcing every absence to point to *what is* in the text (the loci that signal or imply the absence), rather than to an idealised notion of what the paper should contain.

V1/V2/V3 — disciplinary vantage points. Stage 1 extracts absences sequentially from three perspectives: **V1** (primary field) captures absences a primary-field reviewer would identify; **V2** (adjacent field) captures absences visible from a related field's expectations (a chemistry paper might be silent on a question that pharmaceutical chemistry would press); **V3** (methodological perspective) captures absences a methodologically attentive reader of any field would notice (replicate measurement, calibration disclosure, benchmarking comparison). The three perspectives are structural rather than ornamental — they detect different absences. The *cross_field_temperature* metric records the convergence and divergence of severity assessments across V1/V2/V3 as a structural fingerprint.

D-1 / D-2 / D-3 / D-4 — motivational layers. Stage 2 unpacks each absence at four motivational layers: **D-1** (declared motivation) — the paper's stated reasons; **D-2** (inferred motivation) — reasons inferable from the text but not stated; **D-3** (institutional motivation) — reasons traceable to disciplinary or publication-format conventions; **D-4** (unquestioned motivation) — assumptions the paper does not surface as choices.

SRG / RI / EP / VQ — Stage 3 vocabulary. Stage 3 produces a Root-Issue Codebook (**SRG**: Same-Root Group, the structural classification of why each absence matters), identifies Root Issues (**RI**) underlying each SRG, and constructs the **Evidence Pack (EP)** dictionary. EPs are conceptually distinct from absences: absences are *what is missing*, while EPs are *what would be required to add*. Each EP carries a *case-specific implementation* describing what concrete experiment or analysis would constitute filling the EP. The **VQ matrix** is a four-level hierarchy of validation status: **VQ-1** (identity/purity: the product or catalyst is identified, and its purity established, at a granularity that excludes alternative candidates — the RI-02 requirement; the Stage 3 prompt names this level "Observation"), **VQ-2** (methodology: the experimental procedure is reproducible and the conditions are disclosed), **VQ-3** (mechanism: the mechanistic claim is supported by direct evidence rather than analogy), **VQ-4** (quantitative reliability: the claim is supported by replicate measurement and substrate-matched benchmarking).

Era-Norm Annotation. Every VQ judgment is annotated with two values: a *modern reassessment baseline* (the VQ judgment that contemporary evidential standards would render) and an *era-norm-adjusted* value (the VQ judgment in the context of the paper's publication period). The dual annotation preserves both readings, so that a 1972 Communication is documented as not meeting 2024 standards *and* as consistent with 1972 *J. Am. Chem. Soc.* Communication-format conventions, without retroactively penalising the source paper.

Stage 2F vocabulary. Stage 2F integrates a corpus of independent per-paper outputs and produces *recurring_roots* (absence patterns that appear independently across multiple papers), *independent occurrence patterns* (**IPs**), and *subgroup-specific propositions* (**SSPs**) at single-research granularity. SSPs are the primary research-question-brief deliverable of Stage 2F (see Results Part 3 for the seven SSPs surfaced in the present case study).

Evidential requirement and resolution status. In this study, an *evidential requirement* is defined as a form of evidence that would strengthen, delimit, or disambiguate a claim made in the target paper. It should not be equated, by itself, with a presently open research problem. Once an evidential requirement is extracted, its resolution status is assessed separately: it may be satisfied within the target paper, resolved through the paper's cited-reference network, distributed across the target paper and its cited literature, forward-filled by later publications, partially filled, or remain unfilled. This distinction separates the identification of a claim-implied evidential need from the subsequent judgment of where, whether, and how that need has been resolved. For terminological consistency, we reserve *forward-filled* specifically for post-publication resolution. Resolution already available within the target paper or its cited-reference network is treated separately as *target-paper*, *citation-network*, or *distributed paper–citation* resolution.

Reuse-task parameterisation. Each absence claim is bounded by the later reuse task relative to which non-explicitness is judged. In the present chemistry calibration, the recurring reuse tasks are mechanism reconstruction, computation, benchmarking, and data-driven synthesis. In the present implementation no task is declared per run: absences are recorded relative to this fixed set of later uses (Stage 0 records the original communication task), and each absence record is valid only relative to that set and its case-specific implementation. Declaring a specific reuse task per run is a possible extension, not part of the present pipeline. Alternative reuse tasks may yield different absence maps; UQS treats this as a scope condition that is part of the operational specification rather than as instability of the methodology.

SI-11.10: Reviewer navigation guide

The central questions a reviewer may bring to this paper are mapped to their answer locations below.

Reviewer question	Manuscript answer	Evidence location
What is the method?	A staged UQS workflow: positive record, structured absence extraction, Evidence Packs, filling map, and SSP question briefs	Fig. 1; Methods §1–§2
Why Kumada 1972?	A compact, foundational, heavily followed communication providing a stringent deep-case demonstration object, used here as a documentary anchor (not as the historical origin of organonickel chemistry)	Results Parts 1–2
What does the 12-paper corpus show?	Prompt refinement, regression control, and stress testing of the operational form — not broad external validation	Results Part 4; Methods §3
What do model comparisons show?	Internal robustness and consistency-level evidence across model families and runs, not ground-truth validation	Results Part 4; Discussion 2; SI-6
How are non-detections bounded?	By the 1972–2026 published-academic-record retrieval window and pre-specified Complete / Partial / Minimal / Non-filler criteria	Results Part 2; Methods §3; Limitations L1
What is the deliverable?	SSP briefs as research-question outputs requiring downstream domain-expert prioritisation and experimental design, not optimised plans or claims of chemical discovery	Results Part 3; Discussion 3; Limitations L8

SI-11.11: Deposit tier organisation (Tiers 1–9)

The Zenodo deposit is organised in a nine-tier audit-trail structure, from the most reusable methodology specification (Tier 1) to full raw outputs (Tier 9). The full tier-by-tier inventory below is mirrored in the deposit's top-level `README.md`.

Tier 1 — Prompt versions (methodology specification). The full text of the UQS prompt versions v15 through v18.4, the v19 prompts of the reproducibility control (SI-7 rows 24–29), and the Stage 3 / Stage 4 versions listed in SI-7 (earlier exploratory versions are not deposited), with v15, v16, v16.1, and v17 corresponding to the quantitative refinement cycle reported in Results Part 4, and v18.4 constituting the production-run configuration (`prompt_stage0_v184.md`, `prompt_stage1_turn{1–4}_v184.md`, `prompt_stage2_v184_en.md`, `prompt_stage3_v2_en.md`, `prompt_stage4_v2_en.md`, `prompt_stage2f_v71.md`).

Tier 2 — Input metadata. `paperA_21_paper_corpus_manifest.csv` and a README. The manifest lists the 21 papers analysed for Paper A with role, DOI, Crossref title, reference number, the date the Crossref record was checked, and the path of each paper's records in the deposit: the two development-set papers (Kumada 1972, Murai 1993), the ten refinement-set papers in Heck chemistry, the eight historical case-study papers that fill the Kumada 1972 Evidence Packs (Tamao 1976 plus the seven 2009–2024 paradigm-shift papers), and the Tamao 2002 retrospective mini-review analysed as the retrospective comparator for the Results Part 2 mini-review correspondence (the nine Stage 2F integration papers are Kumada 1972, the shared anchor, plus the eight historical case-study papers). Retrieval dates and PDF sources are not tabulated, and the source PDFs are not deposited; execution dates are in the header of each deposited output (Tier 3). The 1973–1982 Layer-1 candidates are not analysed papers; their bibliographic records are in the SI-4.9 table and in `01_response/r33b_candidates/records/stage4_literature_update_20260419.md`.

Tier 3 — Stage-wise output summaries. Complete Stage 0–Stage 4 analysis records for all twenty-one corpus papers identified above, together with Stage 2F integrative-analysis outputs (subgroups G1, G2, G3, and the all-papers integration) produced under prompt versions s2f (v6), s2fv7, and s2fv71. The twenty-one-paper count comprises 2 development-set papers (Kumada 1972, Murai 1993) + 10 refinement-set papers in Heck chemistry + 8 historical case-study papers that fill the Kumada 1972 Evidence Packs (Tamao 1976 plus the seven 2009–2024 paradigm-shift papers) + 1 Tamao 2002 retrospective mini-review (analysed as the retrospective comparator in Results Part 2; not part of the nine Stage 2F integration papers, whose ninth member is the Kumada 1972 anchor).

Tier 4 — Classification tables. EP audit table (10 EPs × per-paper filling-status matrix), the figure source data and rendered figures, the R1 re-aggregation (`reaggregation_R1/`) and the SI-9.10 candidate tally (`stage4_candidates_R1/`). The SSP briefs (7 SSPs with source-paper anchors), the Stage 1 absence classification per paper (A / B / C / U) and the Stage 3 SRG / CLA records are not separate tables: they are the Stage 2F, Stage 1 and Stage 3 outputs in Tier 3, and the tier's README points to each. The tier also covers the four-model cross-comparison (SI-6: Claude Haiku 4.5, Sonnet 4.6, Opus 4.7, GPT-5.4 Thinking on Kumada 1972) including the runs that assigned EP-10a (a label supplied by the Stage 3 prompt; SI-6.5), and the N=6 multi-run aggregation dataset on the development set under v18.4 Sonnet 4.6 EN baseline (six independent Stage 3 and six independent Stage 4 outputs per paper, for both Kumada 1972 and Murai 1993). The Top-3 reinforcement EP SET values (SI-3,

per paper and language cell, three extraction rules) and the Stage 4 re-count of the VQ-hierarchy inversion (SI-3: 12/24 runs at the Complete-or-Partial threshold, 2–16/24 across four thresholds) are derived from this dataset.

Tier 5 — Retrieval logs (Stage 4 + structured bibliographic audit). Stage 4 filling-paper retrieval logs (date, model, candidate paper list, per-EP decisions) for all twelve corpus papers. Structured bibliographic retrieval audit artefacts for the EP-03a pilot reported in SI-9 — five Web-of-Science query variants with raw tagged-format exports, per-candidate direct-lookup records (Tier 1 DOI and Tier 2 triangulation), the consolidated LLM-candidate list, the diagnostic report, and the query-anomaly log. **The primary warrant for EP-07a, EP-08a, and EP-09a is the structured Web of Science Core Collection audit** (four query variants each for EP-08a and EP-09a; EP-08a additionally tested under two syntax-relaxed formulations; two variants for the EP-07a positive control), archived at `tier_5_retrieval_logs/retrieval_audit_ep03a_07a_08a_09a/ep{03a,07a,08a,09a}_wos_llm/` with per-candidate filling-status classification under pre-locked criteria (Complete / Partial / Minimal / Non-filler). Any Scopus, Google Scholar, or Crossref records that appear under these folder paths are supplementary artefacts only and are not part of the main non-detection warrant; they were produced during exploratory comparison and are retained here for completeness, not as evidence supporting the EP-08a / EP-09a status claims.

Tier 6 — Audit decisions and Internal Mini-Audit. Per-EP audit decisions traceable to the criteria locked pre-retrieval in `tier_6_audit_decisions/ep_audit_criteria_20260520.md` (the rationale and audit-target justification for EP-03a, EP-07a, EP-08a, EP-09a). The Internal Mini-Audit Protocol v0.1 — a 12-item sampled evaluation across Stage 1 / 3 / 4 / 2F outputs, with auditor ratings on four axes (source-anchor validity, inference-level appropriateness, classification reasonableness, and chemical actionability for SSPs) and per-item retention / hedge / move-to-SI / removal decisions — is included in the deposit as a non-dependency external-assessment pathway. The protocol design and the 12-item set are released at submission; if the Mini-Audit is executed during revision or post-publication evaluation, the resulting auditor forms and summary table will be added to this tier (private reviewer access for full forms, public summary table). *This planned Mini-Audit is not relied upon as evidence for any claim in the present manuscript.*

Tier 7 — Aggregation scripts and methodology-evolution record. Python scripts for figure / table rendering (Figs. 2–5; the script of the withdrawn SI Fig. S1 is kept for the record); per-version regression log (`improvements/regression_log.md`); per-version changelogs for v17 through v18.4 (`improvements/changelog_v17.md` through `changelog_v184.md`); session-handover memos for v15→v16 (`session_handover_v15_to_v16.md`, `session_handover_v16.1.md`) documenting the regression-controlled expansion narrative summarised in Results Part 4; the v17 twelve-paper aggregate metrics CSV (`v17_twelve_paper_metrics.csv`) auditing the per-paper U / B / D-4 counts (39 U-items, 232 B-items, 94 D-4 candidates across the twelve papers) together with its extraction script (`extract_v17_metrics.sh`); the cross-model verification protocol formalised at v18.3 (`improvements/cross_model_verification_protocol.md`); and the n = 2 cross-model meta-observation (`improvements/cross_model_kumada_vs_murai_meta.md`).

Tier 8 — Redacted public raw outputs. Redacted versions of all Stage 0 / 1 / 2 / 3 / 4 / 2F outputs across the twenty-one-paper corpus, with source-paper textual quotations replaced by location references. Model identifiers, release dates, reliable-knowledge cutoffs, training-data cutoffs, and per-run execution dates are recorded in a manifest file. The models used are: Claude Sonnet 4.6 (2026-02-17 release; reliable-knowledge cutoff August 2025, training-data cutoff January 2026) as the primary model; Claude Opus 4.7 (2026-04-16 release; reliable and training January 2026) for Stage 2F integrative analysis under the v18.4 production configuration (12-output dataset: subgroups G1, G2, G3, and all-papers integration × prompt versions s2f / v6, s2fv7, s2fv71); Claude Haiku 4.5 (2025-10-15 release; reliable February 2025, training July 2025) for the SI-6 four-model cross-comparison on Kumada 1972; and GPT-5.4 Thinking (OpenAI; 2026-03-05 release; knowledge cutoff August 2025 per OpenAI documentation) for the cross-vendor robustness-comparison arm of SI-6. Claude Opus 4.6 (2026-02-05 release; reliable May 2025, training August 2025) ran the v17 Stage 2F integration, from which the seven SSP briefs and Fig. 4 are taken (Results Part 3); its outputs are also cited in the SI-6 historical comparison.

Tier 9 — Full raw LLM execution logs (reviewer-only access). Full raw outputs including source-paper quotations are accessible to peer reviewers and editors via private repository link at submission. Public release of the full raw outputs is conditional on legal review of source-paper-text-fragment redaction; the corresponding author maintains a release schedule for outputs where full publication is legally safe.

SI-11.12: Formal output-object schema and whole-task specification

This subsection gives the formal schemas summarised in Methods 1.1. They restate, in tabular form, the per-stage output objects and the whole-task input/output/constraint/evaluation specification; the main-text workflow-and-claim-status table (Methods 1.1) and the claim-status summary suffice to read the paper, and these tables are provided here for implementers and auditors.

Formal output-object schema. Each stage of UQS produces a structured object with required fields, a defined producer/consumer relation, and a corresponding deposited audit material.

Object	Required fields	Produced by	Used by	Audit material
Stage 1 record	claim/evidence relation, absence class (A/B/C/U), textual locus and quotation (evidence chain), inference tier (T1/T2/T3) for A items	Stage 1 (over V1/V2/V3)	Stage 2 + Stage 3 + Stage 2F	Stage 1 output file per paper (deposited, Tier 3)
EP record	EP id (EP-NNa), evidence needed (case-specific implementation), source anchor (per-paper Stage 1 records); filling status is not part of the record — it is assigned later, by the exploratory Stage 4 labels or, for the audited EPs, against the locked Complete/Partial/Minimal/Non-filler criteria (SI-9.8)	Stage 3	Stage 4	EP dictionary per paper (deposited, Tier 4)
Filling record	candidate paper (citation + DOI), criteria status (Complete/Partial/Minimal/Non-filler under locked criteria), rationale, retrieval source (LLM-parametric / WoS structured / Crossref-verified)	Stage 4	Results Part 2 + Results Part 4	Retrieval log per EP per paper (deposited, Tier 5; `retrieval_audit/`)
SSP brief	recurring pattern (cross-paper anchor), source anchors (per-paper records), question formulation (research-question brief), boundary (illustrative operationalisation, not validated experiment)	Stage 2F	Results Part 3 + Discussion	SSP briefs in the Stage 2F outputs (Tier 3, `tier_3_stage_outputs/stage2f_integration_outputs/`; the seven SSPs of Table 5 are in `kumada1972_all_v17_opus_s2fv71.md`)

Two parallel filling-status category sets. Stage 4 filling status is recorded under two distinct category sets, and the main text marks which one controls each statement. The *locked WoS-audit set* is the controlling classification for every headline non-detection claim; the *historical Stage 4 set* provides exploratory, LLM-parametric descriptive labels (Table 4 entries and Fig. 2 cell colours) that are not used as warrants. Where the two diverge — the two EP-09a *P** cells (Tamao 1976; Monnereau 2009) — the controlling locked-audit classification places them below Partial (SI-9.8).

Category set	Levels	Role in this paper	Provenance
Locked WoS audit (controlling)	Complete / Partial / Minimal / Non-filler	Warrants the EP-08a/EP-09a non-detection claims (EP-07a as positive control) and the classification of the candidates it retrieved; the filling order of the other EPs rests on the historical Stage 4 set	Criteria locked before retrieval; structured Web of Science Core Collection audit (SI-9.8)
Historical Stage 4 (exploratory)	Complete / Partial / Indirect / Attempted / Paradigm-shift (+ Minimal, from the Stage 2F integration)	Descriptive labels only (Table 4, Fig. 2 colours); never a non-detection warrant	LLM-parametric Stage 4 recall + Stage 2F v7.1 integration

Formal UQS task specification. The schema above describes per-stage output objects. The table below specifies the task as a whole — what UQS takes as input, what it produces, what constraints govern the production, what unit it is evaluated against, and what it is *not* evaluated against in the present paper.

UQS task element	Specification
Input	Full text of one or more scientific papers; absences are judged relative to a fixed set of later uses (mechanism reconstruction, computation, benchmarking, data-driven synthesis), not a task declared per run

UQS task element	Specification
Output	Stage 1 evidence-chain records; EP dictionary; bounded filling map under pre-locked criteria; SSP research-question briefs
Constraints	Evidence-chain anchoring to specific textual loci; inference-tier labels (T1/T2/T3) for A items; pre-locked filling criteria (Complete / Partial / Minimal / Non-filler); model/run manifest with prompt version + model identifier + execution date + N
Evaluation unit	Inspectability of records; SET-level rerun consistency under archived configuration; internal robustness across cross-model and multi-run independence checks; retrieval-audit consistency under pre-locked criteria
Not evaluated here	Expert consensus at scale (only a ten-item blind evaluation, SI-8.7, and a one-rater SSP pilot, SI-8.6), precision/recall against ground truth, SSP usefulness as validated experimental plans, experimental priority, head-to-head performance against other systems (only single-prompt baselines, SI-11.20)

SI-11.13: Applicability-boundary calibration-depth gradient

UQS's portability across disciplines is a gradient rather than a binary (Methods 5). Universal components — the triadic interaction of text, field knowledge, and protocol; the anchor principle; the multi-run aggregation governance mechanism; and the falsifiability requirement — transfer across disciplines without modification. Calibration-dependent components — the discipline-specific vocabulary of validation, the period-vocabulary check, the format of the Evidence Pack, and the operational meaning of *filling* — require domain-specific recalibration. Some components (the A/B/C/U taxonomy, the VQ-1 to VQ-4 hierarchy, the conception of *evidence architecture* itself) may require replacement for paradigms that do not generate evidence in the senses these slots presuppose. The expected recalibration effort by domain is:

Domain	Calibration depth	Expected effort
Synthetic organic chemistry	Calibrated (this paper)	—
Adjacent chemistry (organometallic, catalysis subfields)	Minor recalibration	Vocabulary adjustment; n ≈ 5 papers
Biology, materials science	Minor recalibration	n = 1 pilot (biology, Rothlisberger 2008); the cross-vendor anchor-pass value of ~75% given in the submitted SI could not be traced to a record and is withdrawn
Computational chemistry, physical chemistry	Moderate recalibration	Layer-bridging definitions; addressed as an application direction beyond the present paper
Experimental physics, machine learning, empirical CS	Moderate recalibration	Evidence-architecture redefinition required
Theoretical physics, quantitative social science	Substantial redesign	Anchor format and validation hierarchy not directly transferable
Pure mathematics	Major redesign	Validation-by-experiment slot does not apply; replacement required
Humanities	Foundational redesign	Paradigm-level recalibration; collaboration with field methodologists

Cross-domain extension is the subject of subsequent methodological work and is out of scope for the present paper, which reports the strike-zone calibration on a representative chemistry case study; whether domains that need only minor recalibration are tractable awaits systematic study.

SI-11.14: Applicability operating envelope — pre-application screen and in-flight monitoring

This section gives the full operating envelope summarised in Methods 5.

Pre-application 5-check. Whether a paper falls within UQS's effective operating envelope depends on five document-and-domain attributes assessable before pipeline execution. A target passes when at least four of five checks return green; one yellow is admissible, two or more shift the target to borderline status warranting in-flight monitoring; a red on any check signals that UQS is unlikely to produce useful output.

Check	Operational meaning	Green	Yellow	Red
1. Document type	Structured evidence architecture	Full Article in mainstream journal	Communication, Account, Perspective	Review, opinion, textbook, Wikipedia
2. Anchor density	Claims supported by data, methods, citations	Multiple anchors per page	Sparse (one to two per page)	Narrative-only, no anchor structure
3. Domain calibration	Empirical calibration on the field	Synthetic / organometallic chemistry	Biology (single-paper pilot, n = 1), materials, computational chemistry, physical chemistry	Pure mathematics, theoretical proofs, humanities
4. LLM corpus density	Frontier-LLM training-corpus exposure	Major English-language journals	Regional journals, recent niche subfields	Patents, trade-secret records, unpublished theses
5. Citation lineage	Density of follow-up literature for Stage 4 retrieval	Active field with rich descendant lineage	Mature but slow-moving	Standalone paper, no descendant lineage

Red on Check 1 (e.g., a review article) returns absence-categorical artefacts that conflate genre conventions with structural absence; red on Check 3 (e.g., pure mathematics) requires methodology redesign rather than direct application; red on Check 4 (e.g., proprietary literature) blocks Stage 4 filling-paper retrieval.

In-flight 3-signal monitoring. Once execution begins, three operational signals indicate whether UQS is performing within calibrated range: (1) *Stage 2 anchor pass rate* — fraction of Stage 1 candidates carrying a valid evidence_chain at Stage 2 (heuristic range 40–70% observed informally on the twelve-paper refinement corpus, not a deposited calibration; below 30% signals an evidence architecture too sparse for the anchoring rule); (2) *Stage 4 cross-vendor convergence* — fraction of EPs whose filling-paper candidates converge across two or more vendors (no calibrated range: the 60–85% given in the submitted SI could not be traced to a record and is withdrawn; below 30% would indicate a filling pattern not robustly encoded across training corpora); and (3) *multi-run categorical stability* — fraction of EPs whose SET membership stabilises at $\geq$75% inclusion frequency across N$\geq$6 runs at fixed prompt and model (below ~50% marks a borderline target). Per-signal numeric derivations are in SI-11.4.

SI-11.15: Adjacent-system output-class comparison (full seven-dimension table)

This expands the comparison with prior work in Discussion 3. It is an *output-class* comparison, not a performance benchmark: it does not measure accuracy, recall, or ranking quality of any system, and no metric-based head-to-head comparison is claimed.

Operational dimension	Literature synthesis (PaperQA2,[3] OpenScholar[4])	Argumentative zoning (Teufel[9])	Knowledge-gap detection (GAPMAP[5])	Research-idea generation (ResearchAgent[6])	Autonomous discovery (Robin,[7] Coscientist[8])	**UQS (this paper)**
User input	Explicit question	Sentence-classification target	Topic / corpus	Core papers	Hypothesis seed + lab loop	Source paper (no user question required)
Unit of analysis	Document collection	Sentence	Field / topic	Paper set	Experimental run	Individual paper + cross-paper corpus
Output object	Cited synthesised answer	Sentence-level rhetorical labels	List of explicit / implicit gaps	Problems + method ideas	Hypothesis / experiment plan	Per-record evidence-chain non-explicitness records + EP dictionary + filling-status map + SSP briefs
Evidence traceability	Citations to passages used	Per-sentence labels	Field-level evidence	Citation to core papers	Lab notebook / run log	evidence chain to specific textual loci per record; inference tier (T1/T2/T3) for A items
Temporal filling	—	—	Cross-temporal field aggregation	—	—	Cross-temporal filling-status retrieval under pre-locked Complete/Partial/Mini-mal/Non-filler criteria
Question-generation mode	Answers user question	—	Surfaces field-level gaps	Generates ideas from core papers	Generates hypotheses	Generates SSP briefs from cross-paper recurring patterns
Evaluation basis	Answer quality vs source	Inter-annotator agreement on labels	Gap-list utility	Idea novelty / coverage	Experimental success	Auditability + internal robustness; external validation = Phase 2 work

The four adjacent classes emit synthesised answers, gap inventories, idea seeds, or hypothesis-experiment loops, evaluated by their own internal criteria; UQS emits inspectable per-record evidence chains + EP dictionary + bounded filling map + SSP briefs, evaluated by deposited auditability and configuration-pinned SET-level rerun consistency (Results Part 4). The two output classes are complementary rather than competing.

Information extraction from the literature (added in revision). A separate class of work uses LLMs, and before them rule-based tools, to extract what papers state. One line each:

- *Zheng et al.*[10]: a ChatGPT prompt pipeline extracts MOF synthesis conditions from papers into a table, which is then used for prediction. The output is the stated conditions.
- *Dagdelen et al.*[11]: fine-tuned LLMs extract entities and relations from materials text into user-defined schemas. The output is the stated records under a schema.
- *Polak and Morgan* (ChatExtract)[12]: conversational prompting with follow-up questions extracts and checks material–property–value triplets. The output is stated values, with fewer hallucinations.
- *Jablonka et al.*[13]: a hackathon survey of LLM uses in chemistry and materials, including literature extraction. It maps use cases rather than describing one system.
- *Schilling-Wilhelmi et al.*[14]: a review of LLM-based chemical data extraction, covering methods, evaluation and failure modes.
- *ChemDataExtractor*[15]: a rule- and ML-based toolkit that extracts compounds, properties and spectra from chemistry text into a database. The output is stated values.
- *SciFact* (Wadden et al.)[16]: pairs scientific claims with abstracts that support or refute them. The output is labels on stated claims and evidence.
- *Liang et al.*[17]: LLM-generated, reviewer-style feedback on whole manuscripts, compared with human reviews. The output is a critique of a manuscript, not records tied to a later reuse.

All eight target what a paper states, or in the last case how a reviewer would judge it. UQS targets relations that a paper leaves implicit relative to later reuse, with an evidence chain per record. Extraction output is a natural input to UQS rather than a competitor.

SI-11.16: Limitations — extended mitigations and propagation modes

This section gives the per-limitation mitigation enumerations and propagation-mode analyses summarised in the main-text Limitations L1–L8; L9 and L10 are treated in the main text only.

L1 (corpus-boundedness). Any bounded non-detection is best read as an invitation for counter-example reporting rather than a definitive null result. The EP-08a/EP-09a non-detection was stress-tested in May 2026 by a structured Web of Science Core Collection retrieval audit with filling criteria locked pre-retrieval (SI-9.8), calibrated against an EP-07a positive control under identical methodology; the negative claims survive this independent retrieval mode at the WoS-indexed academic-record layer, but corpus-boundedness remains structural.

L2 (cross-vendor independence). All four models share substantial training-corpus overlap (web crawl, academic literature, code) and a common transformer-architecture lineage; convergence is partial-independence evidence at the post-training and tuning levels only. Five mitigation routes — of which the expert-facing route is a Phase 2 protocol, not a completed check: key-quote anchoring (references external PDF text not parametrically encoded), DOI verification against Crossref (training-corpus-independent), web-search augmentation when available, the planned Tier 1 advisor-evaluation layer, and period-vocabulary checks on synchronic absences. Truly independent verification (non-LLM retrieval, disjoint-corpus models, physical experiment) is Phase 2.

L3 (Stage 0 propagation). Two propagation modes: false omission (a claim missed in Stage 0 is routed to U-category as if absent) and false positive (a hallucinated Stage 0 claim induces a spurious counter-claim absence). Five architectural mitigations: (i) PDF context persistence across all stages within one session; (ii) the Stage 1 Turn 4 consolidation step surfacing inter-turn inconsistencies; (iii) cross-vendor and multi-run aggregation; (iv) Stage 4 retrospective consistency checks (filling papers contradicting Stage 0 surface as flags); (v) planned SSP-layer expert prioritisation (Phase 2). A Stage 0 quality self-evaluation field is a v0.2 candidate.

L4 (taxonomy). Ten finer absence types not fully captured: half-articulated unasked, conflated questions, foreclosed questions, reframing absences, counter-evidence absences, process/decision-provenance absences, negative-results absences, statistical/uncertainty absences, reflexive methodological absences, and filling-quality middle ground. Three compensating fields are planned but are not part of the prompts used in this paper: anchor_status, defect_or_opportunity_status, and a Core/Exploratory

tier separation. Candidate set for v0.2: C30 Foreclosed-by-method; C31 Counter-evidence non-engagement; C32 Filling-quality gradation; C33 Process/provenance absences.

L5 (awareness state). Five partial-constraint signals: explicit deferral language ("future work"), period-vocabulary checks, cross-paper contemporary-corpus checks, author retrospective writing where it exists (Tamao 2002), and adjacent-field vocabulary-boundary probing. A planned defect_or_opportunity_status field would require awareness evidence before an absence is classed as a *defect*.

L6 (ground truth). Internal consistency evidence: cross-model convergence (SI-6), the five-failure-mode comparison against structured database queries (SI-9), and the Yamamoto 1978/1981 cross-paper analyses (SI-3.7). (The submitted version also listed a "seven-source EP-10a re-derivation", withdrawn in R1 because the runs had the label in their prompts; SI-6.5.) Indirect evidentiary axes that do not replace a gold standard: retrospective filling, convergent robustness, expert face assessment, pragmatic utility, theoretical coherence, falsifiability. The Phase 2 programme combines human-expert annotation, citation-network analysis on papers citing Kumada 1972, and controlled repeated-run perturbation.

L7 (applicability). The Methods 5 applicability boundary (Full Article + calibrated discipline + major-journal English + mid-to-high anchor density + active citation lineage) defines the envelope; the calibration-depth gradient (SI-11.13) bounds cross-domain claims.

L8 (authorial vantage). Conversion of documentary-resolution findings into experimental work (priority assessment, condition design, scale planning, risk evaluation) is a downstream domain-expert task. *Rational avoidance* — a direction practitioners have tacitly judged unpromising — cannot be distinguished by UQS alone from unresolved difficulty, already-addressed literature, or genuinely overlooked opportunity. The Tier 1 advisor protocol (SI-8) operationalises this calibration in Phase 2.

Reflexive application. UQS has also been applied to the present manuscript, in a corpus kept separate from the chemistry corpora of this paper. That record is not deposited and is not used as evidence here.

SI-11.17: Refinement-set corpus — full table (the ten Heck papers used for v15–v17 prompt refinement; not a held-out test set)

The ten open-access papers of the refinement set (Methods 3), selected from 179 Heck-reaction-focused papers (title or abstract mentioning "Heck") to maximise diversity in evidence architecture. Each of the five categories (homogeneous catalysis, heterogeneous catalysis, computational chemistry, synthetic application, condition screening) contributes two papers, with dispersion of publication years (2016–2025) and journals. HOM-1 and HOM-2 are designated regression-monitoring baselines, re-analysed at each revision milestone to detect regressions; they were used during prompt refinement and are not held-out test papers. Refinement-set selection methodology and the v2 → v3 revision history are documented in `analysis/plan_test_case_selection_v3.md` (project record, not deposited); full Stage 3 analysis logs for each paper are deposited at Zenodo (Data Availability).

id	category	title	journal	year	doi	key_attribute
HOM-1	Homogeneous catalysis	Regiocontrol in the oxidative Heck reaction of indole by ligand-enabled switch of the regioselectivity-determining step	Chem. Sci.	2020	10.1039/d0sc02246b	Aerobic oxidative Heck with ligand-controlled regioselectivity (regression-monitoring baseline)
HOM-2	Homogeneous catalysis	Palladium-Catalyzed Reductive Heck Hydroarylation of Unactivated Alkenes Using Hydrosilane at Room Temperature	J. Org. Chem.	2024	10.1021/acs.joc.3c02488	Room-temperature reductive Heck hydroarylation using hydrosilanes (regression-monitoring baseline)
HET-1	Heterogeneous catalysis	Probing the Evolution of Palladium Species in Pd@MOF Catalysts during the Heck Coupling Reaction: An Operando X-ray Absorption Spectroscopy Study	J. Am. Chem. Soc.	2018	10.1021/jacs.8b03505	Operando XAS study of Pd@MOF catalyst speciation
HET-2	Heterogeneous catalysis	Deconvoluting Substrates, Support, and Temperature Effects on Leaching and Deactivation of Pd Catalysts: An In Situ Study in Flow	ACS Catal.	2024	10.1021/acscatal.4c02028	In situ flow study of leaching / deactivation across three supported Pd catalysts
COMP-1	Computational chemistry	Dichotomous mechanistic behavior in Narasaka–Heck cyclizations: electron rich Pd-catalysts generate iminyl radicals	Chem. Sci.	2016	10.1039/c5sc04037j	DFT mechanism dichotomy for electron-rich Pd catalysts in Narasaka–Heck cyclizations
COMP-2	Computational chemistry	New Mechanistic Insights into the Copper-Free Heck–Cassar–Sonogashira Cross-Coupling Reaction	ACS Catal.	2023	10.1021/acscatal.3c02787	Combined DFT + experimental mechanism study of Cu-free Heck–Cassar–Sonogashira
APP-1	Synthetic application	Palladium-catalyzed enantioselective Heck alkenylation of trisubstituted allylic alkenols: a redox-relay strategy to construct vicinal stereocenters	Chem. Sci.	2017	10.1039/c6sc04585e	Enantioselective Heck alkenylation via redox-relay for vicinal stereocentres
APP-2	Synthetic application	Asymmetric Synthesis of Quaternary Hydantoins via a Palladium-Catalyzed Aza-Heck Cyclization	J. Am. Chem. Soc.	2025	10.1021/jacs.5c16022	Asymmetric synthesis of quaternary hydantoins via Aza-Heck cyclization
SCR-1	Condition screening	Homogeneous Pd-Catalyzed Heck Coupling in γ-Valerolactone as a Green Reaction Medium: A Catalytic, Kinetic, and Computational Study	ACS Sustain. Chem. Eng.	2020	10.1021/acssuschemeng.0c03523	Green solvent (γ-valerolactone); kinetic + DFT study (SCR / COMP border case)
SCR-2	Condition screening	Thermally accelerated Heck reaction under direct mechanocatalysis using palladium milling balls	RSC Mechanochem.	2025	10.1039/d5mr00032g	Air-tolerant mechanocatalytic Heck reaction using Pd milling balls

SI-11.18: How the four control experiments were measured

Methods 6 reports four post-submission control experiments. This section states, for each, what quantity was extracted from the run outputs, by what instrument, and what that instrument cannot decide. The purpose is to let a reader judge the reported numbers rather than accept them: a paper whose claim is that absences should be measured rather than asserted owes the same treatment to its own measurements.

Outcome measures and instruments.

Control experiment	Deposit path	Outcome measure	Instrument
Decontamination control (Stages 0–2, N = 5)	`papers/dev/kumada/decon_2026-07/`	Reproduction of U-1; convergence of the E-proposition	Analyst reading of each run's final taxonomy and E-section against the original text
Reproducibility control (Stages 0–2, N = 5 per prompt version)	`papers/dev/kumada/repro_2026-07/`	A / B / C / U item counts; coverage of seven pre-specified benchmark findings	Script-extracted counts; coverage on two scopes, *strict* (final taxonomy or D-4 section) and *loose* (anywhere in the run's final artefacts), first by a keyword proxy (submitted version) and at R1 by two LLM raters against a written rubric, blind to the arm (SI-11.20)
Model-identity check and quote-provenance gate	`papers/_audits/paperA_quote_gate_R1_2026-09/`	Verbatim `model_info` string and every `key_quote` of the analyses this paper cites	String match against the declared model identifier
Stage 4 exemplar control (N = 5 per arm)	`papers/dev/kumada/s4_exemplar_control_2026-08/`	Robust-core membership; number of recommended candidates per run; bibliographic accuracy of every candidate	Regular-expression match on the bibliographic strings of each run's §C candidate table; per-candidate manual adjudication against Crossref

What the string instruments cannot decide. A regular expression over bibliographic strings can decide whether a specified known paper appears in a candidate table. It cannot decide whether a citation is fabricated, because a fabricated citation and a genuine one may share most of their surface form. We attempted a fabrication-rate comparison between the two Stage 4 arms on such a match and withdrew it: the pattern intended to capture a fabricated author list containing Hayashi also captures the genuine Hayashi, Tajika, Tamao and Kumada paper (*J. Am. Chem. Soc.*, 1976, **98**, 3718, DOI: 10.1021/ja00428a061), which occurs legitimately in the outputs of both arms. The quantities reported for this control are therefore restricted to what a string match can settle — presence of a named paper, and candidate volume — and the fabrication question was moved to per-candidate Crossref adjudication, which resolves each citation on its own coordinates.

Automated Crossref matching was not used as the decision rule. Comparing each candidate against the top hit returned by the Crossref REST API flagged 6 of 21 works as mismatched; manual inspection of the reported coordinates showed 4 of those 6 to be correct. The three causes were a top hit that was a different paper (Tamao *et al.*, *Bull. Chem. Soc. Jpn.*, 1976, **49**, 1958, which exists under DOI 10.1246/bcsj.49.1958), a page-number rendering difference (`144` against `144a`), and a year shift introduced by a collective reprint (an *Org. Synth.* preparation of 1978 indexed by Crossref under the 2003 collective volume). Every mismatch reported in Methods 6 is a manual adjudication against the queried coordinates, not an automated verdict.

Statistical power. With N = 5 per arm, a two-sided Fisher exact test on a binary outcome reaches $p < 0.05$ only for splits at least as extreme as 5/5 against 1/5 or 4/5 against 0/5 ($p = 0.048$). The reported p values (1.000 and 0.52) therefore show that the exemplar material produced no *detectable* difference at this sample size; they do not establish equivalence, and the sample is too small for strong inference either way. The load-bearing observation is the direction and magnitude of the differences (5/5 against 5/5, 4/5 against 2/5, 5.0 against 5.2 candidates per run), not their significance.

A note on pre-registration. Judging conditions for the Stage 4 exemplar control were fixed before the runs were scored, and execution and scoring were carried out in separate sessions so that the design document could not prime the scoring. Two of the four pre-registered conditions nonetheless had to be replaced after the runs: one, the fabrication-rate comparison above, because the quantity was not decidable by the intended instrument; the other, a count of total hallucinations, because Stage 4 output ends at candidate generation and carries no resolution field, so the count had to be produced by a separate Crossref pass. Fixing a criterion before measurement and confirming that the criterion is extractable from the output are separate tasks, and only the first was done at registration time.

SI-11.19: Input-ablation control — extraction versus parametric recall

Motivation. The analysis model was trained on a corpus that plausibly includes the secondary literature around well-known papers — for Kumada 1972, five decades of retellings. UQS outputs might therefore derive from the model's parametric knowledge rather than from the input document. The per-record evidence_chain anchors (Methods 2) constrain this record by record; this control tests it at the pipeline level.

Design (pre-registered). Six papers in two tiers of expected parametric prior: a high-prior tier (Kumada 1972, Murai 1993, and the 2017 synthetic-application paper APP-1) and a low-prior tier of recent, less-cited papers (HOM-2, SCR-2, APP-2). Two arms per paper: **Arm P**, the standard pipeline with the PDF (and Supporting Information where it exists); **Arm W**, the same pipeline with the PDF replaced by a bibliographic stub (title, authors, journal, year, DOI) and the prompts left verbatim-identical. Three independent single-session runs per arm per paper (36 runs), under the configuration reported in this paper (v18.4 prompts, Claude Sonnet 4.6, Stages 0–2). Design, endpoints, thresholds and interpretations were committed to a version-controlled pre-registration before any run (`papers/ablation_provenance/_preregistration_paperA_20260809.md`).

Correction to the pre-registration's premise. The pre-registration described the low-prior tier as published after the model's training cutoff, which would make recall of their content impossible. Against the cutoffs Anthropic states for Sonnet 4.6 (reliable knowledge August 2025, training data January 2026; `zenodo_manifest.md` §1), this holds for none of the three: HOM-2 (online February 2024) and SCR-2 (registered with Crossref in May 2025) predate both cutoffs, and APP-2 (November 2025) postdates only the reliable-knowledge cutoff. The low-prior tier is therefore a tier of recent, less-cited papers, not a tier in which recall is structurally excluded. The conclusion below rests on the high-prior tier — above all on Kumada 1972 — and does not depend on this premise.

Amendments. Three amendments are logged in the pre-registration. Amendments 1 and 2 were made before any run (implementation and calibration of the secondary instrument; the form of the model-identity check). Amendment 3 was made after all runs and before interpretation: the primary instrument initially counted quotes of the bibliographic stub itself (title fragments, which Arm W runs annotated as stub-derived) as found in the paper, and a STUB_ECHO category was added to exclude them. This change lowers Arm W scores and leaves Arm P scores unchanged.

Scoring (mechanical; no LLM judgement). Primary endpoint (E1): **anchor-existence rate**, the fraction of text key_quote strings in a paper's outputs that exist in the source text (paper plus Supporting Information), scored by normalised character-8-gram containment (`scripts/anchor_matcher.py`; threshold 0.75, calibrated on genuine runs before use: genuine-quote minimum 0.756, deviation maximum 0.562). Formula and caption quotes and stub echoes are counted separately and excluded from the denominator. Secondary endpoint (E2): content-specifics match rate for chemical entities and numeric values against the source text (`scripts/specifics_matcher.py`; tokens present in the stub are excluded). Its pre-registered calibration gives entity rates of 0.24–0.39 for genuine runs against their own paper and 0.00–0.07 against a different paper; numeric values have a high null floor (0.36–0.49) because generic values recur across papers. Both endpoints are read only as within-paper P–W contrasts.

Results (Table SI-11.19-1). All values below were recomputed on 2026-09-28 from the 216 deposited per-run files with the two scripts; the deposited stage outputs and stubs, with the source PDFs (not deposited; obtainable from the DOIs), are the only inputs.

Paper	Prior tier	E1 Arm P	E1 Arm W	Arm W stub echoes	E2 entity Arm P	E2 entity Arm W	Arm W numeric tokens
Kumada 1972	high	24/24 = 1.00	0/10 = 0.00	15	60/227 = 0.26	9/79 = 0.11	3
Murai 1993	high	23/24 = 0.96	0/16 = 0.00	5	23/167 = 0.14	6/58 = 0.10	3
APP-1 (2017)	high	24/25 = 0.96	0/3 = 0.00	15	73/204 = 0.36	6/65 = 0.09	9
APP-2 (2025)	low	24/24 = 1.00	— (all 16 quotes are stub echoes)	16	37/193 = 0.19	3/48 = 0.06	3
SCR-2 (2025)	low	26/26 = 1.00	0/8 = 0.00	16	42/115 = 0.37	6/76 = 0.08	4
HOM-2 (2024)	low	26/26 = 1.00	0/18 = 0.00	4	51/242 = 0.21	7/71 = 0.10	12

Without the document, the pipeline produced **no verbatim-anchored quote in any paper (0/55)**, including Kumada 1972; Arm P anchored 0.96–1.00 of its quotes. Arm W runs did not fabricate document quotes; they quoted the stub and annotated the provenance. The Arm W denominator should be read accordingly: of its 55 text quotes, 30 state that the full text is not available and 25 are title fragments, 10 of which carry the same statement in their annotation (the annotation keeps them below the stub-echo threshold); none is an attempted quotation of the paper's body. Because the stub-echo rule only removes quotes from the denominator, a lower threshold would shrink the denominator but could not raise the numerator above zero. At the

content level, the Arm W entity rate was below the Arm P rate in every paper (0.06–0.11 against 0.14–0.37) but, in five of six papers, above the calibrated wrong-paper range (0.00–0.07); Arm W also produced few numeric values (3–12 per paper over three runs, against 137–383 in Arm P).

Earlier summary values. The results memo written at the time (`improvements/_methodology/ablation_results_paperA_20260810.md` (project record, not deposited)) reported the same E1 values but lower Arm W E2 entity rates (0.00–0.11) and fewer Arm W numeric tokens (0–4 per paper). Those E2 values could not be reproduced from the deposited files and scripts and are superseded by the table above.

Interpretation and scope. The pre-registered retrieval prediction — Arm W scores rising with prior density, and the P–W gap narrowing for well-known papers — is rejected at the verbatim-anchor layer: Arm W is at zero for the best-known paper as for the others. We conclude that, at the anchor layer, UQS outputs under this configuration derive from the input document rather than from parametric recall. The content layer is weaker evidence. Arm W's entity rates sit well below Arm P's but slightly above the wrong-paper range, which is consistent either with generic chemical vocabulary shared by related papers or with weak content-level recall; the instrument cannot separate the two, and this control does not exclude content-level leakage of that size. The result is specific to the reported configuration (v18.4 + Sonnet 4.6). This control complements the Stage 4 verification protocol (SI-9, SI-11.2): that protocol governs parametric retrieval where it is intended (Stage 4); the present control tests for unintended parametric leakage where it is not (Stages 0–2). Run outputs, stubs, run instructions and the pre-registration are deposited under `papers/ablation_provenance/`.

SI-11.20: Pre-registered held-out set and single-prompt baselines

Pre-registration and departures. Both checks were registered before any run. The pre-registration file is `preregistration_D3_D4_R1.md`, commit `88e7cc3`, with the B2 arm added before it was run (commit `b89b886`). Every departure is recorded in the file's change log:

1. The pre-registration did not say which quotation kind the quote endpoint counts. We report both: `key_quote` fields and other quoted strings.
2. The benchmark regex has known false positives (see below). The pre-registered regex values are reported, and a rubric-based rating was added.
3. The two-level rubric (original and stricter reading) was designed after arm-level keyword counts had been seen.
4. The held-out driver and structural-check script were committed after the runs had started.
5. **The Evidence Pack anchoring criterion was relaxed after all held-out runs had finished**: item IDs were accepted as anchors, which gave 4/4. This was not recorded at the time; an independent review found it. The criterion was then restored to the pre-registered wording, which gives 1/4. The script as first committed gives 1/4 as well, passing a different paper.
6. Candidate screening applied the "Heck reaction is the subject" criterion inconsistently: nine papers that use a Heck step only instrumentally were admitted. None of the four drawn papers is affected.

Held-out set.

- *Pool.* Crossref records (title or abstract containing "Heck", journal articles, online from 2026-02-01) and the Heck rows of the project corpus. That gave 273 candidates; 75 are dated on or after 2026-02-01 (plus one corpus record whose Crossref lookup failed, excluded), and 53 are eligible research articles. Exclusions and their reasons are listed in `d3_selection/screen_and_draw.py`.
- *Draw.* `random.Random(20260929)` over the eligible papers in DOI order. Drawn:
  - Wang et al., *J. Org. Chem.* 2026 (10.1021/acs.joc.6c01425; received 2026-06-16);
  - Liu et al., *Org. Lett.* 2026 (10.1021/acs.orglett.6c03536; received 2026-08-10);
  - Wu et al., *CCS Chem.* 2026 (10.31635/ccschem.026.202607864; accepted 2026-05-26);
  - Shekarizadeh et al., *J. Saudi Chem. Soc.* 2026 (10.1007/s44442-026-00106-5; received 2026-04-21).
- *Prior exposure.* All four post-date the Sonnet 4.6 training-data cutoff stated by Anthropic (January 2026; `zenodo_manifest.md` §1). A Crossref `posted-content` search found no preprint of any of them.
- *Runs.* One fresh process per paper, S0 → S1 (four turns) → S2 EN → Stage 3 v2 EN in a single context. Prompt hashes match SI-7. Model identity was verified from the CLI usage record and the Stage 0 `model_info`.

Table SI-11.20-1. Held-out endpoints.

paper	output sections complete	SRG/CLA blocks	blocks anchored (quotation or location in the paper)	EP anchoring	S0–S2 quote-gate strings (PASS / WARN / FAIL)	misquotations
Wang 2026	yes	6	4	fail	59 (46 / 8 / 5)	1
Liu 2026	yes	7	5	fail	61 (42 / 15 / 4)	1
Wu 2026	yes	4	4	pass	68 (53 / 7 / 8)	1 (occurs twice)
Shekarizadeh 2026	yes	9	6	fail	63 (54 / 7 / 2)	0

An author checked the seven blocks counted as unanchored (two each in Wang and Liu, three in Shekarizadeh) against the outputs and agreed that each refers to the paper's content without a quotation or a location in the paper (2026-10-05).

Quote adjudication (held-out). Every FAIL and every WARN below partial-match score 85 was classified: 50 gate rows, 48 unique strings. Decisive cases were checked on rendered page images. The classes were:

class	rows
does not faithfully report the source	4 (3 distinct)
word inserted inside quotation marks	1
present in the source (table and scheme text, ligature failures, ellipses)	16
meaning-preserving compression inside quotation marks	9
model-authored text, or text harvested between two quoted terms	20

The three misquotations:

- "asymmetric Ni-phosphine couplings" where the source reads "asymmetric Heck-type couplings" (Wu);
- "anti-cancer agents, pesticide research, and functional materials" where the source reads "drugs, synthetic chemistry, pesticide research, and functional materials" (Wang);
- "The paper assigns the ring-forming step as a 'concerted [4+2] electrocyclization' ", although the source never uses "concerted" (Liu).

Author check of the misquotations. An author checked every misquotation reported in this section — the three above, the three B2 cases and the pipeline case below — and the three boundary cases (the word inserted inside quotation marks above, and two equations transcribed from figures, one with R′ dropped from eq 4 in a pipeline output and one with the ligand dropped from eq 1 in a B2 output) against the source PDF, and agreed with every classification (2026-10-05). The boundary cases are not counted as misquotations.

Single-prompt baselines.

- *Input and design.* Kumada 1972, the same PDF, Sonnet 4.6, and N = 5 fresh processes per arm. The prompts are given verbatim in the pre-registration:
  - B0 asks for unanswered questions and unsupported claims, each with the evidence needed and a verbatim `key_quote`;
  - B1 adds one- or two-sentence definitions of A/B/C/U and same-root groups;
  - B2 adds to B1 only an instruction to be exhaustive (at least 40 items and 45,000 characters). It is a control for output volume and contains none of the pipeline's viewpoint instructions.
- *Comparison arm.* The pipeline arm is the existing v18.4 N = 5 reproducibility set (Methods 6).
- *Transcript recovery.* The CLI result field keeps only the last text block of a reply. For two B2 runs whose output was split into several blocks, the full text was recovered from the session transcripts; no run was repeated.

Scoring. Seven benchmark axes (homo-coupling, black mixture, exotherm, stereo non-specificity, ligand order, GLPC basis, ether/THF), as defined on 2026-07-19 when the reproducibility study was designed.

- *Keyword proxy (`canon.json`).* It has known false positives: "Ni(0)" matches the mechanism label, and "radical" matches any mechanistic discussion.
- *Rubric.* Every cell was therefore rated against a written rubric at two levels:
  - original, following the 2026-07-19 wording;
  - stricter, requiring the output to cite the paper's own observation.

- *Raters.* Two independent LLM raters (Claude), in separate working directories and reading in opposite orders, rated all 175 cells (20 runs × 7 axes, plus the pipeline's strict scope). Each read all 2,680 candidate passages, and arm-identifying text was removed. They agreed on 175/175 cells at both levels. Perfect agreement between raters of the same model family indicates a shared reading of the rubric, not independent validation; it is not a substitute for human inter-rater reliability. An author checked all 100 cells of the four axes on which the arms differ (homo-coupling, black mixture, exotherm, GLPC basis; 25 cells each, arm labels hidden; the other three axes are present in every run of every arm) and agreed with every rating (2026-10-05). A second reading of the deciding passages with an OpenAI GPT model, checked by the author against the PDF, found that two "present" GLPC-basis cells rest on a misreading of Table I. A pipeline output (run 5, strict scope) says the table does not specify which entries are GLPC yields, although footnotes d and e do so and the same output quotes them. A B2 output (run 4) counts at least five GLPC entries; there are four (98, 95, 95 and 89%). Both passages still raise the mixture of GLPC and isolated yields, which is what the rubric frozen before rating asks for; the rubric does not test whether the passage is factually correct. The ratings are therefore kept, and the coverage counts in Table SI-11.20-2 are unchanged. Coverage here means that an output raises the point, not that it raises it correctly.

Table SI-11.20-2. Benchmark coverage (loose scope) and quotation accuracy.

arm	output length (bytes, UTF-8, whole output files)	missed, original definitions	missed, stricter reading	runs with a misquotation
pipeline (v18.4, Stages 1–2)	61,000–68,000	0.0%	11.4%	1/5
pipeline, strict scope (final taxonomy + D-4)	—	8.6%	20.0%	—
B0	7,000–9,000	34.3%	42.9%	0/5
B1	8,000–11,000	42.9%	45.7%	0/5
B2 (length-matched)	62,000–79,000	5.7%	5.7%	3/5

Per axis, stricter reading, findings recovered out of five runs:

- black mixture: pipeline 4, B0 1, B1 1, B2 5;
- exotherm: 2, 0, 0, 5;
- GLPC basis: 5, 1, 0, 3;
- homo-coupling: 5, 3, 3, 5;
- the remaining three axes: 5 in every arm.

The misquotations:

- *B2.* An inverted ligand-activity relation ("dmpe « PPh_3" for the source's "≈"); a footnote attached to the wrong Table I entry; and a misread reagent (n-C_6H_{13}MgCl for n-C_8H_{17}MgCl).
- *Pipeline.* A phrase from the preceding communication printed on the same page, quoted as describing the Kumada catalytic cycle.

Reading.

- Under the pre-registered rule, a difference is claimed only when the ranges do not overlap. On that rule, the pipeline's coverage advantage over B0 and B1 disappears when output length is matched (B2).
- The axes were derived from the pipeline's own analyses and should favour it.
- Quotation accuracy does not separate the pipeline and B2 at this sample size: misquotations in 1/380 against 3/439 harvested quotations (Fisher exact p = 0.63), and key_quote gate failures in 0/43 against 11/262 (p = 0.37; the 95% upper bound for 0/43 is about 7% by the rule of three, 8% by a two-sided exact interval).
- The structural difference (D-4, E section, evidence chains and the EP dictionary, which B2 was not asked for) is a difference of design, not a tested advantage.

Does Stage 3 need the pipeline's Stage 1/2 structure? (B2 → Stage 3; pre-registered, `preregistration_B2_stage3.md`, commit 052a74c.)

- *Design.* Stage 3 v2 EN was run on Sonnet 4.6 on the five B2 records and on the five pipeline repro records, with neutral input IDs and the same adapter sentence. The instruction: where the prompt refers to a field the record lacks, say so.
- *Rating.* Endpoints were rated by two LLM raters with separate directories. Blinding was partial, because the record formats differ.

Table SI-11.20-3. Stage 3 on pipeline input (U) and single-prompt input (B).

endpoint	U	B	rule
execution	5/5 complete	5/5 complete	reported only
SRG/CLA blocks anchored	25/26 (0.86–1.00 per run)	61/73 (0.73–1.00)	no difference
quotations misreporting the source (FAIL + WARN < 85)	0/43 rows	0/31 rows	no difference
unsupported fills (raters A / B, total over 5 runs)	5 / 2 (0–3 per run)	18 / 17 (0–9 per run)	B higher in total; ranges overlap
Table 3 EP classes surfaced	8 in every run	9–10	B higher

The outcome is mixed and is reported per endpoint. The B2 input produced an Evidence Pack record that was as well anchored, and surfaced more of the Table 3 dictionary, but it had more internal-consistency errors: item labels or cross-references contradicting the record, and miscounts. An author checked every unsupported fill listed by either rater (42 rows: 7 on U input, 35 on B input) against the input record and the PDF and agreed with all of them (2026-10-05). The other endpoints were rated by the LLM raters only.

Files.

- `01_response/preregistration_D3_D4_R1.md`;
- `d3_selection/`;
- `d3_heldout/` (driver, check script, gate outputs, `d3_quote_adjudication.md`);
- `d4_baseline/` (driver, prompts, metrics, rubric, packets, rater verdicts, `d4_quote_adjudication.md`);
- raw outputs under `papers/paperA_heldout_R1/` and `papers/dev/kumada/baseline_R1_2026-09/`.

Source PDFs are not deposited.

References (Supplementary Information)

The references below are cited only in the Supplementary Information and are denoted by an "S" prefix (e.g., S1). Superscript citation numbers without an "S" prefix refer to the reference list of the main article.

S1 L. Talavera, C. Odena and R. Martin, Recent advances in Ni-catalyzed functionalization of strong C–O and C–H bonds, *Adv. Catal.*, 2024, **74**, 133–179, DOI: 10.1016/bs.acat.2024.05.002.

S2 J. Clayden, N. Greeves and S. Warren, *Organic Chemistry*, Oxford University Press, Oxford, 2nd edn, 2012, ch. 40 (Organometallic chemistry), pp. 1069–1101.

S3 A. Agrawal, M. Suzgun, L. Mackey and A. T. Kalai, Do Language Models Know When They're Hallucinating References?, in *Findings of the Association for Computational Linguistics: EACL 2024*, Association for Computational Linguistics, St. Julian's, Malta, 2024, pp. 912–928, DOI: 10.18653/v1/2024.findings-eacl.62.

S4 S. Ansari, Compound Deception in Elite Peer Review: A Failure Mode Taxonomy of 100 Fabricated Citations at NeurIPS 2025, *arXiv*, 2026, preprint, DOI: 10.48550/arXiv.2602.05930.

S5 D. Abbonato, CheckIfExist: Detecting Citation Hallucinations in the Era of AI-Generated Content, *arXiv*, 2026, preprint, DOI: 10.48550/arXiv.2602.15871.

S6 T. S. Kuhn, *The Structure of Scientific Revolutions*, University of Chicago Press, Chicago, 1962.

S7 M. Foucault, *L'Archéologie du savoir*, Gallimard, Paris, 1969.

S8 M. Bellomo, *The Common Legal Past of Europe, 1000–1800*, trans. L. G. Cochrane, Catholic University of America Press, Washington, DC, 1995 (Italian original: *L'Europa del diritto comune*, 1988); see esp. Chapter 6 ("Forms of Exposition and Techniques of Diffusion") for the glossator method (*distinctio*, *quaestio*, *casus omissus*, *brocarda*, *summa*) and Chapter 7 ("The System of the Ius commune") for the commentator systematisation.